

WELL ENGINEERS NOTEBOOK

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SHELL INTERNATIONAL EXPLORATION AND PRODUCTION B.V.
EP Learning and Development

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Important - please read

The ownership of this document resides with EPT-HL in SIEP.

It is subject to a process of continuous updating and improvement. This process is only possible if recipients provide critical and constructive feedback. This can refer to :

- amendments to the material included
- inclusion of additional material
- omission of currently included material
- layout

Wherever possible, please be specific about material that is incorrect, missing or in need of improvement.

A – CONVERSION FACTORS

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Think SI

Base units

In SI there are 7 base units from which all the other units can be derived or computed.

The 7 base units are:

Quantity	Name of unit	Symbol
1. Length	metre	m
2. Mass	kilogram	kg
3. Time	second	s
4. Electric current	ampere	A
5. Temperature	kelvin or degree Celsius	K °C
6. Amount of substance	mole	mol
7. Luminous intensity	candela	cd

Supplementary :

Plane angle	radian	rad
Solid angle	steradian	sr

Used alone or in combinations these base units enable us to make any measurement we need in any field of endeavour.

Derived units

Quantity	Name of unit	Symbol
Area	square metre	m ²
Volume	cubic metre	m ³
Velocity	metre per second	m/s
Acceleration	metre per second ²	m/s ²
Density	kilogram per cubic metre	kg/m ³
Frequency	hertz	Hz
Force	newton	N
Pressure	pascal	Pa (N/m ²)
Energy	joule	J (N-m)
Power	watt	W (J/s)
Electric potential	volt	V (W/A)

Think SI

Prefixed

The value of most SI units can be changed by the simple placing of a prefix in front of the unit name.

Prefix	Symbol	Value	Factor
giga	G	1,000,000,000	10^9
mega	M	1,000,000	10^6
kilo	k	1,000	10^3
hecto	h	100	10^2
deca	da	10	10
deci	d	0.1	10^{-1}
centi	c	0.01	10^{-2}
milli	m	0.001	10^{-3}
micro	μ	0.000 001	10^{-6}

Examples

one gigapascal	1 GPa=1,000,000,000 Pa
one kilometre	1 km =1,000 m
one decanewton	1 daN =10 N
one milligram	1 mg =0.001 g
one micrometre	1 mm =0.000 001 m
one square kilometre	1 km ² =10 ⁶ m ²
one cubic megametre	1 Mm ³ =10 ¹⁸ m ³

Force, Work, Torque and Power

Mass is the quantity of matter in an object and is constant on earth as well as in space.

Units of mass: kg kilogram
t metric tonne
1 t=1,000 kg

Force $F = m \times a$

Force = mass x acceleration

Unit of force: N newton
(N = kg.m.s⁻²)

Practical use: daN decanewton
kN kilonewton
MN meganewton

Work Energy is force x distance (N.m)

Unit of work: J Joule

Practical use: kJ kilojoule
MJ megajoule

Torque Unit: N.m newton-metre

Power is the work per unit time

Unit of power: W watt

Practical use: kW kilowatt
MW megawatt

LENGTH (l)

↓ From ↑ To	Metres	Kilometres	Inches	Feet	Miles (statute)	Miles (nautical)
1 metre	1	1×10^{-3}	39.37	3.281	621.4×10^{-6}	539.6×10^{-6}
1 kilometre	1×10^3	1	39.37×10^3	3.281×10^3	621.4×10^{-3}	539.6×10^{-3}
1 inch	25.4×10^{-3}	25.4×10^{-6}	1	83.33×10^{-3}	15.78×10^{-6}	13.71×10^{-6}
1 foot	304.8×10^{-3}	304.8×10^{-6}	12	1	189.4×10^{-6}	164.5×10^{-6}
1 mile (statute)	1.609×10^3	1.609	63.36×10^3	5.28×10^3	1	868.4×10^{-3}
1 mile (nautical)	1.853×10^3	1.853	72.96×10^3	6.08×10^3	1.152	1

CONVERSION FACTORS
VOLUME (l³)

↓ From → To	Cubic metres	Cubic decimetres	Cubic centimetres	Cubic feet	Gallons (imp)	Gallons (US liquid)	Barrels
1 cubic metre	1	1 x 10 ³	1 x 10 ⁶	35.31	220	264.2	6.29
1 cubic decimetre	1 x 10 ⁻³	1	1 x 10 ³	35.31 x 10 ⁻³	220 x 10 ⁻³	264.2 x 10 ⁻³	6.29 x 10 ⁻³
1 cubic centimetre	1 x 10 ⁻⁶	1 x 10 ⁻³	1	35.31 x 10 ⁻⁶	220 x 10 ⁻⁶	264.2 x 10 ⁻⁶	6.29 x 10 ⁻⁶
1 cubic foot	28.32 x 10 ⁻³	28.32	28.32 x 10 ³	1	6.229	7.481	178.1 x 10 ⁻³
1 gallon (imp)	4.546 x 10 ⁻³	4.546	4.536 x 10 ³	160.5 x 10 ⁻³	1	1.201	28.59 x 10 ⁻³
1 gallon (US liquid)	3.785 x 10 ⁻³	3.785	3.785 x 10 ³	133.7 x 10 ⁻³	832.7 x 10 ⁻³	1	23.81 x 10 ⁻³
1 barrel	159 x 10 ⁻³	159	159 x 10 ³	5.615	34.97	42	1

CONVERSION FACTORS

MASS (m)

↓ To ↑ From	Kilograms	Pounds	Tons (metric)	Tons (long)	Tons (short)
1 kilogram	1	2.205	1×10^{-3}	984.2×10^{-6}	1.102×10^{-3}
1 pound	453.6×10^{-3}	1	453.6×10^{-6}	446.4×10^{-6}	500×10^{-6}
1 ton (metric)	1×10^3	2.205×10^3	1	984.2×10^{-3}	1.102
1ton (long)	1.016×10^3	2.240×10^3	1.016	1	1.12
1ton (short)	907.2	2×10^3	907.2×10^{-3}	892.9×10^{-3}	1

CONVERSION FACT

FORCE (m.l.t⁻²)

From To	Newton	Dynes	Kilogram force	Poundals	Pounds force	1 x 10 ³ daN (kdaN)
1 newton	1	100 x 10 ³	102 x 10 ⁻³	7.233	224.8 x 10 ⁻³	0.1 x 10 ⁻³
1 dyne	10 x 10 ⁻⁶	1	1.02 x 10 ⁻⁶	72.33 x 10 ⁻⁶	2.248 x 10 ⁻⁶	1 x 10 ⁻⁹
1 kilogram force	9.807	980.7 x 10 ³	1	70.93	2.205	0.981 x 10 ⁻³
1 poundal	138.4 x 10 ⁻³	13.84 x 10 ³	14.10 x 10 ⁻³	1	31.08 x 10 ⁻³	13.83 x 10 ⁻⁶
1 pound force	4.448	444.8 x 10 ³	453.6 x 10 ⁻³	32.17	1	0.445 x 10 ⁻³
1 x 10 ³ daN	10 x 10 ³	1 x 10 ⁹	1.02 x 10 ³	72.33 x 10 ³	2.248 x 10 ³	1

NOTE: 1) Conversion factors based on $g = 9.807 \text{ m/s}^2 = 32.174 \text{ ft/s}^2$

2) Pound signifies pound (avdp)

CONVERSION FACTORS

PRESSURE

To convert from	To	Multiply by
psi	kPa	6.895
	bar	0.06895
	kg/cm ²	0.07037
	m H ₂ O (15°C)	0.7037*
	ft H ₂ O (39°F)	2.307*
kPa	psi	0.1450
	bar	0.01
	kg/cm ²	0.01020
	m H ₂ O (15°C)	0.1021*
	ft H ₂ O (39°F)	0.3346*
bar	psi	14.50
	kPa	100
	kg/cm ²	1.020
	m H ₂ O (15°C)	10.21*
	ft H ₂ O (39°F)	33.46*
kg/cm ²	psi	14.22
	kPa	98.07
	bar	0.9807
	m H ₂ O (15°C)	10.01*
	ft H ₂ O (39°F)	32.81*
m H ₂ O (15°C)	psi	1.421
	kPa	9.798*
	bar	0.09798*
	kg/cm ²	0.09991*
	ft H ₂ O (39°F)	3.278
ft H ₂ O (39°F)	psi	0.4335*
	kPa	2.989*
	bar	0.02989*
	kg/cm ²	0.03048*
	m H ₂ O (15°C)	0.3051

Notes : * There is no direct conversion between pressure and heights of fluid head. $P = \rho gh$ has been used to obtain multiplication factors indicated by '*'. The fluid density (ρ) depends upon temperature.

Conversion between metres and feet is based on 1 ft = 0.3048 m.
The SI unit of pressure is the Pascal.

PRESSURE GRADIENTS & FLUID DENSITY

To convert from	to	Multiply by
psi/ft	kPa/m	22.62
	bar/10m	2.262
	lb/gal (US)	19.25*
	lb/ft ³	144.0*
	kg/dm ³	2.307*
bar/10m	kPa/m	10.0
	psi/ft	0.4421
	lb/gal (US)	8.51*
	lb/ft ³	63.66*
	kg/dm ³	1.020*
kPa/m	bar/10m	0.10
	psi/ft	0.0442
	lb/gal (US)	0.851*
	lb/ft ³	6.366*
	kg/dm ³	0.1020*
lb/gal(US)	kPa/m	1.175*
	bar/10m	0.1175*
	psi/ft	0.0519*
	lb/ft ³	7.481
	kg/dm ³	0.1198
lb/ft ³	kPa/m	0.1571*
	bar/10m	0.01571*
	psi/ft	0.00694*
	lb/gal (US)	0.1337
	kg/dm ³	0.01602
kg/dm ³	kPa/m	9.807*
	bar/10m	0.9807*
	psi/ft	0.4335*
	lb/gal (US)	8.345
	lb/ft ³	62.43

For the definition of API Gravity see [page A-12](#)

NOTES: * There is no direct conversion between densities and pressure gradients.
The relationship $P = \rho gh$ has been used to obtain multiplication factors indicated by (*).

The SI units of pressure gradient are kPa/m
The SI units of density are kg/m³

In this book we assume that one litre water at 4°C and 1 Atm.
(=101.3 kPa =14.7 psi) equals one dm³ water at 4°C and 1 Atm.
although we know this is not exactly the same . (The difference
between them is less then 0.003%) We take this liberty because it
helps simplifying our calculations.

CONVERSION FACTORS

POWER (m.l².t⁻³)

→ To ↓ From	Watts	Foot-Pounds per second	Horsepower (metric)	Horsepower (british)	British Thermal Units/sec
1 watt	1	0.7376	1.36 x 10 ⁻³	1.341 x 10 ⁻³	948 x 10 ⁻⁶
1 foot-pound/sec	1.356	1	1.843 x 10 ⁻³	1.818 x 10 ⁻³	1.285 x 10 ⁻³
1 horsepower (metric)	735.5	542.5	1	986.3	697.2 x 10 ⁻³
1 horsepower (british)	745.7	550	1.014	1	7.07 x 10 ⁻³
1 British thermal unit/sec	10.055	778	1.434	1.415	1

NOTES : 1) Conversion factors based on g = 9.807 m/sec² = 32.174 ft/sec²
 2) Pound signifies pound (avdp)
 3) Horsepower (metric) = Cheval vapeur

CONVERSION FACTORS
HEAT, ENERGY AND WORK (m.l².t⁻²)

	Joules	Kilowatt hours	Calories	Foot pounds	Horsepower hours (metric)	Horsepower hours (british)	BTU
1 joule	1	277.8 x 10 ⁻⁹	238.8 x 10 ⁻³	737.6 x 10 ⁻³	377.7 x 10 ⁻⁹	372.5 x 10 ⁻⁹	947.8 x 10 ⁻⁶
1 kilowatt hour	3.60 x 10 ⁶	1	859.8 x 10 ³	2.655 x 10 ⁶	1.360	1.341	3.412 x 10 ³
1 calorie	4.187	1.163 x 10 ⁻⁶	1	3.088	1.581 x 10 ⁻⁶	156.0 x 10 ⁻⁶	3.968 x 10 ⁻³
1 foot pound	1.356	376.6 x 10 ⁻⁹	323.8 x 10 ⁻³	1	512.1 x 10 ⁻⁹	505.1 x 10 ⁻⁹	1.285 x 10 ⁻³
1 horsepower hour (metric)	2.648 x 10 ⁶	735.5 x 10 ⁻³	632.4 x 10 ³	1.953 x 10 ⁶	1	986.3 x 10 ⁻³	2.510 x 10 ³
1 horsepower hour (british)	2.685 x 10 ⁶	745.7 x 10 ⁻³	641.2 x 10 ³	1.980 x 10 ⁶	1.014	1	2.544 x 10 ³
1 BTU	1.055 x 10 ³	293.1 x 10 ⁻⁶	252	778.2	398.5 x 10 ⁻⁶	393.0 x 10 ⁻⁶	1

NOTE: 1) Conversion factors based on g = 9.807 m/s² = 32.174 ft/s²
2) Pound signifies pound (avdp)
3) Horsepower (metric) = Cheval vapeur

CONVERSION FACTORS

TEMPERATURE

	water freezing	water boiling
Celcius (C)	0 ----- 100	
Reaumur (Re)	0 ----- 80	
Fahrenheit (F)	32 ----- 212	
Kelvin (K)	273 ----- 373	
Rankine (R)	492 ----- 672	
From:	C° x 0.8	= Re°
C°	C° x 1.8 + 32	= F°
	C° + 273	= K
	C° x 1.8 + 492	= R°
Re°	Re° x 1.25	= C°
	Re° x 2.25 + 32	= F°
	Re° x 1.25 + 273	= K
	Re° x 2.25 + 492	= R°
F°	(F° - 32) / 1.8	= C°
	(F° - 32) x 0.444	= Re°
	(F° - 32) / 1.8 + 273	= K
	F° + 460	= R°
K	K - 273	= C°
	(K - 273) x 0.8	= Re°
	(K - 273) x 1.8 + 32	= F°
	(K - 273) x 1.8 + 492	= R°
R°	(R° - 492) / 1.8	= C°
	(R° - 492) x 0.444	= Re°
	R° - 460	= F°
	(R° - 492) / 1.8 + 273	= K

NOTE : $T_{R^{\circ}} = t_{F^{\circ}} + 459.67$

$T_K = t_{C^{\circ}} + 273.15$

API GRAVITY

As crude is not normally specified in our standard units, but in API gravity, the equations for conversion are as follows :

$$\text{SG, crude} = \frac{141.5}{^{\circ}\text{API} + 131.5} \text{ (kg/litre)}$$

$$\text{Gradient} = \frac{141.5 \times 0.4335}{^{\circ}\text{API} + 131.5} \text{ (psi/ft)}$$

General classification with respect to API gravity:

$^{\circ}\text{API}$	Crude oil
< 20	Heavy
20 - 30	Medium
>30	Light

BUOYANCY FACTORS

Corresponding to drilling fluid densities expressed in various units

In kg/m ³ BF	In ppg BF	In lbs/ft ³ BF	In psi/ft BF	In kPa/m BF
1,000 0.872	8.35 0.872	62.4 0.873	0.434 0.872	9.8 0.872
1,020 0.870	8.40 0.872	63.0 0.871	0.440 0.871	10.0 0.870
1,040 0.867	8.60 0.869	64.0 0.869	0.450 0.868	10.2 0.867
1,060 0.865	8.80 0.866	65.0 0.867	0.460 0.865	10.4 0.865
1,080 0.862	9.00 0.862	66.0 0.865	0.470 0.862	10.6 0.862
1,100 0.860	9.20 0.859	67.0 0.863	0.480 0.859	10.8 0.860
1,120 0.857	9.40 0.856	68.0 0.861	0.490 0.856	11.0 0.857
1,140 0.855	9.60 0.853	69.0 0.859	0.500 0.853	11.2 0.854
1,160 0.852	9.80 0.850	70.0 0.857	0.510 0.850	11.4 0.852
1,180 0.850	10.00 0.847	72.0 0.853	0.520 0.847	11.6 0.849
1,200 0.847	10.20 0.844	74.0 0.849	0.530 0.844	11.8 0.847
1,220 0.844	10.40 0.841	76.0 0.845	0.540 0.841	12.0 0.844
1,240 0.842	10.60 0.838	78.0 0.841	0.550 0.838	12.2 0.841
1,260 0.839	10.80 0.835	80.0 0.837	0.560 0.835	12.4 0.839
1,280 0.837	11.00 0.832	82.0 0.832	0.570 0.832	12.6 0.836
1,300 0.834	11.20 0.829	84.0 0.828	0.580 0.829	12.8 0.834
1,320 0.832	11.40 0.826	86.0 0.824	0.590 0.826	13.0 0.831
1,340 0.829	11.60 0.823	88.0 0.820	0.600 0.824	13.2 0.828
1,360 0.827	11.80 0.820	90.0 0.816	0.610 0.821	13.4 0.826
1,380 0.824	12.00 0.817	92.0 0.812	0.620 0.818	13.6 0.823
1,400 0.821	12.20 0.814	94.0 0.808	0.630 0.815	13.8 0.821
1,420 0.819	12.40 0.810	96.0 0.804	0.640 0.812	14.0 0.818
1,440 0.816	12.60 0.807	98.0 0.800	0.650 0.809	14.2 0.815
1,460 0.814	12.80 0.804	100.0 0.796	0.660 0.806	14.4 0.813
1,480 0.811	13.00 0.801	102.0 0.792	0.670 0.803	14.6 0.810
1,500 0.809	13.20 0.798	104.0 0.788	0.680 0.800	14.8 0.808
1,550 0.802	13.40 0.795	106.0 0.783	0.690 0.797	15.0 0.805
1,600 0.796	13.60 0.792	108.0 0.779	0.700 0.794	15.5 0.798
1,650 0.790	13.80 0.789	110.0 0.775	0.720 0.788	16.0 0.792
1,700 0.783	14.00 0.786	112.0 0.771	0.740 0.782	16.5 0.785
1,750 0.777	14.20 0.783	114.0 0.767	0.760 0.776	17.0 0.779
1,800 0.770	14.40 0.780	116.0 0.763	0.780 0.771	17.5 0.772
1,850 0.764	14.60 0.777	118.0 0.759	0.800 0.765	18.0 0.766
1,900 0.758	14.80 0.774	120.0 0.755	0.820 0.759	18.5 0.759
1,950 0.751	15.00 0.771	122.0 0.751	0.840 0.753	19.0 0.753
2,000 0.745	15.50 0.763	124.0 0.747	0.860 0.747	19.5 0.746
2,050 0.739	16.00 0.755	126.0 0.743	0.880 0.741	20.0 0.740
2,100 0.732	16.50 0.748	128.0 0.739	0.900 0.735	20.5 0.733
2,150 0.726	17.00 0.740	130.0 0.734	0.920 0.729	21.0 0.727
2,200 0.719	17.50 0.733	132.0 0.730	0.940 0.723	21.5 0.720
2,250 0.713	18.00 0.725	134.0 0.726	0.960 0.718	22.0 0.714
2,300 0.707	18.50 0.717	136.0 0.722	0.980 0.712	22.5 0.707
2,350 0.700	19.00 0.710	138.0 0.718	1.000 0.706	23.0 0.701
2,400 0.694	19.50 0.702	140.0 0.714	1.020 0.700	23.5 0.694
2,450 0.688	20.00 0.694	145.0 0.704	1.040 0.694	24.0 0.688
2,500 0.681	20.50 0.687	150.0 0.694	1.060 0.688	24.5 0.681
2,550 0.675	21.00 0.679	155.0 0.683	1.080 0.682	25.0 0.675
2,600 0.668	21.50 0.671	160.0 0.673	1.100 0.676	25.5 0.668
2,650 0.662	22.00 0.664	165.0 0.663	1.150 0.662	26.0 0.662
7,842	65.43	489.5	3.400	76.90

Note: These buoyancy factors are only applicable for **steel** components

Buoyancy factor = $1 - \left\{ \frac{\text{fluid density}}{\text{steel density}} \right\}$ The density of steel in the various units is shown on the bottom line of the tables.

B - DERRICKS, MAST & BLOCK LINE

Clickable list

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DERRICK LOAD CALCULATIONS

(neglecting the weight of the derrick itself and the crown block)

Note:

In all calculations involving hook load, this is by convention taken to include the weight of the hook itself, including also the travelling block.

Thus :

Hook load as shown on weight indicator (Martin-Decker)

= weight of string **in drilling fluid** + weight of travelling block and hook

Static loads

Under static conditions:

load in each line = fast line load = dead line load = $\frac{\text{hook load}}{N}$
where N = number of lines strung

Static derrick load = hook load + fast line load + dead line load
 $= \frac{N+2}{N} \times \text{hook load}$

Dynamic loads

Under dynamic conditions, due to both friction in the sheave bearing and internal friction in the block line, the tension on the fastline side of a given sheave is higher than the tension on the deadline side by a factor "k". This factor is normally taken to be 1.04 for roller bearing sheaves (API RP9B).

The result, for a constant hook load (i.e. no drag) travelling at a constant speed, is that the dynamic fast line tension is higher than the static fast line tension by a certain factor. The factor depends on the number of lines strung and its value for different 'N's are tabulated below. In fact, for these ideal conditions, the dead line load would actually decrease with respect to the static load, and these factors are also shown in the table.

N	2	4	6	8	10	12
dynamic fast line factor	1.060	1.102	1.145	1.188	1.233	1.279
dynamic dead line factor	0.980	0.942	0.905	0.868	0.833	0.799

Also Dynamic derrick load = Hook load + dynamic fast line tension + dynamic dead line tension

Notes :

1. Previous practice was to divide the static load by an "efficiency" factor to give the dynamic fast line tension. The efficiency factor was the reciprocal of the factor tabulated above.
2. The reduction in dead line tension is generally neglected (see note 3 below).
3. In theory, the decrease in dead line tension would cause the hook load indicated on the weight indicator to be too low. In practice the effects of drag, acceleration and shock loads, and the fact that critical hook loads are generally applied in small increments, make this error unimportant.

BLOCK LINE

$$\text{Safety Factor (S.F.)} = \frac{\text{Breaking Strength of Rope}}{\text{Fast Line Load}}$$

Breaking strength of blockline

for 6 x 19 I.P.S. (Improved Plow Steel)

I.W.R.C. (Independent Wire Rope Core)

Rope diameter		Breaking Strength			
inches	mm	short tons	lbs	kg	kN
1	25.4	44.9	89,800	40,726	399.4
1 1/8	28.6	56.5	113,000	51,247	502.6
1 1/4	31.8	69.4	138,800	62,948	617.3
1 3/8	34.9	83.5	167,000	75,737	742.7
1 1/2	38.1	98.9	197,800	89,705	879.7
1 5/8	41.3	114.6	230,000	104,600	1020.0
1 3/4	44.4	133.0	266,000	121,000	1180.0

Shell Safety Factors

Safety factor of 5 is normal for drilling operations

Minimum recommended safety factors are :

3.5 for drilling

2.5 for running casing and fishing operations

A.P.I. Safety Factors

Minimum recommended safety factors are :

3 for drilling

2 for running casing and fishing operations

Note: for 6 x 19 Seale drilling line the recommended Shell Value for sheave diameter factor is 35-40 ratio sheave tread diameter to blockline diameter (Refer A.P.I. RP9B)

BLOCK LINE WORK

Work done during a round trip

$$T_r = \frac{D \cdot W_{dp} \cdot (L_{st} + D) + 4D(M + \frac{1}{2}C_1 + \frac{1}{2}C_2)}{k}$$

Where :

	<u>In oilfield units</u>	<u>in SI units</u>
T_r = Work done during round trip	(short) ton-miles	megajoules
D = Depth of hole, or trip	ft	m
L_{st} = Length of drill pipe stand	ft	m
W_{dp} = Approximate weight of DP (see page C-2), adjusted for drilling fluid density	lbs/ft	N/m
M = weight of block, hook, elevator, etc	lbs	N
C_1 = Excess weight of DCs in drilling fluid*	lbs	N
C_2 = Excess weight of HWDP in drilling fluid*	lbs	N
k = a constant	10,560,000	1,000,000

* Excess weight of tubulars = weight of tubulars less weight of same length of DP

Work done while running casing

$$T_c = \frac{D \cdot W_c \cdot (L_c + D) + 4DM}{2k}$$

Where :

	<u>In oilfield units</u>	<u>in SI units</u>
T_c = Work done while running casing	(short)ton-miles	megajoules
D = Setting depth of casing	ft	m
L_c = Length of average casing joint	ft	m
W_c = Effective weight/unit length of casing in drilling fluid	lbs/ft	N/m

and other symbols are as given above

Work done while drilling an interval

$$T_d = 2(T_2 - T_1) \text{ if hole drilled without reaming}$$

$$T_d = 3(T_2 - T_1) \text{ if hole reamed once}$$

$$T_d = 4(T_2 - T_1) \text{ if hole reamed twice}$$

Where :

T_1 = T_r at top of interval

T_2 = T_r at bottom of interval

Work done while coring

$$T_{co} = 2(T_2 - T_1)$$

Where :

T_1, T_2 are as above

RECOMMENDED CUT-OFF LENGTHS FOR ROTARY DRILLING LINES (API RP9B)

Derrick or mast height		Drum diameter												ins	
														mm	
ft.	m.	11	13	14	16	18	20	22	24	26	28	30	32	34	36
		279	330	357	406	457	508	559	610	660	711	762	813	864	914
		cut-off length in number of drum laps													
		12	11												
			17	14	12	11									
			19	17	14	12	11	10	9	9	8				
					17	15	14	12	12	11	10	9	9		
							15	14	12	11	11	10	9		
									13	12	11	11	10		
												15	14	13	12

NOTE: Add 1/4 lap for counterbalanced groove drums
Add 1/2 lap for all other types of drum

CONVERSION OF DRUM LAPS TO CUT-OFF LENGTH

In order to ensure a change of the point of drum crossover, where the wear and crushing is very severe, either 1/4 or 1/2 lap should be added to the number of laps listed on [page B-4](#).

Add 1/4 lap for counterbalanced groove drums.

Add 1/2 lap for all other types of drum.

Conversion of laps to length is simply: Cut-off length = $\pi \times d \times \text{no. of laps}$

EXAMPLE: What is the recommended number of laps and cut-off length for the block line on a rig with a derrick of 138 ft (42m) and a drum of 30" (762 mm) diameter. The drum is counterbalanced.

From the table on [page B-4](#) the number of laps = $10 + 1/4$

In field units: Cut-off length = $\pi \times 30/12 \times 10^{1/4} = 80.5 \text{ ft}$

In S.I. units: Cut-off length = $\pi \times 0.762 \times 10^{1/4} = 24.5 \text{ m}$

WORK PER UNIT LENGTH CUT WHEN OPERATING AT A SAFETY FACTOR OF 5

Size of rope	Ton miles between cuts for each foot of rope cut	Megajoules between cuts for each metre of rope cut
1"	8	375.3
1 1/8"	12	562.9
1 1/4"	16	750.6
1 3/8"	20	938.2
1 1/2"	24	1,125.8

NOTE: 1 ton-mile = 14.30 MJ

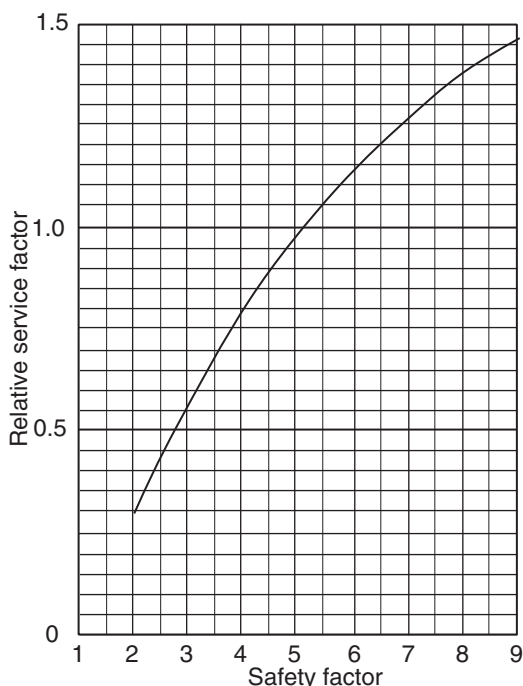
WHEN SAFETY FACTOR IS OTHER THAN "5"

Safety Factors will certainly be other than 5 for most operations. The block line work should therefore be adjusted by the relative service factor.

Note: adjustments should only be made to the drilling block line work. Given the high variations in the safety factors during casing and round trips block line work during these operations should be calculated on a safety factor of 5.

From the graph below, obtain the RELATIVE SERVICE FACTOR.

The calculated work must be divided by this factor to obtain the ADJUSTED WORK.



EXAMPLE: A safety factor of 3.86 is calculated when drilling a section of hole. The block line work calculated for drilling this section is 146 TM (6,849 MJ). Referring to the graph an S.F. of 3.86 gives a Relative Service Factor of 0.76. The adjusted work is therefore $146/0.76 = 192$ TM or $6,849/0.76 = 9,012$ MJ

BLOCK LINE

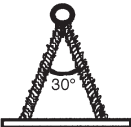
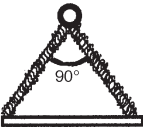
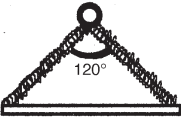
BLOCK LINE WEIGHT

Construction classification		Nominal diameter		Approximate weight	
		mm	inch	kg/m	lbs.ft
6 x 37	6 x 19	26	1	2.75	1.85
		29	1 ¹ / ₈	3.48	2.34
		32	1 ¹ / ₄	4.30	2.89
		35	1 ³ / ₈	5.21	3.50
		38	1 ¹ / ₂	6.19	4.16
		42	1 ⁵ / ₈	7.26	4.88
		45	1 ³ / ₄	8.44	5.67
		48	1 ⁷ / ₈	9.67	6.50
		51	2	11.0	7.39
		54	2 ¹ / ₈	12.4	8.35
	6 x 61	57	2 ¹ / ₄	13.9	9.36
		61	2 ³ / ₈	15.5	10.4
		64	2 ¹ / ₂	17.3	11.6
		67	2 ⁵ / ₈	19.0	12.8
		70	2 ³ / ₄	20.8	14.0
		74	2 ⁷ / ₈	22.8	15.3
		77	3	24.7	16.6
		80	3 ¹ / ₈	26.8	18.0
		83	3 ¹ / ₄	29.0	19.5
		86	3 ³ / ₈	31.3	21.0
		90	3 ¹ / ₂	33.8	22.7
		96	3 ³ / ₄	38.7	26.0
		103	4	44.0	29.6
		109	4 ¹ / ₄	49.6	33.3
		115	4 ¹ / ₂	55.7	37.4
		122	4 ³ / ₄	62.1	41.7
		128	5	68.8	46.2

WIRE ROPE SLINGS

SAFE LOADS

Safe loads for single and double 6 x 37 improved plow steel wire rope slings under different loading conditions

Diameter		Single vertical rope		Two ropes used at 30°		Two ropes used at 90°		Two ropes used at 120°	
									
Inch	mm	lbs	kg	lbs	kg	lbs	kg	lbs	kg
3/8	9.5	1,500	680	2,600	1,180	2,000	910	1,500	680
1/2	12.7	3,000	1,360	5,000	2,270	4,200	1,910	3,000	1,360
5/8	15.9	5,000	2,270	8,000	3,630	7,000	3,180	5,000	2,270
3/4	19.1	7,000	3,180	12,000	5,440	10,000	4,540	7,000	3,180
7/8	22.2	10,000	4,540	17,000	7,710	14,000	6,350	10,000	4,540
1	25.4	13,000	5,900	22,000	9,980	18,000	8,160	13,000	5,900
1 1/8	28.6	16,000	7,260	28,000	12,700	22,000	9,980	16,000	7,260
1 1/4	31.8	19,000	8,620	32,000	14,520	27,000	12,250	19,000	8,620
1 3/8	34.9	23,000	10,430	40,000	18,140	32,000	14,520	23,000	10,430
1 1/2	38.1	27,000	12,250	46,000	20,870	38,000	17,240	27,000	12,250
1 5/8	41.3	32,000	14,520	55,000	24,950	45,000	20,410	32,000	14,520
1 3/4	44.5	36,000	16,330	62,000	28,120	51,000	23,130	36,000	16,330
1 7/8	47.6	42,000	19,050	73,000	33,110	59,000	26,760	42,000	19,050
2	50.8	48,000	21,770	83,000	37,650	68,000	30,840	48,000	21,770

SLING CHAINS

SAFE WORKING LOADS (based on 62,5 % of proof test)

Alloy sling chains

Size of chain	inch	mm	Single sling chain		Double sling chain used at 60° angle		Double sling chain used at 90° angle		Double sling chain used at 120° angle		Double sling chain used at 140° angle		Double sling chain used at 150° angle		Double sling chain used at 160° angle		Double sling chain used at 170° angle	
			pnds	kgs	pnds	kgs	pnds	kgs	pnds	kgs	pnds	kgs	pnds	kgs	pnds	kgs	pnds	kgs
$\frac{9}{32}$	7.1	3,425	1,554	2,692	5,935	2,198	4,845	2,198	3,425	1,554	3,760	1,061	1,775	805	1,185	538	600	272
$\frac{5}{16}$	7.9	5,500	2,495	9,525	4,321	7,775	7,775	3,527	5,500	2,495	3,760	1,061	2,840	1,288	1,905	864	960	435
$\frac{3}{8}$	9.5	8,250	3,742	14,290	6,482	11,665	5,291	8,250	8,250	3,742	5,645	2,561	4,275	1,939	2,860	1,297	1,445	655
$\frac{7}{16}$	11.1	11,000	4,990	19,050	8,641	15,555	7,056	11,000	11,000	4,990	7,525	3,413	5,695	2,583	3,815	1,730	1,925	873
$\frac{1}{2}$	12.7	14,000	6,350	24,250	11,000	19,800	8,981	14,000	14,000	6,350	9,575	4,343	7,250	3,289	4,855	2,202	2,450	1,111
$\frac{9}{16}$	14.3	17,150	7,779	29,700	13,472	24,250	11,000	17,150	17,150	7,779	11,730	5,321	8,885	4,030	5,950	2,699	3,000	1,361
$\frac{5}{8}$	15.9	20,600	9,344	35,680	16,184	29,130	13,213	20,600	20,600	9,344	14,090	6,391	10,670	4,840	7,150	3,243	3,600	1,633
$\frac{3}{4}$	19.1	28,750	13,041	49,800	22,589	40,655	18,441	28,750	28,750	13,041	19,665	8,920	14,895	8,457	9,970	4,522	5,035	2,284
$\frac{7}{8}$	22.2	36,000	16,330	62,350	28,282	50,900	23,088	36,000	36,000	16,330	24,625	11,170	18,645	11,170	12,490	5,665	6,300	2,858
1	25.4	48,400	21,954	83,830	38,025	68,440	31,044	48,400	48,400	21,954	33,100	15,014	25,070	15,014	16,795	7,618	8,470	3,842

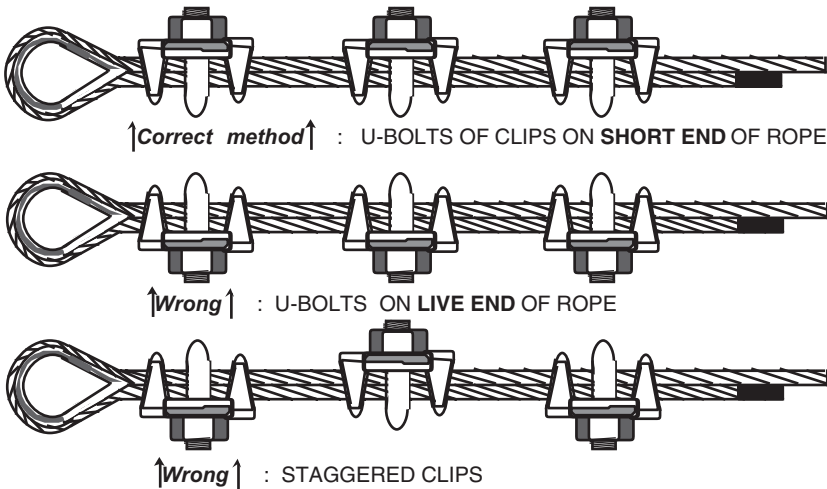
Wrought iron sling chains

$\frac{3}{8}$	9.5	2,700	1,225	4,700	2,132	3,800	1,724	2,700	2,700	1,225	1,850	839	1,450	658	940	426	470	213
$\frac{7}{16}$	11.1	3,450	1,565	5,900	2,676	4,900	2,222	3,450	3,450	1,565	2,350	1,066	1,750	794	1,200	544	600	272
$\frac{1}{2}$	12.7	4,500	2,041	7,800	3,538	6,350	2,880	4,500	4,500	2,041	3,100	1,406	2,300	1,043	1,570	712	780	354
$\frac{5}{8}$	15.9	6,900	3,130	12,000	5,443	9,750	4,423	6,900	6,900	3,130	4,700	2,132	3,550	1,610	2,400	1,089	1,200	544
$\frac{3}{4}$	19.1	10,100	4,581	17,500	7,938	14,000	6,350	10,100	10,100	4,581	6,900	3,130	5,200	2,359	3,500	1,588	1,750	794
$\frac{7}{8}$	22.2	14,000	6,350	24,000	10,886	19,500	8,845	14,000	14,000	6,350	9,600	4,355	7,250	3,289	4,900	2,223	2,400	1,089
1	25.4	18,600	8,437	32,000	14,515	26,000	11,794	18,600	18,600	8,437	12,700	5,761	9,650	4,377	6,500	2,948	3,200	1,452
$\frac{1}{8}$	28.6	23,400	10,614	40,000	18,144	33,000	14,969	23,400	23,400	10,614	16,000	7,258	12,000	5,443	8,000	3,629	4,000	1,814
$\frac{1}{4}$	31.8	28,800	13,064	50,000	22,680	40,500	18,371	28,800	28,800	13,064	19,700	8,936	15,000	6,804	10,000	4,536	5,000	2,268
$\frac{3}{8}$	34.9	34,500	15,649	60,000	27,216	49,000	22,226	34,500	34,500	15,649	23,500	10,660	17,800	8,074	12,000	5,443	6,000	2,722
$\frac{1}{2}$	38.1	40,800	18,507	70,000	31,752	57,500	26,082	40,800	40,800	18,507	28,000	12,701	21,000	9,526	14,000	6,350	7,000	3,175
$\frac{5}{8}$	41.3	46,500	21,092	80,000	36,288	66,000	29,938	46,500	46,500	21,092	31,800	14,424	24,000	10,886	16,000	7,258	8,000	3,629
$\frac{3}{4}$	44.5	52,500	23,814	91,000	41,278	74,000	33,566	52,500	52,500	23,814	36,000	16,330	27,000	12,247	18,000	8,165	9,100	4,128
2	50.8	66,600	30,210	115,000	52,164	94,000	42,638	66,600	66,600	30,210	45,600	20,684	34,500	15,649	23,000	10,433	11,500	5,216

WIRE ROPE CLIPS

METHOD OF ATTACHMENT AND NUMBER REQUIRED

Distance between clips should be equal to six rope diameters



Number of clips needed for safety

Diameter of rope		Number of clips	Space between clips		Length of rope turned back exclusive of eye	
inch	mm		inch	mm	inch	mm
$\frac{3}{8}$	10	2	$2 \frac{1}{4}$	57	5	127
$\frac{1}{2}$	13	3	3	76	9	229
$\frac{5}{8}$	16	3	$3 \frac{3}{4}$	95	11	279
$\frac{3}{4}$	19	4	$4 \frac{1}{2}$	114	18	457
$\frac{7}{8}$	22	4	$5 \frac{1}{4}$	133	21	533
1	25	4	6	152	24	610
$1 \frac{1}{8}$	29	5	7	178	35	889
$1 \frac{1}{4}$	32	5	8	203	40	1,016
$1 \frac{3}{8}$	35	6	9	229	54	1,372
$1 \frac{1}{2}$	38	6	10	254	60	1,524

NOTE : When clips are properly applied efficiency is approximately 80 %

FIBRE ROPE

FIBRE ROPE FOR GENERAL USE

Manila Rope, Grade 2 Standard Quality



Material : New genuine long fibre manila,
i.e. Abaca or approved equivalent.

Construction : 3-strand, plain laid.

Lay : right hand

Circ. of rope		Approx. diameter of rope		Minimum breaking strength (For 3 stand fibre rope.)		Approx. weight	
inch	mm	inch	mm	lbs	kg	lbs/ft	kg/m
7/8	21.9	1/4	6.4	720	330	0.023	0.036
1	25.4	5/16	7.9	1,060	480	0.035	0.053
1 1/4	31.8	3/8	9.6	1,400	630	0.046	0.067
1 1/2	38.1	1/2	12.7	2,100	950	0.070	0.106
2	50.8	5/8	15.9	3,970	1,800	0.13	0.19
2 1/4	57.2	3/4	19.1	4,760	2,150	0.15	0.23
2 3/4	69.9	7/8	21.9	7,500	3,400	0.23	0.35
3	76.2	1	25.4	8,960	4,060	0.28	0.41
3 1/2	88.9	1 1/8	28.6	11,920	5,400	0.38	0.57
3 3/4	95.3	1 1/4	31.8	13,600	6,170	0.43	0.63
4 3/4	120.7	1 1/2	38.1	21,000	9,520	0.71	1.04
6	152.4	2	50.8	32,700	14,830	1.12	1.66
7	177.8	2 1/4	57.2	43,900	19,910	1.52	2.26
8	203.2	2 1/2	63.5	56,440	25,600	2.00	2.95
10	254.0	3 1/4	82.6	86,460	39,210	3.20	4.61
12	304.8	3 3/4	95.3	123,200	55,880	4.46	6.63
14	355.6	4 1/2	114.3	165,760	75,180	6.08	9.02

C – TUBULARS & DRILL STRING DESIGN

Clickable list

(Use the expanded list under "Bookmarks" to access individual tables)

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BHA CONNECTION FATIGUE FAILURE PREVENTION

Historically the majority of drill string failures are attributable to BHA connection fatigue. What can **YOU** do to help reduce these failures ?

P ROPERTIES	- Specify BHA material that is very resistant to crack growth. (Toughness) - Connection stress relief. (Boreback box, stress relief pin, cold rolled threads)
R IG OPERATIONS	- Specify proper make-up. (Dope friction factor, torque, tong angle, calibrated torque gauge) - Avoid BHA vibration. (Apply vibration control guidelines) - Washout detection. (Twist-offs are ten times more expensive than washouts)
I NSPECTION	- Inspect according to a formal schedule - Look for cracks in thread roots. - Measure ID and OD to determine BSR.
D ESIGN	- Select proper connection BSR. - Stabilise BHA in enlarged holes. - Dampen vibration. - Design low stiffness ratios. (All these steps lower stress and lengthen fatigue life)
E NVIRONMENT	- Enlarged hole at BHA accelerates attack. - Control drilling fluid corrosion rate.
HAVE ‘PRIDE’ IN YOUR DRILL STRING !	Drill crew checks warn of possible BHA connection fatigue ! - Look for dry or muddy connection on break-out - Make-up torque should be adjusted if dope friction factor is not 1.0 - Is there a calibration sticker on the torque indicator ? - Check that numbers on calibration stickers agree with serial numbers on the equipment - Look out for small or missing bevels on BHA connections - Look out for unusual OD or ID on any BHA component - Look out for missing or oddly sized stress relief features on any BHA connection - Look out for any flat bottomed thread roots on BHA connections - Look out for any evidence of overtorque on a connection

This table has been adapted from an original in Shell Expro's "Drillstring Failure Prevention Quality Improvement Project (WEIN 687)"

DRILL PIPE BASICS

RANGE

Drill pipe is furnished in the following length ranges, which include the upsets but not the tooljoints :-

Range 1:	18 - 22 ft (5.49 - 6.71 m) — this is rarely seen
Range 2:	27 - 30 ft (8.23 - 9.14 m)
Range 3:	38 - 45 ft (11.58 - 13.72 m)

DIAMETER

Drill pipe is furnished in diameters ranging from $2\frac{3}{8}$ " to $6\frac{5}{8}$ ". The designated, or nominal, size of drill pipe is the actual outside diameter in inches of the pipe body when new.

WEIGHT

Drill pipe is furnished in different "weights", i.e. **weight per unit length**, corresponding to different wall thicknesses. This term "weight" is used to describe several different properties of a length of drill pipe, as follows:

Nominal weight: The designated, or nominal, weight does not now have a physical significance; it is used only for the purpose of identifying the drill pipe referred to. It is actually the theoretical weight per foot of a 20 ft length of threaded and coupled pipe based on the dimensions of the joint in use for the class of product when that particular diameter and wall thickness was introduced.

Plain end weight: Otherwise known as pipe body weight. This is the weight per unit length of pipe having the nominal dimensions given in the specification. It is the nominal cross-sectional area multiplied by the density.

Adjusted weight: This is the average weight per unit length of a length of drill pipe including the end finish (upsets), but excluding the tool joints, based on a total length (excluding the tool joints) of 29.4 ft.

Approximate weight: This is the average weight per unit length of the drill pipe including both upsets and tool joints, again based on a joint length (excluding the tool joints) of 29.4 ft. It varies with the type of tool joint used. **This is the weight which must be used for the calculation of the total weight of a string of drill pipe in air.**

MANUFACTURING TOLERANCES

For drill pipe up to and including 4" the tolerance on the OD is ± 0.031 ". For sizes of $4\frac{1}{2}$ " and above the tolerance on the OD is (+1%,-0.5%).

The most significant tolerance is that on wall thickness, with a value of (+0%,-12.5%). The strength of new drill pipe is always based on nominal OD with a wall thickness of 87.5% of nominal.

There is a tolerance of (+6.5%,-3.5%) on the weight of a single joint of drill pipe which defines the limits of average ID and wall thickness for a single joint. For the total weight of a large number of joints, as used in a string, the tolerance on the low side is reduced to -1.75%

YIELD STRENGTH

Each size and weight of drill pipe is furnished in a range of up to four standard strengths, known as grades. These grades are known as E-75, X-95, G-105 and S-135. The steel from which these are manufactured has the following yield strengths:

Given that strength is a critical property it is always assumed that the yield strength has its minimum allowable value. This is referred to as the minimum yield strength.

It must be emphasised that the "yield strength" of the steels is not the elastic limit - it is the tensile stress at which a specified extension has occurred. This

latter is 0.5% for E-75 and X-95 grades, 0.6% for G-105 and 0.7% for S-135, and is such that after removal of the stress a permanent deformation remains of the order of 0.2%.

Grade	Yield strength			
	Minimum		Maximum	
	psi	MPa	psi	MPa
E-75	75,000	517	105,000	724
X-95	95,000	655	125,000	862
G-105	105,000	724	135,000	931
S-135	135,000	931	165,000	1,138

USED DRILL PIPE

The API has established a classification for used drill pipe, according to the amount of wear on the pipe wall. This is reproduced on [page C-4](#). Note that drill pipe does not remain "new" for very long, and that Class 2 is rarely used within Shell (Class 3 never), thus the majority of drill pipe strings in use within the group fall into the category of "Premium Class".

DESIGN FACTORS

Given the fact that taking drill pipe up to its minimum yield stress will result in permanent deformation, it is recommended that this should be avoided and that a design factor should be applied when calculating allowable loads. The API recommends a factor of 10% applied to the yield strength, but the usual practice within Shell is to use 15% (this equates to a design factor of 1.18).

For checking resistance to collapse under the loads caused by external pressure a design factor is normally applied to the calculated collapse load. A value of 1.1 is usually used.

No design factor is required for torsion, as the torque applied is always limited to the make-up torque of the tool joints, being either 50% or 60% of the tool joint torsional yield strength. Since tool joints are almost always weaker in torsion than the tubes to which they are attached, the latter never approach their limiting strength in torsion. In case of doubt, or critical cases, compare the torsional strength of the pipe as tabulated on page C-21 with the tool joint make-up torque tabulated on pages C-30/34.

Drill pipe specifications have been taken from API Spec 5D, 4th Edition, August 1999.

The definitions of drill pipe weights are based on API RP 7G 16th Edition, August 1998 (Appendix A Para 13).

CLASSIFICATION OF USED DRILLPIPE

Applicable to all sizes, weights and grades. Nominal dimension is the basis for all calculations.

PIPE CONDITION	PREMIUM CLASS Two White Bands One centre punch mark ¹	CLASS 2 Yellow Bands Two centre punch marks ¹	CLASS 3 Orange Bands Three centre punch marks ¹
I. EXTERIOR CONDITIONS²			
A. OD Wear Wall	Remaining wall not less than 80%	Remaining wall not less than 70%	Any imperfections or damages exceeding CLASS 2
B. Dents & mashs	Diameter reduction not over 3% of OD	Diameter reduction not over 4% of OD	
Crushing, necking	Diameter reduction not over 3% of OD	Diameter reduction not over 4% of OD	
C. Slip area Mechanical damage Cuts ³ , gouges ³	Depth not to exceed 10% of the average adjacent wall ⁴	Depth not to exceed 20% of the average adjacent wall ⁴	
D. Stress induced diameter variations			
1. Stretched	Diameter reduction not over 3% of OD	Diameter reduction not over 4% of OD	
2. String Shot	Diameter increase not over 3% of OD	Diameter increase not over 4% of OD	
E. Corrosion, cuts & gouges			
1. Corrosion	Remaining wall not less than 80%	Remaining wall not less than 70%	
2. Cuts & Gouges Longitudinal	Remaining wall not less than 80%	Remaining wall not less than 70%	
Transverse	Remaining wall not less than 80%	Remaining wall not less than 80%	
F. Cracks ⁵	None	None	None
II INTERIOR CONDITIONS			
A. Corrosive Pitting Wall	Remaining wall not less than 80%, measured from base of deepest pit	Remaining wall not less than 70%, measured from base of deepest pit	
B. Erosion & Wear Wall	Remaining wall not less than 80%	Remaining wall not less than 70%	
C. Cracks	None	None	None

1. The centre punch marks are made on the 35° or 18° shoulder of the pin end tool joint.
2. An API Recommended Practice 7G inspection cannot be made with drill pipe rubbers on the pipe.
3. Remaining wall shall not be less than the value in 1E2. Defects may be ground out providing the remaining wall is not reduced below the value shown in 1E1 of this table and such grinding to be approximately faired into outer contour of the pipe.
4. Average adjacent wall is determined by measuring the wall thickness on each side of the cut or gouge adjacent to the deepest penetration.
5. In any classification where cracks or washouts appear, the pipe will be identified with the red band and considered unfit for further drilling service.
6. The drill pipe manufacturing date can be found on the pin.

This Table has been taken from API RP 7G, 16th Edition, August 1998 (Table 24) and is reproduced by courtesy of the API.

NOTES ON THE DRILL PIPE TABLES

The following notes apply to the drill pipe tables on pages C-6 to C-23

The strength of drill pipe is determined by the strength of the weakest point, thus the "worst case" of major dimensional tolerances has been assumed for calculating the tensile and torsional strengths, and burst and collapse resistance, of drill pipe.

In particular:

- The "minimum yield strength" has been used in all calculations.
- For all calculations for **new drill pipe** the **nominal OD** and **minimum allowable wall thickness** have been used.
- For the calculation of the **tensile and torsional strengths of used drill pipe** it has been assumed that the ID has its nominal value, that there has been the maximum wear allowable under the classification scheme, and that the wear has taken place uniformly on the outside of the pipe. This minimises the cross-sectional area of steel, thus producing the least tension/torsion resistance allowable within the class.
- For the calculation of the **burst and collapse resistances of used drill pipe** it has been assumed that the OD has its nominal value, that there has been the maximum wear allowable under the classification scheme and that the wear has taken place uniformly on the inside of the pipe. This maximises the diameter over which burst or collapse pressures act, thus producing the least theoretical burst/collapse resistance allowable within the class.
- No **design factors** have been used in the calculations

Weights, displacements and capacities are not governed by a critical value in the same way that a strength is. These parameters are normally applied to a **string** of drill pipe as opposed to a single joint. Under these circumstances manufacturing tolerances on wall thickness average out over the length of the string and need not be taken into account.* Furthermore, given that it is not necessary to adopt a "worst case" approach, it is acceptable to base the calculations on the more practical assumption that all the wear is on the outside of the string.

- For calculations relating to new drill pipe the **nominal OD** and **nominal wall thickness** have been used.
- For used drill pipe, given that the classifications Premium Class and Class 2 can be applied to a range of different degrees of wear, no specific single dimensions can be assumed. The approach that has been taken is to make the calculation assuming that there has been the maximum wear allowable under the classification scheme, and that this has taken place uniformly on the **outside** of the pipe. The value quoted is then a range between that value and the equivalent one corresponding to a "less worn" classification.

In particular

- For the calculation of the average weight, closed-ended and open-ended displacement of Premium Class pipe the quoted range is based on the calculated value and the corresponding value for new pipe. For Class 2 pipe the range is between the calculated values for Class 2 and Premium pipe.
- For the calculation of the capacity of all classes of drill pipe the ID is taken to be the nominal ID. It follows that the capacity of a string is taken to have the same value, whatever the class.
- As the drill pipe body wears, the tool joints also wear. In the calculation of weight and displacement of used pipe it is assumed that the thickness of metal worn from the tool joints is equal to the thickness of metal worn from the pipe body, but that the external upsets are protected by the tool joints and are not significantly worn.

* Strictly speaking, this is not correct as the specifications contain a tolerance on the total weight of a shipment (see [page C-2](#)). However the tolerance is such that it has no practical effect on field operations.

DIMENSIONS AND WEIGHTS OF DRILL PIPE

OILFIELD UNITS

DP specification			Nominal dimensions of pipe body (new)			Approximate weight (in air) of a string of drill pipe, including tool joints		
Size/style Tool joint	Nominal weight	Grade	OD	Wall thickness	ID	New pipe	Premium class	Class 2
	lbs/ft		inches	inches	inches	lbs/ft	lbs/ft	lbs/ft
2 ³ / ₈ " EU NC26	6.65	E75	2.375	0.280	1.815	7.02	6.31 ± 0.71	5.26 ± 0.34
		X95	2.375	0.280	1.815	7.11	6.40 ± 0.71	5.35 ± 0.34
		G105	2.375	0.280	1.815	7.11	6.40 ± 0.71	5.35 ± 0.34
2 ⁷ / ₈ " EU NC31	10.40	E75	2.875	0.362	2.151	10.89	9.78 ± 1.11	8.14 ± 0.53
		X95	2.875	0.362	2.151	11.08	9.97 ± 1.11	8.33 ± 0.53
		G105	2.875	0.362	2.151	11.08	9.97 ± 1.11	8.33 ± 0.53
		S135	2.875	0.362	2.151	11.55	10.43 ± 1.12	8.78 ± 0.54
3 ¹ / ₂ " EU NC38	9.50	E75	3.500	0.254	2.992	10.59	9.64 ± 0.96	8.21 ± 0.47
		X95	3.500	0.254	2.992	10.59	9.64 ± 0.96	8.21 ± 0.47
	13.30	E75	3.500	0.368	2.764	13.95	12.57 ± 1.38	10.53 ± 0.67
		X95	3.500	0.368	2.764	14.61	13.23 ± 1.38	11.18 ± 0.67
		G105	3.500	0.368	2.764	14.71	13.32 ± 1.38	11.27 ± 0.67
		S135	3.500	0.368	2.764	14.92	13.54 ± 1.38	11.48 ± 0.67
	15.50	E75	3.500	0.449	2.602	16.57	14.89 ± 1.68	12.40 ± 0.81
		X95	3.500	0.449	2.602	16.83	15.16 ± 1.68	12.67 ± 0.81
		G105	3.500	0.449	2.602	17.05	15.37 ± 1.68	12.88 ± 0.81
3 ¹ / ₂ " EU NC40	15.50	S135	3.500	0.449	2.602	17.59	15.90 ± 1.69	13.39 ± 0.81
4" IU NC40	14.00	E75	4.000	0.330	3.340	15.05	13.64 ± 1.41	11.53 ± 0.69
		X95	4.000	0.330	3.340	15.28	13.87 ± 1.41	11.76 ± 0.69
		G105	4.000	0.330	3.340	15.85	14.43 ± 1.42	12.32 ± 0.69
		S135	4.000	0.330	3.340	16.13	14.71 ± 1.42	12.59 ± 0.69
4" EU NC46	14.00	E75	4.000	0.330	3.340	15.89	14.46 ± 1.43	12.34 ± 0.70
		X95	4.000	0.330	3.340	16.19	14.77 ± 1.43	12.64 ± 0.70
		G105	4.000	0.330	3.340	16.19	14.77 ± 1.43	12.64 ± 0.70
		S135	4.000	0.330	3.340	16.42	14.99 ± 1.43	12.87 ± 0.70
4 ¹ / ₂ " IU NC46	13.75	E75	4.500	0.271	3.958	15.11	13.80 ± 1.31	11.84 ± 0.65
4 ¹ / ₂ " EU NC50	13.75	E75	4.500	0.271	3.958	15.88	14.56 ± 1.32	12.59 ± 0.65
		X95	4.500	0.271	3.958	15.88	14.56 ± 1.32	12.59 ± 0.65
	16.60	E75	4.500	0.337	3.826	18.47	16.83 ± 1.64	14.39 ± 0.80
		X95	4.500	0.337	3.826	18.85	17.21 ± 1.64	14.77 ± 0.80
		G105	4.500	0.337	3.826	18.85	17.21 ± 1.64	14.77 ± 0.80
		S135	4.500	0.337	3.826	19.11	17.47 ± 1.64	15.03 ± 0.80
	20.00	E75	4.500	0.430	3.640	22.11	20.03 ± 2.08	16.93 ± 1.01
		X95	4.500	0.430	3.640	22.58	20.49 ± 2.08	17.40 ± 1.01
		G105	4.500	0.430	3.640	22.58	20.49 ± 2.08	17.40 ± 1.01
		S135	4.500	0.430	3.640	23.06	20.97 ± 2.09	17.86 ± 1.02
4 ¹ / ₂ " IEU NC46	16.60	E75	4.500	0.337	3.826	18.37	16.74 ± 1.63	14.31 ± 0.80
		X95	4.500	0.337	3.826	18.62	16.98 ± 1.63	14.55 ± 0.80
		G105	4.500	0.337	3.826	18.62	16.98 ± 1.63	14.55 ± 0.80
		S135	4.500	0.337	3.826	18.83	17.19 ± 1.64	14.76 ± 0.80
	20.00	E75	4.500	0.430	3.640	22.12	20.04 ± 2.08	16.96 ± 1.01
		X95	4.500	0.430	3.640	22.62	20.54 ± 2.08	17.45 ± 1.01
		G105	4.500	0.430	3.640	22.81	20.73 ± 2.08	17.64 ± 1.01
		S135	4.500	0.430	3.640	22.98	20.90 ± 2.08	17.81 ± 1.01

The nominal dimensions and weights of the body, and upsets, of new drill pipe have been taken from API Spec 5D, 4th Edition, August 1999. Approximate weights have been calculated by the method specified in API RP 7G 16th Edition, August 1998 using tool joint dimensions as specified in API Spec 7, 39th Edition, December 1997.

DP specification			Nominal dimensions of pipe body (new)			Approximate weight (in air) of a string of drill pipe, including tool joints		
Size/style Tool joint	Nominal weight	Grade	OD	Wall thickness	ID	New pipe	Premium class	Class 2
	lbs/ft		inches	inches	inches	lbs/ft	lbs/ft	lbs/ft
5" IEU NC50	19.50	E75	5.000	0.362	4.276	21.35	19.40 ± 1.94	16.51 ± 0.95
		X95	5.000	0.362	4.276	21.87	19.93 ± 1.94	17.03 ± 0.95
		G105	5.000	0.362	4.276	22.24	20.29 ± 1.95	17.39 ± 0.95
		S135	5.000	0.362	4.276	22.56	20.61 ± 1.95	17.70 ± 0.95
	25.60	E75	5.000	0.500	4.000	27.35	24.68 ± 2.67	20.72 ± 1.29
		X95	5.000	0.500	4.000	28.07	25.40 ± 2.67	21.43 ± 1.30
		G105	5.000	0.500	4.000	28.28	25.60 ± 2.67	21.63 ± 1.30
5" IEU 5 1/2" FH	19.50	E75	5.000	0.362	4.276	22.30	20.35 ± 1.95	17.44 ± 0.96
		X95	5.000	0.362	4.276	22.56	20.61 ± 1.95	17.71 ± 0.96
		G105	5.000	0.362	4.276	22.56	20.61 ± 1.95	17.71 ± 0.96
		S135	5.000	0.362	4.276	23.43	21.47 ± 1.96	18.55 ± 0.96
	25.60	E75	5.000	0.500	4.000	28.30	25.62 ± 2.68	21.64 ± 1.30
		X95	5.000	0.500	4.000	28.54	25.86 ± 2.68	21.88 ± 1.30
		G105	5.000	0.500	4.000	29.11	26.42 ± 2.69	22.43 ± 1.31
		S135	5.000	0.500	4.000	29.38	26.69 ± 2.69	22.69 ± 1.31
5 1/2" IEU 5 1/2" FH	21.90	E75	5.500	0.361	4.778	23.79	21.66 ± 2.13	18.48 ± 1.05
		X95	5.500	0.361	4.778	24.41	22.28 ± 2.13	19.10 ± 1.05
		G105	5.500	0.361	4.778	25.26	23.12 ± 2.14	19.93 ± 1.05
		S135	5.500	0.361	4.778	26.37	24.22 ± 2.15	21.02 ± 1.06
	24.70	E75	5.500	0.415	4.670	26.31	23.87 ± 2.45	20.22 ± 1.20
		X95	5.500	0.415	4.670	27.74	25.29 ± 2.46	21.63 ± 1.20
		G105	5.500	0.415	4.670	27.74	25.29 ± 2.46	21.63 ± 1.20
		S135	5.500	0.415	4.670	28.85	26.39 ± 2.47	22.71 ± 1.21
6 5/8" IEU 6 5/8" FH	25.20	E75	6.625	0.330	5.965	27.55	25.20 ± 2.35	21.69 ± 1.16
		X95	6.625	0.330	5.965	27.55	25.20 ± 2.35	21.69 ± 1.16
		G105	6.625	0.330	5.965	28.60	26.24 ± 2.36	22.72 ± 1.16
		S135	6.625	0.330	5.965	30.03	27.66 ± 2.37	24.13 ± 1.17
	27.70	E75	6.625	0.362	5.901	29.40	26.82 ± 2.58	22.98 ± 1.27
		X95	6.625	0.362	5.901	30.45	27.87 ± 2.59	24.01 ± 1.27
		G105	6.625	0.362	5.901	30.45	27.87 ± 2.59	24.01 ± 1.27
		S135	6.625	0.362	5.901	31.88	29.28 ± 2.60	25.41 ± 1.28

Note that there is no single figure that can be quoted for the weight per unit length of a string of used drill pipe - it depends on the amount of wear.

The drill pipe used in Shell operations will almost always be premium class which has, **by definition**, an amount of wear that can be anywhere between 0% and 20% of the wall thickness (Refer to the Classification table on [page C-4](#)). The ranges quoted in these tables take account of this possible variation - the high end is equal to the value for new pipe, the low end is the average weight per unit length that a string would have if every joint were worn to the maximum allowable degree - i.e. just before the joint would have to be reclassified as Class 2.

Note also that although the Classification scheme allows for pipe that is eroded and worn on the inside, that is in practice rare, and these tables assume that all wear is on the OD of the pipe and tool joints.

The quoted mid-point of the range (which is equivalent to just under 10% wear) will be sufficiently accurate for most cases. If you know, or can estimate, the actual wear, a linear interpolation within the tolerances quoted can be used to improve accuracy.

For completeness the data for Class 2 drill pipe is included - also as a range rather than a single value, for the same reasons.

DIMENSIONS AND WEIGHTS OF DRILL PIPE

SI UNITS

DP specification			Nominal dimensions of pipe body (new)			Approximate weight (in air) of a string of drill pipe, including tool joints		
Size (mm) Style & Tool joint	Nominal weight	Grade	OD	Wall thickness	ID	New pipe	Premium class	Class 2
	lbs/ft (kg/m)		mm	mm	mm	kg/m	kg/m	kg/m
2 ³ / ₈ " EU 60.3 NC26	6.65	E75	60.3	7.11	46.1	10.44	9.39 ± 1.05	7.83 ± 0.51
		X95	60.3	7.11	46.1	10.57	9.52 ± 1.05	7.96 ± 0.51
	9.90	G105	60.3	7.11	46.1	10.57	9.52 ± 1.05	7.96 ± 0.51
2 ⁷ / ₈ " EU 73.0 NC31		E75	73.0	9.19	54.6	16.21	14.56 ± 1.65	12.12 ± 0.79
		X95	73.0	9.19	54.6	16.49	14.84 ± 1.65	12.40 ± 0.79
	10.40	G105	73.0	9.19	54.6	16.49	14.84 ± 1.65	12.40 ± 0.79
	15.48	S135	73.0	9.19	54.6	17.18	15.52 ± 1.66	13.07 ± 0.80
3 ¹ / ₂ " EU 88.9 NC38	9.50	E75	88.9	6.45	76.0	15.76	14.34 ± 1.42	12.22 ± 0.70
	14.14							
		E75	88.9	9.35	70.2	20.76	18.71 ± 2.05	15.67 ± 0.99
		X95	88.9	9.35	70.2	21.75	19.69 ± 2.06	16.64 ± 1.00
	13.30	G105	88.9	9.35	70.2	21.89	19.83 ± 2.06	16.77 ± 1.00
	19.79	S135	88.9	9.35	70.2	22.21	20.14 ± 2.06	17.09 ± 1.00
	15.50	E75	88.9	11.40	66.1	24.65	22.16 ± 2.50	18.46 ± 1.20
	23.07	X95	88.9	11.40	66.1	25.05	22.55 ± 2.50	18.85 ± 1.20
		G105	88.9	11.40	66.1	25.37	22.87 ± 2.50	19.16 ± 1.20
3 ¹ / ₂ " EU 88.9 NC40	15.50	S135	88.9	11.40	66.1	26.17	23.66 ± 2.51	19.93 ± 1.21
	23.07							
4" IU 101.6 NC40		E75	101.6	8.38	84.8	22.39	20.29 ± 2.10	17.16 ± 1.03
	14.00	X95	101.6	8.38	84.8	22.74	20.63 ± 2.10	17.50 ± 1.03
	20.83	G105	101.6	8.38	84.8	23.59	21.48 ± 2.11	18.34 ± 1.03
		S135	101.6	8.38	84.8	24.00	21.89 ± 2.12	18.74 ± 1.03
4" EU 101.6 NC46		E75	101.6	8.38	84.8	23.65	21.53 ± 2.12	18.37 ± 1.04
	14.00	X95	101.6	8.38	84.8	24.10	21.98 ± 2.12	18.82 ± 1.04
	20.83	G105	101.6	8.38	84.8	24.10	21.98 ± 2.12	18.82 ± 1.04
		S135	101.6	8.38	84.8	24.44	22.31 ± 2.13	19.15 ± 1.04
4 ¹ / ₂ " IU 114.3 NC46	13.75	E75	114.3	6.88	100.5	22.48	20.53 ± 1.95	17.62 ± 0.96
	20.46							
4 ¹ / ₂ " EU 114.3 NC50	13.75	E75	114.3	6.88	100.5	23.63	21.67 ± 1.97	18.73 ± 0.97
	20.46							
		E75	114.3	8.56	97.2	27.49	25.05 ± 2.44	21.42 ± 1.19
		X95	114.3	8.56	97.2	28.05	25.61 ± 2.44	21.97 ± 1.19
	16.60	G105	114.3	8.56	97.2	28.05	25.61 ± 2.44	21.97 ± 1.19
	24.70	S135	114.3	8.56	97.2	28.44	26.00 ± 2.44	22.36 ± 1.19
	20.00	E75	114.3	10.92	92.5	32.91	29.81 ± 3.10	25.20 ± 1.51
	29.76	X95	114.3	10.92	92.5	33.60	30.50 ± 3.10	25.89 ± 1.51
		G105	114.3	10.92	92.5	33.60	30.50 ± 3.10	25.89 ± 1.51
		S135	114.3	10.92	92.5	34.31	31.20 ± 3.11	26.59 ± 1.51
4 ¹ / ₂ " IEU 114.3 NC46		E75	114.3	8.56	97.2	27.34	24.91 ± 2.43	21.29 ± 1.19
	16.60	X95	114.3	8.56	97.2	27.71	25.27 ± 2.43	21.65 ± 1.19
	24.70	G105	114.3	8.56	97.2	27.71	25.27 ± 2.43	21.65 ± 1.19
		S135	114.3	8.56	97.2	28.02	25.58 ± 2.43	21.96 ± 1.19
	20.00	E75	114.3	10.92	92.5	32.92	29.83 ± 3.09	25.24 ± 1.50
	29.76	X95	114.3	10.92	92.5	33.66	30.57 ± 3.09	25.98 ± 1.50
		G105	114.3	10.92	92.5	33.95	30.85 ± 3.09	26.26 ± 1.50
		S135	114.3	10.92	92.5	34.20	31.11 ± 3.10	26.51 ± 1.50

The nominal dimensions and weights of the body, and upsets, of new drill pipe have been taken from API Spec 5D, 4th Edition, August 1999. Approximate weights have been calculated by the method specified in API RP 7G 16th Edition, August 1998 using tool joint dimensions as specified in API Spec 7, 39th Edition, December 1997.

DP specification			Nominal dimensions of pipe body (new)			Approximate weight (in air) of a string of drill pipe, including tool joints		
Size (mm) Style & Tool joint	Nominal weight	Grade	OD	Wall thickness	ID	New pipe	Premium class	Class 2
	lbs/ft (kg/m)		mm	mm	mm	kg/m	kg/m	kg/m
5" IEU 127.0 NC50	19.50 29.02	E75	127.0	9.19	108.6	31.77	28.88 ± 2.89	24.57 ± 1.41
		X95	127.0	9.19	108.6	32.55	29.66 ± 2.89	25.35 ± 1.42
		G105	127.0	9.19	108.6	33.09	30.19 ± 2.90	25.88 ± 1.42
		S135	127.0	9.19	108.6	33.57	30.67 ± 2.90	26.35 ± 1.42
	25.60 38.10	E75	127.0	12.70	101.6	40.70	36.73 ± 3.97	30.84 ± 1.93
		X95	127.0	12.70	101.6	41.77	37.79 ± 3.98	31.89 ± 1.93
5" IEU 127.0 5½" FH	19.50 29.02	E75	127.0	9.19	108.6	33.19	30.28 ± 2.90	25.96 ± 1.42
		X95	127.0	9.19	108.6	33.58	30.67 ± 2.90	26.35 ± 1.42
		G105	127.0	9.19	108.6	33.58	30.67 ± 2.90	26.35 ± 1.42
		S135	127.0	9.19	108.6	34.86	31.94 ± 2.92	27.60 ± 1.43
	25.60 38.10	E75	127.0	12.70	101.6	42.11	38.13 ± 3.99	32.20 ± 1.94
		X95	127.0	12.70	101.6	42.48	38.49 ± 3.99	32.56 ± 1.94
5½" IEU 139.7 5½" FH	21.90 32.59	G105	127.0	12.70	101.6	43.32	39.32 ± 4.00	33.38 ± 1.94
		S135	127.0	12.70	101.6	43.73	39.72 ± 4.00	33.77 ± 1.94
		E75	139.7	9.17	121.4	35.41	32.23 ± 3.17	27.50 ± 1.56
		X95	139.7	9.17	121.4	36.33	33.16 ± 3.18	28.42 ± 1.56
	24.70 36.76	G105	139.7	9.17	121.4	37.59	34.40 ± 3.19	29.65 ± 1.56
		S135	139.7	9.17	121.4	39.25	36.05 ± 3.20	31.28 ± 1.57
6⅝" IEU 168.3 6⅝" FH	25.20 37.50	E75	168.3	8.38	151.5	41.00	37.50 ± 3.50	32.27 ± 1.73
		X95	168.3	8.38	151.5	41.00	37.50 ± 3.50	32.27 ± 1.73
		G105	168.3	8.38	151.5	42.56	39.05 ± 3.51	33.81 ± 1.73
		S135	168.3	8.38	151.5	44.70	41.17 ± 3.53	35.91 ± 1.74
	27.70 41.22	E75	168.3	9.19	149.9	43.75	39.92 ± 3.84	34.20 ± 1.89
		X95	168.3	9.19	149.9	45.32	41.47 ± 3.85	35.73 ± 1.89
G105		168.3	9.19	149.9	45.32	41.47 ± 3.85	35.73 ± 1.89	
S135		168.3	9.19	149.9	47.44	43.58 ± 3.86	37.82 ± 1.90	

Note that there is no single figure that can be quoted for the weight per unit length of a string of used drill pipe - it depends on the amount of wear.

The drill pipe used in Shell operations will almost always be premium class which has, **by definition**, an amount of wear that can be anywhere between 0% and 20% of the wall thickness (Refer to the Classification table on Page C-4). The ranges quoted in these tables take account of this possible variation - the high end is equal to the value for new pipe, the low end is the average weight per unit length that a string would have if every joint were worn to the maximum allowable degree - i.e. just before the joint would have to be reclassified as Class 2.

Note also that although the Classification scheme allows for pipe that is eroded and worn on the inside, that is in practice rare, and these tables assume that all wear is on the OD of the pipe and tool joints.

The quoted mid-point of the range (which is equivalent to just under 10% wear) will be sufficiently accurate for most cases. If you know, or can estimate, the actual wear, a linear interpolation within the tolerances quoted can be used to improve accuracy.

For completeness the data for Class 2 drill pipe is included - also as a range rather than a single value, for the same reasons.

THE DISPLACEMENT AND CAPACITY OF A STRING OF NEW DRILL PIPE, INCLUDING TOOL JOINTS

Size inches mm	Style Tool Joint	Weight lbs/ft kg/m	Grade	Closed-end Displ.			Open-ended Displ.			Capacity		
				l/m	bbls per 1,000 ft	gals per 1,000 ft	l/m	bbls per 1,000 ft	gals per 1,000 ft	l/m	bbls per 1,000 ft	gals per 1,000 ft
2 3/8" 60.3	EU NC26	6.65	E75	2.99	5.74	241	1.33	2.55	107	1.66	3.19	134
			X95	3.00	5.75	242	1.35	2.59	109	1.65	3.17	133
		9.90	G105	3.00	5.75	242	1.35	2.59	109	1.65	3.17	133
2 7/8" 73.0	EU NC31	10.40 15.48	E75	4.41	8.45	355	2.07	3.96	166	2.34	4.49	189
			X95	4.42	8.48	356	2.10	4.03	169	2.32	4.44	187
			G105	4.42	8.48	356	2.10	4.03	169	2.32	4.44	187
			S135	4.47	8.58	360	2.19	4.20	176	2.28	4.38	184
3 1/2" 88.9	EU NC38	9.50	E75	6.51	12.5	524	2.01	3.85	162	4.50	8.62	362
		14.14										
		13.30 19.79	E75	6.51	12.5	524	2.65	5.08	213	3.86	7.40	311
			X95	6.60	12.6	531	2.77	5.32	223	3.82	7.33	308
			G105	6.60	12.6	531	2.79	5.35	225	3.80	7.29	306
			S135	6.60	12.6	531	2.83	5.43	228	3.76	7.22	303
		15.50 23.07	E75	6.58	12.6	529	3.14	6.03	253	3.43	6.58	276
			X95	6.60	12.6	531	3.19	6.12	257	3.40	6.52	274
			G105	6.60	12.6	531	3.24	6.20	261	3.36	6.44	271
3 1/2" 88.9	EU NC40	15.50 23.07	S135	6.71	12.9	541	3.34	6.40	269	3.38	6.47	272
4" 101.6	IU NC40	14.00 20.83	E75	8.41	16.1	677	2.86	5.48	230	5.55	10.6	447
			X95	8.42	16.1	678	2.90	5.56	233	5.52	10.6	444
			G105	8.49	16.3	684	3.01	5.77	242	5.48	10.5	441
			S135	8.49	16.3	684	3.06	5.87	246	5.43	10.4	437
4" 101.6	EU NC46	14.00 20.83	E75	8.66	16.6	697	3.02	5.78	243	5.64	10.8	454
			X95	8.68	16.7	699	3.07	5.89	247	5.61	10.8	452
			G105	8.68	16.7	699	3.07	5.89	247	5.61	10.8	452
			S135	8.68	16.7	699	3.12	5.97	251	5.57	10.7	448
4 1/2" 114.3	IU NC46	13.75 20.46	E75	10.7	20.5	860	2.87	5.50	231	7.81	15.0	629
4 1/2" 114.3	EU NC50	13.75	E75	10.9	20.9	879	3.01	5.78	243	7.90	15.1	636
		20.46										
		16.60 24.70	E75	10.8	20.6	866	3.49	6.68	281	7.27	13.9	585
			X95	10.8	20.6	866	3.53	6.77	284	7.22	13.8	582
			G105	10.8	20.6	866	3.53	6.77	284	7.22	13.8	582
			S135	10.8	20.6	866	3.57	6.85	288	7.18	13.8	578
		20.00 29.76	E75	10.9	20.9	879	3.51	6.72	282	7.41	14.2	596
			X95	11.0	21.0	882	3.58	6.86	288	7.37	14.1	594
			G105	11.0	21.0	882	3.58	6.86	288	7.37	14.1	594
			S135	11.0	21.0	882	3.63	6.95	292	7.32	14.0	590
4 1/2" 114.3	IEU NC46	16.60 24.70	E75	10.8	20.6	866	4.20	8.05	338	6.56	12.6	528
			X95	10.8	20.7	867	4.29	8.23	346	6.48	12.4	522
			G105	10.8	20.7	867	4.33	8.30	349	6.44	12.4	519
			S135	10.8	20.7	867	4.36	8.36	351	6.41	12.3	516
		20.00 29.76	E75	10.9	20.9	879	4.20	8.05	338	6.72	12.9	541
			X95	11.0	21.0	882	4.28	8.21	345	6.67	12.8	537
			G105	11.0	21.0	882	4.28	8.21	345	6.67	12.8	537
			S135	11.0	21.0	882	4.38	8.39	352	6.58	12.6	529

The nominal dimensions and weights of the body, and upsets, of new drill pipe have been taken from API Spec 5D, 4th Edition, August 1999. The dimensions of new tool joints have been taken from API Spec 7, 39th Edition, December 1997.

Size	Style	Weight	Grade	Closed-end Displ.			Open-ended Displ.			Capacity		
inches mm	Tool Joint	lbs/ft kg/m		l/m	bbls per 1,000 ft	gals per 1,000 ft	l/m	bbls per 1,000 ft	gals per 1,000 ft	l/m	bbls per 1,000 ft	gals per 1,000 ft
5" 127.0	IEU NC50	19.50 29.02	E75	13.2	25.2	1,060	4.05	7.77	326	9.11	17.5	733
			X95	13.2	25.2	1,060	4.15	7.96	334	9.02	17.3	726
			G105	13.2	25.2	1,060	4.22	8.09	340	8.95	17.2	721
			S135	13.2	25.2	1,060	4.28	8.21	345	8.89	17.0	716
		25.60 38.10	E75	13.3	25.6	1,074	4.23	8.11	341	9.10	17.4	733
			X95	13.3	25.6	1,074	4.28	8.21	345	9.06	17.4	729
			G105	13.3	25.6	1,074	4.28	8.21	345	9.06	17.4	729
			S135	13.3	25.6	1,074	4.28	8.21	345	9.06	17.4	729
5" 127.0	IEU 5 1/2"FH	19.50 29.02	E75	13.4	25.8	1,083	4.45	8.52	358	9.00	17.3	725
			X95	13.2	25.2	1,060	5.19	9.95	418	7.97	15.3	642
			G105	13.2	25.2	1,060	5.33	10.2	429	7.84	15.0	632
			S135	13.2	25.2	1,060	5.37	10.3	432	7.80	15.0	628
		25.60 38.10	E75	13.3	25.6	1,074	5.37	10.3	432	7.96	15.3	641
			X95	13.3	25.6	1,074	5.42	10.4	436	7.93	15.2	638
			G105	13.4	25.8	1,083	5.52	10.6	445	7.92	15.2	638
			S135	13.4	25.8	1,083	5.58	10.7	449	7.87	15.1	634
5 1/2" 139.7	IEU 5 1/2"FH	21.90 32.59	E75	15.8	30.4	1,276	4.52	8.66	364	11.3	21.7	912
			X95	15.9	30.4	1,277	4.63	8.88	373	11.2	21.5	904
			G105	16.0	30.6	1,285	4.79	9.19	386	11.2	21.4	899
			S135	16.1	30.8	1,294	5.00	9.60	403	11.1	21.2	891
		24.70 36.76	E75	15.8	30.4	1,276	4.99	9.57	402	10.9	20.8	874
			X95	16.0	30.6	1,285	5.27	10.1	424	10.7	20.5	861
			G105	16.0	30.6	1,285	5.27	10.1	424	10.7	20.5	861
			S135	16.1	30.8	1,294	5.48	10.5	441	10.6	20.3	853
6 5/8" 168.3	IEU 6 5/8"FH	25.20 37.50	E75	22.9	43.8	1,840	5.23	10.0	421	17.6	33.8	1,419
			X95	22.9	43.8	1,840	5.23	10.0	421	17.6	33.8	1,419
			G105	23.0	44.0	1,850	5.43	10.4	437	17.5	33.6	1,413
			S135	23.1	44.3	1,860	5.70	10.9	459	17.4	33.4	1,401
		27.70 41.22	E75	22.9	43.8	1,840	5.58	10.7	449	17.3	33.1	1,391
			X95	23.0	44.0	1,850	5.78	11.1	465	17.2	33.0	1,385
			G105	23.0	44.0	1,850	5.78	11.1	465	17.2	33.0	1,385
			S135	23.1	44.3	1,860	6.05	11.6	487	17.1	32.7	1,373

Note 1

The displacements quoted in this table are based on the nominal dimensions of the drill pipe and tool joints. The manufacturing tolerances applicable to the total weight of large batches of drill pipe (i.e. "string-length" quantities as opposed to single joints) have been ignored.

Note 2

In this edition of the WENB a different approach to capacities has been taken to the one taken for Editions 2 & 3. In those, the possibility of wear on the ID of the drill pipe was taken into account, as allowed for in the pipe classification scheme; hence it was necessary to quote a range of capacities for each size/weight combination. In this edition it has been acknowledged that internal wear rarely occurs (internal coating) and the tables have been simplified by not taking it into account. With this assumption the capacities of both Premium grade and Class 2 pipes are equal to the capacity of new drill pipe.

THE DISPLACEMENT AND CAPACITY OF A STRING OF PREMIUM CLASS DRILL PIPE, INCLUDING TOOL JOINTS

OILFIELD UNITS

Size/Style Tool joint	Weight	Grade	Closed ended		Open ended		Capacity	
	lbs/ft		bbls/1000 ft	Gals/1000 ft	bbls/1000 ft	Gals/1000 ft	bbls/1000 ft	Gals/1000 ft
2 ³ / ₈ " EU NC26	6.65	E75	5.48 ± 0.26	230 ± 11	2.29 ± 0.26	96.4 ± 10.8	3.19	134
		X95	5.50 ± 0.26	231 ± 11	2.33 ± 0.26	97.7 ± 10.8	3.17	133
		G105	5.50 ± 0.26	231 ± 11	2.33 ± 0.26	97.7 ± 10.8	3.17	133
2 ⁷ / ₈ " EU NC31	10.40	E75	8.05 ± 0.40	338 ± 17	3.56 ± 0.40	149 ± 17	4.49	189
		X95	8.07 ± 0.40	339 ± 17	3.63 ± 0.40	152 ± 17	4.44	187
		G105	8.07 ± 0.40	339 ± 17	3.63 ± 0.40	152 ± 17	4.44	187
		S135	8.17 ± 0.41	343 ± 17	3.79 ± 0.41	159 ± 17	4.38	184
3 ¹ / ₂ " EU NC38	9.50	E75	12.1 ± 0.3	509 ± 15	3.50 ± 0.35	147 ± 15	8.62	362
	13.30	E75	12.0 ± 0.5	503 ± 21	4.57 ± 0.50	192 ± 21	7.40	311
		X95	12.1 ± 0.5	510 ± 21	4.81 ± 0.50	202 ± 21	7.33	308
		G105	12.1 ± 0.5	510 ± 21	4.85 ± 0.51	204 ± 21	7.29	306
		S135	12.1 ± 0.5	510 ± 21	4.92 ± 0.51	207 ± 21	7.22	303
	15.50	E75	12.0 ± 0.6	504 ± 26	5.41 ± 0.61	227 ± 26	6.58	276
		X95	12.0 ± 0.6	505 ± 26	5.51 ± 0.61	231 ± 26	6.52	274
G105	12.0 ± 0.6	505 ± 26	5.59 ± 0.61	235 ± 26	6.44	271		
3 ¹ / ₂ " EU NC40	15.50	S135	12.3 ± 0.6	515 ± 26	5.78 ± 0.62	243 ± 26	6.47	272
4" IU NC40	14.00	E75	15.6 ± 0.5	656 ± 22	4.96 ± 0.52	208 ± 22	10.65	447
		X95	15.6 ± 0.5	656 ± 22	5.04 ± 0.52	212 ± 22	10.58	444
		G105	15.8 ± 0.5	662 ± 22	5.25 ± 0.52	220 ± 22	10.51	441
		S135	15.8 ± 0.5	662 ± 22	5.35 ± 0.52	225 ± 22	10.41	437
4" EU NC46	14.00	E75	16.1 ± 0.5	675 ± 22	5.26 ± 0.52	221 ± 22	10.82	454
		X95	16.1 ± 0.5	677 ± 22	5.37 ± 0.52	226 ± 22	10.76	452
		G105	16.1 ± 0.5	677 ± 22	5.37 ± 0.52	226 ± 22	10.76	452
		S135	16.1 ± 0.5	677 ± 22	5.45 ± 0.52	229 ± 22	10.68	448
4 ¹ / ₂ " IU NC46	13.75	E75	20.0 ± 0.5	839 ± 20	5.02 ± 0.48	211 ± 20	14.97	629
4 ¹ / ₂ " EU NC50	13.75	E75	20.4 ± 0.5	858 ± 20	5.30 ± 0.48	222 ± 20	15.14	636
	16.60	E75	20.3 ± 0.6	854 ± 25	6.12 ± 0.60	257 ± 25	14.20	596
		X95	20.4 ± 0.6	857 ± 25	6.26 ± 0.60	263 ± 25	14.14	594
		G105	20.4 ± 0.6	857 ± 25	6.26 ± 0.60	263 ± 25	14.14	594
		S135	20.4 ± 0.6	857 ± 25	6.35 ± 0.60	267 ± 25	14.04	590
	20.00	E75	20.2 ± 0.8	847 ± 32	7.28 ± 0.76	306 ± 32	12.88	541
		X95	20.2 ± 0.8	850 ± 32	7.45 ± 0.76	313 ± 32	12.78	537
G105		20.2 ± 0.8	850 ± 32	7.45 ± 0.76	313 ± 32	12.78	537	
S135		20.2 ± 0.8	850 ± 32	7.63 ± 0.76	320 ± 32	12.61	529	
4 ¹ / ₂ " IEU NC46	16.60	E75	20.0 ± 0.6	841 ± 25	6.09 ± 0.60	256 ± 25	13.94	585
		X95	20.0 ± 0.6	841 ± 25	6.18 ± 0.60	259 ± 25	13.85	582
		G105	20.0 ± 0.6	841 ± 25	6.18 ± 0.60	259 ± 25	13.85	582
		S135	20.0 ± 0.6	841 ± 25	6.25 ± 0.60	263 ± 25	13.77	578
	20.00	E75	19.9 ± 0.8	834 ± 32	7.29 ± 0.76	306 ± 32	12.58	528
		X95	19.9 ± 0.8	835 ± 32	7.47 ± 0.76	314 ± 32	12.42	522
		G105	19.9 ± 0.8	835 ± 32	7.54 ± 0.76	317 ± 32	12.35	519
		S135	19.9 ± 0.8	835 ± 32	7.60 ± 0.76	319 ± 32	12.29	516

The nominal dimensions of new drill pipe have been taken from API Spec 5D, 4th Edition, August 1999.

The dimensions of new tool joints have been taken from API Spec 7, 39th Edition, December 1997.

Premium Class and Class 2 drill pipe are as defined in API RP 7G 16th Edition, August 1998, Table 24 (reproduced on page C-4).

Size/Style Tool joint	Weight	Grade	Closed ended		Open ended		Capacity	
	lbs/ft		bbls/1000 ft	Gals/1000 ft	bbls/1000 ft	Gals/1000 ft	bbls/1000 ft	Gals/1000 ft
5" IEU NC50	19.50	E75	24.5 ± 0.7	1030 ± 30	7.06 ± 0.71	296 ± 30	17.46	733
		X95	24.5 ± 0.7	1031 ± 30	7.25 ± 0.71	304 ± 30	17.29	726
		G105	24.5 ± 0.7	1031 ± 30	7.38 ± 0.71	310 ± 30	17.16	721
		S135	24.5 ± 0.7	1031 ± 30	7.49 ± 0.71	315 ± 30	17.04	716
	25.60	E75	24.3 ± 1.0	1019 ± 41	8.98 ± 0.98	377 ± 41	15.28	642
		X95	24.3 ± 1.0	1019 ± 41	9.23 ± 0.98	388 ± 41	15.04	632
5" IEU 5 1/2" FH	19.50	E75	24.8 ± 0.7	1044 ± 30	7.40 ± 0.71	311 ± 30	17.45	733
		X95	24.9 ± 0.7	1044 ± 30	7.50 ± 0.71	315 ± 30	17.37	729
		G105	24.9 ± 0.7	1044 ± 30	7.50 ± 0.71	315 ± 30	17.37	729
		S135	25.1 ± 0.7	1053 ± 30	7.81 ± 0.72	328 ± 30	17.26	725
	25.60	E75	24.6 ± 1.0	1032 ± 41	9.32 ± 0.98	391 ± 41	15.26	641
		X95	24.6 ± 1.0	1033 ± 41	9.40 ± 0.98	395 ± 41	15.19	638
		G105	24.8 ± 1.0	1041 ± 41	9.61 ± 0.98	404 ± 41	15.19	638
		S135	24.8 ± 1.0	1041 ± 41	9.71 ± 0.98	408 ± 41	15.09	634
	21.90	E75	29.6 ± 0.8	1243 ± 33	7.88 ± 0.78	331 ± 33	21.72	912
		X95	29.6 ± 0.8	1244 ± 33	8.10 ± 0.78	340 ± 33	21.52	904
		G105	29.8 ± 0.8	1252 ± 33	8.41 ± 0.78	353 ± 33	21.41	899
		S135	30.0 ± 0.8	1261 ± 33	8.81 ± 0.79	370 ± 33	21.21	891
5 1/2" IEU 5 1/2" FH	24.70	E75	29.5 ± 0.9	1238 ± 38	8.68 ± 0.89	365 ± 38	20.80	874
		X95	29.7 ± 0.9	1247 ± 38	9.20 ± 0.90	386 ± 38	20.50	861
		G105	29.7 ± 0.9	1247 ± 38	9.20 ± 0.90	386 ± 38	20.50	861
		S135	29.9 ± 0.9	1256 ± 38	9.60 ± 0.90	403 ± 38	20.31	853
	25.20	E75	43.0 ± 0.9	1804 ± 36	9.16 ± 0.86	385 ± 36	33.79	1419
		X95	43.0 ± 0.9	1804 ± 36	9.16 ± 0.86	385 ± 36	33.79	1419
		G105	43.2 ± 0.9	1814 ± 36	9.54 ± 0.86	401 ± 36	33.64	1413
		S135	43.4 ± 0.9	1824 ± 36	10.1 ± 0.9	423 ± 36	33.36	1401
	27.70	E75	42.9 ± 0.9	1800 ± 40	9.76 ± 0.94	410 ± 40	33.11	1391
		X95	43.1 ± 0.9	1810 ± 40	10.1 ± 0.9	426 ± 40	32.97	1385
		G105	43.1 ± 0.9	1810 ± 40	10.1 ± 0.9	426 ± 40	32.97	1385
		S135	43.3 ± 0.9	1820 ± 40	10.7 ± 0.9	447 ± 40	32.69	1373

Note that there is no single figure that can be quoted for the displacement of a string of used drill pipe - it depends on the amount of wear. It is however assumed that the capacity remains unchanged from that of new pipe.

The drill pipe used in Shell operations will almost always be premium class which has, **by definition**, a wall thickness that can be anywhere between 80% and 100% of the nominal wall thickness (Refer to the classification table on [page C-4](#)). The ranges quoted in these tables take account of this possible variation - the high ends of the displacements of Premium pipe are equal to the values for new pipe, the low ends are the displacements that a string would have if every joint were worn (on the OD) to the maximum allowable degree - i.e. just before the joint would have to be reclassified as Class 2.

The quoted mid-point of the range (which is equivalent to just under 10% wear) will be sufficiently accurate for most cases. If you know, or can estimate, the actual wear, a linear interpolation within the tolerances quoted can be used to improve accuracy.

For completeness the displacement and capacity of Class 2 drill pipe can be found in the table on the following pages - also as a range rather than a single value, for the same reasons.

THE DISPLACEMENT AND CAPACITY OF A STRING OF PREMIUM CLASS DRILL PIPE, INCLUDING TOOL JOINTS

SI UNITS

Size/Style Tool joint	Weight		Grade	Displacement in litres/metre		Capacity in litres/metre
	lbs/ft	kg/m		Closed ended	Open ended	
2 3/8" EU 60.3 mm NC26	6.65 9.90	E75 X95 G105		2.86 ± 0.13	1.20 ± 0.13	1.66
				2.87 ± 0.13	1.21 ± 0.13	1.65
				2.87 ± 0.13	1.21 ± 0.13	1.65
2 7/8" EU 73.0 mm NC31	10.40 15.48	E75		4.20 ± 0.21	1.86 ± 0.21	2.34
		X95		4.21 ± 0.21	1.89 ± 0.21	2.32
		G105		4.21 ± 0.21	1.89 ± 0.21	2.32
		S135		4.26 ± 0.21	1.98 ± 0.21	2.28
3 1/2" EU 88.9 mm NC38	9.50 14.14	E75		6.32 ± 0.18	1.83 ± 0.18	4.50
	13.30 19.79	E75		6.24 ± 0.26	2.39 ± 0.26	3.86
		X95		6.33 ± 0.26	2.51 ± 0.26	3.82
		G105		6.33 ± 0.26	2.53 ± 0.26	3.80
		S135		6.33 ± 0.26	2.57 ± 0.26	3.76
	15.50 23.07	E75		6.26 ± 0.32	2.82 ± 0.32	3.43
		X95		6.28 ± 0.32	2.87 ± 0.32	3.40
3 1/2" EU 88.9 mm NC40	15.50 23.07	G105		6.28 ± 0.32	2.91 ± 0.32	3.36
		S135				
4" IU 101.6 mm NC40	14.00 20.83	E75		8.14 ± 0.27	2.59 ± 0.27	5.55
		X95		8.15 ± 0.27	2.63 ± 0.27	5.52
		G105		8.22 ± 0.27	2.74 ± 0.27	5.48
		S135		8.22 ± 0.27	2.79 ± 0.27	5.43
4" EU 101.6 mm NC46	14.00 20.83	E75		8.39 ± 0.27	2.74 ± 0.27	5.64
		X95		8.41 ± 0.27	2.80 ± 0.27	5.61
		G105		8.41 ± 0.27	2.80 ± 0.27	5.61
		S135		8.41 ± 0.27	2.84 ± 0.27	5.57
4 1/2" IU 114.3 mm NC46	13.75 20.46	E75		10.4 ± 0.2	2.62 ± 0.25	7.81
4 1/2" EU 114.3 mm NC50	13.75 20.46	E75		10.7 ± 0.3	2.76 ± 0.25	7.90
	16.60 24.70	E75		10.6 ± 0.3	3.19 ± 0.31	7.41
		X95		10.6 ± 0.3	3.26 ± 0.31	7.37
		G105		10.6 ± 0.3	3.26 ± 0.31	7.37
		S135		10.6 ± 0.3	3.31 ± 0.31	7.32
	20.00 29.76	E75		10.5 ± 0.4	3.80 ± 0.40	6.72
		X95		10.6 ± 0.4	3.89 ± 0.40	6.67
		G105		10.6 ± 0.4	3.89 ± 0.40	6.67
4 1/2" IEU 114.3 mm NC46	16.60 24.70	S135		10.6 ± 0.4	3.98 ± 0.40	6.58
		E75		10.4 ± 0.3	3.18 ± 0.31	7.27
		X95		10.4 ± 0.3	3.22 ± 0.31	7.22
		G105		10.4 ± 0.3	3.22 ± 0.31	7.22
		S135		10.4 ± 0.3	3.26 ± 0.31	7.18
	20.00 29.76	E75		10.4 ± 0.4	3.80 ± 0.40	6.56
		X95		10.4 ± 0.4	3.90 ± 0.40	6.48
		G105		10.4 ± 0.4	3.93 ± 0.40	6.44
		S135		10.4 ± 0.4	3.97 ± 0.40	6.41

The nominal dimensions of new drill pipe have been taken from API Spec 5D, 4th Edition, August 1999.

The dimensions of new tool joints have been taken from API Spec 7, 39th Edition, December 1997.

Premium Class and Class 2 drill pipe are as defined in API RP 7G 16th Edition, August 1998, Table 24 (reproduced on page C-4).

Size/Style Tool joint	Weight	Grade	Displacement in litres/metre		Capacity in litres/metre
	kg/m (lbs/ft)		Closed ended	Open ended	
5" IEU 127.0 mm NC50	19.50 29.02	E75	12.8 ± 0.4	3.68 ± 0.37	9.11
		X95	12.8 ± 0.4	3.78 ± 0.37	9.02
		G105	12.8 ± 0.4	3.85 ± 0.37	8.95
		S135	12.8 ± 0.4	3.91 ± 0.37	8.89
	25.60 38.10	E75	12.7 ± 0.5	4.68 ± 0.51	7.97
		X95 G105	12.7 ± 0.5 12.7 ± 0.5	4.82 ± 0.51 4.86 ± 0.51	7.84 7.80
5" IEU 127.0 mm 5½" FH	19.50 29.02	E75	13.0 ± 0.4	3.86 ± 0.37	9.10
		X95	13.0 ± 0.4	3.91 ± 0.37	9.06
		G105	13.0 ± 0.4	3.91 ± 0.37	9.06
		S135	13.1 ± 0.4	4.07 ± 0.37	9.00
	25.60 38.10	E75	12.8 ± 0.5	4.86 ± 0.51	7.96
		X95 G105 S135	12.8 ± 0.5 12.9 ± 0.5 12.9 ± 0.5	4.91 ± 0.51 5.01 ± 0.51 5.06 ± 0.51	7.93 7.92 7.87
5½" IEU 139.7 mm 5½" FH	21.90 32.59	E75	15.4 ± 0.4	4.11 ± 0.41	11.33
		X95	15.5 ± 0.4	4.23 ± 0.41	11.23
		G105	15.6 ± 0.4	4.39 ± 0.41	11.17
		S135	15.7 ± 0.4	4.59 ± 0.41	11.06
	24.70 36.76	E75	15.4 ± 0.5	4.53 ± 0.47	10.85
		X95 G105 S135	15.5 ± 0.5 15.5 ± 0.5 15.6 ± 0.5	4.80 ± 0.47 4.80 ± 0.47 5.01 ± 0.47	10.69 10.69 10.59
6½" IEU 168.3 mm 6½" FH	25.20 37.50	E75	22.4 ± 0.4	4.78 ± 0.45	17.62
		X95	22.4 ± 0.4	4.78 ± 0.45	17.63
		G105	22.5 ± 0.4	4.98 ± 0.45	17.55
		S135	22.6 ± 0.5	5.25 ± 0.45	17.40
	27.70 41.22	E75	22.4 ± 0.5	5.09 ± 0.49	17.27
		X95 G105 S135	22.5 ± 0.5 22.5 ± 0.5 22.6 ± 0.5	5.29 ± 0.49 5.29 ± 0.49 5.56 ± 0.49	17.20 17.20 17.05

Note that there is no single figure that can be quoted for the displacement of a string of used drill pipe - it depends on the amount of wear. It is however assumed that the capacity remains unchanged from that of new pipe.

The drill pipe used in Shell operations will almost always be premium class which has, **by definition**, a wall thickness that can be anywhere between 80% and 100% of the nominal wall thickness (Refer to the classification table on [Page C-4](#)). The ranges quoted in these tables take account of this possible variation - the high ends of the displacements of Premium pipe are equal to the values for new pipe, the low ends are the displacements that a string would have if every joint were worn (on the OD) to the maximum allowable degree - i.e. just before the joint would have to be reclassified as Class 2.

The quoted mid-point of the range (which is equivalent to just under 10% wear) will be sufficiently accurate for most cases. If you know, or can estimate, the actual wear, a linear interpolation within the tolerances quoted can be used to improve accuracy.

For completeness the displacement and capacity of Class 2 drill pipe can be found in the table on the following pages - also as a range rather than a single value, for the same reasons.

THE DISPLACEMENT AND CAPACITY OF A STRING OF CLASS 2 DRILL PIPE, INCLUDING TOOL JOINTS OILFIELD UNITS

Size/Style Tool joint	Weight	Grade	Closed ended		Open ended		Capacity	
	lbs/ft		bbls/1000 ft	Gals/1000 ft	bbls/1000 ft	Gals/1000 ft	bbls/1000 ft	Gals/1000 ft
2 ³ / ₈ " EU NC26	6.65	E75	5.10 ± 0.12	214 ± 5	1.91 ± 0.12	80.3 ± 5.2	3.19	134
		X95	5.11 ± 0.12	215 ± 5	1.94 ± 0.12	81.7 ± 5.2	3.17	133
		G105	5.11 ± 0.12	215 ± 5	1.94 ± 0.12	81.7 ± 5.2	3.17	133
2 ⁷ / ₈ " EU NC31	10.40	E75	7.45 ± 0.19	313 ± 8	2.96 ± 0.19	124.3 ± 8.2	4.49	189
		X95	7.47 ± 0.19	314 ± 8	3.03 ± 0.19	127.1 ± 8.2	4.44	187
		G105	7.47 ± 0.19	314 ± 8	3.03 ± 0.19	127.1 ± 8.2	4.44	187
		S135	7.57 ± 0.20	318 ± 8	3.19 ± 0.20	134.0 ± 8.2	4.38	184
3 ¹ / ₂ " EU NC38	9.50	E75	11.6 ± 0.2	487 ± 7	2.99 ± 0.17	125.4 ± 7.2	8.62	362
	13.30	E75	11.2 ± 0.2	471 ± 10	3.83 ± 0.24	161 ± 10	7.40	311
		X95	11.4 ± 0.2	478 ± 10	4.06 ± 0.24	171 ± 10	7.33	308
		G105	11.4 ± 0.2	478 ± 10	4.10 ± 0.24	172 ± 10	7.29	306
		S135	11.4 ± 0.2	478 ± 10	4.17 ± 0.24	175 ± 10	7.22	303
	15.50	E75	11.1 ± 0.3	466 ± 12	4.51 ± 0.29	189 ± 12	6.58	276
		X95	11.1 ± 0.3	467 ± 12	4.60 ± 0.29	193 ± 12	6.52	274
3 ¹ / ₂ " EU NC40	15.50	G105	11.1 ± 0.3	467 ± 12	4.68 ± 0.30	197 ± 12	6.44	271
		S135	11.3 ± 0.3	476 ± 12	4.87 ± 0.30	204 ± 12	6.47	272
4" IU NC40	14.00	E75	14.8 ± 0.3	623 ± 11	4.19 ± 0.25	176 ± 11	10.65	447
		X95	14.9 ± 0.3	624 ± 11	4.28 ± 0.25	180 ± 11	10.58	444
		G105	15.0 ± 0.3	630 ± 11	4.48 ± 0.25	188 ± 11	10.51	441
		S135	15.0 ± 0.3	629 ± 11	4.58 ± 0.25	192 ± 11	10.41	437
4" EU NC46	14.00	E75	15.3 ± 0.3	643 ± 11	4.49 ± 0.25	188 ± 11	10.82	454
		X95	15.4 ± 0.3	645 ± 11	4.60 ± 0.25	193 ± 11	10.76	452
		G105	15.4 ± 0.3	645 ± 11	4.60 ± 0.25	193 ± 11	10.76	452
		S135	15.4 ± 0.3	645 ± 11	4.68 ± 0.25	196 ± 11	10.68	448
4 ¹ / ₂ " IU NC46	13.75	E75	19.3 ± 0.2	809 ± 10	4.30 ± 0.24	181 ± 10	14.97	629
4 ¹ / ₂ " EU NC50	13.75	E75	19.7 ± 0.2	828 ± 10	4.58 ± 0.24	192 ± 10	15.14	636
	16.60	E75	19.4 ± 0.3	816 ± 12	5.23 ± 0.29	220 ± 12	14.20	596
		X95	19.5 ± 0.3	819 ± 12	5.37 ± 0.29	225 ± 12	14.14	594
		G105	19.5 ± 0.3	819 ± 12	5.37 ± 0.29	225 ± 12	14.14	594
		S135	19.5 ± 0.3	819 ± 12	5.46 ± 0.29	229 ± 12	14.04	590
	20.00	E75	19.0 ± 0.4	799 ± 16	6.15 ± 0.37	258 ± 16	12.88	541
		X95	19.1 ± 0.4	802 ± 16	6.32 ± 0.37	265 ± 16	12.78	537
		G105	19.1 ± 0.4	802 ± 16	6.32 ± 0.37	265 ± 16	12.78	537
		S135	19.1 ± 0.4	802 ± 16	6.49 ± 0.37	273 ± 16	12.61	529
4 ¹ / ₂ " IEU NC46	16.60	E75	19.1 ± 0.3	804 ± 12	5.20 ± 0.29	218 ± 12	13.94	585
		X95	19.1 ± 0.3	804 ± 12	5.29 ± 0.29	222 ± 12	13.85	582
		G105	19.1 ± 0.3	804 ± 12	5.29 ± 0.29	222 ± 12	13.85	582
		S135	19.1 ± 0.3	804 ± 12	5.36 ± 0.29	225 ± 12	13.77	578
	20.00	E75	18.7 ± 0.4	787 ± 15	6.16 ± 0.37	259 ± 15	12.58	528
		X95	18.8 ± 0.4	788 ± 15	6.34 ± 0.37	266 ± 15	12.42	522
		G105	18.8 ± 0.4	788 ± 15	6.41 ± 0.37	269 ± 15	12.35	519
		S135	18.8 ± 0.4	788 ± 16	6.47 ± 0.37	272 ± 16	12.29	516

Size/Style Tool joint	Weight	Grade	Closed ended		Open ended		Capacity	
	lbs/ft		bbls/1000 ft	Gals/1000 ft	bbls/1000 ft	Gals/1000 ft	bbls/1000 ft	Gals/1000 ft
5" IEU NC50	19.50	E75	23.5 ± 0.3	986 ± 15	6.00 ± 0.35	252 ± 15	17.46	733
		X95	23.5 ± 0.3	986 ± 15	6.19 ± 0.35	260 ± 15	17.29	726
		G105	23.5 ± 0.3	986 ± 15	6.32 ± 0.35	265 ± 15	17.16	721
		S135	23.5 ± 0.3	986 ± 15	6.43 ± 0.35	270 ± 15	17.04	716
	25.60	E75	22.8 ± 0.5	958 ± 20	7.53 ± 0.47	316 ± 20	15.28	642
		X95	22.8 ± 0.5	958 ± 20	7.78 ± 0.47	327 ± 20	15.04	632
		G105	22.8 ± 0.5	958 ± 20	7.86 ± 0.47	330 ± 20	14.96	628
5" IEU 5 1/2" FH	19.50	E75	23.8 ± 0.3	999 ± 15	6.34 ± 0.35	266 ± 15	17.45	733
		X95	23.8 ± 0.3	1000 ± 15	6.43 ± 0.35	270 ± 15	17.37	729
		G105	23.8 ± 0.3	1000 ± 15	6.43 ± 0.35	270 ± 15	17.37	729
		S135	24.0 ± 0.4	1008 ± 15	6.74 ± 0.35	283 ± 15	17.26	725
	25.60	E75	23.1 ± 0.5	971 ± 20	7.86 ± 0.48	330 ± 20	15.26	641
		X95	23.1 ± 0.5	972 ± 20	7.95 ± 0.48	334 ± 20	15.19	638
		G105	23.3 ± 0.5	980 ± 20	8.15 ± 0.48	342 ± 20	15.19	638
		S135	23.3 ± 0.5	980 ± 20	8.24 ± 0.48	346 ± 20	15.09	634
5 1/2" IEU 5 1/2" FH	21.90	E75	28.4 ± 0.4	1194 ± 16	6.72 ± 0.38	282 ± 16	21.72	912
		X95	28.5 ± 0.4	1195 ± 16	6.94 ± 0.38	292 ± 16	21.52	904
		G105	28.6 ± 0.4	1203 ± 16	7.24 ± 0.38	304 ± 16	21.41	899
		S135	28.8 ± 0.4	1212 ± 16	7.64 ± 0.39	321 ± 16	21.21	891
	24.70	E75	28.2 ± 0.4	1182 ± 18	7.35 ± 0.44	309 ± 18	20.80	874
		X95	28.4 ± 0.4	1191 ± 18	7.86 ± 0.44	330 ± 18	20.50	861
		G105	28.4 ± 0.4	1191 ± 18	7.86 ± 0.44	330 ± 18	20.50	861
		S135	28.6 ± 0.4	1199 ± 18	8.25 ± 0.44	347 ± 18	20.31	853
6 5/8" IEU 6 5/8" FH	25.20	E75	41.7 ± 0.4	1750 ± 18	7.88 ± 0.42	331 ± 18	33.79	1419
		X95	41.7 ± 0.4	1750 ± 18	7.88 ± 0.42	331 ± 18	33.79	1419
		G105	41.9 ± 0.4	1760 ± 18	8.26 ± 0.42	347 ± 18	33.64	1413
		S135	42.1 ± 0.4	1769 ± 18	8.77 ± 0.43	368 ± 18	33.36	1401
	27.70	E75	41.5 ± 0.5	1741 ± 19	8.35 ± 0.46	351 ± 19	33.11	1391
		X95	41.7 ± 0.5	1751 ± 20	8.73 ± 0.46	366 ± 20	32.97	1385
		G105	41.7 ± 0.5	1751 ± 20	8.73 ± 0.46	366 ± 20	32.97	1385
		S135	41.9 ± 0.5	1761 ± 20	9.24 ± 0.47	388 ± 20	32.69	1373

Note that there is no single figure that can be quoted for the displacement of a string of used drill pipe - it depends on the amount of wear. It is however assumed that the capacity remains unchanged from that of new pipe.

The drill pipe used in Shell operations will almost always be premium class but these tables for Class 2 pipe are included for completeness.

Class 2 pipe has, **by definition**, a wall thickness that can be anywhere between 70% and 80% of the nominal wall thickness (Refer to the classification table on [page C-4](#)). The ranges quoted in these tables take account of this possible variation - the high ends of the displacements of Class 2 pipe are equal to the values for Premium grade pipe, the low ends are the displacements that a string would have if every joint were worn (on the OD) to the maximum allowable degree - i.e. just before the joint would have to be reclassified as Class 3, which is not used in Shell operations.

THE DISPLACEMENT AND CAPACITY OF A STRING OF CLASS 2 DRILL PIPE, INCLUDING TOOL JOINTS

SI UNITS

Size/Style Tool joint	Weight	Grade	Displacement in litres/metre		Capacity in litres/metre
	lbs/ft kg/m		Closed ended	Open ended	
2 3/8" EU 60.3 mm NC26	6.65	E75	2.66 ± 0.06	1.00 ± 0.06	1.66
	9.90	X95	2.67 ± 0.06	1.01 ± 0.06	1.65
		G105	2.67 ± 0.06	1.01 ± 0.06	1.65
2 7/8" EU 73 mm NC31	10.40	E75	3.89 ± 0.10	1.54 ± 0.10	2.34
	15.48	X95	3.90 ± 0.10	1.58 ± 0.10	2.32
		G105	3.90 ± 0.10	1.58 ± 0.10	2.32
		S135	3.95 ± 0.10	1.66 ± 0.10	2.28
3 1/2" EU 88.9 mm NC38	9.50	E75	6.05 ± 0.09	1.56 ± 0.09	4.50
	14.14				
	13.30	E75	5.86 ± 0.13	2.00 ± 0.13	3.86
	19.79	X95	5.94 ± 0.13	2.12 ± 0.13	3.82
		G105	5.94 ± 0.13	2.14 ± 0.13	3.80
		S135	5.94 ± 0.13	2.18 ± 0.13	3.76
	15.50	E75	5.78 ± 0.15	2.35 ± 0.15	3.43
3 1/2" mm 88.9 mm NC40	23.07	X95	5.80 ± 0.15	2.40 ± 0.15	3.40
		G105	5.80 ± 0.15	2.44 ± 0.15	3.36
4" IU 101.6 mm NC40	15.50	S135	5.92 ± 0.15	2.54 ± 0.15	3.38
	23.07				
4" EU 101.6 mm NC46	14.00	E75	7.74 ± 0.13	2.19 ± 0.13	5.55
	20.83	X95	7.75 ± 0.13	2.23 ± 0.13	5.52
		G105	7.82 ± 0.13	2.34 ± 0.13	5.48
		S135	7.82 ± 0.13	2.39 ± 0.13	5.43
4" EU 101.6 mm NC46	20.83	E75	7.98 ± 0.13	2.34 ± 0.13	5.64
	20.83	X95	8.01 ± 0.13	2.40 ± 0.13	5.61
		G105	8.01 ± 0.13	2.40 ± 0.13	5.61
		S135	8.01 ± 0.13	2.44 ± 0.13	5.57
4 1/2" IU 114.3 mm NC46	13.75	E75	10.1 ± 0.1	2.24 ± 0.12	7.81
	20.46				
4 1/2" EU 114.3 mm NC50	13.75	E75	10.3 ± 0.1	2.39 ± 0.12	7.90
	20.46				
	16.60	E75	10.1 ± 0.2	2.73 ± 0.15	7.41
	24.70	X95	10.2 ± 0.2	2.80 ± 0.15	7.37
		G105	10.2 ± 0.2	2.80 ± 0.15	7.37
		S135	10.2 ± 0.2	2.85 ± 0.15	7.32
4 1/2" IEU 114.3 mm NC46	20.00	E75	9.93 ± 0.19	3.21 ± 0.19	6.72
	29.76	X95	10.0 ± 0.2	3.30 ± 0.19	6.67
		G105	10.0 ± 0.2	3.30 ± 0.19	6.67
		S135	10.0 ± 0.2	3.39 ± 0.19	6.58
	16.60	E75	10.0 ± 0.2	2.71 ± 0.15	7.27
	24.70	X95	10.0 ± 0.2	2.76 ± 0.15	7.22
		G105	10.4 ± 0.3	3.22 ± 0.31	7.22
		S135	10.0 ± 0.2	2.80 ± 0.15	7.18
	20.00	E75	9.78 ± 0.19	3.21 ± 0.19	6.56
	29.76	X95	9.79 ± 0.19	3.31 ± 0.19	6.48
		G105	9.79 ± 0.19	3.34 ± 0.19	6.44
		S135	9.79 ± 0.19	3.38 ± 0.19	6.41

Size/Style Tool joint	Weight	Grade	Displacement in litres/metre		Capacity in litres/metre
	lbs/ft kg/m		Closed ended	Open ended	
5" IEU 127.0 mm NC50	19.50 29.02	E75	12.2 ± 0.2	3.13 ± 0.18	9.11
		X95	12.2 ± 0.2	3.23 ± 0.18	9.02
		G105	12.2 ± 0.2	3.30 ± 0.18	8.95
		S135	12.2 ± 0.2	3.36 ± 0.18	8.89
	25.60 38.10	E75	11.9 ± 0.2	3.93 ± 0.25	7.97
		X95	11.9 ± 0.2	4.06 ± 0.25	7.84
5" IEU 127.0 mm 5 1/2" FH	19.50 29.02	G105	11.9 ± 0.2	4.10 ± 0.25	7.80
		S135	12.4 ± 0.2	3.31 ± 0.18	9.10
		X95	12.4 ± 0.2	3.36 ± 0.18	9.06
		G105	12.4 ± 0.2	3.36 ± 0.18	9.06
	25.60 38.10	S135	12.5 ± 0.2	3.52 ± 0.18	9.00
		E75	12.1 ± 0.2	4.10 ± 0.25	7.96
		X95	12.1 ± 0.2	4.15 ± 0.25	7.93
		G105	12.2 ± 0.2	4.25 ± 0.25	7.92
5 1/2" IEU 139.7 mm 5 1/2" FH	21.90 32.59	S135	12.2 ± 0.2	4.30 ± 0.25	7.87
		E75	14.8 ± 0.2	3.50 ± 0.20	11.33
		X95	14.8 ± 0.2	3.62 ± 0.20	11.23
		G105	14.9 ± 0.2	3.78 ± 0.20	11.17
	24.70 36.76	S135	15.0 ± 0.2	3.98 ± 0.20	11.06
		E75	14.7 ± 0.2	3.83 ± 0.23	10.85
		X95	14.8 ± 0.2	4.10 ± 0.23	10.69
		G105	14.8 ± 0.2	4.10 ± 0.23	10.69
		S135	14.9 ± 0.2	4.31 ± 0.23	10.59
	25.20 37.50	E75	21.7 ± 0.2	4.11 ± 0.22	17.62
		X95	21.7 ± 0.2	4.11 ± 0.22	17.63
		G105	21.9 ± 0.2	4.31 ± 0.22	17.55
		S135	22.0 ± 0.2	4.57 ± 0.22	17.40
6 5/8" IEU 168.3 mm 6 5/8" FH	27.70 41.22	E75	21.6 ± 0.2	4.36 ± 0.24	17.27
		X95	21.7 ± 0.2	4.55 ± 0.24	17.20
		G105	21.7 ± 0.2	4.55 ± 0.24	17.20
		S135	21.9 ± 0.2	4.82 ± 0.24	17.05

Note that there is no single figure that can be quoted for the displacement of a string of used drill pipe - it depends on the amount of wear. It is however assumed that the capacity remains unchanged from that of new pipe.

The drill pipe used in Shell operations will almost always be premium class but these tables for Class 2 pipe are included for completeness.

Class 2 pipe has, **by definition**, a wall thickness that can be anywhere between 70% and 80% of the nominal wall thickness (Refer to the classification table on [page C-4](#)). The ranges quoted in these tables take account of this possible variation - the high ends of the displacements of Class 2 pipe are equal to the values for Premium grade pipe, the low ends are the displacements that a string would have if every joint were worn (on the OD) to the maximum allowable degree - i.e. just before the joint would have to be reclassified as Class 3, which is not used in Shell operations.

TENSILE STRENGTH OF DRILL PIPE

Nominal OD weight inches lbs/ft		Yield strength in tension, in lbs											
		New drill pipe				Premium class drill pipe				Class 2 drill pipe			
		E75	X95	G105	S135	E75	X95	G105	S135	E75	X95	G105	S135
2 ³ / ₈	4.85	86,520	109,600	121,100	155,700	76,890	97,400	107,700	138,400	66,690	84,470	93,360	120,000
	6.65	123,000	155,700	172,100	221,300	107,600	136,300	150,700	193,700	92,870	117,600	130,000	167,200
2 ⁷ / ₈	6.85	120,100	152,200	168,200	216,200	106,900	135,500	149,700	192,500	92,800	117,500	129,900	167,000
	10.40	190,900	241,800	267,300	343,700	166,500	210,900	233,100	299,800	143,600	181,800	201,000	258,400
3 ¹ / ₂	9.50	171,600	217,400	240,300	309,000	153,000	193,800	214,200	275,400	132,800	168,200	185,900	239,000
	13.30	241,100	305,400	337,600	434,000	212,200	268,700	297,000	381,900	183,400	232,300	256,800	330,100
	15.50	287,600	364,300	402,700	517,700	250,600	317,500	350,900	451,100	216,000	273,600	302,400	388,700
4	11.85	203,700	258,000	285,200	366,600	182,000	230,600	254,800	327,600	158,100	200,300	221,400	284,600
	14.00	252,500	319,800	353,500	454,500	224,200	284,000	313,900	403,500	194,400	246,200	272,100	349,900
	15.70	287,300	363,900	402,300	517,200	253,900	321,500	355,400	456,900	219,700	278,300	307,600	395,500
4 ¹ / ₂	13.75	238,200	301,700	333,400	428,700	213,300	270,100	298,600	383,900	185,400	234,800	259,500	333,700
	16.60	292,200	370,100	409,000	525,900	260,200	329,500	364,200	468,300	225,800	286,000	316,100	406,400
	20.00	365,600	463,100	511,800	658,000	322,900	409,000	452,100	581,200	279,500	354,000	391,300	503,100
	22.82	418,800	530,500	586,300	753,800	367,600	465,600	514,600	661,600	317,500	402,200	444,500	571,500
5	16.25	289,300	366,500	405,100	520,800	259,200	328,300	362,800	466,500	225,300	285,400	315,400	405,600
	19.50	349,500	442,700	489,300	629,100	311,500	394,600	436,100	560,800	270,400	342,500	378,600	486,800
	25.60	470,300	595,700	658,400	846,600	414,700	525,300	580,600	746,400	358,700	454,400	502,200	645,700
5 ¹ / ₂	19.20	328,000	415,500	459,300	590,500	294,300	372,700	412,000	529,700	256,000	324,200	358,300	460,700
	21.90	385,800	488,700	540,200	694,500	344,800	436,700	482,700	620,600	299,500	379,400	419,300	539,200
	24.70	439,500	556,700	615,300	791,100	391,300	495,600	547,800	704,300	339,500	430,100	475,300	611,200
6 ⁵ / ₈	25.20	431,100	546,000	603,500	776,000	387,500	490,800	542,500	697,400	337,200	427,200	472,100	607,000
	27.70	470,800	596,300	659,100	847,400	422,400	535,100	591,400	760,400	367,500	465,400	514,400	661,400

Nominal OD weight mm kg/m		Yield strength in tension, in kdaNs											
		New drill pipe				Premium class drill pipe				Class 2 drill pipe			
		E75	X95	G105	S135	E75	X95	G105	S135	E75	X95	G105	S135
60.3	7.22	38.49	48.75	53.88	69.28	34.20	43.32	47.89	61.57	29.66	37.57	41.53	53.39
	9.90	54.69	69.28	76.57	98.45	47.87	60.64	67.02	86.17	41.31	52.33	57.84	74.36
73.0	10.19	53.44	67.68	74.81	96.18	47.57	60.26	66.60	85.63	41.28	52.29	57.79	74.30
	15.48	84.93	107.6	118.9	152.9	74.08	93.83	103.7	133.3	63.86	80.89	89.40	114.9
88.9	14.14	76.35	96.71	106.9	137.4	68.05	86.19	95.3	122.5	59.07	74.82	82.70	106.3
	19.79	107.3	135.9	150.2	193.1	94.37	119.5	132.1	169.9	81.58	103.3	114.2	146.8
	23.07	127.9	162.1	179.1	230.3	111.5	141.2	156.1	200.7	96.07	121.7	134.5	172.9
101.6	17.64	90.60	114.8	126.8	163.1	80.96	102.6	113.4	145.7	70.34	89.10	98.48	126.6
	20.83	112.3	142.3	157.2	202.2	99.72	126.3	139.6	179.5	86.46	109.5	121.0	155.6
	23.36	127.8	161.9	178.9	230.1	112.9	143.0	158.1	203.3	97.74	123.8	136.8	175.9
114.3	20.46	105.9	134.2	148.3	190.7	94.86	120.2	132.8	170.8	82.47	104.5	115.5	148.4
	24.70	130.0	164.6	181.9	233.9	115.7	146.6	162.0	208.3	100.4	127.2	140.6	180.8
	29.76	162.6	206.0	227.7	292.7	143.6	181.9	201.1	258.6	124.3	157.5	174.1	223.8
	33.96	186.3	236.0	260.8	335.3	163.5	207.1	228.9	294.3	141.2	178.9	197.7	254.2
127.0	24.18	128.7	163.0	180.2	231.7	115.3	146.0	161.4	207.5	100.2	127.0	140.3	180.4
	29.02	155.5	196.9	217.7	279.9	138.6	175.5	194.0	249.4	120.3	152.4	168.4	216.5
	38.10	209.2	265.0	292.9	376.6	184.5	233.7	258.2	332.0	159.6	202.1	223.4	287.2
139.7	28.57	145.9	184.8	204.3	262.7	130.9	165.8	183.3	235.6	113.9	144.2	159.4	204.9
	32.59	171.6	217.4	240.3	308.9	153.4	194.3	214.7	276.1	133.2	168.8	186.5	239.8
	36.76	195.5	247.6	273.7	351.9	174.1	220.5	243.7	313.3	151.0	191.3	211.4	271.9
168.3	37.50	191.8	242.9	268.5	345.2	172.4	218.3	241.3	310.2	150.0	190.0	210.0	270.0
	41.22	209.4	265.3	293.2	377.0	187.9	238.0	263.1	338.2	163.5	207.0	228.8	294.2

Notes

The tensile strength of drill pipe is based on the "worst case" combination of dimensions allowable under the classification scheme (see page C-4). For new pipe this is the nominal OD with the minimum allowable wall thickness as defined in the specifications. For used pipe it is nominal ID with the maximum allowable wear having occurred on the OD. See also the notes on page C-5

No safety factors have been included in these tabulated values.

The nominal dimensions and wall thickness tolerances of new drill pipe have been taken from API Spec 5D, 4th Edition, August 1999. Premium Class and Class 2 drill pipe are as defined in API RP 7G 16th Edition, August 1998, Table 24 (reproduced on page C-4).

TORSIONAL STRENGTH OF DRILL PIPE

Nominal		Torsional yield strength*, in lbs-ft											
OD inches	weight lbs/ft	New drill pipe				Premium class drill pipe				Class 2 drill pipe			
		E75	X95	G105	S135	E75	X95	G105	S135	E75	X95	G105	S135
2 ³ / ₈	4.85	4,297	5,443	6,016	7,735	3,725	4,719	5,215	6,705	3,224	4,083	4,513	5,802
	6.65	5,722	7,247	8,010	10,300	4,811	6,093	6,735	8,659	4,130	5,232	5,782	7,434
2 ⁷ / ₈	6.85	7,279	9,220	10,190	13,100	6,332	8,020	8,865	11,400	5,484	6,946	7,677	9,871
	10.40	10,610	13,440	14,850	19,100	8,858	11,220	12,400	15,940	7,591	9,615	10,630	13,660
3 ¹ / ₂	9.50	12,730	16,120	17,820	22,910	11,090	14,050	15,530	19,970	9,612	12,180	13,460	17,300
	13.30	16,900	21,410	23,660	30,420	14,360	18,190	20,110	25,850	12,370	15,660	17,310	22,260
	15.50	19,380	24,550	27,130	34,880	16,150	20,450	22,600	29,060	13,830	17,520	19,360	24,890
4	11.85	17,470	22,130	24,460	31,450	15,310	19,390	21,430	27,560	13,280	16,820	18,590	23,910
	14.00	21,030	26,640	29,440	37,850	18,200	23,050	25,470	32,750	15,740	19,940	22,030	28,330
	15.70	23,420	29,660	32,790	42,150	20,070	25,420	28,090	36,120	17,310	21,930	24,240	31,170
4 ¹ / ₂	13.75	23,190	29,380	32,470	41,750	20,400	25,840	28,560	36,720	17,720	22,440	24,800	31,890
	16.60	27,740	35,130	38,830	49,930	24,140	30,580	33,790	43,450	20,910	26,480	29,270	37,630
	20.00	33,990	42,420	46,890	60,280	28,680	36,330	40,160	51,630	24,750	31,350	34,650	44,540
	22.82	37,350	47,310	52,290	67,240	31,590	40,010	44,220	56,860	27,160	34,400	38,030	48,890
5	16.25	31,360	39,730	43,910	56,450	27,610	34,970	38,650	49,690	23,970	30,370	33,560	43,150
	19.50	37,030	46,900	51,840	66,650	32,290	40,890	45,200	58,110	27,980	35,440	39,170	50,360
	25.60	47,510	60,180	66,510	85,510	40,540	51,360	56,760	72,980	34,950	44,270	48,930	62,910
5 ¹ / ₂	19.20	39,380	49,890	55,140	70,890	34,760	44,030	48,670	62,580	30,210	38,260	42,290	54,370
	21.90	45,500	57,630	63,690	81,890	39,860	50,490	55,810	71,750	34,580	43,800	48,410	62,250
	24.70	50,950	64,530	71,330	91,710	44,320	56,140	62,050	79,780	38,380	48,620	53,740	69,090
6 ⁵ / ₈	25.20	62,940	79,720	88,110	113,300	55,770	70,640	78,070	100,400	48,500	61,430	67,900	87,290
	27.70	68,160	86,340	95,420	122,700	60,190	76,240	84,270	108,300	52,310	66,260	73,230	94,150

Nominal		Torsional yield strength*, in daN-m											
OD mm	weight kg/m	New drill pipe				Premium class drill pipe				Class 2 drill pipe			
		E75	X95	G105	S135	E75	X95	G105	S135	E75	X95	G105	S135
60.3	7.22	583	738	816	1,049	505	640	707	909	437	554	612	787
	9.90	776	983	1,086	1,396	652	826	913	1,174	560	709	784	1,008
73.0	10.19	987	1,250	1,382	1,776	858	1,087	1,202	1,545	743	942	1,041	1,338
	15.48	1,438	1,822	2,014	2,589	1,201	1,521	1,681	2,162	1,029	1,304	1,441	1,852
88.9	14.14	1,725	2,185	2,415	3,106	1,504	1,905	2,106	2,707	1,303	1,651	1,825	2,346
	19.79	2,291	2,902	3,208	4,124	1,947	2,466	2,726	3,505	1,677	2,124	2,347	3,018
101.6	23.07	2,627	3,328	3,678	4,729	2,189	2,773	3,065	3,940	1,875	2,375	2,625	3,375
	17.64	2,369	3,000	3,316	4,264	2,076	2,629	2,906	3,736	1,801	2,281	2,521	3,241
114.3	20.83	2,851	3,611	3,992	5,132	2,467	3,125	3,454	4,441	2,134	2,703	2,987	3,841
	23.36	3,175	4,022	4,445	5,715	2,721	3,446	3,809	4,897	2,348	2,974	3,287	4,226
127.0	20.46	3,145	3,983	4,403	5,661	2,766	3,504	3,873	4,979	2,402	3,042	3,363	4,323
	24.70	3,761	4,764	5,265	6,769	3,273	4,146	4,582	5,891	2,835	3,591	3,969	5,102
139.7	29.76	4,541	5,751	6,357	8,173	3,889	4,926	5,445	7,000	3,355	4,250	4,697	6,039
	33.96	5,064	6,415	7,090	9,116	4,283	5,425	5,996	7,709	3,683	4,665	5,156	6,629
168.3	24.18	4,252	5,386	5,953	7,654	3,743	4,741	5,240	6,737	3,250	4,117	4,551	5,851
	29.02	5,021	6,359	7,029	9,037	4,377	5,545	6,128	7,879	3,793	4,804	5,310	6,827
199.8	38.10	6,441	8,159	9,018	11,590	5,497	6,963	7,696	9,895	4,738	6,002	6,634	8,529
	28.57	5,340	6,764	7,476	9,611	4,713	5,970	6,599	8,484	4,096	5,188	5,734	7,372
229.0	32.59	6,168	7,813	8,636	11,100	5,405	6,846	7,567	9,729	4,689	5,939	6,564	8,440
	36.76	6,908	8,750	9,671	12,430	6,009	7,611	8,413	10,820	5,204	6,592	7,286	9,367
271.4	37.50	8,533	10,810	11,950	15,360	7,561	9,577	10,590	13,610	6,575	8,329	9,205	11,840
	41.22	9,241	11,710	12,940	16,630	8,161	10,337	11,430	14,690	7,092	8,983	9,929	12,770

Notes

The strength of drill pipe in torsion is based on the "worst case" combination of dimensions allowable under the classification scheme (see page C-4). For new pipe this is the nominal OD with the minimum allowable wall thickness as defined in the specifications. For used pipe it is nominal ID with the maximum allowable wear having occurred on the OD. See also the notes on page C-5.

The calculation of the torsional strength data is based on the shear strength of the material being equal to 57.7% of its minimum yield strength, as per API RP 7G.

No safety factors have been included in these tabulated values.

The nominal dimensions and wall thickness tolerances of new drill pipe have been taken from API Spec 5D, 4th Edition, August 1999. Premium Class and Class 2 drill pipe are as defined in API RP 7G 16th Edition, August 1998, Table 24 (reproduced on page C-4).

BURST RESISTANCE OF DRILL PIPE

Nominal OD weight inches lbs/ft		Minimum burst pressure, in psi											
		New drill pipe				Premium class drill pipe				Class 2 drill pipe			
		E75	X95	G105	S135	E75	X95	G105	S135	E75	X95	G105	S135
2 ³ / ₈	4.85	10,500	13,300	14,700	18,900	9,600	12,160	13,440	17,280	8,400	10,640	11,760	15,120
	6.65	15,470	19,600	21,660	27,850	14,150	17,920	19,810	25,470	12,380	15,680	17,330	22,280
2 ⁷ / ₈	6.85	9,907	12,550	13,870	17,830	9,057	11,470	12,680	16,300	7,925	10,040	11,100	14,270
	10.40	16,530	20,930	23,140	29,750	15,110	19,140	21,150	27,200	13,220	16,750	18,510	23,800
3 ¹ / ₂	9.50	9,525	12,070	13,340	17,150	8,709	11,030	12,190	15,680	7,620	9,652	10,670	13,720
	13.30	13,800	17,480	19,320	24,840	12,620	15,980	17,660	22,710	11,040	13,980	15,460	19,870
4	15.50	16,840	21,330	23,570	30,310	15,390	19,500	21,550	27,710	13,470	17,060	18,860	24,250
	11.85	8,597	10,890	12,040	15,470	7,860	9,956	11,000	14,150	6,878	8,712	9,629	12,380
4 ¹ / ₂	14.00	10,830	13,720	15,160	19,490	9,900	12,540	13,860	17,820	8,663	10,970	12,130	15,590
	15.70	12,470	15,790	17,460	22,440	11,400	14,440	15,960	20,520	9,975	12,640	13,970	17,960
5	13.75	7,904	10,010	11,070	14,230	7,227	9,154	10,120	13,010	6,323	8,010	8,853	11,380
	16.60	9,829	12,450	13,760	17,690	8,987	11,380	12,580	16,180	7,863	9,960	11,010	14,150
5 ¹ / ₂	20.00	12,540	15,890	17,560	22,580	11,470	14,520	16,050	20,640	10,030	12,710	14,050	18,060
	22.82	14,580	18,470	20,420	26,250	13,330	16,890	18,670	24,000	11,670	14,780	16,330	21,000
6 ⁵ / ₈	16.25	7,770	9,842	10,880	13,990	7,104	8,998	9,946	12,790	6,216	7,874	8,702	11,190
	19.50	9,503	12,040	13,300	17,100	8,688	11,000	12,160	15,640	7,602	9,629	10,640	13,680
6 ⁵ / ₈	25.60	13,130	16,630	18,380	23,630	12,000	15,200	16,800	21,600	10,500	13,300	14,700	18,900
	19.20	7,255	9,189	10,160	13,060	6,633	8,401	9,286	11,940	5,804	7,351	8,125	10,450
6 ⁵ / ₈	21.90	8,615	10,910	12,060	15,510	7,876	9,977	11,030	14,180	6,892	8,730	9,649	12,410
	24.70	9,903	12,540	13,860	17,830	9,055	11,470	12,680	16,300	7,923	10,040	11,090	14,260
6 ⁵ / ₈	25.20	6,538	8,281	9,153	11,770	5,977	7,571	8,368	10,760	5,230	6,625	7,322	9,414
	27.70	7,172	9,084	10,040	12,910	6,557	8,306	9,180	11,800	5,737	7,267	8,032	10,330

Nominal OD weight mm kg/m		Minimum burst pressure, in MPa											
		New drill pipe				Premium class drill pipe				Class 2 drill pipe			
		E75	X95	G105	S135	E75	X95	G105	S135	E75	X95	G105	S135
60.3	7.22	72.39	91.70	101.4	130.3	66.19	83.84	92.67	119.1	57.92	73.36	81.08	104.2
	9.90	106.7	135.1	149.4	192.0	97.54	123.6	136.6	175.6	85.35	108.1	119.5	153.6
73.0	10.19	68.30	86.52	95.62	122.9	62.45	79.10	87.43	112.4	54.64	69.21	76.50	98.36
	15.48	113.9	144.3	159.5	205.1	104.2	132.0	145.8	187.5	91.15	115.5	127.6	164.1
88.9	14.14	65.67	83.19	91.94	118.2	60.04	76.06	84.06	108.1	52.54	66.55	73.55	94.57
	19.79	95.15	120.5	133.2	171.3	86.99	110.2	121.8	156.6	76.12	96.42	106.6	137.0
101.6	23.07	116.1	147.0	162.5	209.0	106.1	134.4	148.6	191.1	92.87	117.6	130.0	167.2
	17.64	59.27	75.08	82.98	106.7	54.19	68.64	75.87	97.55	47.42	60.06	66.39	85.35
114.3	20.83	74.66	94.57	104.5	134.4	68.26	86.46	95.56	122.9	59.73	75.65	83.62	107.5
	23.36	85.97	108.9	120.4	154.7	78.60	99.56	110.0	141.5	68.78	87.12	96.29	123.8
127.0	20.46	54.50	69.03	76.30	98.10	49.83	63.11	69.76	89.69	43.60	55.22	61.04	78.48
	24.70	67.77	85.84	94.88	122.0	61.96	78.48	86.75	111.5	54.22	68.67	75.90	97.59
139.7	29.76	86.47	109.5	121.1	155.6	79.06	100.1	110.7	142.3	69.18	87.62	96.85	124.5
	33.96	100.5	127.4	140.8	181.0	91.93	116.4	128.7	165.5	80.44	101.9	112.6	144.8
168.3	24.18	53.57	67.86	75.00	96.43	48.98	62.04	68.57	88.16	42.86	54.29	60.00	77.14
	29.02	65.52	82.99	91.72	117.9	59.90	75.88	83.86	107.8	52.41	66.39	73.38	94.35
168.3	38.10	90.49	114.6	126.7	162.9	82.74	104.8	115.8	148.9	72.39	91.70	101.4	130.3
	28.57	50.02	63.36	70.03	90.03	45.73	57.93	64.02	82.32	40.01	50.69	56.02	72.03
168.3	32.59	59.40	75.24	83.16	106.9	54.31	68.79	76.03	97.75	47.52	60.19	66.52	85.53
	36.76	68.28	86.49	95.59	122.9	62.43	79.08	87.40	112.4	54.63	69.19	76.48	98.33
168.3	37.50	45.08	57.10	63.11	81.14	41.21	52.20	57.70	74.18	36.06	45.68	50.49	64.91
	41.22	49.45	62.63	69.23	89.00	45.21	57.26	63.29	81.38	39.56	50.11	55.38	71.20

Notes

The burst resistance of drill pipe is based on the "worst case" combination of dimensions allowable under the classification scheme (see page C-4). For new pipe this is the nominal OD with the minimum allowable wall thickness as defined in the specifications. For used pipe it is nominal OD with maximum allowable wear having occurred on the ID. In practice it is unlikely that there will be very much wear on the ID compared with the wear on the OD, but the "worst case" scenario has to be taken into account for critical strengths. See also the notes on page C-5

No safety factors have been included in these tabulated values.

The nominal dimensions and wall thickness tolerances of new drill pipe have been taken from API Spec 5D, 4th Edition, August 1999. Premium Class and Class 2 drill pipe are as defined in API RP 7G 16th Edition, August 1998, Table 24 (reproduced on page C-4).

COLLAPSE RESISTANCE OF DRILL PIPE

Nominal OD weight inches lbs/ft		Minimum collapse pressure, in psi											
		New drill pipe				Premium class drill pipe				Class 2 drill pipe			
		E75	X95	G105	S135	E75	X95	G105	S135	E75	X95	G105	S135
2 3/8	4.85	9,412	11,310	12,200	14,610	8,038	9,533	10,210	11,960	6,206	7,158	7,554	8,415
	6.65	13,880	17,580	19,430	24,980	12,810	16,230	17,940	23,060	11,360	14,390	15,900	20,150
2 7/8	6.85	8,506	10,140	10,890	12,860	7,209	8,459	9,009	10,350	5,481	6,219	6,503	7,014
	10.40	14,710	18,630	20,590	26,470	13,590	17,210	19,020	24,460	12,060	15,270	16,880	21,700
3 1/2	9.50	7,923	9,384	10,040	11,730	6,677	7,769	8,237	9,325	5,015	5,615	5,828	6,321
	13.30	12,530	15,870	17,540	22,550	11,560	14,640	16,180	20,800	10,230	12,380	13,400	16,200
	15.50	14,950	18,930	20,930	26,910	13,810	17,500	19,340	24,870	12,260	15,530	17,160	22,070
4	11.85	6,506	7,548	7,990	8,995	5,381	6,090	6,359	6,822	3,881	4,252	4,472	4,896
	14.00	9,913	11,960	12,930	15,580	8,496	10,130	10,870	12,840	6,606	7,678	8,135	9,189
	15.70	11,430	14,480	16,010	20,420	10,530	13,090	14,190	17,260	8,610	10,270	11,040	13,060
4 1/2	13.75	5,448	6,177	6,457	6,952	4,414	4,837	4,974	5,566	3,154	3,540	3,676	3,834
	16.60	8,388	9,986	10,720	12,630	7,101	8,319	8,853	10,150	5,386	6,096	6,366	6,831
	20.00	11,490	14,560	16,090	20,630	10,590	13,230	14,340	17,460	8,699	10,390	11,170	13,230
	22.82	13,170	16,680	18,430	23,700	12,150	15,390	17,010	21,870	10,760	13,620	14,780	18,050
5	16.25	5,244	5,912	6,160	6,609	4,227	4,594	4,797	5,330	3,048	3,402	3,521	3,636
	19.50	7,889	9,340	9,994	11,670	6,645	7,728	8,192	9,264	4,987	5,579	5,788	6,286
	25.60	11,980	15,170	16,770	21,560	11,040	13,980	15,460	19,030	9,412	11,310	12,200	14,610
5 1/2	19.20	4,457	4,892	5,019	5,619	3,507	3,938	4,120	4,426	2,637	2,873	2,929	2,943
	21.90	6,533	7,583	8,029	9,048	5,406	6,122	6,395	6,870	3,903	4,271	4,492	4,923
	24.70	8,501	10,130	10,880	12,850	7,205	8,454	9,003	10,350	5,477	6,214	6,498	7,007
6 5/8	25.20	3,368	3,816	3,984	4,243	2,810	3,096	3,178	3,223	2,067	2,137	2,137	2,137
	27.70	4,330	4,728	4,895	5,460	3,392	3,840	4,011	4,280	2,571	2,787	2,833	2,840

Nominal OD weight mm kg/m		Minimum collapse pressure, in MPa											
		New drill pipe				Premium class drill pipe				Class 2 drill pipe			
		E75	X95	G105	S135	E75	X95	G105	S135	E75	X95	G105	S135
60.3	7.22	64.89	78.01	84.13	100.7	55.42	65.73	70.40	82.42	42.79	49.36	52.08	58.02
	9.90	95.68	121.2	134.0	172.2	88.34	111.9	123.7	159.0	78.31	99.19	109.6	138.9
73.0	10.19	58.65	69.91	75.07	88.66	49.71	58.32	62.12	71.39	37.79	42.88	44.84	48.36
	15.48	101.4	128.4	141.9	182.5	93.68	118.7	131.2	168.6	83.12	105.3	116.4	149.6
88.9	14.14	54.63	64.70	69.25	80.90	46.03	53.56	56.79	64.29	34.57	38.71	40.18	43.58
	19.79	86.39	109.4	121.0	155.5	79.67	100.9	111.5	143.4	70.52	85.37	92.37	111.7
	23.07	103.1	130.5	144.3	185.5	95.25	120.6	133.3	171.4	84.53	107.1	118.3	152.2
101.6	17.64	44.86	52.04	55.09	62.02	37.10	41.99	43.84	47.03	26.76	29.32	30.83	33.75
	20.83	68.35	82.48	89.14	107.4	58.58	69.82	74.97	88.53	45.55	52.94	56.09	63.36
	23.36	78.82	99.84	110.4	140.8	72.63	90.28	97.86	119.0	59.37	70.84	76.12	90.05
114.3	20.46	37.57	42.59	44.52	47.93	30.43	33.35	34.29	38.37	21.75	24.41	25.34	26.43
	24.70	57.83	68.85	73.89	87.09	48.96	57.36	61.04	69.95	37.14	42.03	43.89	47.10
	29.76	79.24	100.4	110.9	142.3	73.02	91.19	98.88	120.4	59.98	71.64	77.01	91.24
	33.96	90.77	115.0	127.1	163.4	83.76	106.1	117.3	150.8	74.18	93.92	101.9	124.5
127.0	24.18	36.15	40.76	42.47	45.57	29.14	31.67	33.08	36.75	21.01	23.46	24.28	25.07
	29.02	54.39	64.40	68.91	80.44	45.82	53.28	56.48	63.87	34.39	38.47	39.91	43.34
	38.10	82.58	104.6	115.6	148.6	76.12	96.42	106.6	131.2	64.89	78.01	84.13	100.7
139.7	28.57	30.73	33.73	34.61	38.74	24.18	27.15	28.41	30.51	18.18	19.81	20.19	20.29
	32.59	45.05	52.29	55.36	62.38	37.27	42.21	44.09	47.37	26.91	29.44	30.97	33.94
	36.76	58.61	69.87	75.02	88.60	49.68	58.29	62.07	71.33	37.76	42.84	44.80	48.31
168.3	37.50	23.22	26.31	27.47	29.26	19.37	21.34	21.91	22.22	14.25	14.73	14.73	14.73
	41.22	29.86	32.60	33.75	37.65	23.38	26.48	27.66	29.51	17.73	19.22	19.54	19.58

Notes

The collapse resistance of drill pipe is based the "worst case" combination of dimensions allowable under the classification scheme (see page C-4). For new pipe this is the nominal OD with the minimum allowable wall thickness as defined in the specifications. For used pipe it is nominal OD with maximum allowable wear having occurred on the ID. In practice it is unlikely that there will be very much wear on the ID compared with the wear on the OD, but the "worst case" scenario has to be taken into account for critical strengths. See also the notes on page C-5

No safety factors have been included in these tabulated values.

The nominal dimensions and wall thickness tolerances of new drill pipe have been taken from API Spec 5D, 4th Edition, August 1999. Premium Class and Class 2 drill pipe are as defined in API RP 7G, 16th Edition, August 1998, Table 24 (reproduced on page C-4). The collapse pressures have been calculated by the method described in API Bulletin 5C3, 6th Edition, October 1994 (Supplement 1 of April 1999)

CALCULATING THE MAXIMUM LENGTH OF A SECTION OF DRILL PIPE

Single grade in use

$$L = \frac{T_1 / DF_{ts} - MOP}{W \times K_b} - \frac{W_c \times L_c}{W} \quad \text{and / or} \quad L = \frac{T_1}{DF_{ts} \times DF_{sc} \times W \times K_b} - \frac{W_c \times L_c}{W}$$

String with two grades of DP in use

L_1 - as above

$$L_2 = \frac{T_2 / DF_{ts} - MOP}{W_2 \times K_b} - \frac{(W_1 \times L_1) + (W_c \times L_c)}{W_2}$$

$$\text{and / or} \quad L_2 = \frac{T_2}{DF_{ts} \times DF_{sc} \times W_2 \times K_b} - \frac{(W_1 \times L_1) + (W_c \times L_c)}{W_2}$$

Where :

- T = Tensile strength of the DP in question as tabulated on [page C-20](#)
- DF_{ts} = Design factor applied to tensile strength (normally 1.18, see [page C-3](#))
- W = Approximate weight of DP (see [page C-2](#))
- W_c = Drill collar weight/unit length in air
- L_c = Length of drill collars
- K_b = Buoyancy factor
- DF_{sc} = Design factor for slip crushing, as shown in the table below.
- MOP = Margin of overpull, which is the chosen value of excess tensile capacity, above the normal working load, to account for factors such as hole drag and to give the ability to pull on stuck pipe.

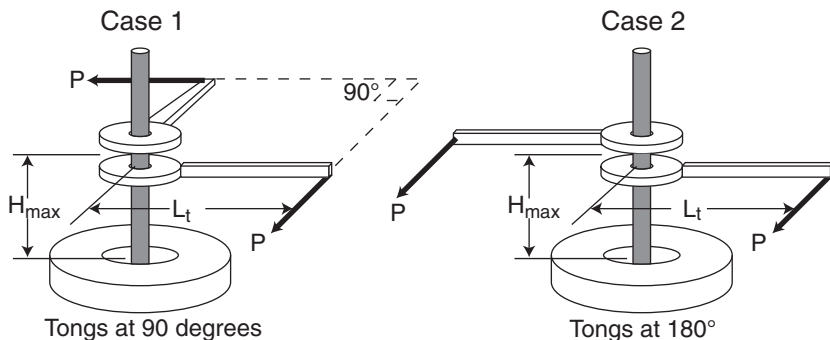
The maximum length of the section should be calculated for each of the two cases given, and the lesser of the two results used for the string design.

Slip length	Coeff. of friction*	Lateral load factor**	Design factor for slip crushing							
			Pipe size							
			2 3/8"	2 7/8"	3 1/2"	4"	4 1/2"	5"	5 1/2"	6 5/8"
12"	0.06	4.37	1.27	1.34	1.43	1.50	1.58	1.65	1.73	1.91
	0.08	4.00	1.25	1.31	1.39	1.45	1.52	1.59	1.66	1.82
	0.10	3.69	1.22	1.28	1.35	1.41	1.47	1.54	1.60	1.75
	0.12	3.42	1.21	1.26	1.32	1.38	1.43	1.49	1.55	1.68
	0.14	3.18	1.19	1.24	1.30	1.35	1.40	1.45	1.50	1.63
	0.16	2.98	1.18	1.22	1.27	1.32	1.37	1.42	1.47	1.58
16"	0.06	4.37	1.20	1.24	1.31	1.36	1.41	1.47	1.52	1.65
	0.08	4.00	1.18	1.22	1.28	1.32	1.37	1.42	1.47	1.59
	0.10	3.69	1.16	1.20	1.25	1.29	1.34	1.38	1.43	1.53
	0.12	3.42	1.15	1.18	1.23	1.27	1.31	1.35	1.39	1.49
	0.14	3.18	1.14	1.17	1.21	1.25	1.28	1.32	1.36	1.45
	0.16	2.98	1.13	1.16	1.20	1.23	1.26	1.30	1.33	1.41

* The friction factor is normally taken to be 0.08 when using standard pipe dope to lubricate the slips.

** This is the total horizontal force exerted by the slips on the pipe, which is distributed over their contact area. It is expressed as a multiple of the tension in the pipe.

MAXIMUM HEIGHT OF TOOL JOINT ABOVE SLIPS TO PREVENT BENDING DURING TONGING



The maximum height above the slips is given by:

	in field units	in S.I. units
Case 1 :	$H_{\max} = \frac{0.059 Y_m \cdot L_t \cdot (I/C)}{T}$	$H_{\max} = \frac{0.707 Y_m \cdot L_t \cdot (I/C)}{10^9 \cdot T}$
Case 2 :	$H_{\max} = \frac{0.042 Y_m \cdot L_t \cdot (I/C)}{T}$	$H_{\max} = \frac{0.500 Y_m \cdot L_t \cdot (I/C)}{10^9 \cdot T}$

Where :

$H_{\max}$ = Height of tool joint above slips

Y_m = Minimum tensile yield stress of pipe

L_t = Tong arm length

P = Line pull (load)

T = Make-up torque (= $P \times L_t$)

I/C = Section modulus of pipe

where :

$$I = \frac{\pi}{64} (D^4 - d^4)$$

$$C = \frac{D}{2}$$

D = outside diameter of pipe

d = inside diameter of pipe

field unit	S.I. unit
ft	m
psi	Pa
ft	m
lbs	N
lbs-ft	N.m
in ³	mm ³
ins	mm
ins	mm

No safety or design factors have been included in the constants in the above equations.

Section modulus values have been calculated for commonly used drill pipe sizes and are shown in the table on the [following page](#).

Values of recommended make-up torque values can be found in the tables on pages C-30 to C-34.

SECTION MODULUS VALUES

FOR NEW AND PREMIUM PIPE

Nominal O.D.		Nominal weight		I/C for new pipe		I/C for premium pipe	
inches	mm	lbs/ft	kg/m	inches ³	mm ³	inches ³	mm ³
2 ³ / ₈ "	60.33	4.85	7.22	0.6604	10,820	0.5165	8,464
		6.65	9.89	0.8666	14,200	0.6670	10,930
2 ⁷ / ₈ "	73.03	6.85	10.19	1.1210	18,360	0.8779	14,390
		10.40	15.48	1.6020	26,250	1.2280	20,130
3 ¹ / ₂ "	88.90	9.50	14.14	1.9610	32,140	1.5380	25,210
		13.30	19.79	2.5720	42,150	1.9910	32,630
		15.50	23.07	2.9230	47,910	2.2390	36,680
4"	101.60	11.85	17.63	2.7000	44,250	2.1230	34,780
		14.00	20.83	3.2290	52,910	2.5230	41,340
4 ¹ / ₂ "	114.30	13.75	20.46	3.5920	58,860	2.8290	46,360
		16.60	24.70	4.2710	69,990	3.3470	54,850
		20.00	29.76	5.1160	83,840	3.9770	65,170
5"	127.00	16.25	24.18	4.8590	79,620	3.8280	62,720
		19.50	29.02	5.7080	93,530	4.4760	73,350
		25.60	38.10	7.2450	118,700	5.6210	92,120
5 ¹ / ₂ "	139.70	21.90	32.59	7.0310	115,200	5.5270	90,570
		24.70	36.75	7.8440	128,500	6.1450	100,700
6 ⁵ / ₈ "	168.28	25.20	37.50	9.7860	160,400	7.7320	126,700

Note:

In the calculation of I/C for premium pipe it has been assumed that the amount of drill pipe wear is the maximum allowable under the classification scheme (see page C-4). In practice the value will be intermediate between the one shown and the corresponding value for new pipe.

ROTARY SHOULDERED CONNECTION INTERCHANGE LIST

Common Name		Same As or Interchanges With			
Style	Size				
Internal Flush (I.F.)	2 $\frac{3}{8}$	2 $\frac{7}{8}$ SH	NC 26		
	2 $\frac{7}{8}$	3 $\frac{1}{2}$ SH	NC 31		
	3 $\frac{1}{2}$	4 $\frac{1}{2}$ SH	NC 38		
	4	4 $\frac{1}{2}$ XH	NC 46		
	4 $\frac{1}{2}$	5 XH	NC 50	5 $\frac{1}{2}$ DSL	
Full Hole (F.H.)	4	4 $\frac{1}{2}$ DSL	NC 40		
Extra Hole (X.H.) (E.H.)	2 $\frac{7}{8}$	3 $\frac{1}{2}$ DSL			
	3 $\frac{1}{2}$	4 SH	4 $\frac{1}{2}$ EF		
	4 $\frac{1}{2}$	4 IF	NC 46		
	5	4 $\frac{1}{2}$ IF	NC 50	5 $\frac{1}{2}$ DSL	
Slim Hole (S.H.)	2 $\frac{7}{8}$	2 $\frac{3}{8}$ IF	NC 26		
	3 $\frac{1}{2}$	2 $\frac{7}{8}$ IF	NC 31		
	4	3 $\frac{1}{2}$ XH	4 $\frac{1}{2}$ EF		
	4 $\frac{1}{2}$	3 $\frac{1}{2}$ IF	NC 38		
Double Streamline (DSL)	3 $\frac{1}{2}$	2 $\frac{7}{8}$ XH			
	4 $\frac{1}{2}$	4 FH	NC 40		
	5 $\frac{1}{2}$	4 $\frac{1}{2}$ IF	5 XH	NC 50	
Numbered Connection (N.C.)	NC 26	2 $\frac{3}{8}$ IF	2 $\frac{7}{8}$ SH		
	NC 31	2 $\frac{7}{8}$ IF	3 $\frac{1}{2}$ SH		
	NC 38	3 $\frac{1}{2}$ IF	4 $\frac{1}{2}$ SH		
	NC 40	4 FH	4 $\frac{1}{2}$ DSL		
	NC 46	4 IF	4 $\frac{1}{2}$ XH		
	NC 50	4 $\frac{1}{2}$ IF	5 XH	5 $\frac{1}{2}$ DSL	
External Flush (E.F.)	4 $\frac{1}{2}$	4 SH	3 $\frac{1}{2}$ XH		

The data in this table have been taken from the similarly titled Table 12 in API RP 7G 16th Edition, August 1998. It is reproduced by courtesy of the API.

ELONGATION OF THE DRILL STRING

The elongation due to a tensile load

The elongation of a section of tubulars of uniform composition due to an applied tensile load is given by the equations:

In field units	In SI units
$e = L \times T / 735,000 W$	$e = L \times T / 26.7 W$

Where : e = elongation	in inches	in mm
L = length of section	in feet	in metres
T = tensile load	in pounds	in kN
W = unit weight of tubulars	in lbs/ft	in kg/m

The unit weight for drill pipe can be taken as the "approximate weight" as tabulated on [pages C-6/9](#), for Hevi-Wate drill pipe take the "approximate weight" as tabulated on [page C-29](#) and for drill collars use the values tabulated on [pages C-48/49](#).

The elongation of the string due to an applied tensile load should be obtained by adding together the individual elongations of each section due to that same load. This load does not change along the string (excluding the effect of friction), unlike the self-load dealt with below.

The elongation of a string due to its own weight

The elongation of a suspended section of tubulars of uniform composition due to its own weight is given by the equations:

In field units	In SI units
$e = \frac{2 L^2}{10^7} (3.40 - 1.44 \rho_{df})$	$e = \frac{L^2}{41.2 \times 10^4} (77.0 - 1.44 \rho_{df})$

Where e = elongation	in inches	in mm
L = length of section	in feet	in metres
ρ_{df} = drilling fluid gradient	in psi/ft	in kPa/m

Note that this cannot be done in one step if the string contains more than one section. The following procedure must be followed (numbering the sections from the bottom up):

- 1) Calculate the elongation of Section 1 due to its own weight.
- 2a) Calculate the elongation of Section 2 due to its own weight
- 2b) Calculate the elongation of Section 2 due to the applied load that is the buoyant weight of Section 1, as described above.
- 2c) The total elongation of Section 2 is the sum of those obtained in 2a and 2b
- 3a) Calculate the elongation of Section 3 due to its own weight
- 3b) Calculate the elongation of Section 3 due to the applied load that is the buoyant weight of Section 1 plus Section 2, as described above.
- 3c) The total elongation of Section 3 is the sum of those obtained in 3a and 3b
- etc. etc.

The above equations are taken from API RP 7G 16th Edition, August 1998.

PHYSICAL PROPERTIES OF HEVI-WATE DRILLPIPE (Range II)

	Field units			S.I. Units		
Nominal size	ins	3 ¹ / ₂	5	mm	88.9	127
Nominal weight	lbs/ft	26	50	kg/m	38.7	74.4
Approximate weight	lbs/ft	25.3	49.3	kg/m	37.6	73.4
Pipe O.D.	ins	3 ¹ / ₂	5	mm	88.9	127.0
Pipe I.D.	ins	2 ¹ / ₁₆	3	mm	52.4	76.2
Tool joint		NC38	NC50		NC38	NC50
Tool joint O.D.	ins	4 ³ / ₄	6 ¹ / ₂	mm	120.7	165.1
Tool joint I.D.	ins	2 ³ / ₁₆	3 ¹ / ₁₆	mm	55.16	77.73
Pipe min. tensile yield	lbs	345,400	691,185	kdaN	154.8	309.8
Tool joint tensile yield	lbs	748,750	1,266,000	kdaN	335.6	567.4
Pipe torsional yield	lbs-ft	19,575	56,495	daN.m	2,652	7,655
Tool joint torsional yield	lbs-ft	17,575	51,375	daN.m	2,381	6,961
Make-up torque	lbs-ft	9,900	29,400	daN.m	1,341	3,984
Capacity	bbls/ft	0.0042	0.0087	l/m	2.19	4.53
O.E.displacement (including tool joints)	bbls/ft	0.0092	0.018	l/m	4.79	9.35

TOOL JOINT MAKE-UP TORQUE (lbs-ft)

Box-weak connections are shown in bold type

NC26

ID (in.)	OD (in.)							
	3-5/8	3-1/2	3-3/8	3-1/4	3-1/8	3	2-31/32	2-15/16
1-3/4	4330	4300	4130	3010	1940	940	700	470
1-7/8	3720	3690	3660	3010	1940	940	700	470
2	3070	3040	3020	2990	1940	940	700	470
2-1/8	2380	2360	2340	2320	1940	940	700	470
2-5/32	2200	2180	2160	2140	1940	940	700	470
2-1/4	1640	1630	1610	1600	1580	940	700	470

HT26

ID(in.)	OD (in.)						
	3-5/8	3-1/2	3-3/8	3-1/4	3-1/8	3	2-7/8
1-1/2	7850	7280	6090	4970	3910	2910	1970
1-3/4	5890	5890	5260	4310	3070	2070	1130
2	3590	3590	3590	3120	2060	1060	

NC31

ID (in.)	OD (in.)							
	4-3/8	4-1/4	4-1/8	4	3-7/8	3-3/4	3-5/8	3-1/2
1-5/8	10170	10090	8510	6890	5350	3870	2460	1120
1-3/4	9510	9440	8510	6890	5350	3870	2460	1120
1-7/8	8800	8730	8510	6890	5350	3870	2460	1120
2	8040	7980	7920	6890	5350	3870	2460	1120
2-1/8	7230	7180	7120	6890	5350	3870	2460	1120

2-7/8 SL-H90

ID (in.)	OD (in.)							
	4-1/4	4-1/8	4	3-7/8	3-3/4	3-5/8	3-1/2	3-7/16
2	8040	7990	7930	7880	6500	4910	3400	4453
2-1/8	7150	7100	7050	7000	6500	4910	3400	4453
2-5/32	6920	6870	6820	6770	6500	4910	3400	4453
2-1/4	6200	6160	6110	6070	6030	4910	3400	4453
2-3/8	5190	5160	5120	5090	5050	4910	3400	4453
2-7/16	4670	4630	4600	4570	4540	4510	3400	4453

2-7/8 PAC

ID (in.)	OD (in.)							
	3-1/8	3-1/16	3	2-15/16	2-7/8	2-13/16	2-3/4	2-11/16
1-1/4	4090	3590	3130	2670	2220	1780	1370	960
1-3/8	3880	3590	3130	2670	2220	1780	1370	960
1-1/2	3440	3420	3130	2670	2220	1780	1370	960
1-5/8	2960	2950	2930	2670	2220	1780	1370	960
1-3/4	2450	2440	2420	2670	2220	1780	1370	960

Note : To obtain the torque in N-m multiply the above figures by 1.356

The tables on this and the following four pages have been taken from Shell Expro's "Drillstring Failure Prevention - Drillstring Design Manual", Revision 0, October 1994.

TOOL JOINT MAKE-UP TORQUE (lbs-ft)

Box-weak connections are shown in bold type

NC38

ID (in.)	OD (in.)							
	5	4-15/16	4-7/8	4-13/16	4-3/4	4-11/16	4-5/8	4-9/16
2-1/8	15900	14900	13740	12610	11500	10420	9350	8290
2-1/4	14920	14870	13740	12610	11500	10420	9350	8290
2-3/8	13870	13830	13740	12610	11500	10420	9350	8290
2-7/16	13320	13280	13240	12610	11500	10420	9350	8290
2-1/2	12770	12730	12680	12610	11500	10420	9350	8290
2-9/16	12200	12160	12120	12080	11500	10420	9350	8290
2-5/8	11610	11570	11530	11500	11460	10420	9350	8290
2-11/16	11000	10970	10930	10900	10860	10420	9350	8290

ID (in.)	OD (in.)						
	4-1/2	4-7/16	4-3/8	4-5/16	4-1/4	4-3/16	4-1/8
2-1/8	7280	6280	5280	4310	3370	2450	1540
2-1/4	7280	6280	5280	4310	3370	2450	1540
2-3/8	7280	6280	5280	4310	3370	2450	1540
2-7/16	7280	6280	5280	4310	3370	2450	1540
2-1/2	7280	6280	5280	4310	3370	2450	1540
2-9/16	7280	6280	5280	4310	3370	2450	1540
2-5/8	7280	6280	5280	4310	3370	2450	1540
2-11/16	7280	6280	5280	4310	3370	2450	1540

HT38

ID (in.)	OD (in.)							
	5	4-7/8	4-3/4	4-5/8	4-1/2	4-3/8	4-1/4	4-1/8
2-1/8	23630	21290	19030	16860	14770	12770	10840	9000
2-7/16	19740	19310	17050	14880	12790	10790	8860	7020
2-9/16	17660	17660	16160	13990	11900	9890	7970	6120
2-11/16	15470	15470	15190	13020	10930	8930	7000	5160

NC40

ID (in.)	OD (in.)							
	5-1/2	5-3/8	5-1/4	5-1/8	5	4-7/8	4-3/4	4-5/8
2-1/4	19760	19640	17620	15050	12570	10180	7880	5660
2-3/8	18650	18530	17620	15050	12570	10180	7880	5660
2-7/16	18070	17950	17620	15050	12570	10180	7880	5660
2-1/2	17470	17360	17260	15050	12570	10180	7880	5660
2-9/16	16870	16760	16660	15050	12570	10180	7880	5660
2-5/8	16240	16130	16040	15050	12570	10180	7880	5660
2-11/16	15590	15490	15400	15050	12570	10180	7880	5660
2-3/4	14940	14840	14750	14660	12570	10180	7880	5660
2-13/16	14270	14180	14100	14010	12570	10180	7880	5660

3-1/2 SL-H90

ID (in.)	OD (in.)							
	5	4-7/8	4-3/4	4-5/8	4-1/2	4-3/8	4-1/4	4-1/8
2-1/2	13310	13240	13160	13080	11000	8740	6580	4490
2-5/8	12010	11940	11870	11800	11000	8740	6580	4490
2-11/16	11330	11260	11200	11130	11000	8740	6580	4490
2-3/4	10640	10580	10520	10460	10400	8740	6580	4490
2-7/8	9210	9160	9110	9050	9000	8740	6580	4490
3	7720	7670	7630	7580	7540	7490	6580	4490

Note : To obtain the torque in N-m multiply the above figures by 1.356

TOOL JOINT MAKE-UP TORQUE (lbs-ft)

Box-weak connections are shown in bold type

3-1/2 XH

ID (in.)	OD (in.)							
	5	4-7/8	4-3/4	4-5/8	4-1/2	4-3/8	4-1/4	4-1/8
2-1/4	11930	11850	11770	11690	9770	7820	5950	4150
2-3/8	10910	10840	10770	10700	9770	7820	5950	4150
2-7/16	10380	10310	10240	10180	9770	7820	5950	4150
2-1/2	9850	9780	9710	9650	9580	7820	5950	4150
2-9/16	9290	9230	9170	9110	9040	7820	5950	4150
2-5/8	8720	8660	8600	8540	8480	7820	5950	4150

4 H90

ID (in.)	OD (in.)										
	6-1/4	6-1/8	6	5-7/8	5-3/4	5-5/8	5-1/2	5-5/32	5-1/32	4-7/8	4-27/32
2-3/4	22700	22580	22460	22350	22230	22120	22000	13910	11060	7630	6970
2-13/16	21910	21790	21680	21570	21460	21340	21230	13910	11060	7630	6970
2-7/8	21080	20970	20860	20750	20650	20540	20430	13910	11060	7630	6970
3	19390	19290	19190	19090	19000	18890	18790	13910	11060	7630	6970
3-1/8	17630	17540	17450	17360	17270	17180	17090	13910	11060	7630	6970

NC46

ID (in.)	OD (in.)									
	6-1/2	6-7/16	6-3/8	6-5/16	6-1/4	6-3/16	6-1/8	6-1/16	6	5-15/16
2-1/2	30100	30020	29940	29860	29780	29700	29620	28670	26980	25320
2-5/8	28690	28610	28540	28460	28390	28310	28230	28150	26980	25320
2-3/4	27220	27140	27070	26990	26920	26850	26780	26710	26630	25320
2-7/8	25670	25600	25530	25460	25390	25330	25250	25190	25120	25320
3	24050	23990	23920	23860	23800	23730	23660	23600	23540	23470
3-1/8	22370	22310	22250	22190	22130	22070	22010	21950	21890	21830
3-1/4	20620	20560	20510	20450	20390	20340	20290	20230	20180	20120
3-3/8	18800	18740	18700	18640	18590	18550	18490	18440	18400	18340
3-1/2	16910	16860	16820	16770	16730	16680	16630	16590	16540	16500

ID (in.)	OD (in.)									
	5-7/8	5-13/16	5-3/4	5-11/16	5-5/8	5-9/16	5-3/8	5-5/16	5-1/4	5-5/32
2-1/2	23660	22020	20440	18880	17310	15770	11360	9920	8530	6470
2-5/8	23660	22020	20440	18880	17310	15770	11360	9920	8530	6470
2-3/4	23660	22020	20440	18880	17310	15770	11360	9920	8530	6470
2-7/8	23660	22020	20440	18880	17310	15770	11360	9920	8530	6470
3	23410	22020	20440	18880	17310	15770	11360	9920	8530	6470
3-1/8	21770	21710	20440	18880	17310	15770	11360	9920	8530	6470
3-1/4	20060	20010	19960	18880	17310	15770	11360	9920	8530	6470
3-3/8	18290	18240	18190	18140	17310	15770	11360	9920	8530	6470
3-1/2	16450	16410	16360	16320	16270	15770	11360	9920	8530	6470

Note : To obtain the torque in N-m multiply the above figures by 1.356

TOOL JOINT MAKE-UP TORQUE (lbs-ft)

Box-weak connections are shown in bold type

4-1/2 FH

ID (in.)	OD (in.)								
	6-1/4	6-1/8	6	5-7/8	5-3/4	5-5/8	5-1/2	5-3/8	5-9/32
2-1/2	26860	26710	26560	23890	20870	17860	14950	12130	10070
2-5/8	25520	25380	25240	23890	20870	17860	14950	12130	10070
2-3/4	24110	23980	23840	23710	20870	17860	14950	12130	10070
2-7/8	22640	22520	22390	22260	20870	17860	14950	12130	10070
3	21110	20990	20870	20750	20630	17860	14950	12130	10070

NC50

ID (in.)	OD (in.)							
	6-5/8	6-9/16	6-1/2	6-7/16	6-3/8	6-5/16	6-1/4	6-3/16
2-3/4	38040	36170	34190	32240	30290	28370	26500	24660
2-7/8	36400	36170	34190	32240	30290	28370	26500	24660
3	34680	34590	34190	32240	30290	28370	26500	24660
3-1/16	33800	33710	33630	32240	30290	28370	26500	24660
3-1/8	32890	32810	32720	32240	30290	28370	26500	24660
3-1/4	31020	30950	30870	30790	30290	28370	26500	24660
3-3/8	29090	29020	28940	28870	28790	28370	26500	24660
3-1/2	27080	27010	26940	26870	26800	26740	26500	24660
3-5/8	24990	24930	24870	24800	24740	24680	24620	24550
3-3/4	22840	22780	22720	22660	22610	22550	22490	22430

ID (in.)	OD (in.)							
	6-1/8	6-1/16	6	5-15/16	5-7/8	5-13/16	5-3/4	5-11/16
2-3/4	22820	21020	19240	17510	15770	14070	12410	10790
2-7/8	22820	21020	19240	17510	15770	14070	12410	10790
3	22820	21020	19240	17510	15770	14070	12410	10790
3-1/16	22820	21020	19240	17510	15770	14070	12410	10790
3-1/8	22820	21020	19240	17510	15770	14070	12410	10790
3-1/4	22820	21020	19240	17510	15770	14070	12410	10790
3-3/8	22820	21020	19240	17510	15770	14070	12410	10790
3-1/2	22820	21020	19240	17510	15770	14070	12410	10790
3-5/8	22820	21020	19240	17510	15770	14070	12410	10790
3-3/4	22370	21020	19240	17510	15770	14070	12410	10790

HT50

ID (in.)	OD (in.)							
	6-5/8	6-1/2	6-3/8	6-1/4	6-1/8	6	5-7/8	5-3/4
2-3/4	54080	50080	46190	42410	38750	35190	31730	28380
3	49530	47590	43710	39930	36260	32700	29250	25900
3-1/4	42950	42950	40920	37140	33470	29910	26460	23110
3-1/2	35790	35790	35790	34040	30370	26810	23350	20000
3-3/4	28050	28050	28050	28050	26940	23380	19930	16580

Note : To obtain the torque in N-m multiply the above figures by 1.356

TOOL JOINT MAKE-UP TORQUE (lbs-ft)

Box-weak connections are shown in bold type

4-1/2 H90

ID (in.)	OD (in.)									
	6-3/4	6-5/8	6-1/2	6-3/8	6-1/4	6-1/8	6	5-7/8	5-3/4	5-1/2
2-3/4	31430	31280	31130	30980	30830	30680	30340	26660	23090	16260
2-7/8	29700	29560	29420	29270	29130	28990	28850	26660	23090	16260
3	27890	27760	27620	27490	27360	27220	27090	26660	23090	16260
3-1/8	26010	25880	25760	25630	25510	25390	25260	25130	23090	16260
3-1/4	24050	23930	23810	23700	23590	23470	23360	23240	23090	16260
3-3/8	22010	21900	21800	21690	21590	21480	21380	21270	21160	16260
3-1/2	19890	19790	19700	19610	19510	19420	19320	19220	19130	16260

5-1/2 FH

ID (in.)	OD (in.)									
	7-1/2	7-3/8	7-1/4	7-1/8	7	6-7/8	6-3/4	6-5/8	6-17/32	6-11/32
3	52300	52060	47230	42430	37740	33180	28740	22410	21240	17120
3-1/16	51330	51100	47230	42430	37740	33180	28740	22410	21240	17120
3-1/8	50320	50090	47230	42430	37740	33180	28740	22410	21240	17120
3-1/4	48260	48040	47230	42430	37740	33180	28740	22410	21240	17120
3-3/8	46120	45910	45700	42430	37740	33180	28740	22410	21240	17120
3-1/2	43890	43690	43490	42430	37740	33180	28740	22410	21240	17120
3-5/8	41590	41390	41210	41020	37740	33180	28740	22410	21240	17120
3-3/4	39200	39020	38840	38660	37740	33180	28740	22410	21240	17120
3-7/8	36730	36560	36400	36230	36060	33180	28740	22410	21240	17120
4	34180	34030	33870	33710	33560	33180	28740	22410	21240	17120

NC56

ID (in.)	OD (in.)							
	8	7-7/8	7-3/4	7-5/8	7-1/2	7-3/8	7-1/4	7-1/8
3	53780	53540	53300	53060	52810	51520	46660	41900
3-1/8	51790	51560	51320	51090	50860	50620	46660	41900
3-1/4	49720	49490	49270	49040	48820	48590	46660	41900
3-3/8	47560	47350	47130	46910	46700	46480	46270	41900
3-1/2	45320	45120	44910	44710	44500	44300	44090	41900
3-5/8	43000	42810	42610	42420	42200	42030	41620	41640
3-3/4	40600	40420	40240	40050	39870	39680	39500	39320
3-7/8	38120	37940	37780	37600	37430	37260	37090	36910
4	35560	35390	35230	35080	34910	34750	34590	34430

6-5/8 FH

ID (in.)	OD (in.)									
	8-3/4	8-5/8	8-1/2	8-3/8	8-1/4	8-1/8	8	7-3/4	7-1/2	7-1/8
4	72230	71950	71670	71380	65120	58850	52720	40860	29550	24100
4-1/2	59500	59270	59030	58800	58570	58330	52720	40860	29550	24100
4-3/4	52570	52360	52160	51950	51740	51530	51330	40860	29550	24100
5	45260	45080	44910	44730	44550	44380	44200	40860	29550	24100
5-1/4	37850	37440	37290	37140	37000	36850	36700	36400	29550	24100

Note : To obtain the torque in N-m multiply the above figures by 1.356

ALLOWABLE TORQUE AND PULL ON API DRILLPIPE

Torque during drilling

The torque in the drill pipe while drilling is given by the following equations:

$$\text{Torque} = \frac{5,250 \times \text{Horsepower}}{\text{RPM}} \quad \text{lbs-ft}$$

$$\text{Torque} = \frac{9,542 \times \text{Power}}{\text{RPM}} \quad \text{kN.m}$$

Torque under tension

When pulling on stuck pipe, or fishing, combined tension and torsion loads of high magnitudes are common. However the simultaneous application of both loads reduces the capacity of drill pipe to carry either.

The following equations may be used to determine the maximum allowable torque for drillpipe under a tension P :

In field units :

$$Q = \frac{0.09617 \times J}{D} \times \sqrt{(Y_m / DF)^2 - \frac{P^2}{A^2}}$$

In S.I. units :

$$Q = \frac{1.154 \times J}{D} \times \sqrt{(Y_m / DF)^2 - \frac{P^2}{A^2}}$$

Where

Q = Minimum torsional yield strength under tension. lb-ft (N.m)

J = Polar moment of inertia - inches⁴ (m⁴)

= $\pi/32(D^4 - d^4)$ for tubes

D = Outside diameter - inches (m)

d = Inside diameter- inches (m)

Y_m = Minimum yield stress - psi (Pa)

P = Total tensile load - pounds (N)

A = Cross-section area - inches² (m²)

DF is the design factor applied to the required strength (commonly 1.15)

It is assumed that the minimum yield stress in shear is 57.7% of the minimum yield stress in tension, as per API practice.

These equations have been used to construct a set of graphs to allow the determination of the maximum allowable torque that can be applied in combination with a given tensile load, and vice versa, for the common drill pipe sizes, using design factors of both 1.0 and 1.15. These are presented on the following pages.

The minimum yield strength values depend on the percentage of wear on the pipe, therefore graphs are included for the various API drill pipe classifications. For the latter refer to [page C-4](#).

A method is provided on [page G-7](#) for determining the number of turns required in order to achieve a certain torque under conditions of combined tension and torsion.

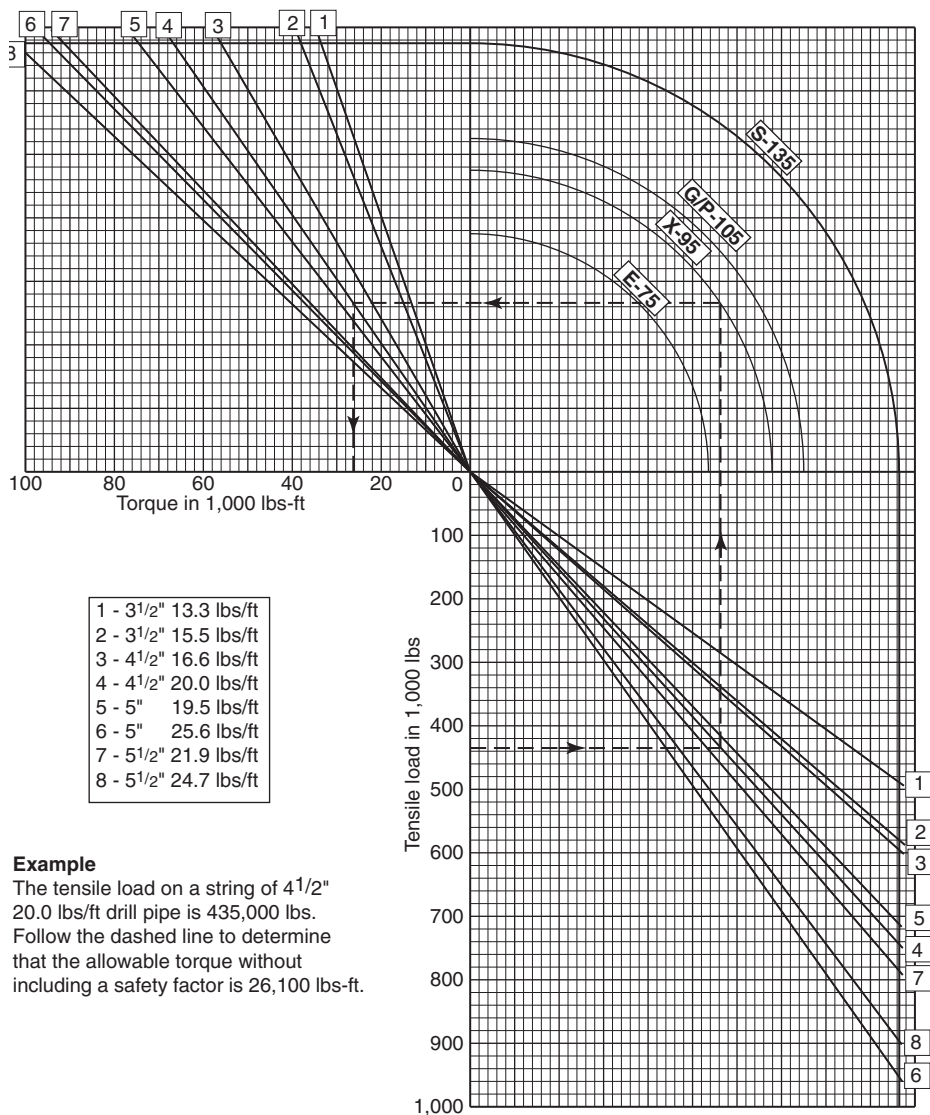
Tool joint strength

It should be realised that when applying torque to a drill string the torsional strength of the pipe is often not the limiting factor. The limit may be determined by the torsional strength of the tool joint. This depends on the type of tool joint and the percentage of wear of the pipe. In critical cases compare the torsional strength of the pipe as tabulated on page C-21 with the tool joint make-up torque tabulated on pages C-30/34.

The equation for maximum torque (in field units) in the presence of a tensile load is taken from API RP 7G 16th Edition, August 1998. The definition of yield strength is taken from API Spec 5D, 4th Edition, August 1999. Both are reproduced by courtesy of the API.

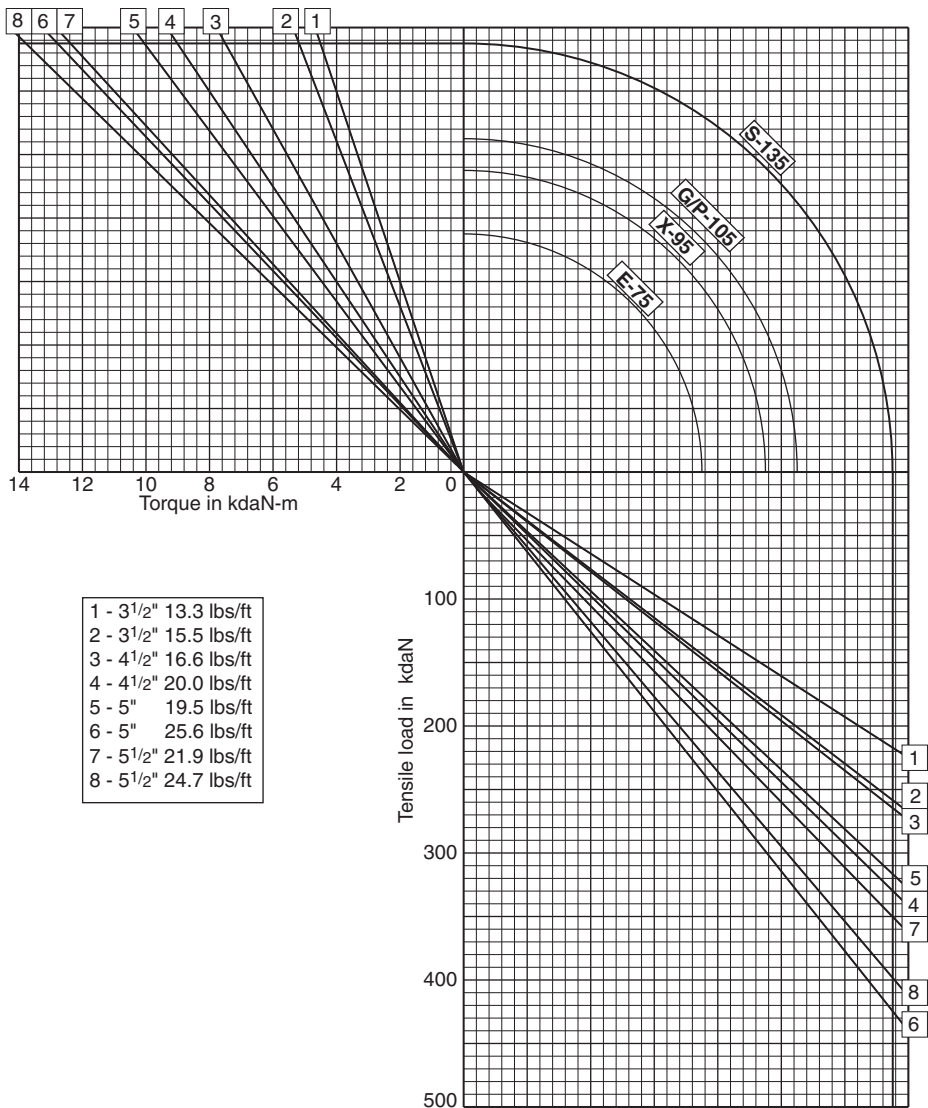
ALLOWABLE TORQUE AND TENSION ON API “NEW” DRILL PIPE

DESIGN FACTOR OF 1.0 - FIELD UNITS



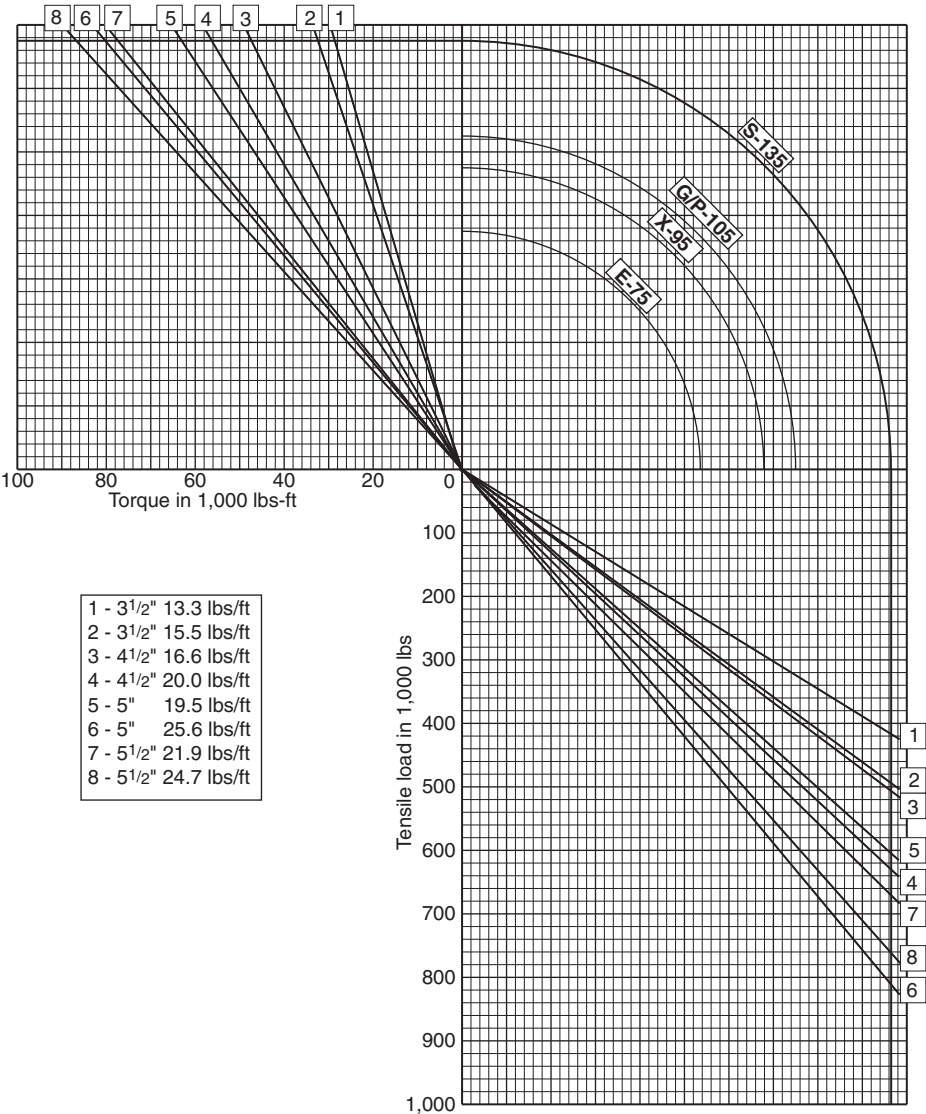
ALLOWABLE TORQUE AND TENSION ON API “NEW” DRILL PIPE

DESIGN FACTOR OF 1.0 - SI UNITS



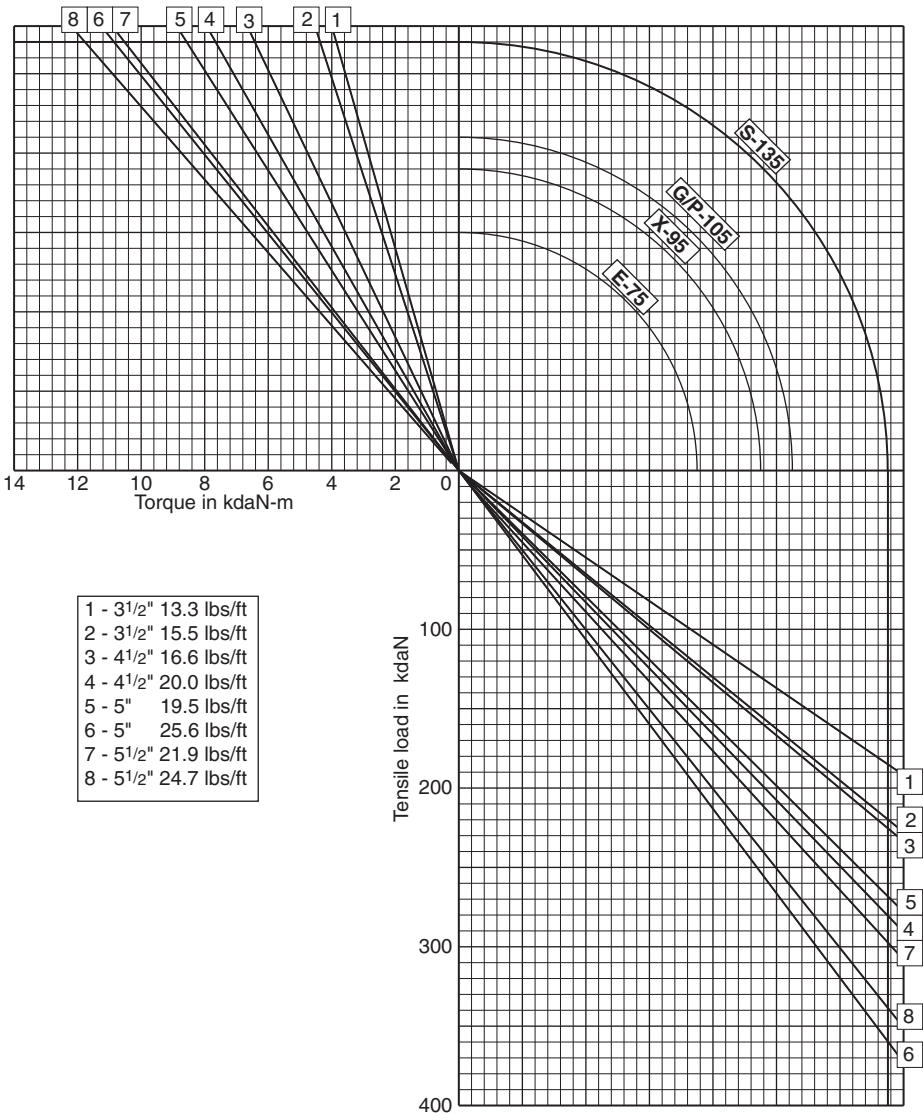
ALLOWABLE TORQUE AND TENSION ON API “NEW” DRILL PIPE

DESIGN FACTOR OF 1.15 - FIELD UNITS



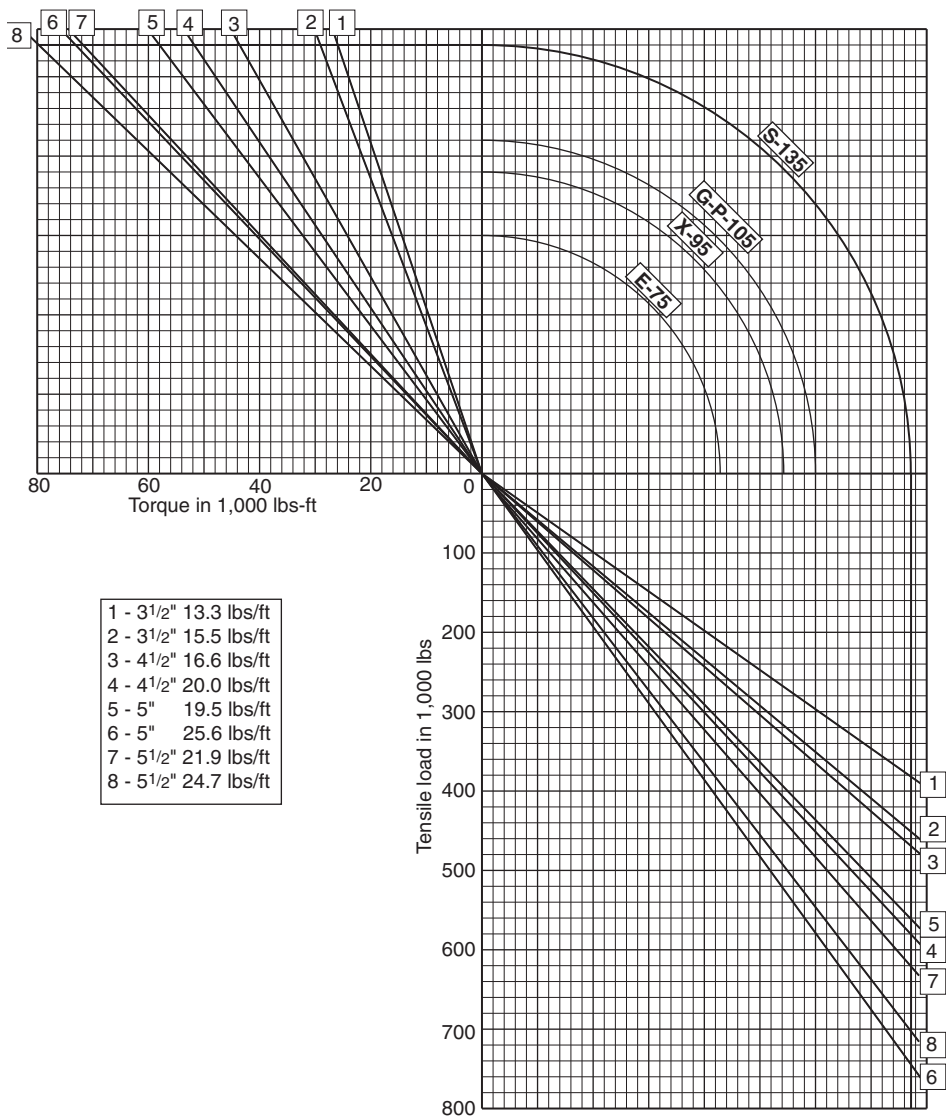
ALLOWABLE TORQUE AND TENSION ON API “NEW” DRILL PIPE

DESIGN FACTOR OF 1.15 - SI UNITS



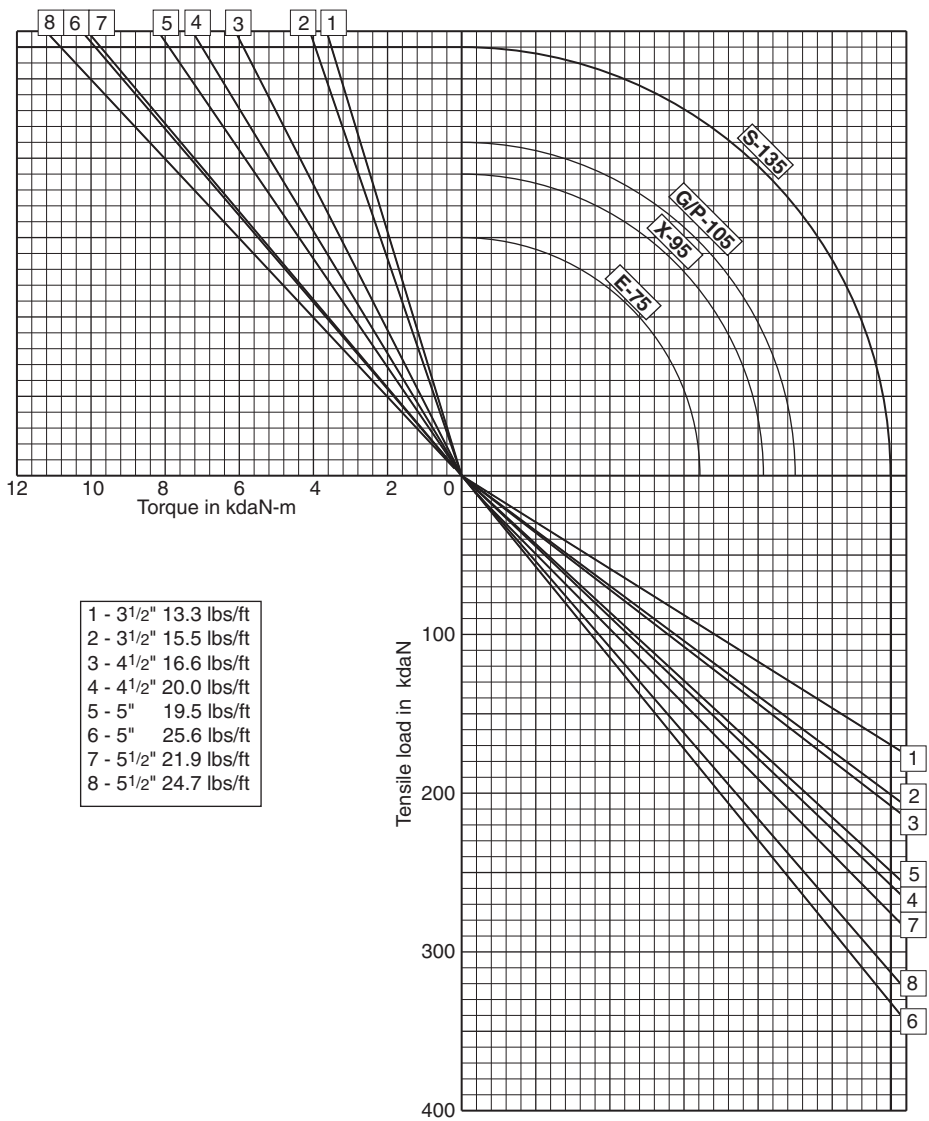
ALLOWABLE TORQUE AND TENSION ON API “PREMIUM” DRILL PIPE

DESIGN FACTOR OF 1.0 - FIELD UNITS



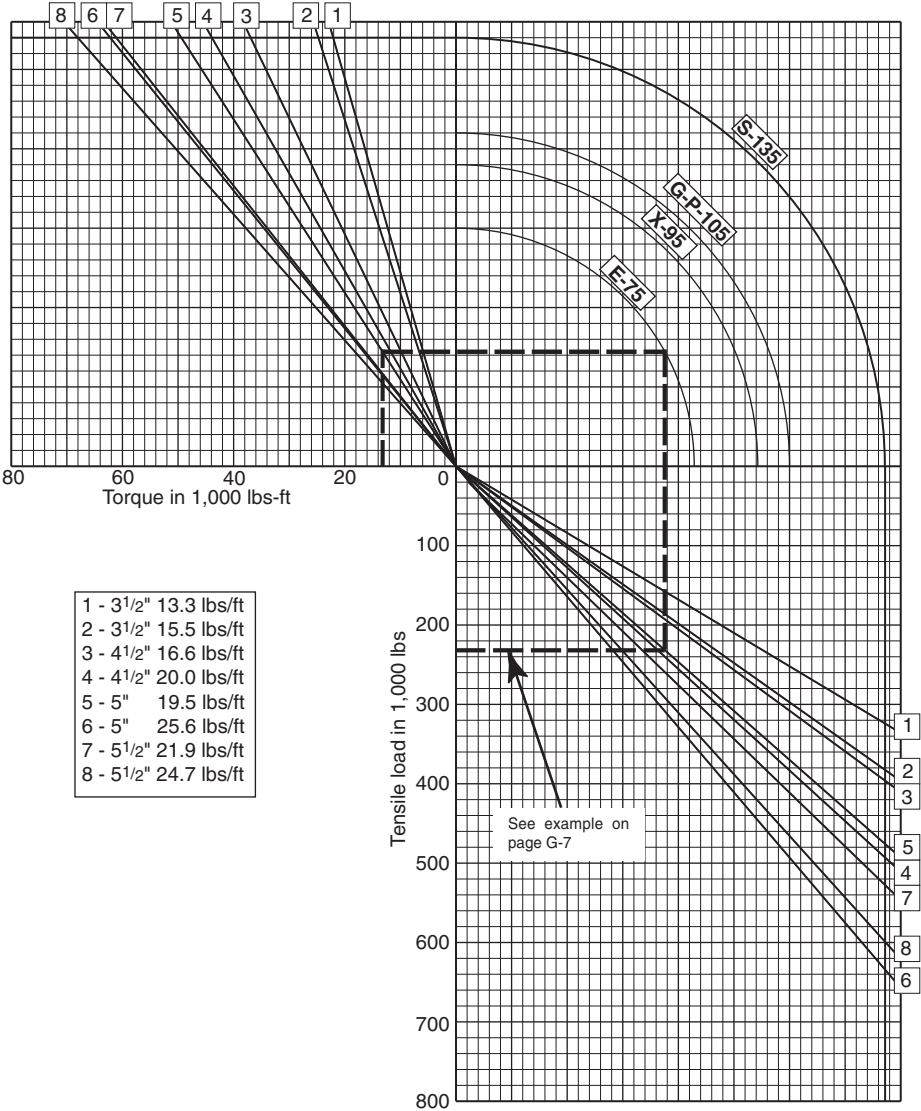
ALLOWABLE TORQUE AND TENSION ON API “PREMIUM” DRILL PIPE

DESIGN FACTOR OF 1.0 - SI UNITS



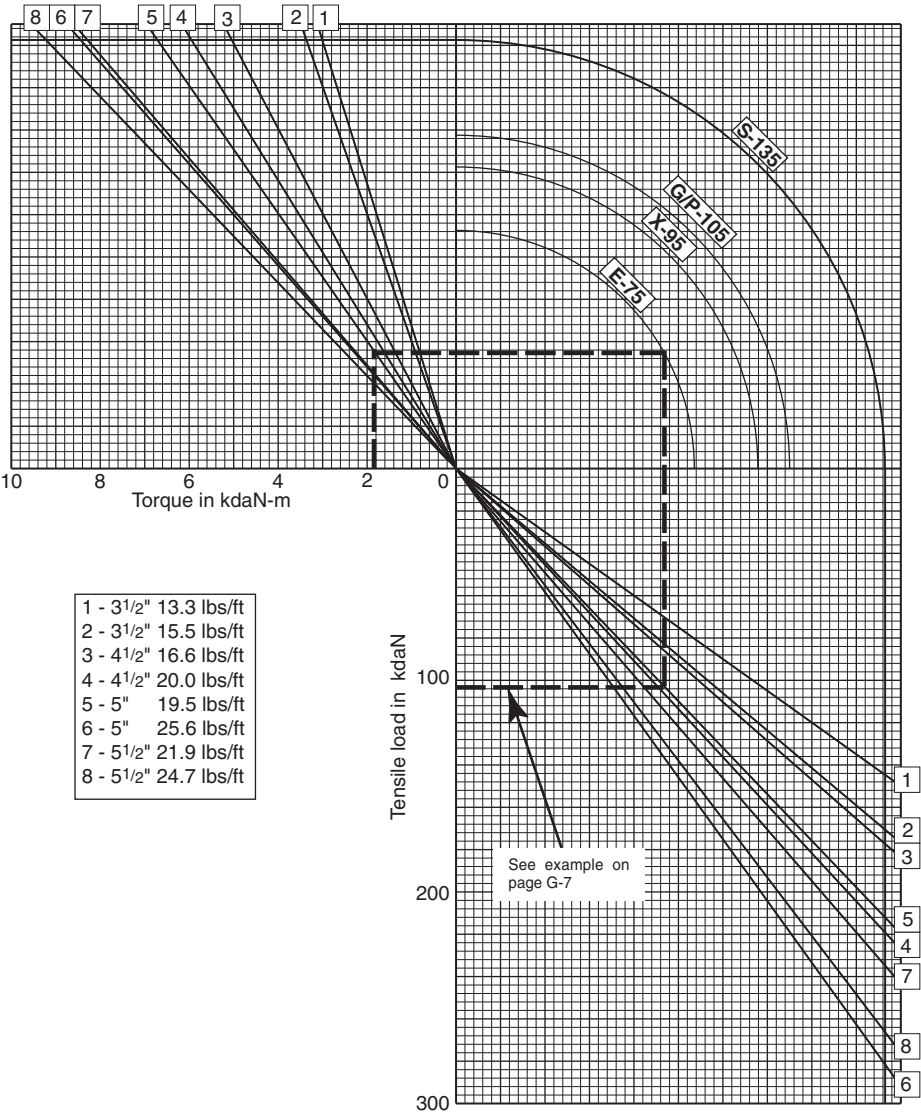
ALLOWABLE TORQUE AND TENSION ON API “PREMIUM” DRILL PIPE

DESIGN FACTOR OF 1.15 - FIELD UNITS



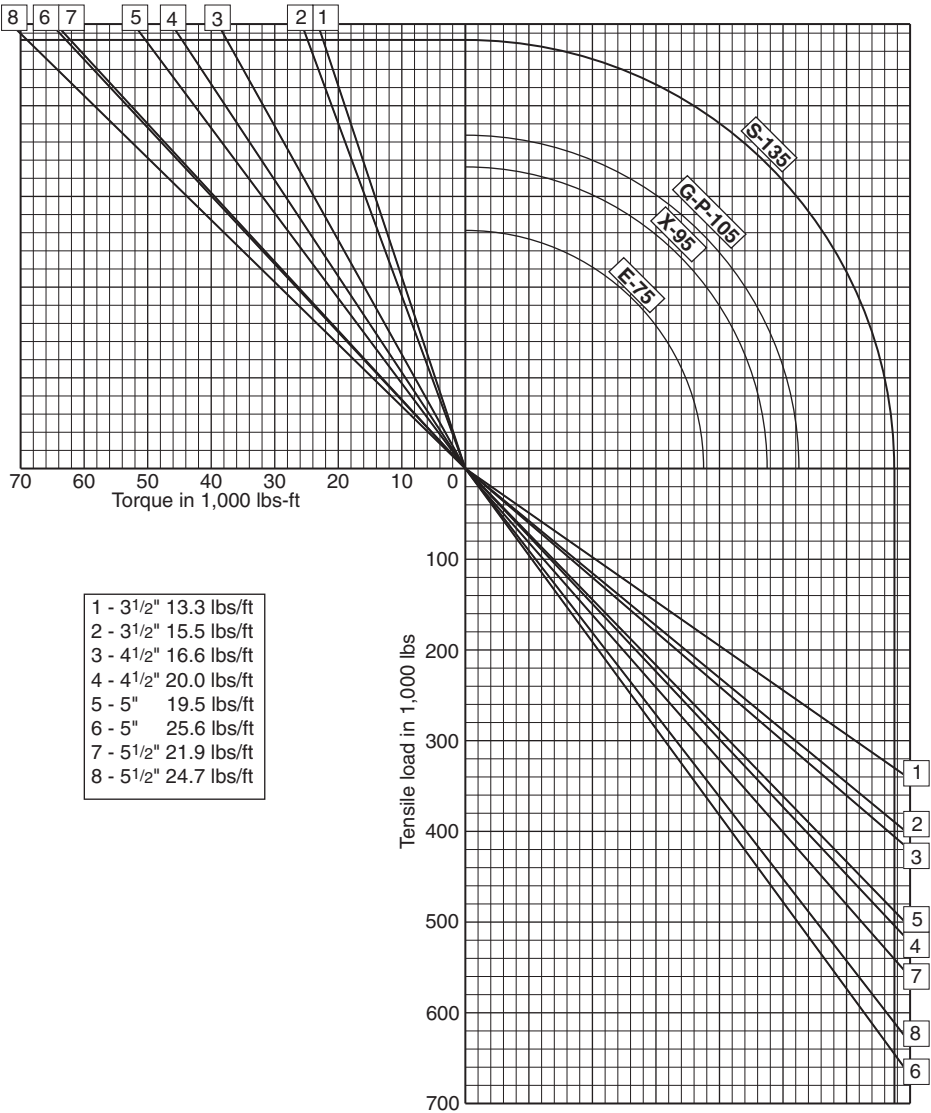
ALLOWABLE TORQUE AND TENSION ON API “PREMIUM” DRILL PIPE

DESIGN FACTOR OF 1.15 - SI UNITS

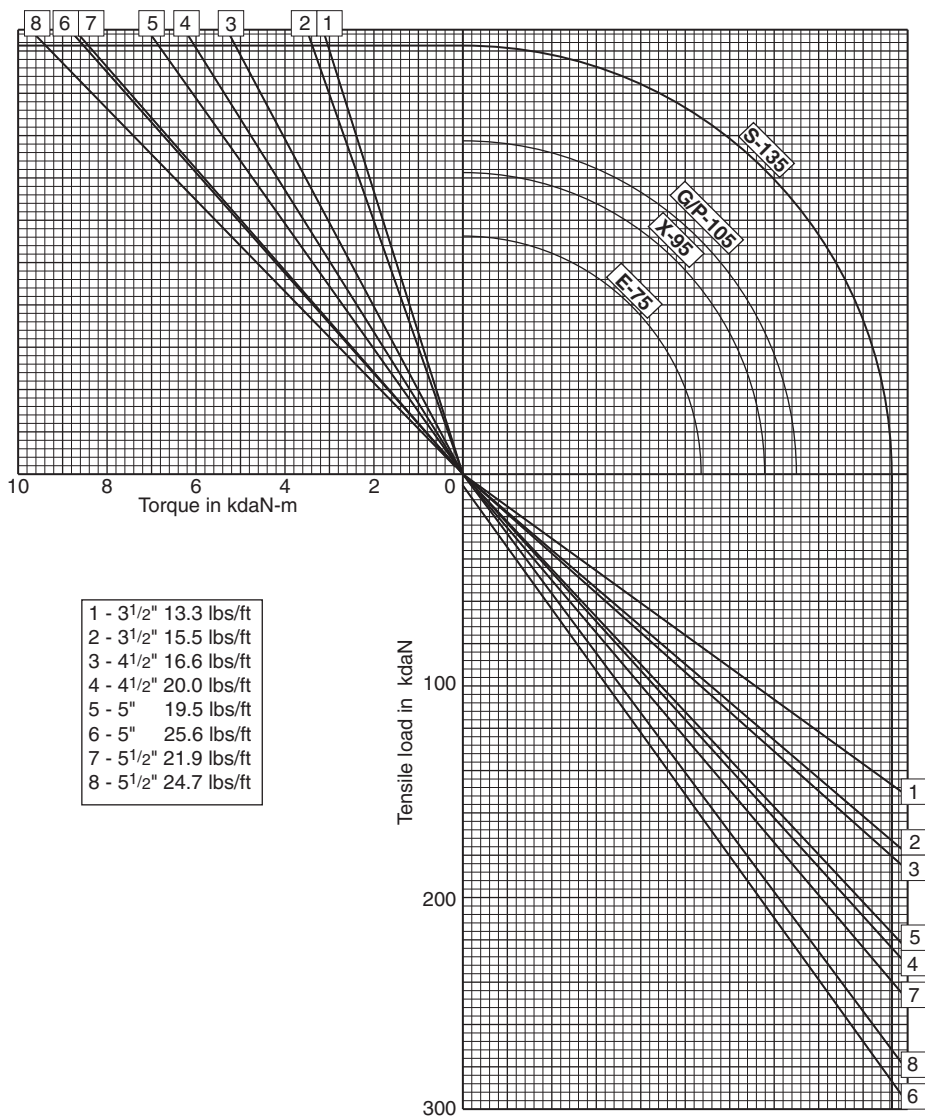


ALLOWABLE TORQUE AND TENSION ON API “CLASS 2” DRILL PIPE

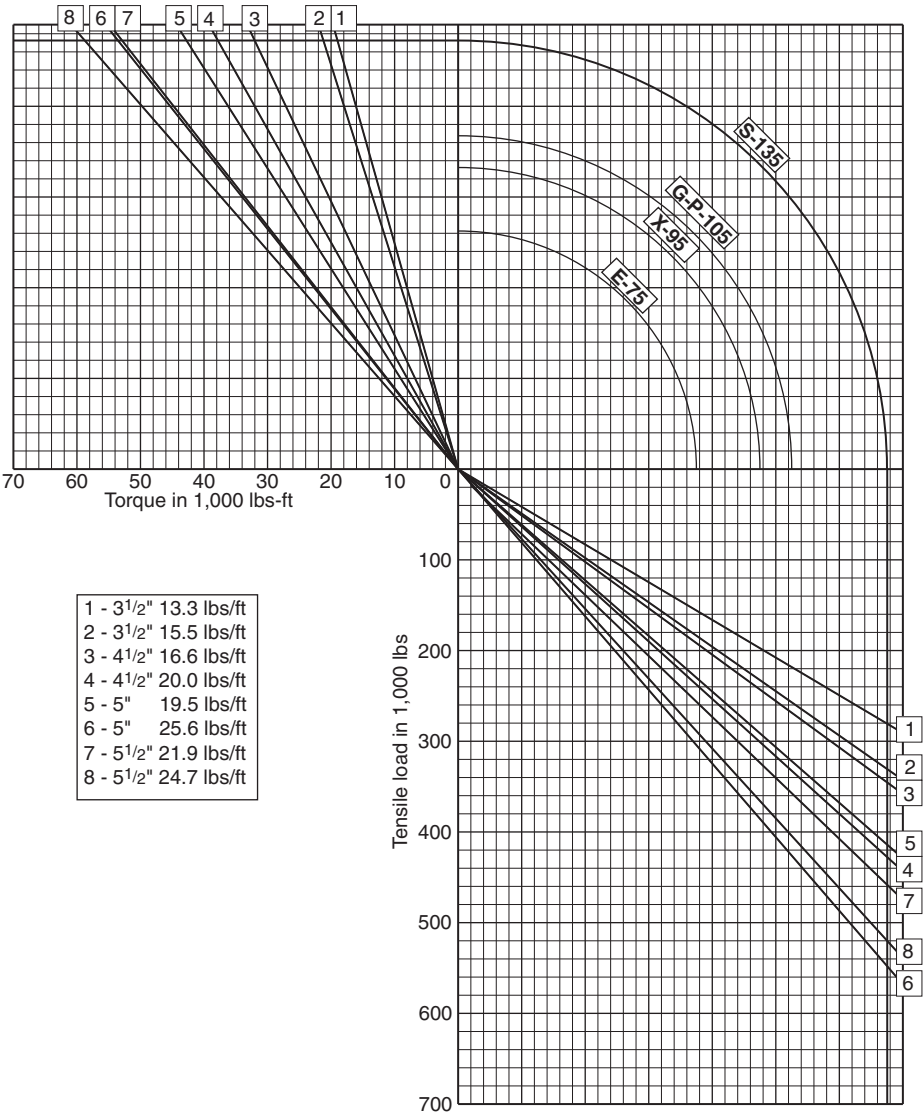
DESIGN FACTOR OF 1.0 - FIELD UNITS



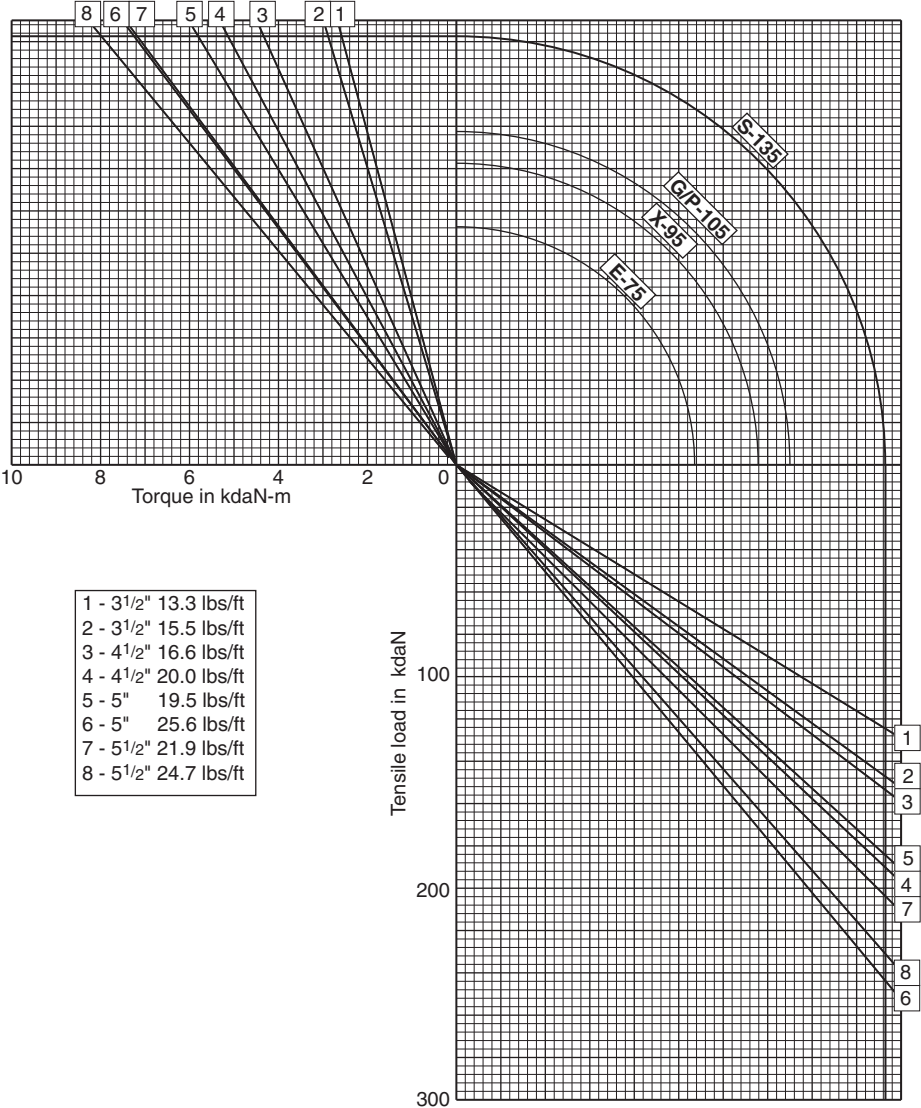
DESIGN FACTOR OF 1.0 – SI UNITS



ALLOWABLE TORQUE AND TENSION ON API “CLASS 2” DRILL PIPE
DESIGN FACTOR OF 1.15 - FIELD UNITS



ALLOWABLE TORQUE AND TENSION ON API “CLASS 2” DRILL PIPE
DESIGN FACTOR OF 1.15 – SI UNITS



STEEL DRILL COLLAR WEIGHTS IN KILOGRAMMES PER METRE

Drill collar OD mm (inches)	Drill collar ID in millimetres (inches)												
	25.4 (1)	31.8 (1 1/4)	38.1 (1 1/2)	44.5 (1 3/4)	50.8 (2)	57.2 (2 1/4)	63.5 (2 1/2)	71.4 (2 13/16)	76.2 (3)	82.6 (3 1/4)	88.9 (3 1/2)	95.3 (3 3/4)	101.6 (4)
73.0 (27/8)	28.9	26.6	23.9										
76.2 (3)	31.8	29.6	26.8										
79.4 (3 1/8)	34.8	32.6	29.9										
82.6 (3 1/4)	38.0	35.8	33.0										
88.9 (3 1/2)	44.7	42.5	39.7										
95.3 (3 3/4)	51.9	49.7	46.9										
101.6 (4)	59.6	57.4	54.6	51.4	47.7	43.5							
104.8 (4 1/8)	63.6	61.4	58.7	55.4	51.7	47.5							
108.0 (4 1/4)	67.8	65.6	62.8	59.6	55.9	51.7							
114.3 (4 1/2)	76.5	74.3	71.5	68.3	64.6	60.3							
120.7 (4 3/4)			80.7	77.5	73.8	69.5	64.8						
127.0 (5)			90.4	87.2	83.4	79.2	74.5						
133.4 (5 1/4)			100.6	97.3	93.6	89.4	84.7						
139.7 (5 1/2)			111.3	108.0	104.3	100.1	95.4	88.8					
146.1 (5 3/4)			122.4	119.2	115.5	111.3	106.5	99.9	95.6	89.4			
152.4 (6)			134.1	130.9	127.1	122.9	118.2	111.6	107.3	101.1			
158.8 (6 1/4)			146.3	143.0	139.3	135.1	130.4	123.8	119.5	113.2	106.5		
165.1 (6 1/2)			158.9	155.7	152.0	147.8	143.0	136.4	132.1	125.9	119.2		
171.5 (6 3/4)			172.1	168.9	165.1	160.9	156.2	149.6	145.3	139.1	132.4		
177.8 (7)			185.8	182.5	178.8	174.6	169.9	163.3	158.9	152.7	146.0	138.8	131.1
184.2 (7 1/4)			199.9	196.7	193.0	188.7	184.0	177.4	173.1	166.9	160.2	153.0	145.3
190.5 (7 1/2)			214.6	211.3	207.6	203.4	198.7	192.1	187.7	181.5	174.8	167.6	159.9
196.9 (7 3/4)			229.7	226.5	222.8	218.5	213.8	207.2	202.9	196.7	190.0	182.8	175.1
203.2 (8)			245.4	242.1	238.4	234.2	229.5	222.9	218.5	212.3	205.6	198.4	190.7
209.6 (8 1/4)			261.5	258.3	254.5	250.3	245.6	239.0	234.7	228.5	221.8	214.6	206.9
215.9 (8 1/2)			278.1	274.9	271.2	267.0	262.2	255.6	251.3	245.1	238.4	231.2	223.5
228.6 (9)			312.9	309.7	306.0	301.7	297.0	290.4	286.1	279.9	273.2	266.0	258.3
241.3 (9 1/2)			349.7	346.4	342.7	338.5	333.8	327.2	322.8	316.6	309.9	302.7	295.0
247.7 (9 3/4)			368.8	365.6	361.8	357.6	352.9	346.3	342.0	335.8	329.0	321.8	314.1
254.0 (10)			388.4	385.2	381.4	377.2	372.5	365.9	361.6	355.4	348.7	341.5	333.8
279.4 (11)			471.8	468.6	464.9	460.7	455.9	449.4	445.0	438.8	432.1	424.9	417.2
304.8 (12)			563.2	560.0	556.3	552.1	547.3	540.7	536.4	530.2	523.5	516.3	508.6

Note:
For spiral drill collars, subtract
4% from these weights

STEEL DRILL COLLAR WEIGHTS IN POUNDS PER FOOT

Drill collar OD inches	1	1 1/4	1 1/2	1 3/4	2	2 1/4	2 1/2	2 13/16	3	3 1/4	3 1/2	3 3/4	4
27/8 3	19.4 21.4	17.9 19.9	16.1 18.0										
31/8	23.4	21.9	20.1										
31/4	25.5	24.0	22.2										
31/2	30.0	28.5	26.7										
33/4	34.9	33.4	31.5										
4	40.0	38.5	36.7	34.5	32.0	29.2							
41/8	42.8	41.3	39.4	37.3	34.8	31.9							
41/4	45.6	44.1	42.2	40.0	37.5	34.7							
41/2	51.4	49.9	48.1	45.9	43.4	40.6							
43/4			54.2	52.1	49.6	46.7	43.6						
5			60.7	58.6	56.1	53.2	50.1						
51/4			67.6	65.4	62.9	60.1	56.9						
51/2			74.8	72.6	70.1	67.3	64.1	59.6					
53/4			82.3	80.1	77.6	74.8	71.6	67.2	64.2	60.1			
6			90.1	87.9	85.4	82.6	79.4	75.0	72.1	67.9			
61/4			98.3	96.1	93.6	90.8	87.6	83.2	80.3	76.1	71.6		
61/2			106.8	104.6	102.1	99.3	96.1	91.7	88.8	84.6	80.1		
63/4			115.6	113.5	111.0	108.1	105.0	100.5	97.6	93.4	88.9		
7			124.8	122.7	120.1	117.3	114.1	109.7	106.8	102.6	98.1	93.3	88.1
71/4			134.3	132.2	129.7	126.8	123.7	119.2	116.3	112.1	107.6	102.8	97.6
71/2			144.2	142.0	139.5	136.7	133.5	129.1	126.2	122.0	117.5	112.6	107.5
73/4			154.4	152.2	149.7	146.8	143.7	139.2	136.3	132.2	127.7	122.8	117.6
8			164.9	162.7	160.2	157.4	154.2	149.8	146.8	142.7	138.2	133.3	128.2
81/4			175.7	173.5	171.0	168.2	165.0	160.6	157.7	153.5	149.0	144.2	139.0
81/2			186.9	184.7	182.2	179.4	176.2	171.8	168.9	164.7	160.2	155.4	150.2
9			210.3	208.1	205.6	202.8	199.6	195.1	192.2	188.1	183.6	178.7	173.5
91/2			235.0	232.8	230.3	227.5	224.3	219.8	216.9	212.8	208.3	203.4	198.2
93/4			247.8	245.6	243.1	240.3	237.1	232.7	229.8	225.6	221.1	216.3	211.1
10			261.0	258.8	256.3	253.5	250.3	245.9	243.0	238.8	234.3	229.5	224.3
11			317.1	314.9	312.4	309.6	306.4	301.9	299.0	294.9	290.4	285.5	280.3
12			378.5	376.3	373.8	371.0	367.8	363.4	360.4	356.3	351.8	346.9	341.8

Note:
For spiral drill collars, subtract
4% from these weights

CYLINDRICAL DRILL COLLAR CONNECTIONS AND MAKE - UP TORQUES

OUTSIDE DIAMETER		INSIDE DIAMETER		THREAD TYPE	RECOMMENDED MAKE - UP TORQUE	
inch	mm	inch	mm		lbs-ft	daN-m
3 1/2	88.9	1 1/4 - 1 1/2	31.8 - 38.1	2 3/8 IF NC 26	4,610	624
3 3/4	95.3	1 1/4	31.8		5,500	746
3 3/4	95.3	1 1/2	38.1		4,670	633
3 7/8	98.4	1 1/4 - 2	31.8 - 50.8	2 7/8 IF NC 31	4,640	629
4 1/8	104.8	1 1/4 - 1 3/4	31.8 - 44.5		7,390	1,002
4 3/4	120.7	1 3/4 - 2 1/2	44.5 - 63.5	3 1/2 IF NC 38	9,990	1,354
5	127	1 3/4 - 2	44.5 - 50.8		13,950	1,891
5	127	2 1/4	57.2		12,910	1,750
5 1/2	139.7	2 - 2 13/16	50.8 - 71.4	4 1/2 REG	15,580	2,110
5 3/4	146.1	2 - 2 1/4	50.8 - 57.2		20,610	2,790
5 3/4	146.1	2 1/2	63.5		19,600	2,660
6	152.4	2	50.8		23,700	3,210
6	152.4	2 1/4	57.2		21,700	2,950
6	152.4	2 1/2	63.5		19,600	2,660
6	152.4	2 13/16	71.4		16,630	2,260
6	152.4	2 1/4 - 2 1/2	57.2 - 63.5	4 IF NC 46	23,400	3,180
6	152.4	2 13/16	71.4		22,400	3,040
6 1/4	158.8	2 1/4	57.2		28,000	3,800
6 1/4	158.8	2 1/2	63.5		25,700	3,480
6 1/4	158.8	2 13/16	71.4		22,400	3,040
6 1/4	158.8	2 1/4 - 2 13/16	57.2 - 71.4	4 1/2 IF NC 50	23,000	3,120
6 1/2	165.1	2 1/4 - 3	57.2 - 76.2		29,700	4,020
6 3/4	171.5	2 1/4	57.2		36,700	4,980
6 3/4	171.5	2 1/2	63.5		35,800	4,860
6 3/4	171.5	2 13/16	71.4		32,300	4,380
6 3/4	171.5	3	76.2		30,000	4,060
7 - 7 1/4	177.8-184.2	2 1/4	57.2		38,400	5,200
7 - 7 1/4	177.8-184.2	2 1/2	63.5		35,800	4,800
7 - 7 1/4	177.8-184.2	2 13/16	71.4		32,300	4,380

Note 1:

In the calculation of these recommended make-up torque values a coefficient of friction is assumed corresponding to the use of a thread compound conforming to API specifications (containing 40-60% by weight of finely powdered metallic zinc or 60% by weight of finely powdered metallic lead). If a pipe dope not conforming to API specifications is in use it will have a friction factor greater or less than 1 (by definition dope to API specifications has a friction factor equal to 1). The recommended make up torque then becomes the tabulated value multiplied by the actual friction factor.

Note 2:

The normal recommended torque range is the tabulated value -0%/+10%. Higher torque values may be used under extreme conditions.

CYLINDRICAL DRILL COLLAR CONNECTIONS AND MAKE - UP TORQUES

OUTSIDE DIAMETER		INSIDE DIAMETER		THREAD TYPE	RECOMMENDED MAKE - UP TORQUE	
inch	mm	inch	mm		lbs-ft	daN-m
6 ³ / ₄	171.5	2 ¹ / ₄ - 3	57.2 - 76.2	5 ¹ / ₂ REG	31,900	4,330
7	177.8	2 ¹ / ₄ - 2 ¹ / ₂	57.2 - 63.5		39,400	5,340
7 ¹ / ₄ - 7 ¹ / ₂	184.2-190.5	2 ¹ / ₄	57.2		42,500	5,760
7 ¹ / ₄ - 7 ¹ / ₂	184.2-190.5	2 ¹ / ₂	63.5		39,900	5,410
7 ¹ / ₄ - 7 ¹ / ₂	184.2-190.5	2 ³ / ₁₆	71.4		36,200	4,930
7 ¹ / ₂	190.5	2 ¹ / ₂ - 3 ¹ / ₄	63.5 - 82.6	6 ⁵ / ₈ REG	46,400	6,290
7 ³ / ₄	196.9	2 ¹ / ₂	63.5		55,600	7,540
7 ³ / ₄	196.9	2 ¹³ / ₁₆	71.4		53,300	7,230
7 ³ / ₄	196.9	3	76.2		50,700	6,880
8 - 8 ¹ / ₄	203-210	2 ¹ / ₂	63.5		57,400	7,780
8 - 8 ¹ / ₄	203-210	2 ¹³ / ₁₆	71.4		53,300	7,230
8 - 8 ¹ / ₄	203-210	3	76.2		50,700	6,880
8 ¹ / ₂	216	3 - 3 ³ / ₄	76.2 - 95.3	7 ⁵ / ₈ REG	60,400	8,190
8 ³ / ₄	222	3 - 3 ³ / ₄	76.2 - 95.3		72,200	9,790
9	229	3	76.2		84,400	11,450
9	229	3 ¹ / ₄	82.6		84,200	11,420
9	229	3 ¹ / ₂	88.9		79,500	10,780
9 ¹ / ₄ - 9 ¹ / ₂	235-241	3	76.2		88,600	12,010
9 ¹ / ₄ - 9 ¹ / ₂	235-241	3 ¹ / ₄	82.6		84,200	11,420
9 ¹ / ₄ - 9 ¹ / ₂	235-241	3 ¹ / ₂	88.9		79,500	10,780
9	229	3 - 3 ³ / ₄	76.2 - 95.3	7 ⁵ / ₈ H90	73,000	9,900
9 ¹ / ₄	235	3 - 3 ³ / ₄	76.2 - 95.3		86,000	11,660
9 ¹ / ₂	241	3 - 3 ¹ / ₂	76.2 - 88.9		99,500	13,490
10	254	3 - 3 ³ / ₄	76.2 - 95.3	8 ⁵ / ₈ REG	109,300	14,830
10 ¹ / ₄	260	3 - 3 ¹ / ₂	76.2 - 88.9		125,300	16,980
10 ¹ / ₄	260	3 ³ / ₄	95.3		125,000	16,950
10 ¹ / ₂	267	3	76.2		141,100	19,140
10 ¹ / ₂	267	3 ¹ / ₄	82.6		136,100	18,460
10 ¹ / ₂	267	3 ¹ / ₂	88.9		130,800	17,730
10 ¹ / ₂	267	3 ³ / ₄	95.3		125,000	16,950

CAPACITY OF CASING (1)

Size O.D.		Weight with couplings		Inside diameter		Drift diameter		Bbls per foot	m ³ /m	Feet per bbl	m/m ³
inch	mm	lbs/ft	kg/m	inch	mm	inch	mm				
4.500	114.30	9.50	14.14	4.090	103.89	3.965	100.71	0.0163	0.0085	61.54	117.93
		10.50	15.63	4.052	102.92	3.927	99.75	0.0159	0.0083	62.70	120.15
		11.60	17.26	4.000	101.60	3.875	98.43	0.0155	0.0081	64.34	123.30
		13.50	20.09	3.920	99.57	3.795	96.39	0.0149	0.0078	66.99	128.38
		15.10*	22.47*	3.826	97.18	3.701	94.01	0.0142	0.0074	70.32	134.77
		16.60*	24.70*	3.754	95.35	3.629	92.18	0.0137	0.0071	73.05	139.99
		18.80*	27.98*	3.640	92.46	3.515	89.28	0.0129	0.0067	77.69	148.89
4.750*	120.65*	16.00	23.81	4.082	103.68	3.957	100.51	0.0162	0.0084	61.78	118.39
5.000	127.00	11.50	17.11	4.560	115.82	4.435	112.65	0.0202	0.0105	49.51	94.87
		13.00	19.35	4.494	114.15	4.369	110.97	0.0196	0.0102	50.97	97.68
		15.00	22.32	4.408	111.96	4.283	108.79	0.0189	0.0098	52.98	101.53
		18.00	26.79	4.276	108.61	4.151	105.44	0.0178	0.0093	56.30	107.89
		20.30*	30.21*	4.184	106.27	4.059	103.10	0.0170	0.0089	58.80	112.69
		20.80*	30.95*	4.156	105.56	4.031	102.39	0.0168	0.0088	59.60	114.22
		23.20*	34.53*	4.044	102.72	3.919	99.54	0.0159	0.0083	62.95	120.63
		24.20*	36.01*	4.000	101.60	3.875	98.43	0.0155	0.0081	64.34	123.30
5.500	139.70	13.00*	19.35*	5.044	128.12	4.919	124.94	0.0247	0.0129	40.46	77.54
		14.00	20.83	5.012	127.31	4.887	124.13	0.0244	0.0127	40.98	78.53
		15.00*	22.32*	4.974	126.34	4.849	123.16	0.0240	0.0125	41.61	79.74
		15.50	23.07	4.950	125.73	4.825	122.56	0.0238	0.0124	42.01	80.51
		17.00	25.30	4.892	124.26	4.767	121.08	0.0232	0.0121	43.01	82.43
		20.00	29.76	4.778	121.36	4.653	118.19	0.0222	0.0116	45.09	86.41
		23.00	34.23	4.670	118.62	4.545	115.44	0.0212	0.0111	47.20	90.46
		26.00*	38.69*	4.548	115.52	4.423	112.34	0.0201	0.0105	49.77	95.37
5.750*	146.05*	14.00	20.83	5.290	134.37	5.165	131.19	0.0272	0.0142	36.79	70.50
		17.00	25.30	5.190	131.83	5.065	128.65	0.0262	0.0137	38.22	73.24
		19.50	29.02	5.090	129.29	4.965	126.11	0.0252	0.0131	39.73	76.14
		22.50	33.48	4.990	126.75	4.865	123.57	0.0242	0.0126	41.34	79.23
6.000*	152.40*	15.00	22.32	5.524	140.31	5.399	137.13	0.0296	0.0155	33.74	64.65
		16.00	23.81	5.500	139.70	5.375	136.53	0.0294	0.0153	34.03	65.22
		17.00	25.30	5.450	138.43	5.325	135.26	0.0289	0.0151	34.66	66.42
		18.00	26.79	5.424	137.77	5.299	134.59	0.0286	0.0149	34.99	67.06
		20.00	29.76	5.352	135.94	5.227	132.77	0.0278	0.0145	35.94	68.87
		23.00	34.23	5.240	133.10	5.115	129.92	0.0267	0.0139	37.49	71.85
		26.00	38.69	5.140	130.56	5.015	127.38	0.0257	0.0134	38.96	74.67
6.625	168.28	17.00*	25.30*	6.135	155.83	6.010	152.65	0.0366	0.0191	27.35	52.41
		20.00	29.76	6.049	153.64	5.924	150.47	0.0355	0.0185	28.13	53.91
		22.00*	32.74*	5.989	152.12	5.864	148.95	0.0348	0.0182	28.70	55.00
		24.00	35.72	5.921	150.39	5.796	147.22	0.0341	0.0178	29.36	56.27
		26.00*	38.69*	5.855	148.72	5.730	145.54	0.0333	0.0174	30.03	57.55
		28.00	41.67	5.791	147.09	5.666	143.92	0.0326	0.0170	30.70	58.83
		29.00*	43.16*	5.761	146.33	5.636	143.15	0.0322	0.0168	31.02	59.44
		32.00	47.62	5.675	144.15	5.550	140.97	0.0313	0.0163	31.96	61.26
7.000	177.80	17.00*	25.30*	6.538	166.07	6.413	162.89	0.0415	0.0217	24.08	46.15
		20.00	29.76	6.456	163.98	6.331	160.81	0.0405	0.0211	24.70	47.33
		22.00*	32.74*	6.398	162.10	6.273	159.33	0.0398	0.0207	25.15	48.19
		23.00	34.23	6.366	161.70	6.241	158.52	0.0394	0.0205	25.40	48.68
		24.00*	35.72*	6.336	160.93	6.211	157.76	0.0390	0.0203	25.64	49.14
		26.00	38.69	6.276	159.41	6.151	156.24	0.0383	0.0200	26.14	50.09
		28.00*	41.67*	6.214	157.84	6.089	154.66	0.0375	0.0196	26.66	51.09
		29.00	43.16	6.184	157.07	6.059	153.90	0.0371	0.0194	26.92	51.59
		30.00*	44.64*	6.154	156.31	6.029	153.14	0.0368	0.0192	27.18	52.09
		32.00	47.62	6.094	154.79	5.969	151.61	0.0361	0.0188	27.72	53.12
		33.70*	50.15*	6.048	153.62	5.923	150.44	0.0355	0.0185	28.14	53.93
		35.00	52.09	6.004	152.50	5.879	149.33	0.0350	0.0183	28.56	54.73
		38.00	56.55	5.920	150.37	5.795	147.19	0.0340	0.0178	29.37	56.29
		40.00*	59.53*	5.836	148.23	5.711	145.06	0.0331	0.0173	30.22	57.92

* Not API standard. Shown for information only

CAPACITY OF CASING (2)

Size O.D.		Weight with couplings		Inside diameter		Drift diameter		Bbls per foot	m ³ /m	Feet per bbl	m/m ³
inch	mm	lbs/ft	kg/m	inch	mm	inch	mm				
7.625	193.68	20.00*	29.76*	7.125	180.98	7.000	177.80	0.0493	0.0257	20.28	38.86
		24.00	35.72	7.025	178.44	6.900	175.26	0.0479	0.0250	20.86	39.97
		26.40	39.29	6.969	177.01	6.844	173.84	0.0472	0.0246	21.20	40.62
		29.70	44.20	6.875	174.63	6.750	171.45	0.0459	0.0240	21.78	41.74
		33.70	50.15	6.765	171.83	6.640	168.66	0.0445	0.0232	22.49	43.11
		39.00	58.04	6.625	168.28	6.500	165.10	0.0426	0.0222	23.45	44.95
		45.30*	67.41*	6.435	163.45	6.310	160.27	0.0402	0.0210	24.86	47.64
7.750*	196.85*	46.10	68.60	6.560	166.62	6.500	165.10	0.0418	0.0218	23.92	45.84
8.000*	203.20*	26.00	38.69	7.386	187.60	7.261	184.43	0.0530	0.0277	18.87	36.16
8.125*	206.38*	28.00	41.67	7.485	190.12	7.360	186.94	0.0644	0.0284	18.37	35.21
		32.00	47.62	7.385	187.58	7.260	184.40	0.0530	0.0276	18.88	36.17
		35.50	52.83	7.285	185.04	7.160	181.86	0.0516	0.0269	19.40	37.17
		39.50	58.78	7.185	182.50	7.060	179.32	0.0501	0.0262	19.94	38.21
8.625	219.08	20.00*	29.76*	8.191	208.05	8.066	204.88	0.0652	0.0340	15.34	29.40
		24.00	35.72	8.097	205.66	7.972	202.49	0.0637	0.0332	15.70	30.09
		28.00	41.67	8.017	203.63	7.892	200.46	0.0624	0.0326	16.02	30.69
		32.00	47.62	7.921	201.19	7.796	198.00	0.0609	0.0318	16.41	31.44
		36.00	53.57	7.825	198.76	7.700	195.58	0.0595	0.0310	16.81	32.22
		38.00*	56.55*	7.775	197.49	7.650	194.31	0.0587	0.0306	17.03	32.63
		40.00	59.53	7.725	196.22	7.600	193.04	0.0580	0.0302	17.25	33.06
		44.00*	63.99*	7.651	194.34	7.526	191.16	0.0569	0.0297	17.59	33.70
		44.00	65.48	7.625	193.68	7.500	190.50	0.0565	0.0295	17.71	33.93
		49.00	72.92	7.611	190.78	7.386	187.60	0.0548	0.0286	18.25	34.97
8.750*	222.25*	49.70	73.96	7.636	193.95	7.500	190.50	0.0566	0.0296	17.65	33.83
9.000*	228.60*	34.00	50.60	8.290	210.57	8.134	206.60	0.0668	0.0348	14.98	28.71
		38.00	56.55	8.196	208.18	8.040	204.22	0.0653	0.0341	15.32	29.37
		40.00	59.53	8.150	207.01	7.994	203.05	0.0645	0.0337	15.50	29.70
		45.00	66.97	8.032	204.01	7.876	200.05	0.0627	0.0327	15.96	30.58
		55.00	81.85	7.812	198.43	7.656	194.46	0.0593	0.0309	16.87	32.33
9.625	244.48	29.30*	43.60*	9.063	230.20	8.907	226.24	0.0798	0.0416	12.53	24.02
		32.30	48.07	9.001	228.63	8.845	224.66	0.0787	0.0411	12.71	24.35
		36.00	53.57	8.921	226.59	8.765	222.63	0.0773	0.0403	12.93	24.79
		38.00*	56.55*	8.885	225.68	8.760	222.50	0.0767	0.0400	13.04	24.99
		40.00	59.53	8.835	224.41	8.679	220.45	0.0758	0.0396	13.19	25.27
		43.50	64.74	8.755	222.38	8.599	218.42	0.0745	0.0389	13.43	25.74
		47.00	69.64	8.681	220.50	8.525	216.54	0.0732	0.0382	13.66	26.18
		53.50	79.62	8.535	216.79	8.379	212.83	0.0708	0.0369	14.13	27.08
		58.40*	86.91	8.435	214.25	8.279	210.29	0.0691	0.0361	14.47	27.73
		61.10*	90.93	8.375	212.73	8.219	208.76	0.0681	0.0356	14.68	28.13
		71.80*	106.85*	8.125	206.38	7.969	202.41	0.0641	0.0335	15.59	29.88
9.750*	247.65*	59.20	88.10	8.560	217.42	8.500	215.90	0.0712	0.0371	14.05	26.92
9.875*	250.83*	62.80	93.46	8.625	219.08	8.500	215.90	0.0723	0.0377	13.84	26.52
10.000*	254.00*	33.00	49.11	9.384	238.35	9.228	234.39	0.0855	0.0446	11.69	22.40
10.750	273.05	32.75	48.74	10.192	258.88	10.036	254.91	0.1009	0.0527	9.91	18.99
		35.75*	53.20*	10.136	257.45	9.980	253.49	0.0998	0.0521	10.02	19.20
		40.50	60.27	10.050	255.27	9.894	251.31	0.0981	0.0512	10.19	19.53
		45.50	67.71	9.950	252.73	9.794	248.77	0.0862	0.0502	10.40	19.93
		51.00	75.90	9.850	250.19	9.694	246.23	0.0943	0.0492	10.61	20.33
		54.00*	80.36*	9.784	248.51	9.628	244.55	0.0930	0.0485	10.75	20.61
		55.50	82.59	9.760	247.90	9.604	243.94	0.0925	0.0483	10.81	20.71
		60.70*	90.33*	9.660	245.36	9.504	241.40	0.0906	0.0473	11.03	21.14
		65.70	97.77*	9.560	242.82	9.404	238.86	0.0888	0.0463	11.26	21.59
		71.10*	105.81*	9.450	240.03	9.294	236.07	0.0868	0.0453	11.53	22.09

* Not API standard. Shown for information only

CAPACITY OF CASING (3)

Size O.D.		Weight with couplings		Inside diameter		Drift diameter		Bbls per foot	m ³ /m	Feet per bbl	m/m ³
inch	mm	lbs/ft	kg/m	inch	mm	inch	mm				
11.750	298.45	38.00*	56.55*	11.150	283.21	10.994	279.25	0.1208	0.0630	8.28	15.87
		42.00	62.50	11.084	281.53	10.928	277.57	0.1193	0.0623	8.38	16.06
		47.00	69.94	11.000	279.40	10.844	275.44	0.1175	0.0613	8.51	16.30
		54.00	80.36	10.880	276.35	10.724	272.39	0.1150	0.0600	8.70	16.67
		60.00	89.29	10.772	273.61	10.616	269.65	0.1127	0.0588	8.87	17.00
		65.00*	96.73*	10.682	271.32	10.526	267.36	0.1108	0.0578	9.02	17.29
		71.00*	105.66*	10.586	268.88	10.430	264.92	0.1089	0.0568	9.19	17.60
11.875	301.63	71.80	106.85	10.711	272.06	10.625	269.88	0.1114	0.0582	8.97	17.20
12.000*	304.80*	40.00	59.53	11.384	289.15	11.228	285.19	0.1259	0.0657	7.94	15.22
13.000*	330.20*	40.00	50.53	12.438	315.93	12.282	311.96	0.1503	0.0784	6.65	12.75
		45.00	66.97	12.360	313.94	12.204	309.98	0.1484	0.0774	6.74	12.91
		50.00	74.41	12.282	311.96	12.126	308.00	0.1465	0.0765	6.82	13.08
		54.00	80.36	12.220	310.39	12.064	306.43	0.1451	0.0757	6.89	13.21
13.375	339.73	48.00	71.43	12.715	322.96	12.559	319.00	0.1571	0.0820	6.37	12.20
		54.50	81.10	12.615	320.42	12.459	316.46	0.1546	0.0807	6.47	12.40
		61.00	90.78	12.515	317.88	12.359	313.92	0.1521	0.0794	6.57	12.60
		68.00	101.20	12.415	315.34	12.259	311.38	0.1497	0.0781	6.68	12.80
		72.00	107.15	12.347	313.61	12.191	309.65	0.1481	0.0773	6.75	12.94
		77.00*	114.59*	12.275	311.79	12.119	307.82	0.1464	0.0764	6.83	13.09
		80.70*	120.09*	12.215	310.26	12.059	306.30	0.1449	0.0756	6.90	13.22
		83.00*	123.52*	12.175	309.25	12.019	305.28	0.1440	0.0751	6.94	13.31
		85.00*	126.49*	12.159	308.84	12.003	304.88	0.1436	0.0749	6.96	13.34
		86.00*	127.98*	12.125	307.98	11.969	304.01	0.1428	0.0745	7.00	13.42
		92.00*	136.91*	12.031	305.59	11.875	301.63	0.1406	0.0734	7.11	13.63
		98.00*	145.84*	11.937	303.20	11.781	299.24	0.1384	0.0722	7.22	13.84
13.500*	342.90*	81.40	121.14	12.340	313.44	12.250	311.15	0.1479	0.0772	6.76	12.96
13.625*	346.08*	88.20	131.26	12.375	314.33	12.250	311.15	0.1488	0.0776	6.72	12.88
14.000*	355.60*	50.00	74.41	13.344	338.94	13.156	334.16	0.1730	0.0903	5.78	11.08
16.000	406.40	55.00*	81.85*	15.375	390.53	15.187	385.75	0.2296	0.1198	4.35	8.35
		65.00	96.73	15.250	387.35	15.062	382.58	0.2259	0.1179	4.43	8.48
		70.00*	104.17*	15.198	386.03	15.010	381.25	0.2244	0.1171	4.46	8.54
		75.00	111.61	15.124	384.15	14.936	379.38	0.2222	0.1159	4.50	8.62
		84.00	125.01	15.010	381.25	14.822	376.48	0.2189	0.1142	4.57	8.76
		109.00*	162.21*	14.688	373.08	14.500	368.30	0.2096	0.1094	4.77	9.14
18.625	473.08	78.00*	116.08*	17.855	453.52	17.667	448.74	0.3097	0.1616	3.23	6.19
		87.50	130.21	17.755	450.98	17.567	446.20	0.3062	0.1598	3.27	6.26
		96.50*	143.61*	17.655	448.44	17.467	443.66	0.3028	0.1580	3.30	6.33
20.000	508.00	94.00	139.89	19.124	485.75	18.936	480.98	0.3553	0.1854	2.81	5.39
		106.50	158.49	19.000	482.60	18.812	477.83	0.3507	0.1830	2.85	5.46
		133.00	197.93	18.730	475.74	18.542	470.97	0.3408	0.1778	2.93	5.62
21.500*	546.10*	92.50	137.66	20.710	526.04	20.522	521.26	0.4166	0.2174	2.40	4.60
		103.00	153.28	20.610	523.50	20.422	518.72	0.4126	0.2153	2.42	4.64
		114.00	169.65	20.510	520.96	20.322	516.18	0.4086	0.2132	2.45	4.69
24.000*	609.60*	94.62	140.81	23.250	590.55	--:--	--:--	0.5251	0.2740	1.90	3.65
		125.49	186.75	23.000	584.20	--:--	--:--	0.5139	0.2682	1.95	3.73
		156.03	232.20	22.750	577.85	--:--	--:--	0.5028	0.2624	1.99	3.81
		186.23	277.14	22.500	571.50	--:--	--:--	0.4918	0.2566	2.03	3.90
24.500*	622.30*	100.50	149.56	23.750	603.25	23.562	598.48	0.5479	0.2859	1.83	3.50
		113.00	168.16	23.650	600.71	23.462	595.94	0.5433	0.2835	1.84	3.53
30.000*	762.00*	118.65	176.57	29.250	742.95	--:--	--:--	0.8311	0.4337	1.20	2.31
		157.53	234.43	29.000	736.60	--:--	--:--	0.8170	0.4263	1.22	2.35
		196.08	291.80	28.750	730.25	--:--	--:--	0.8029	0.4190	1.25	2.39
		234.29	348.66	28.500	723.90	--:--	--:--	0.7890	0.4117	1.27	2.43

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CAPACITY OF TUBING

size O.D.		Weight With Couplings						Inside Diameter		Drift Diameter		bbls per lin. ft.	m³/m	lin. ft. per bbls	m/m³
		Non Upset		Upset		Integral Joint									
inch	mm	lb/ft	kg/m	lb/ft	kg/m	lb/ft	kg/m	inch	mm	inch	mm				
1.050	26.67	1.14	1.70	1.20	1.79	1.20	1.79	0.824	20.93	0.730	18.54	0.0007	0.0003	1516.13	2905.49
1.315	33.40	-----	-----	-----	-----	1.30*	1.93*	1.125	28.58	0.955	24.26	0.0012	0.0006	813.36	1558.72
		-----	-----	-----	-----	1.43*	2.13*	1.097	27.86	0.955	24.26	0.0012	0.0006	855.42	1639.31
		-----	-----	-----	-----	1.63*	2.43*	1.065	27.05	0.955	24.26	0.0011	0.0006	907.59	1739.30
		1.70	2.53	1.80	2.68	1.70	2.53	1.049	26.64	0.955	24.26	0.0011	0.0006	935.49	1792.77
1.660	42.16	-----	-----	-----	-----	2.25*	3.35*	0.957	24.31	0.848	21.54	0.0009	0.0005	1124.00	2154.02
		-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
		2.30	3.42	2.33	3.57	2.10	3.13	1.410	35.81	1.286	32.66	0.0019	0.0010	517.79	992.28
		-----	-----	-----	-----	2.33	3.47	1.380	35.05	1.286	32.66	0.0018	0.0010	540.55	1035.90
1.900	48.26	-----	-----	-----	-----	3.02*	4.49*	1.278	32.46	1.184	30.07	0.0016	0.0008	630.27	1207.85
		-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
		2.75	4.09	2.90	4.32	2.40	3.57	1.650	41.91	1.516	38.51	0.0026	0.0014	378.11	724.61
		-----	-----	-----	-----	2.75	4.09	1.610	40.89	1.516	38.51	0.0025	0.0013	397.14	761.07
2.000	50.80	-----	-----	-----	-----	3.64*	5.42*	1.500	38.10	1.406	35.71	0.0022	0.0011	457.52	876.78
		-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
		-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
		-----	-----	-----	-----	3.30	4.91	1.670	42.42	1.576	40.03	0.0027	0.0014	369.11	707.36
2.063	52.40	-----	-----	-----	-----	2.66*	3.96*	1.813	46.05	1.656	42.06	0.0032	0.0017	313.18	600.18
		3.20	4.76	-----	-----	3.25	4.84	1.751	44.48	1.656	42.06	0.0030	0.0016	335.75	643.43
2.375	60.33	-----	-----	-----	-----	3.10*	4.61*	2.125	53.98	1.901	48.29	0.0044	0.0023	227.97	436.87
		-----	-----	-----	-----	3.32*	4.94*	2.107	53.52	1.901	48.29	0.0043	0.0023	231.88	444.37
		4.00	5.95	-----	-----	-----	-----	2.041	51.84	1.947	49.45	0.0040	0.0021	247.12	473.57
		4.60	6.85	4.70	6.99	4.70	6.99	1.995	50.67	1.901	48.29	0.0039	0.0020	258.65	495.67
2.875	73.03	-----	-----	5.30	7.89	5.30	7.89	1.939	49.25	1.845	46.86	0.0037	0.0019	273.80	524.71
		5.80	8.63	5.95	8.86	5.95	8.86	1.867	47.42	1.773	45.03	0.0034	0.0018	295.33	565.96
		-----	-----	-----	-----	6.20*	9.23*	1.853	47.07	1.759	44.68	0.0033	0.0017	299.81	574.54
		-----	-----	-----	-----	7.70*	11.46*	1.703	43.26	1.609	40.87	0.0028	0.0015	354.94	680.21
3.500	88.90	-----	-----	-----	-----	4.36*	6.49*	2.579	65.51	2.485	63.12	0.0065	0.0034	154.77	296.60
		-----	-----	-----	-----	4.64*	6.91*	2.563	65.10	2.347	59.61	0.0064	0.0033	156.71	300.32
		6.40	9.52	6.50	9.67	6.50	9.67	2.441	62.00	2.347	59.61	0.0058	0.0030	172.76	331.08
		7.90	11.76	7.90	11.76	7.90	11.76	2.323	59.00	2.229	56.62	0.0052	0.0027	190.76	365.57
4.000	101.60	8.60	12.80	8.70	12.95	8.70	12.95	2.259	57.38	2.165	54.99	0.0050	0.0026	201.72	386.58
		-----	-----	-----	-----	8.90*	13.25*	2.243	56.97	2.149	54.58	0.0049	0.0026	204.61	392.12
		-----	-----	9.50	14.14	9.50	14.14	2.195	55.75	2.101	53.37	0.0047	0.0024	213.66	409.45
		-----	-----	-----	-----	10.40*	15.48*	2.151	54.64	2.057	52.25	0.0045	0.0023	222.49	426.38
4.500	114.30	-----	-----	-----	-----	10.70*	15.92*	2.091	53.11	1.997	50.72	0.0042	0.0022	235.44	451.20
		-----	-----	-----	-----	11.00*	16.37*	2.065	52.45	1.971	50.06	0.0041	0.0022	241.41	462.63
		-----	-----	-----	-----	11.65*	17.34*	1.995	50.67	1.901	48.29	0.0039	0.0020	258.65	495.67
		-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	
5.000	127.00	-----	-----	-----	-----	5.63*	8.38*	3.188	80.98	3.063	77.80	0.0099	0.0052	101.29	194.11
		7.70	11.46	-----	-----	7.70	11.46	3.068	77.93	2.943	74.75	0.0091	0.0048	109.37	209.59
		9.20	13.69	9.30	13.84	9.30	13.84	2.992	76.00	2.867	72.82	0.0087	0.0045	114.99	220.37
		10.20	15.18	10.30	15.33	10.30	15.33	2.922	74.22	2.797	71.04	0.0083	0.0043	120.57	231.05
5.500	141.30	-----	-----	-----	-----	12.80*	19.05*	2.764	70.21	2.639	67.03	0.0074	0.0039	134.75	258.23
		12.70	18.90	12.95	19.27	12.95	19.27	2.750	69.85	2.625	66.68	0.0073	0.0038	136.12	260.86
		-----	-----	-----	-----	13.30*	19.79*	2.764	70.21	2.639	67.03	0.0074	0.0039	134.75	258.23
		-----	-----	-----	-----	15.80*	23.51*	2.548	64.72	2.423	61.54	0.0063	0.0033	158.56	303.86
6.000	165.10	-----	-----	-----	-----	16.70*	24.85*	2.480	62.99	2.355	59.82	0.0060	0.0031	167.37	320.75
		-----	-----	-----	-----	17.05*	25.37*	2.440	61.98	2.315	58.80	0.0058	0.0030	172.91	331.36
		-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
		-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	
6.500	182.90	-----	-----	-----	-----	9.40	13.99	3.548	90.12	3.423	86.94	0.0122	0.0064	81.78	156.71
		-----	-----	-----	-----	11.00*	16.37*	3.476	88.29	3.351	85.12	0.0117	0.0061	85.20	163.27
		-----	-----	-----	-----	11.60*	17.26*	3.428	87.07	3.303	83.90	0.0114	0.0060	87.60	167.88
		-----	-----	-----	-----	13.40*	19.94*	3.340	84.84	3.215	81.66	0.0108	0.0057	92.28	176.84
7.000	203.20	-----	-----	-----	-----	22.80*	33.93*	2.780	70.61	2.655	67.44	0.0075	0.0039	133.20	255.26
		-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
		-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
		-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	
7.500	224.40	12.60	18.75	12.75	18.98	12.75	18.98	3.958	100.53	3.833	97.36	0.0152	0.0079	65.71	125.93
		-----	-----	-----	-----	13.50*	20.09*	3.920	99.57	3.795	96.39	0.0149	0.0078	66.99	128.38
		-----	-----	-----	-----	15.50*	23.07*	3.826	97.18	3.701	94.01	0.0142	0.0074	70.32	134.77
		-----	-----	-----	-----	16.90*	25.15*	3.754	95.35	3.629	92.18	0.0137	0.0071	73.05	139.99
8.000	254.00	-----	-----	-----	-----	19.20*	28.57*	3.640	92.46	3.515	89.28	0.0129	0.0067	77.69	148.89
		-----	-----	-----	-----	21.60*	32.15*	3.500	88.90	3.375	85.73	0.0119	0.0062	84.03	161.04
		-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
		-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	

* Not API standard. Shown for information only

THE VOLUME OF A CYLINDER

I.D. or O.D.		Bbls per lin. foot	m ³ /m	Lin. feet per bbl.	m/m ³	I.D. or O.D.		Bbls per lin. foot	m ³ /m	Lin. feet per bbl.	m/m ³
inches	mm					inches	mm				
2.000	50.80	0.00389	0.00203	257.4	493.4	10.000	254.0	0.097	0.0507	10.29	19.74
2.125	53.98	0.00439	0.00229	228.0	437.0	10.125	257.2	0.100	0.0519	10.04	19.25
2.250	57.15	0.00492	0.00257	203.3	389.8	10.250	260.4	0.102	0.0532	9.80	18.78
2.375	60.33	0.00548	0.00286	182.5	349.9	10.375	263.5	0.105	0.0545	9.56	18.33
2.500	63.50	0.00607	0.00317	164.7	315.8	10.500	266.7	0.107	0.0559	9.34	17.90
2.625	66.68	0.00669	0.00349	149.4	286.4	10.625	269.9	0.110	0.0572	9.12	17.48
2.750	69.85	0.00735	0.00383	136.1	261.0	10.750	273.1	0.112	0.0586	8.91	17.08
2.875	73.03	0.00803	0.00419	124.5	238.8	10.875	276.2	0.115	0.0599	8.70	16.69
3.000	76.20	0.00874	0.00456	114.4	219.3	11.000	279.4	0.118	0.0613	8.51	16.31
3.125	79.38	0.00949	0.00495	105.4	202.1	11.125	282.6	0.120	0.0627	8.32	15.95
3.250	82.55	0.0103	0.00535	97.5	186.8	11.250	285.8	0.123	0.0641	8.13	15.59
3.375	85.73	0.0111	0.00577	90.4	173.3	11.375	288.9	0.126	0.0656	7.96	15.25
3.500	88.90	0.0119	0.00621	84.0	161.1	11.500	292.1	0.128	0.0670	7.78	14.92
3.625	92.08	0.0128	0.00666	78.3	150.2	11.625	295.3	0.131	0.0685	7.62	14.60
3.750	95.25	0.0137	0.00713	73.2	140.3	11.750	298.5	0.134	0.0700	7.46	14.29
3.875	98.43	0.0146	0.00761	68.6	131.4	11.875	301.6	0.137	0.0715	7.30	14.00
4.000	101.6	0.0155	0.00811	64.3	123.3	12.000	304.8	0.140	0.0730	7.15	13.71
4.125	104.8	0.0165	0.00862	60.5	116.0	12.125	308.0	0.143	0.0745	7.00	13.45
4.250	108.0	0.0175	0.00915	57.0	109.3	12.250	311.2	0.146	0.0760	6.86	13.12
4.375	111.1	0.0186	0.00970	53.8	103.1	12.375	314.3	0.149	0.0776	6.72	12.89
4.500	114.3	0.0197	0.0103	50.8	97.5	12.500	317.5	0.152	0.0792	6.59	12.63
4.625	117.5	0.0208	0.0108	48.1	92.3	12.625	320.7	0.155	0.0808	6.46	12.38
4.750	120.7	0.0219	0.0114	45.6	87.5	12.750	323.9	0.158	0.0824	6.33	12.14
4.875	123.8	0.0231	0.0120	43.3	83.0	12.875	327.0	0.161	0.0840	6.21	11.91
5.000	127.0	0.0243	0.0127	41.2	78.9	13.000	330.2	0.164	0.0856	6.09	11.68
5.125	130.2	0.0255	0.0133	39.2	75.1	13.125	333.4	0.167	0.0873	5.98	11.46
5.250	133.4	0.0268	0.0140	37.3	71.6	13.250	336.6	0.171	0.0890	5.86	11.24
5.375	136.5	0.0281	0.0146	35.6	68.3	13.375	339.7	0.174	0.0906	5.75	11.03
5.500	139.7	0.0294	0.0153	34.0	65.2	13.500	342.9	0.177	0.0923	5.65	10.83
5.625	142.9	0.0307	0.0160	32.5	62.4	13.625	346.1	0.180	0.0941	5.55	10.63
5.750	146.1	0.0321	0.0168	31.1	59.7	13.750	349.3	0.184	0.0958	5.44	10.44
5.875	149.2	0.0335	0.0175	29.8	57.2	13.875	352.4	0.187	0.0975	5.35	10.25
6.000	152.4	0.0350	0.0182	28.6	54.8	14.000	355.6	0.190	0.0993	5.25	10.07
6.125	155.6	0.0364	0.0190	27.4	52.6	14.125	358.8	0.194	0.1011	5.16	9.89
6.250	158.8	0.0379	0.0198	26.4	50.5	14.250	362.0	0.197	0.1029	5.07	9.72
6.375	161.9	0.0395	0.0206	25.3	48.6	14.375	365.1	0.201	0.1047	4.98	9.55
6.500	165.1	0.0410	0.0214	24.4	46.7	14.500	368.3	0.204	0.1065	4.90	9.39
6.625	168.3	0.0426	0.0222	23.5	45.0	14.625	371.5	0.208	0.1084	4.81	9.23
6.750	171.5	0.0443	0.0231	22.6	43.3	14.750	374.7	0.211	0.1102	4.73	9.07
6.875	174.6	0.0459	0.0239	21.8	41.8	14.875	377.8	0.215	0.1121	4.65	8.92
7.000	177.8	0.0476	0.0248	21.0	40.3	15.000	381.0	0.219	0.1140	4.58	8.77
7.125	181.0	0.0493	0.0257	20.3	38.9	15.125	384.2	0.222	0.1159	4.50	8.63
7.250	184.2	0.0511	0.0266	19.6	37.5	15.250	387.4	0.226	0.1178	4.43	8.49
7.375	187.3	0.0528	0.0276	18.9	36.3	15.375	390.5	0.230	0.1198	4.35	8.35
7.500	190.5	0.0546	0.0285	18.3	35.1	15.500	393.7	0.233	0.1217	4.28	8.21
7.625	193.7	0.0565	0.0295	17.7	33.9	15.625	396.9	0.237	0.1237	4.22	8.08
7.750	196.9	0.0583	0.0304	17.1	32.9	15.750	400.1	0.241	0.1257	4.15	7.96
7.875	200.0	0.0602	0.0314	16.6	31.8	15.875	403.2	0.245	0.1277	4.08	7.83
8.000	203.2	0.0622	0.0324	16.1	30.8	16.000	406.4	0.249	0.1297	4.02	7.71
8.125	206.4	0.0641	0.0335	15.6	29.9	16.125	409.6	0.253	0.1318	3.96	7.59
8.250	209.6	0.0661	0.0345	15.1	29.0	16.250	412.8	0.257	0.1338	3.90	7.47
8.375	212.7	0.0681	0.0355	14.7	28.1	16.375	415.9	0.260	0.1359	3.84	7.36
8.500	215.9	0.0702	0.0366	14.2	27.3	16.500	419.1	0.264	0.1380	3.78	7.25
8.625	219.1	0.0723	0.0377	13.8	26.5	16.625	422.3	0.268	0.1400	3.72	7.14
8.750	222.3	0.0744	0.0388	13.4	25.8	16.750	425.5	0.273	0.1422	3.67	7.03
8.875	225.4	0.0765	0.0399	13.1	25.1	16.875	428.6	0.277	0.1443	3.61	6.93
9.000	228.6	0.0787	0.0410	12.7	24.4	17.000	431.8	0.281	0.1464	3.56	6.83
9.125	231.8	0.0809	0.0422	12.4	23.7	17.125	435.0	0.285	0.1486	3.51	6.73
9.250	235.0	0.0831	0.0434	12.0	23.1	17.250	438.2	0.289	0.1508	3.46	6.63
9.375	238.1	0.0854	0.0445	11.7	22.5	17.375	441.3	0.293	0.1530	3.41	6.54
9.500	241.3	0.0877	0.0457	11.4	21.9	17.500	444.5	0.298	0.1552	3.36	6.44
9.625	244.5	0.0900	0.0469	11.1	21.3	17.625	447.7	0.302	0.1574	3.31	6.35
9.750	247.7	0.0923	0.0482	10.8	20.8	17.750	450.9	0.306	0.1596	3.27	6.26
9.875	250.8	0.0947	0.0494	10.6	20.2	17.875	454.0	0.310	0.1619	3.22	6.18

THE VOLUME OF A CYLINDER

I.D. or O.D.		Bbls per lin. foot	m³/m	Lin. feet per bbl.	m/m³
inches	mm				
18.00	457.2	0.315	0.164	3.18	6.09
18.25	463.6	0.324	0.169	3.09	5.93
18.50	469.9	0.332	0.173	3.01	5.77
18.75	476.3	0.342	0.178	2.93	5.61
19.00	482.6	0.351	0.183	2.85	5.47
19.25	489.0	0.360	0.188	2.78	5.33
19.50	495.3	0.369	0.193	2.71	5.19
19.75	501.7	0.379	0.198	2.64	5.06
20.00	508.0	0.389	0.203	2.574	4.93
20.25	514.4	0.398	0.208	2.510	4.81
20.50	520.7	0.408	0.213	2.449	4.70
20.75	527.1	0.418	0.218	2.391	4.58
21.00	533.4	0.428	0.223	2.334	4.48
21.25	539.8	0.439	0.229	2.280	4.37
21.50	546.1	0.449	0.234	2.227	4.27
21.75	552.5	0.460	0.240	2.176	4.17
22.00	558.8	0.470	0.245	2.127	4.08
22.25	565.2	0.481	0.251	2.079	3.99
22.50	571.5	0.492	0.257	2.033	3.90
22.75	577.9	0.503	0.262	1.989	3.81
23.00	584.2	0.514	0.268	1.946	3.73
23.25	590.6	0.525	0.274	1.904	3.65
23.50	596.9	0.536	0.280	1.864	3.57
23.75	603.3	0.548	0.286	1.825	3.50
24.00	609.6	0.560	0.292	1.787	3.43
24.25	616.0	0.571	0.298	1.750	3.36
24.50	622.3	0.583	0.304	1.715	3.29
24.75	628.7	0.595	0.310	1.680	3.22
25.00	635.0	0.607	0.317	1.647	3.16
25.25	641.4	0.619	0.323	1.615	3.10
25.50	647.7	0.632	0.329	1.583	3.04
25.75	654.1	0.644	0.336	1.552	2.98
26.00	660.4	0.657	0.343	1.523	2.92
26.25	666.8	0.669	0.349	1.494	2.86
26.50	673.1	0.682	0.356	1.466	2.81
26.75	679.5	0.695	0.363	1.439	2.76
27.00	685.8	0.708	0.369	1.412	2.71
27.25	692.2	0.721	0.376	1.386	2.66
27.50	698.5	0.735	0.383	1.361	2.61
27.75	704.9	0.748	0.390	1.337	2.56
28.00	711.2	0.762	0.397	1.313	2.52
28.25	717.6	0.775	0.404	1.290	2.47
28.50	723.9	0.789	0.412	1.267	2.43
28.75	730.3	0.803	0.419	1.245	2.39
29.00	736.6	0.817	0.426	1.224	2.35
29.25	743.0	0.831	0.434	1.203	2.31
29.50	749.3	0.845	0.441	1.183	2.27
29.75	755.7	0.860	0.448	1.163	2.23
30.00	762.0	0.874	0.456	1.144	2.19
30.25	768.4	0.889	0.464	1.125	2.16
30.50	774.7	0.904	0.471	1.107	2.12
30.75	781.1	0.919	0.479	1.089	2.09
31.00	787.4	0.934	0.487	1.071	2.05
31.25	793.8	0.949	0.495	1.054	2.02
31.50	800.1	0.964	0.503	1.037	1.99
31.75	806.5	0.979	0.511	1.021	1.96
32.00	812.8	0.995	0.519	1.005	1.93
32.25	819.2	1.010	0.527	0.990	1.90
32.50	825.5	1.026	0.535	0.975	1.87
32.75	831.9	1.042	0.543	0.960	1.84
33.00	838.2	1.058	0.552	0.945	1.81
33.25	844.6	1.074	0.560	0.931	1.79
33.50	850.9	1.090	0.569	0.917	1.76
33.75	857.3	1.107	0.577	0.904	1.73
I.D. or O.D.		Bbls per lin. foot	m³/m	Lin. feet per bbl.	m/m³
inches	mm				
34.00	863.6	1.12	0.586	0.890	1.707
34.25	870.0	1.14	0.594	0.878	1.682
34.50	876.3	1.16	0.603	0.865	1.658
34.75	882.7	1.17	0.612	0.852	1.634
35.00	889.0	1.19	0.621	0.840	1.611
35.25	895.4	1.21	0.630	0.828	1.588
35.50	901.7	1.22	0.639	0.817	1.566
35.75	908.1	1.24	0.648	0.805	1.544
36.00	914.4	1.26	0.657	0.794	1.523
36.25	920.8	1.28	0.666	0.783	1.502
36.50	927.1	1.29	0.675	0.773	1.481
36.75	933.5	1.31	0.684	0.762	1.461
37.00	939.8	1.33	0.694	0.752	1.442
37.25	946.2	1.35	0.703	0.742	1.422
37.50	952.5	1.37	0.713	0.732	1.403
37.75	958.9	1.38	0.722	0.722	1.385
38.00	965.2	1.40	0.732	0.713	1.367
38.25	971.6	1.42	0.741	0.704	1.349
38.50	977.9	1.44	0.751	0.694	1.331
38.75	984.3	1.46	0.761	0.686	1.314
39.00	990.6	1.48	0.771	0.677	1.298
39.25	997.0	1.50	0.781	0.668	1.281
39.50	1,003	1.52	0.791	0.660	1.265
39.75	1,009	1.54	0.801	0.651	1.249
40.0	1,016	1.55	0.811	0.643	1.233
40.5	1,029	1.59	0.831	0.628	1.203
41.0	1,041	1.63	0.852	0.612	1.174
41.5	1,054	1.67	0.873	0.598	1.146
42.0	1,067	1.71	0.894	0.584	1.119
42.5	1,080	1.75	0.915	0.570	1.093
43.0	1,092	1.80	0.937	0.557	1.067
43.5	1,105	1.84	0.959	0.544	1.043
44.0	1,118	1.88	0.981	0.532	1.019
44.5	1,130	1.92	1.003	0.520	0.997
45.0	1,143	1.97	1.026	0.508	0.975
45.5	1,156	2.01	1.049	0.497	0.953
46.0	1,168	2.06	1.072	0.486	0.933
46.5	1,181	2.10	1.096	0.476	0.913
47.0	1,194	2.15	1.119	0.466	0.893
47.5	1,207	2.19	1.143	0.456	0.875
48.0	1,219	2.24	1.167	0.447	0.857
48.5	1,232	2.29	1.192	0.438	0.839
49.0	1,245	2.33	1.217	0.429	0.822
49.5	1,257	2.38	1.242	0.420	0.805
50.0	1,270	2.43	1.267	0.412	0.789
50.5	1,283	2.48	1.292	0.404	0.774
51.0	1,295	2.53	1.318	0.396	0.759
51.5	1,308	2.58	1.344	0.388	0.744
52.0	1,321	2.63	1.370	0.381	0.730
52.5	1,334	2.68	1.397	0.373	0.716
53.0	1,346	2.73	1.423	0.366	0.703
53.5	1,359	2.78	1.450	0.360	0.690
54.0	1,372	2.83	1.478	0.353	0.677
54.5	1,384	2.89	1.505	0.347	0.664
55.0	1,397	2.94	1.533	0.340	0.652
55.5	1,410	2.99	1.561	0.334	0.641
56.0	1,422	3.05	1.589	0.328	0.629
56.5	1,435	3.10	1.618	0.322	0.618
57.0	1,448	3.16	1.646	0.317	0.607
57.5	1,461	3.21	1.675	0.311	0.597
58.0	1,473	3.27	1.705	0.306	0.587
58.5	1,486	3.32	1.734	0.301	0.577
59.0	1,499	3.38	1.764	0.296	0.567
59.5	1,511	3.44	1.794	0.291	0.557
60.0	1,524	3.50	1.824	0.286	0.548

D – BITS

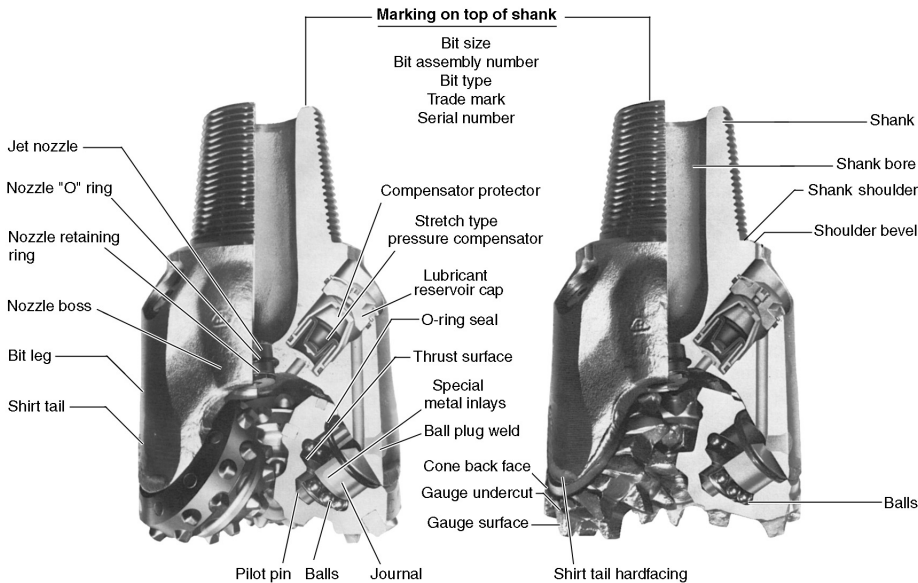
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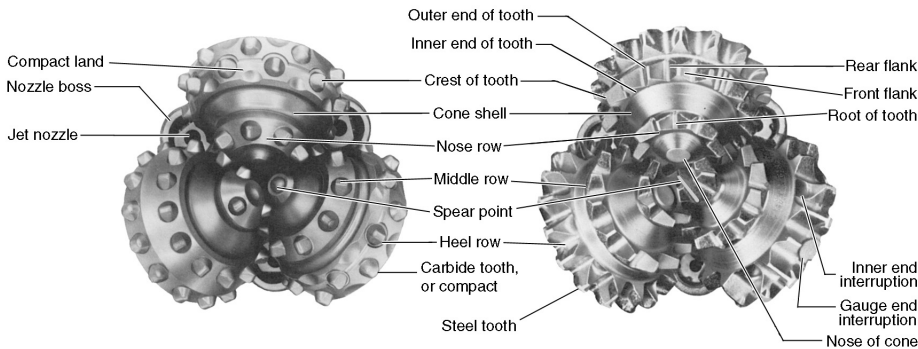
ROCK BIT NOMENCLATURE

TRI-CONE ROLLER BITS



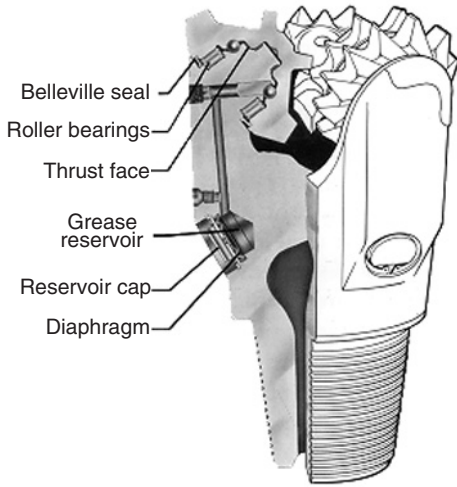
TUNGSTEN CARBIDE BIT
with sealed journal bearings

STEEL TOOTH BIT
with sealed journal bearings

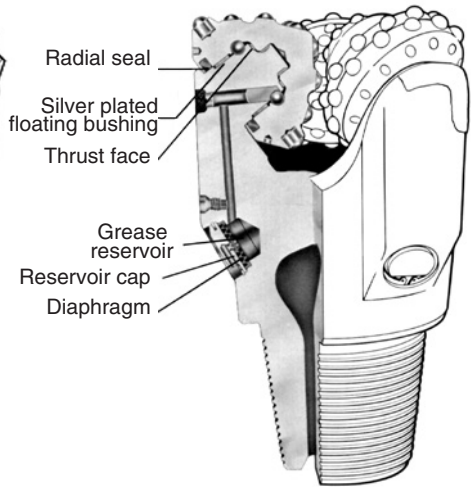


ROCK BIT NOMENCLATURE

DETAILS OF SEALED BEARINGS



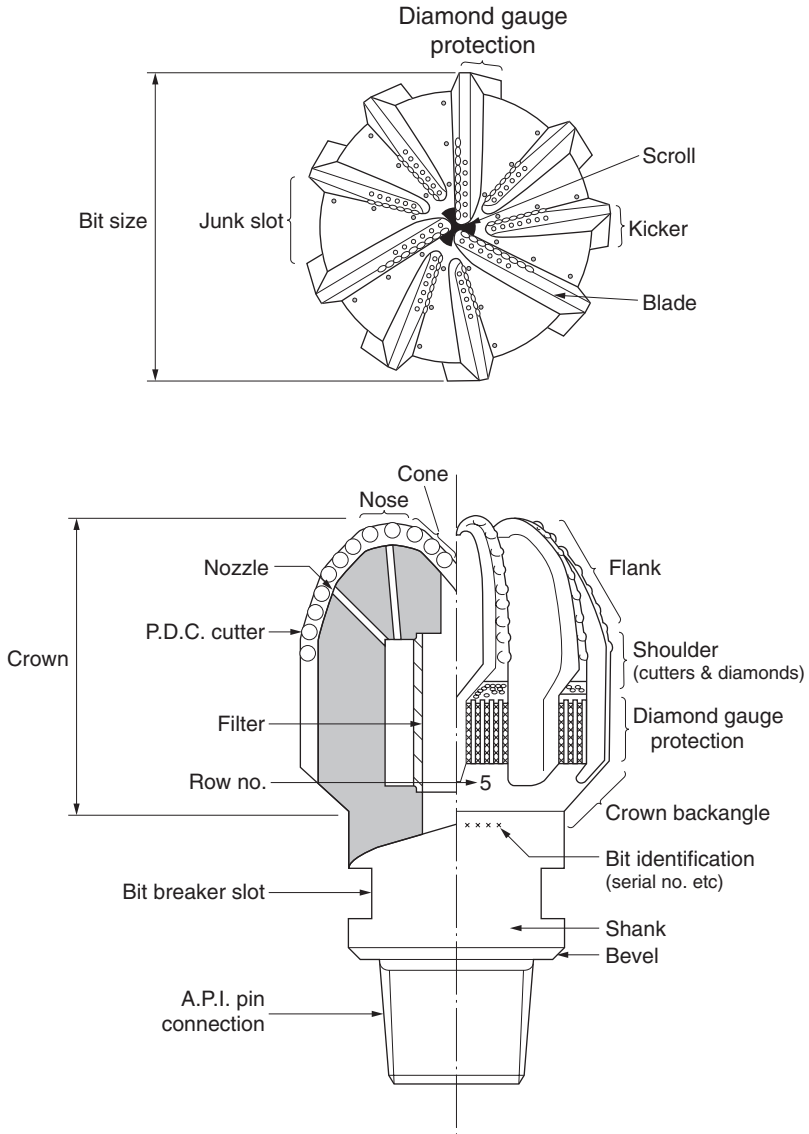
Roller bearings



Journal bearings

ROCK BIT NOMENCLATURE

PDC (Polycrystalline Diamond Compact) BITS

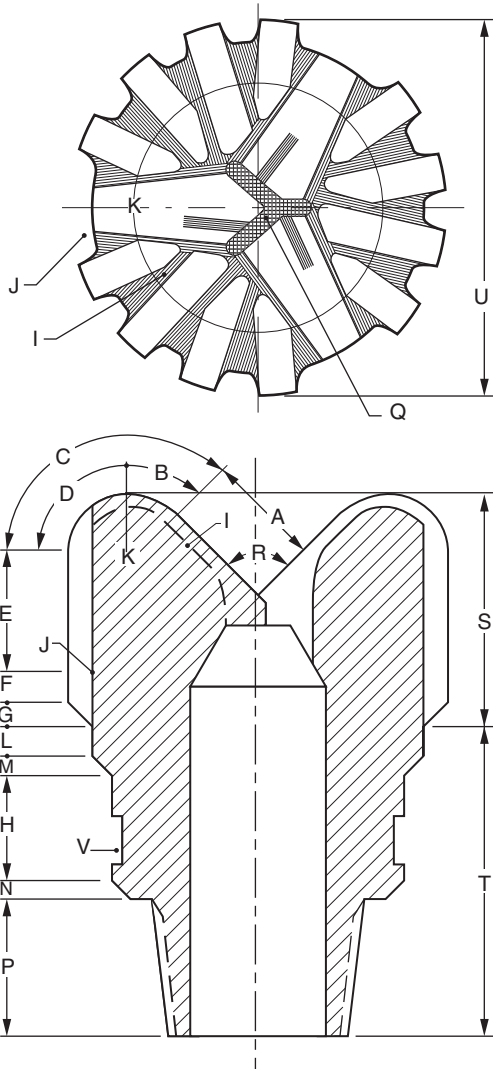


Typical part section through centre of bit

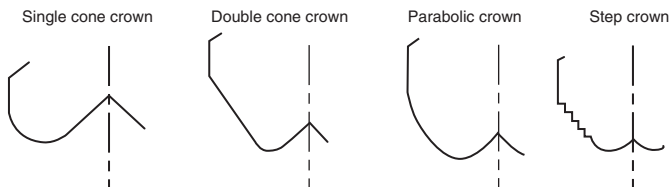
ROCK BIT NOMENCLATURE

DIAMOND BITS

- A - Throat
- B - I.D. Radius
- C - Nose
- D - O.D. Radius
- E - O.D. Gage
- F - O.D. Above Diamonds
- G - O.D. Angle
- H - Steel Shank
- I - Fluid Courses
- J - Junk Slots/Slabs
- K - Contact Point/Diameter
- L - Shoulder
- M - Shoulder Angle
- N - Shank Angle
- P - Thread Connection
- Q - Fluid Entrance
- R - Cone Angle
- S - Crown
- T - Pin Shank
- V - Breaker Slot



Diamond bit profiles



ROCK BIT CLASSIFICATION SCHEMES

INTRODUCTION

There are two classification schemes, one for roller cone bits and one for fixed cutter (diamond) bits. The current versions of each were introduced in 1987 jointly by the SPE (Society of Petroleum Engineers) and the IADC (International Association of Drilling Contractors).

The classifications schemes both make use of four characters. For roller cone bits this consists of three numbers and a letter, whereas diamond bits use a letter and three numbers (diamond bits may also use a letter as the third character). The basis for the classification is however slightly different as explained below, even though the end result is the same.

Roller cone bits

The system is based primarily on the formation characteristics with the first two characters indicating the hardness of the formation for which the bit is designed/suited, and also indicating whether it has milled teeth or tungsten carbide inserts. The second character is used to sub-divide the hardness classes defined by the first character.

The third and fourth characters indicate the general features of the bit itself, such as the type of bearing, whether there is gauge protection or not (which also reflect the type of formation for which it is intended) and whether the bit has any special features or whether it is intended for any special applications, such as air drilling.

The significance of these four characters is summarised in the boxes on [page D-6](#).

As an example a bit classified 6.3.5.Y would be a tungsten carbide insert bit with sealed roller bearings and gauge protection. It would have conical inserts intended for hard formations.

Diamond bits

The classification system of diamond bits is based much more on the construction and geometry of the bit than on the explicit formation type. For this reason the manufacturers sometimes quote not only the classification code for the diamond bit itself, but also the code for the tri-cone bit which would be appropriate for the same formations.

The first character indicates the cutter type and the body material. The second character indicates the profile of the cutting face of the bit. The third character indicates the design of the bit with regard to the flow of drilling fluid across its face. The fourth and last character indicates the size and density of the cutters.

The meanings of the four characters are shown in the boxes on [page D-7](#).

Charts

Each bit manufacturer produces a classification chart for tri-cone bits showing how their own and their competitors' bits fit into the system. The roller cone bits of four major manufacturers and one smaller one have been listed in the tabulations on pages D-32 to D-39 giving the manufacturers own type codes with the bits arranged according to the IADC classification

Equivalent classification charts for diamond bits do not exist, probably because the designs, and thus the type names, are changing much more rapidly than tri-cone bits, and any comparative chart would become out of date as soon as it was printed. The choice of diamond bits is made from the individual manufacturers catalogue and often in discussion with his representative.

ROCK BIT CLASSIFICATION SCHEMES

TRICONE BITS

First digit :

Tooth material and length

The numbers 1, 2 and 3 designate steel tooth bits and correspond to increasing formation hardness.

The numbers 4, 5, 6, 7 and 8 designate bits with tungsten carbide inserts and also correspond to increasing formation hardness.

Second digit :

Formation hardness (finer grading)

The numbers 1, 2, 3 and 4 denote a sub-classification of the formation hardness in each of the eight classes determined by the first digit.

Third digit :

Bearings and gauge protection

The numbers 1 to 7 define the type of bearing and specify the presence or absence of gage protection by tungsten carbide inserts, on the leading flanks of the bit cones:

- 1 = standard roller bearing
- 2 = roller bearing, air-cooled
- 3 = roller bearing, gage protected
- 4 = sealed roller bearing
- 5 = sealed roller bearing, gage protected
- 6 = sealed friction bearing
- 7 = sealed friction bearing, gage protected

The numbers 8 and 9 are reserved for future use. However, some bit manufacturers use this space to show their directional bits (8) and special application bits (9).

Additional letter :

Miscellaneous characteristics

- A = air application : journal bearing bits with air circulation nozzles.
- C = centre jet
- D = deviation control
- E = extended jets
- G = extra gauge/body protection
- J = jet deflection
- R = reinforced welds (for percussion applications)
- S = standard steel tooth model
- X = chisel insert
- Y = conical insert
- Z = other insert shape

ROCK BIT CLASSIFICATION SCHEMES

DIAMOND BITS

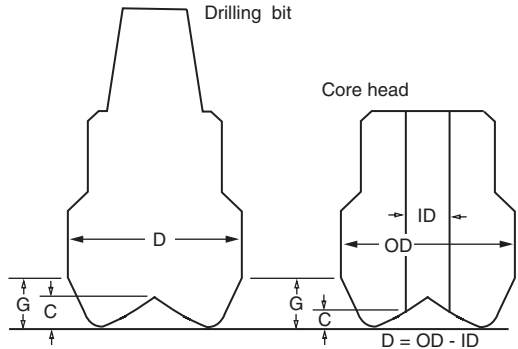
First character :

Cutter type & body material

D: Matrix body / Natural diamonds
M: Matrix body / PDC cutters
S: Steel body / PDC cutters
T: Matrix body / TSP cutters
O: Other

Second character:

Bit Profile Codes



G: Gage height	C : Cone height		
	High (C > 1/4 D)	Medium (1/8 D ≤ C ≤ 1/4 D)	Low (C < 1/8 D)
High G (> 3/8 D)	1	2	3
Medium G (1/8 D ≤ G ≤ 3/8 D)	4	5	6
Low G (< 1/8 D)	7	8	9

Third character :

Hydraulic design

Cutter distribution	Fluid exit		
	Changeable jets	Fixed ports	Open throat
Bladed(1)	1	2	3
Ribbed(2)	4	5	6
Open-faced	7	8	9
Alternate codes: R = Radial flow X = Feeder/collector flow O = Other These letters are used in preference to Numbers 6 & 9 for most natural diamond and TSP bits.			
(1) Bladed refers to raised, continuous flow restrictors with a standoff distance from the bit body of more than 1" (25.4 mm). (2) Ribbed refers to raised, continuous flow restrictors with a stand-off distance from the bit body of 1" (25.4 mm) or less.			

Fourth character :

Cutter size and density

Size	Density		
	Light	Medium	Heavy
Large	1	2	3
Medium	4	5	6
Small	7	8	9
0 = impregnated			
Cutter size range	Natural diamonds: stones/carat	Synthetic diamonds: usable cutter height	
Large	< 3	> 5/8"	
Medium	3 - 7	3/8" - 5/8"	
Small	> 7	< 3/8"	

CORRELATIONS OF FORMATIONS TO IADC CODES FOR TRI-CONE BITS

	Series	Type
Milled tooth	1. Soft formations having low compressive strength and high drillability	1. Very soft shale 2. Soft shales 3. Medium soft shale/lime 4. Medium lime shale
	2. Medium to medium hard formations with high compressive strengths	1. & 2. Medium lime/shale 3. Medium hard lime/sand/slate
	3. Hard semi-abrasive or abrasive formations	1. Hard lime 2. Hard lime/dolomite 3. Hard dolomite
Insert	4. Soft formations having low compressive strength and high drillability	1. Very soft shale 2. Soft shales 3. Medium soft shale/lime 4. Medium lime shale
	5. Soft to medium formations of high compressive strength	1. Very soft shale/sand 2. Soft shale/sand 3. Medium soft shale/lime
	6. Medium hard formations of high compressive strength	1. Medium lime/shale 2. Medium hard lime/sand 3. Medium hard lime/sand/slate
	7. Hard semi-abrasive and abrasive formations	1. Hard lime/dolomite 2. Hard sand/dolomite 3. Hard dolomite
	8. Extremely hard and abrasive formations	1. Hard chert 2. Very hard chert 3. Very hard granite

GENERAL DATA

ROCK BITS

IADC Bearing code* type		Formation				WOB (lbs/inch bit diam).	RPM
		Soft	Med.soft				
			Med.hard Hard				
437	Friction	x				1,500-3,500	120-60
517	Friction	x				2,000-4,500	100-50
527	Friction	x				2,000-5,000	110-60
537	Friction		x			2,500-5,000	75-45
547	Friction		x			2,500-5,500	0-50
617	Friction		x			2,500-5,500	65-45
617	Friction			x		2,000-6,000	70-40
627	Friction			x		2,000-6,000	65-40
627	Friction			x		3,000-6,000	65-40
637	Friction			x		3,000-6,500	55-40
727	Friction				x	2,500-6,000	60-40
737	Friction				x	3,000-6,500	55-35
837	Friction				x	3,000-7,000	50-30
519	Friction	x				1,500-2,500	100-55
515	Sealed*	x				2,000-4,500	100-50
535	Sealed*		x			2,500-5,000	75-45
615	Sealed*		x			2,500-5,500	65-45
612	Air		x			2,000-5,000	70-45
622	Air			x		2,000-5,500	70-45
732	Air				x	2,500-6,000	65-45
832	Air				x	2,500-6,500	60-40
116	Friction	x				2,000-5,000	140-70
126	Friction	x				2,000-6,000	120-60
136	Friction		x			3,000-7,000	110-60
216	Friction			x		3,000-8,000	80-50
114	Sealed*	x				2,000-6,000	250-75
124	Sealed*	x				2,000-6,000	250-75
134	Sealed*		x			3,000-7,000	175-60
214	Sealed*			x		3,000-8,000	120-50
314	Sealed*				x	4,000-9,000	70-45
111	Open	x				2,000-6,000	250-75
121	Open	x				2,000-6,000	250-75
131	Open		x			3,000-7,000	175-60
211	Open			x		3,000-8,000	90-50
311	Open				x	4,000-9,000	70-45
231	Open			x		3,000-8,000	80-45
118	Open	x				2,000-6,000	250-75
128	Open	x				1,000-4,000	250-75

Pin connections - rock bits

Size range in inches	API Connection
55/8-63/4	31/2 Reg.
75/8-83/4	41/2 Reg.
91/2-141/2	65/8 Reg.
143/4-20	75/8 Reg.
22	65/8 Reg. or 75/8 Reg.
24 - 26	75/8 Reg. or 85/8 Reg.
28	75/8 Reg.

Note:

Certain sizes can be supplied with 75/8" or 85/8" API Reg. connections on special order

Recommended make-up torque - rock bits

	Minimum	Don't Exceed
	lbs-ft	
31/2 Reg.	7,000	9,000
41/2 Reg.	12,000	16,000
65/8 Reg.	28,000	32,000
75/8 Reg.	34,000	40,000
85/8 Reg.	40,000	60,000
	N-m	
31/2 Reg.	9,500	12,200
41/2 Reg.	16,300	21,700
65/8 Reg.	38,000	43,400
75/8 Reg.	46,100	54,200
85/8 Reg.	54,200	81,400

GENERAL DATA

DIAMOND BITS

Diamond bit drilling parameters

Bit size		RPM	Weight on bit		Flow Rate	
inches	mm		pounds	kdaN	gpm	dm ³ /sec
5 ⁷ / ₈	149.2	100-140	6-10,000	2.67-4.45	160-220	10-14
6	152.4	100-140	6-10,000	2.67-4.45	160-220	10-14
6 ¹ / ₂	165.1	100-140	6-12,000	2.67-5.34	160-220	10-14
6 ³ / ₄	171.5	100-140	6-14,000	2.67-6.23	160-220	10-14
7 ⁷ / ₈	200.0	80-120	8-16,000	3.56-7.12	200-300	13-19
8 ¹ / ₂	215.9	80-120	8-18,000	3.56-8.00	300-400	19-25
8 ³ / ₄	222.3	80-120	8-18,000	3.56-8.00	300-400	19-25
9 ⁷ / ₈	225.4	60-120	10-23,000	4.45-10.2	350-500	22-32
10 ⁵ / ₈	269.9	60-120	12-28,000	5.34-12.5	500-600	32-38
12 ¹ / ₄	311.2	60-100	15-39,000	6.67-17.4	550-700	34-44

Recommended make-up torque - PDC & diamond bits

Bit size		API pin size		Recommended torque	
inches	mm	inches	mm	lbs-ft	N-m
3 ³ / ₄ - 4 ¹ / ₂	95.2 - 114.3	2 ³ / ₈ Reg.	60.3	3,000 - 3,500	4,000 - 4,800
4 ⁵ / ₈ - 5	117.5 - 127.0	2 ⁷ / ₈ Reg.	73.0	6,000 - 7,000	8,000 - 9,500
5 ¹ / ₈ - 7 ³ / ₈	136.5 - 187.3	3 ¹ / ₂ Reg.	88.9	7,000 - 9,000	9,500 - 12,200
7 ⁵ / ₈ - 9	193.7 - 228.6	4 ¹ / ₂ Reg.	114.3	12,000 - 16,000	16,300 - 21,700
9 ¹ / ₂ - 26	241.3 - 660.4	6 ⁵ / ₈ Reg.	168.3	28,000 - 32,000	38,000 - 43,400
14 ³ / ₄ - 26	374.6 - 660.4	7 ⁵ / ₈ Reg.	193.7	34,000 - 40,000	46,100 - 54,200

GENERAL DATA

API TOLERANCES

on new rock bits

Size		Tolerance	
ins	mm	ins	mm
3 ³ / ₈ - 13 ³ / ₄	85.7 - 349.3	+ ¹ / ₃₂ ,0	+0.79 ,0
14 - 17 ¹ / ₂	355.6 - 444.5	+ ¹ / ₁₆ ,0	+1.59 ,0
17 ⁵ / ₈ or more	447.7 or more	+ ³ / ₃₂ ,0	+2.38 ,0

on new diamond and PDC bits

Size		Tolerance	
ins	mm	ins	mm
6 ³ / ₄ or less	171.5 or less	0, -0.015	0, -0.38
6 ²⁵ / ₃₂ - 9	172.2 - 228.6	0, -0.020	0, -0.51
9 ¹ / ₃₂ - 13 ³ / ₄	229.4 - 349.3	0, -0.030	0, -0.76
13 ²⁵ / ₃₂ - 17 ¹ / ₂	350.0 - 444.5	0, -0.045	0, -1 14
17 ¹⁷ / ₃₂ or more	445.3 or more	0, -0.063	0, -1.60

on casing drift mandrels

Size ins	Weight lbs/ft	Length		Tolerance	
		ins	mm	ins	mm
7	23.0	6	152	6.250	159
7	32.0	6	152	6.000	152
8 ⁵ / ₈	32.0	6	152	7.875	200
8 ⁵ / ₈	40.0	6	152	7.625	194
9 ⁵ / ₈	40.0	6	152	8.625	220
9 ⁵ / ₈	53.5	6	152	8.375	213
10 ³ / ₄	45.5	12	305	9.750	248
10 ³ / ₄	55.5	12	305	9.625	244
11 ³ / ₄	42.0	12	305	10.625	270
11 ³ / ₄	60.0	12	305	10.875	276
13 ³ / ₈	72.0	12	305	12.000	305

THE DULLNESS GRADING SYSTEM FOR USED BITS

The following information is recorded:

- Distance drilled
- Time taken
- Averaged drilling parameters (WOB, RPM, Pump speed)
- Average drilling fluid properties (type, density, viscosity, fluid loss)
- The condition of the bit when pulled.

The first four of these are objective measurements which can be obtained by reference to the standard daily reports. The condition however is a very subjective assessment made by the driller. In order to provide a measure of consistency between bit condition reports made by all drillers, world wide, a grading system has been introduced. This is the IADC system which applies to roller cone bits, diamond bits and core heads. It uses code characters for describing six categories of wear, grouped into the three sections cutters, bearings and gauge, and adds two codes for remarks.

If a standard bit report form is being completed there are eight boxes in which the individual codes are entered. If the bit condition is being discussed, or described in "free-format" text the three sections containing the description of the wear are each identified by a letter, or in one case a phrase.

Cutting Structure				Bearing or seal (B)	Gauge mm or 16ths (G)	Remarks	
Inner rows (I)	Outer rows (O)	Dullness character (D)	Location (L)			Other character (O)	Reason pulled (R)

The cutting structure

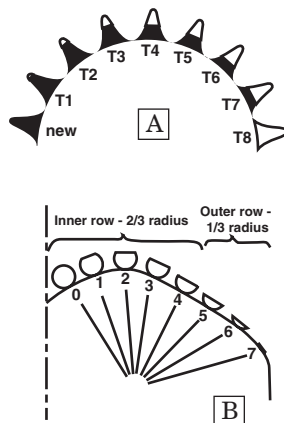
Four codes are used to describe the cutting structure - the teeth/inserts on a roller cone bit, or the cutting elements of a diamond bit. These are entered into the first four boxes of a standard report, otherwise they are identified by the letter "T" for roller cone bits or "cutting structure" for diamond bits.

The first two codes define the wear on the cutters using a scale of 0 to 8, where 0 represents no wear and 8 indicates that no usable cutting structure is left. The first code represents the average wear of the cutters in the inner two thirds of the bit radius, the second refers to the average wear of those in the outer third. Note that in the case of core bits the "radius" is to be interpreted as the distance from the ID to the OD of the core head, i.e. in Figure B the centre line shown would be the ID of the core head/OD of the core. For a roller cone bit the worst cone is taken for the grading.

The wear of milled teeth and PDC cutters is graded in eighths of the original tooth height - see Figures A & B.

For a bit worn as shown in figure B the first two codes would be $(0+1+2+3+4)/5=2$ and $(5+6+7)/3=6$.

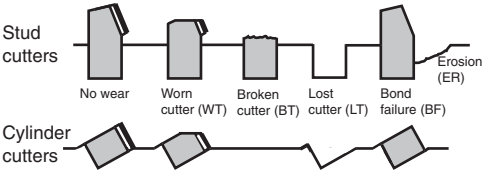
For roller cone insert bits and for cutting element wear on natural diamond and TSP bits the number of inserts or diamonds broken or missing is more relevant than the



actual wear on the individual inserts or cutting elements. It is a combination of wear and broken or missing inserts/diamonds which determines the amount of wear to be reported. The same scale of 1 to 8 is used with T1 representing 1/8 of the cutting elements lost or broken and T8 representing all the cutting elements lost or broken. Some experience is required to do this correctly.

The third box is for the code describing the primary wear characteristic of the cutting structure, chosen from the list in Table 1. The figure below shows how these “tooth” wear terms are applied to PDC cutters.

Wear characteristics - PDC cutters



The fourth part of the cutting structure code defines the basic location of the primary wear characteristic described in the third box of the eight. This can range from a specific part of the bit face to the entire bit. The codes are chosen from the list given in the second figure, opposite; in the case of a tri-cone bit the number(s) of the affected cone(s) is/are included.

The bearing/seals

The letter B is used to identify the code used for bearing/seal reporting. If a standard report form is being used the code is entered into the fifth box.

If no seals are used then the bearing wear is reported on a scale of 1 to 8, as for the teeth, with 0 representing “as new” condition, and 8 indicating that all bearing life has been used up (i.e. the bearings have failed). The condition of the worst bearing is reported.

For sealed bearings, only the condition of the seals is reported with either E indicating that the seals are still effective or F indicating that the seals have failed.

For fixed cutter bits without bearings or seals the code “X” is entered into standard report forms.

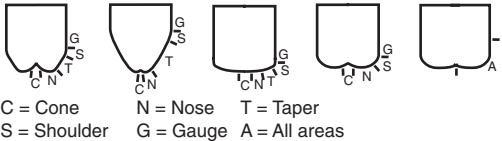
Table 1

*BC:	Broken Cone
BF:	Bond Failure
BT:	Broken Teeth/Cutters
BU:	Balled Up
*CC:	Cracked Cone
*CD:	Cone Dragged
CI:	Cone Interference
CR:	Cored
CT:	Chipped Teeth/Cutters
ER:	Erosion
FC:	Flat Crested Wear
HC:	Heat Checking (and spalling)
JD:	Junk Damage
*LC:	Lost Cone
LN:	Lost Nozzle
LT:	Lost Teeth/Cutters
NR:	Not Re-Runnable
OC:	Off-CentreWear
PB:	Pinched Bit
PN:	Plugged Nozzle/Flow Passage
RG:	Rounded Gauge
RO:	Ring Out
RR:	Re-Runnable
SD:	Shirt-tail Damage
SS:	Self Sharpening Wear
TR:	Tracking
WO:	Washed Out Bit
WT:	Worn Teeth/Cutters
NO:	No Major/Other Dull Characteristics
* Show cone number(s) under "Location".	

Roller cone location codes

N = Nose row, M = Middle row, G = Gauge row, A = All rows (followed by the number of the cone)

Fixed cutter location codes



The gauge

The letter G is used to identify the code used for reporting the condition of the bit gauge. If a standard report form is being used the code is entered into the sixth box.

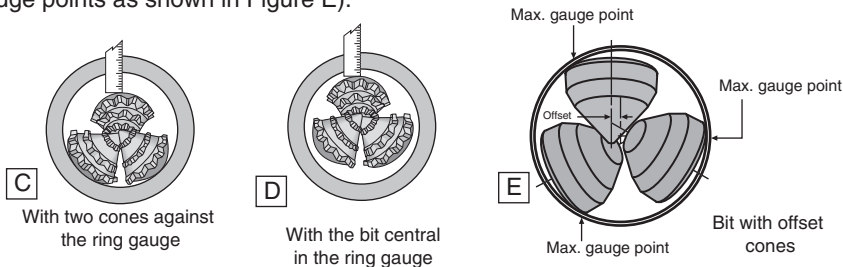
The reduction in diameter is measured in millimetres or in sixteenths of an inch. If the bit is full gauge an “1” indicates that it is in gauge. Working in SI units a “1” indicates that it is 0 to 1 mm under gauge. A “2” indicates that it is 1-2 mm under gauge. A “3” indicates that it is 2-3 mm under gauge, and so on.

In oilfield units the wear would be reported as “1/16”, indicating that the bit is zero to 1/16" under gauge. “2/16” would indicate that it is 1/16" to 1/8" under gauge, etc.

Gauge wear can be determined by using a ring gauge and ruler. This should be done with the bit standing on its cones so that they take up the position they had when cutting the hole.

There are two methods used to measure the wear. In the first, most common, method the ring gauge is pulled against the gauge points of two cones, and the space between the ring and third cone is measured (Figure C). Usually, this measurement is used for the amount of wear; however, to be exact, the measurement should be multiplied by 2/3.

In the second method, the bit is centred in the gauge ring and the ruler is used to measure the distance from the ring to the outermost cutting surface (gauge surface) (Figure D). This measurement must be multiplied by 2 to give the loss in diameter and thus the total amount of wear. Offset bits should be measured at one of the maximum gauge points as shown in Figure E).



Remarks

In the first of the two places available for remarks at the end of the wear code more information is given on the state of the cutting structure and flow passage(s), chosen from the same list as for the primary characteristic of tooth/cutter wear.

In the last place the reason for pulling the bit is given. This is taken from the following list.

BHA: To change Bottom Hole Assembly	HR: Hours on Bit (estimated maximum usable hours reached)
DMF: Down-hole Motor Failure	LIH: Left in hole
DSF: Drill String Failure	PP: Unexpected variation in Pump Pressure
DST: To perform a Drill Stem Test	PR: Insufficient Penetration Rate
DTF: Down-hole Tool Failure	RIG: To carry out Rig Repairs
LOG: To run Logs	TD: Total Depth/Casing Depth reached
CM: To condition Mud	TQ: Unexpected variation in Torque
CP: Core Point reached	TW: Twist Off
DP: To Drill Plug	WC: Weather Conditions
FM: Formation Change	WO: Washout in drill string
HP: Hole Problems	

DRILLING PRACTICES

ROCK BITS

Rules of thumb for bit selection

- Shale has a better drilling response to RPM.
- Limestone has a better drilling response to bit weight.
- Bits with roller bearings can be run at a higher RPM than bits with journal bearings.
- Bits with sealed bearings can give longer life than bits with open bearings.
- Milled tooth bits with journal bearings can be run at higher weights than milled tooth bits with roller bearings.
- Diamond bits can run at higher RPM than tri-cone bits.
- Bits with high offset may wear more on gauge.
- Bits with high offset may cause more hole deviation.
- Cost per foot analysis can help you decide which bit to use.
- Examination of used bits can help you decide which bit to use.

Running in

- Make the bit up to proper torque.
- Hoist and lower the bit slowly through ledges and dog legs.
- Hoist and lower the bit slowly at liner tops.
- Rock bits are not designed for reaming. If you do ream, do it with light weight and low RPM.
- Protect nozzles from plugging with Smith Tool jet plugs.

Establish a bottom hole pattern

- Rotate the bit and circulate when approaching bottom. This will prevent plugged nozzles and clear out fill.
- Lightly tag the bottom with low RPM.
- Gradually increase the RPM.
- Gradually increase the weight.

If the expected penetration rate is not achieved

- Drilling fluid density may be too high with respect to formation pressure.
- Drilling fluid solids may need to be controlled.
- Pump pressure or pump volume may be too low.
- The bit used may be too hard for the formation.
- Formation hardness may have increased.
- RPM and weight may not be the best for bit type and formation — carry out a drill off test.
- Ensure that the bit is stabilised.

Before re-running green bits

- Make sure the bit is in gauge.
- Check any bit for complete cutting structure.
- Check any sealed bearing bit for effective seals .
- Soak any sealed bearing bit in water or diesel to loosen formation packed in the reservoir cap equalisation ports.
- Regrease 14³/₄" diameter and larger open bearing bits.

DRILLING PRACTICES

DIAMOND BITS - APPLICATIONS

Economics and Application

Regardless of how well designed or manufactured a bit, the situation in which it is used, and the applied practices, ultimately determine the success or failure of the run.

When to use a diamond bit

- When economics dictate
- Generally, the rate of penetration ultimately determines the economics of the bit run
- When on-bottom times are important
- When oil-phase mud systems are used (preferred)
- When water-phase systems are used in non-hydrating formations (preferred)
- When rotating at high speeds (turbine or PDM)
- When high bottom hole temperatures are encountered (approx. 300° F and higher)
- When drilling in deviated hole section requiring light bit weight -When drilling significantly overbalanced

Why use a diamond bit

- Economics
- To reduce the number of trips in order to :
 - minimise running through dangerous hole sections
 - minimise rig wear
- To avoid tripping in bad weather

Where to use a diamond bit (Formations in which diamond bits are normally beneficial)

- Polycrystalline Diamond Compact (PDC) Bits
 - Very weak, poorly consolidated, brittle, shallow sediments (e.g. Miocene sands, silts, clays)
 - Low strength, poorly compacted, brittle, non-abrasive, relatively shallow sediments, precipitates and evaporites (e.g. salt, anhydrite, marls, chalk – Devonian/Muschelkalk)
 - Moderately strong, somewhat abrasive and ductile, indurated medium-depth sediments, precipitates and evaporites (e.g. silty claystone, siliceous shales, porous carbonates, anhydrite — Eocene)
- Natural/Thermally Stable Diamond Bits
 - Moderately strong, somewhat abrasive and ductile, indurated medium-depth sediments, precipitates and evaporites (e.g. siliceous shales, porous carbonates, anhydrite, silty claystone— deep Miocenes)
 - Strong and abrasive indurated, very ductile deep sediments, precipitates and evaporites (e.g. sandy shales, calcareous sandstones, dolomites, limestone — Pennsylvanian/Mississippian)
 - Very strong and abrasive, indurated ductile and non-ductile sediments, precipitates and evaporites (e.g. Bunter sandstone, bromides, etc.)

Formations detrimental to diamond bits

- Polycrystalline Diamond Compact (PDC) Bits
 - Hard, cemented abrasive sandstone (e.g. sedimentary quartzite)
 - Hard dolomites (sedimentary or metamorphic)
 - Iron (e.g. pyrite — metamorphic or igneous)
 - Chert (metamorphic or sedimentary)
 - Granite and basalt (igneous)
- Natural/Thermally Stable Diamond Bits
 - Hard, cemented quartzitic sands that are highly fractured and abrasive

How to determine the application

- Use geological information to determine which offset wells in the area are likely to be representative for the well to be drilled.
- Review the bit records from the offset wells, in particular their condition when pulled and the calculated economics.
- Review the wireline logging data from the offset wells

DRILLING PRACTICES

DIAMOND BITS - FIELD OPERATIONS

Prior to running the bit

- Reach an agreement with the operator/contractor on what is expected of the bit including, if appropriate, its suitability for drilling float equipment
- Reach an agreement on what mechanical and hydraulic requirements are available and/or necessary to achieve optimum or expected performance
- Before running the diamond bit into the hole, have a junk basket run on the previous bit
- After the previous bit is pulled, inspect it for junk damage and other wear, then gauge it
- If the previous bit appears OK, the bit may be readied to run into the hole
- Check O-ring and install nozzles, if appropriate
- Check for cutter damage
- Check that the bit is within tolerance on diameter and that there is no foreign material inside it
- Recommend the use of drill pipe screens

Running the bit (rotary assembly)

- Handle the diamond bit with care. DO NOT set the bit down without placing wood or a rubber pad beneath the diamond cutters
- A correct bit breaker should be used and the bit should be made up to the correct torque as determined by the pin connection size
- Care should be taken in running the bit through the rotary table and through any known tight spots. Hitting ledges or running through tight spots carelessly may damage the bit gauge
- Reaming is not recommended, however, if necessary, pick up the kelly and run the maximum fluid possible. Rotate at about 60 RPM. Advance bit through tight spot with no more than 4000 pounds weight on bit (WOB) at any time
- As hole bottom is approached, the last three joints should be washed down slowly at full flow and with 40 to 60 RPM to avoid plugging the bit with fill
- The bottom is found by observing the rotary torque indicator as well as the weight indicator. The first on bottom indication is usually an increase in rotary torque
- Once the bottom is located, the bit should be lifted just off bottom (0 to 1 foot if possible) and full volume circulated while slowly rotating for about 5 to 10 minutes
- After circulating, ease back to bottom and be patient in establishing the bottom hole pattern
- When ready to start drilling, increase the rotary speed to about 100 RPM and start cutting a new bottom hole pattern with approx. 1000 to 4000 pounds WOB
- Cut at least one foot in this manner before determining optimum bit weight and RPM for drilling
- Determine optimum ROP through a drill-off test

Running the bit (PDM & turbine)

- Start the pumps and increase to the desired flow rate when approaching bottom
- After a short cleaning period, lower the bit to bottom and increase WOB slowly
- After establishing a bottom hole pattern, additional weight may be slowly added
- As weight is increased, pump pressure will increase, so the differential pressure and WOB must be kept within the recommended downhole motor specifications
- Drill pipe should be slowly rotated to prevent differential sticking
- All other operating practices are as per standard practices

Pull the bit when

- The bit stops drilling
- The bit ceases to be economical as shown by cost/foot calculations
- When very high on-bottom torque with little WOB and a decrease in ROP occurs - the bit may be undergauge in a tough formation
- There is a dramatic decrease in ROP and on-bottom torque
- If there are changes in stand pipe pressure
 - if it goes up there is probably a cutter structure failure
 - if it goes down there is probably washout or a nozzle has been lost

DRILLING PRACTICES

DIAMOND BITS - COMMON PROBLEMS

Problem	Probable cause	Preferred action
Difficulty going to bottom	Previous bit undergauge New bottom hole assembly Collapsed casing Out of drift Bit oversized Stabiliser oversized	Ream with roller cone bit When reaming to bottom, pick up and ream section again. If difficulty remains, check stabilisers. Roll casing with smaller bit Use bi centre bit or reduce bit size Gauge bit with API gauge; if not in tolerance, replace bit Replace with correct size stabiliser
Low pressure differential across nozzles or bit face	Flow area too large Flow area too small Different drilling parameters than designed for Washout in drill string	Increase flow rate and correct on next bit Increase flow rate/strokes Change liners Attempt to optimise flow area on next bit change Check bit pressure drop, drop softline, trip to check pipe and collars
High pressure differential across nozzles or bit face	Flow area too small Excessive flow rate Diamonds too small for formation Bit partially plugged (formation impaction) Formation change Ring out Downhole motor stalled	Reduce flow rate, change flow area on next bit Reduce flow rate If ROP acceptable, change on next bit If ROP unacceptable, pull bit and use bit with correct diamond size Check off bottom standpipe pressure Let bit drill off, circulate full volume for 10 minutes while rotating. Check off bottom pressure again Pick up, circulate, resume drilling at higher RPM, reset, drill off test On and off bottom pressure test, pull bit Refer to manufacturer's handbook
Fluctuating standpipe pressure	Drilling through fractured formation Formation breaking up beneath bit Stabilisers hanging up Equipment failure	If ROP acceptable, continue If ROP acceptable, continue Check equipment Try combination of lighter weight and higher RPM Check overpull Check stabilisers on next trip Repair equipment
Bit won't drill	Bottom not reached Stabilisers hanging up or too large Formation too plastic Establishing bottom hole pattern Core stump left Bit balled	Check tally Check torque, overpull Check pressure – increase flow rate, decrease/increase bit weight, RPM Can take up to an hour Attempt to carefully drill ahead with low bit weight Back off and increase flow rate, then slug with detergent or oil
Slow rate of penetration	Not enough weight on bit: hydraulic lift RPM too low/high Plastic formation Change in formation Overbalanced Diamonds flattened off Cutters flattened Pressure drop too low Wrong bit selection	Increase weight on bit Increase/decrease RPM Reset drill off Reset weight Reset drill off Accept ROP Pull bit Compare beginning and present pressure drops – new bit may be required Increase weight Pull bit Increase flow rate – new bit may be required Pull bit
Excessive torque	Excessive weight on bit Slow rotary speed Stabilisers too large Collars packing off Bit undergauge	Reduce weight and RPM Increase RPM Decrease weight Check bottom hole assembly, stabilisers should be 1/32" to 1/16" under hole size Increase flow rate and work up and down Pull bit
Bit bouncing	Slip-stick action Broken formation Pump off force	Change RPM/weight combination Reduce RPM and weight Increase weight Decrease volume

DRILL OFF TESTS

Drill off tests are performed in order to ascertain the optimum combination of weight on bit and rotary speed to maximize penetration rate. They should be done:

- At the start of a bit run
- On encountering a new formation
- If a reduction in ROP occurs

Procedure

1. Maintain a constant RPM. Select a WOB (near maximum allowable).
2. Record the time to drill off a weight increment, ie 5,000 lbs.
3. Re-apply the starting weight and record the length of pipe drilled during step 2.
4. From steps 2 and 3 the penetration rate may be found.
5. Repeat steps 2 and 3 at least four times. The last test should be at the same value as the first. This repeat test will determine if the formation has changed or not.
6. Plot seconds to drill off versus bit weight.
7. Plot penetration rate versus bit weight.
8. Select the bit weight which produced the fastest ROP. Maintain this WOB constant and repeat the above but varying the RPM.
9. Plot ROP versus RPM and select the RPM which resulted in the fastest ROP. This is the optimum rotary speed.
10. These values for WOB and RPM obtained will result in optimum progress for the particular formation and bit type.

ROCK BITS

EVALUATION AND COMPARISON

Rock bit evaluation

The Cost per Unit Length method can be used:

- To determine when to pull the bit, and
- To compare different types of bit

$$\text{Cost per Unit length} = \frac{\text{Bit cost} + \{\text{Rig cost/hour} \times (\text{Trip time} + \text{Drilling time})\}}{\text{Length of drilled interval}}$$

When to pull bit

$$\text{Ratio } \alpha = \frac{F}{D + T + B}$$

Where :

F = Distance drilled during present bit run

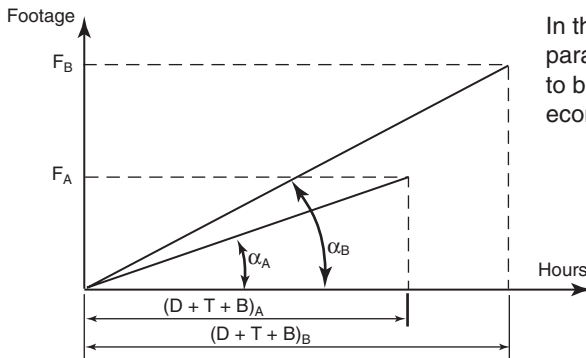
D = Drilling time during present bit run in hours

T = Trip time in hours

B = Cost of bit/rig cost per hour

The bit should be pulled when the ratio α starts to decrease after reaching a maximum

Bit comparison



In this example, assuming all other parameters including the formation to be constant, bit B was more economical than bit A.

Breakeven calculation

By analysis of good offset bit runs, the average cost per foot for a particular section can be calculated. This value can be used to find the breakeven line to which a more expensive bit must reach before it can be considered a good economical run.

1. Calculate cost per foot of comparison bit.
2. Draw breakeven line

$$F_0 = \frac{\text{cost of proposed bit} + (\text{trip time} \times \text{rig cost/hr})}{\text{cost per foot of comparison bit}}$$

$$F_{100} = \frac{\text{rig cost/hr} \times 100}{\text{cost per foot of comparison bit}} + F(0)$$

Notes :

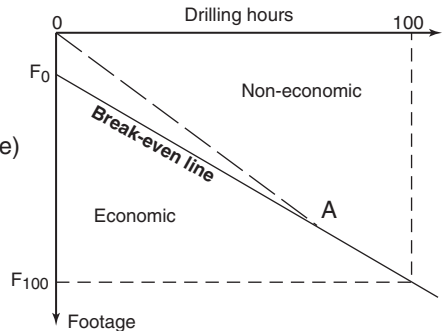
F_0 and F_{100} are breakeven footage at 0 hours and 100 hours.

The line drawn from $(0, F_0)$ to $(100, F_{100})$ represents the target cost per foot to breakeven.

As drilling progresses with the more expensive bit :

Plot footage drilled v. drilling hours (broken line)

Bit breaks even at point A.



Other factors which should be taken into account when doing a breakeven analysis include:

- turbine and PDM rentals
- clean up trips prior to running a stratapax bit
- wiper trips on long bit runs
- wear and tear of rig

EVALUATION OF USED TRI-CONE BITS

DESIRABLE CONDITIONS

Condition of bit	Comments/possible causes	Possible remedies
Soft and medium formation milled tooth bits Hardfacing has held cutting edge on tooth flanks (self-sharpening wear) Figs. 1 & 2	Good bit selection Good drilling practices	NA
Hard formation milled tooth bits Ragged teeth (not blunt) Fig. 3	As above	
Insert bits Balanced wear on bearings and teeth	As above	

UNDESIRABLE CONDITIONS

Condition of bit	Comments/possible causes	Possible remedies
All bits Excessive bearing wear	Excessive RPM Excessive rotating time Excessive W.O.B. Excessive sand in mud Unstabilised drillcollars Improper bit type	Decrease RPM Decrease rotating hours Decrease W.O.B. Decrease sand Stabilise drillcollars Use harder formation bit having larger bearing structure

USED BIT CONDITIONS
DESIRABLE

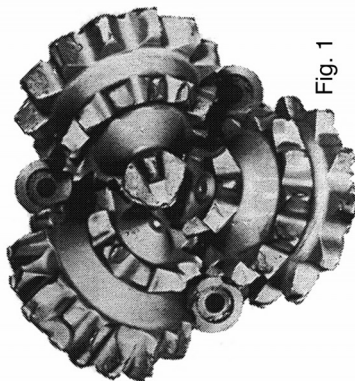


Fig. 1

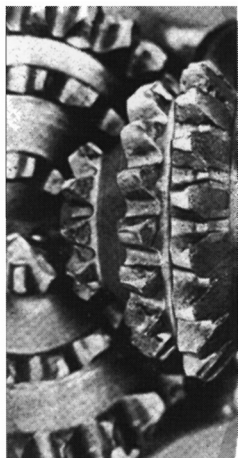


Fig. 2

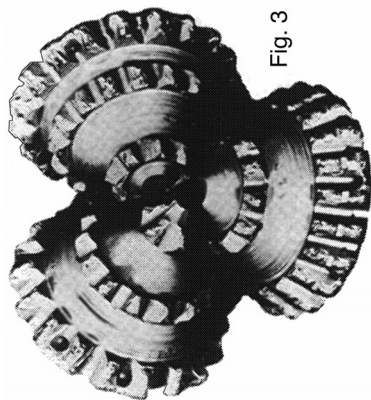


Fig. 3

EVALUATION OF USED TRI-CONE BITS

MILLED TOOTH BITS - UNDESIRABLE CONDITIONS (1)

Condition of bit	Comments/possible causes	Possible remedies
Rounded-off/blunt teeth (hard formation bit)	Formation too soft Insufficient W.O.B.	Use softer-formation bit Increase W.O.B.
Brinell marks Fig. 4	Seen under rollers and/or balls Impact loads	Caution while running-in, making connection etc.
Cones skidded, bearings good Fig. 5	Bit balling up Improper bit type Excessive W.O.B. Cones locking while drilling float shoe	Increase circulating rate (esp. in 'sticky' formations) Evaluation of hydraulics Use bit with wider spaced and longer teeth Decrease W.O.B. Decrease RPM (see torque more easily)
Excessive tooth breakage	Improper break-in Junk in hole Improper bit type Excessive W.O.B. and/or RPM	Drill a few feet before applying desired W.O.B. Wash on bottom before drilling. Run junk basket Use bit with shorter teeth Decrease W.O.B. and/or RPM
Excessive shirt-tail wear Fig. 6	Cuttings 'milling' around bit	Increase circulating rate Review bit hydraulic horsepower
Bradded teeth Fig. 7	Occurs more in soft-formation bits Excessive W.O.B. Failing to pull dull bit Formation too hard for bit type	- Decrease W.O.B. Replace bit when R.O.P becomes excessively slow Use harder formation bit
Heavy gauge wear and inner bearing loose	Improper bit type Excessive rotary speed and/or time Unstabilised drill collars	Use bit with less offset Use bit with more gauge protection Reduce rotary speed and/or time Stabilise drill collars

USED BIT CONDITIONS UNDESIRABLE



Fig. 4



Fig. 5



Inadequate cleaning

Fig. 6

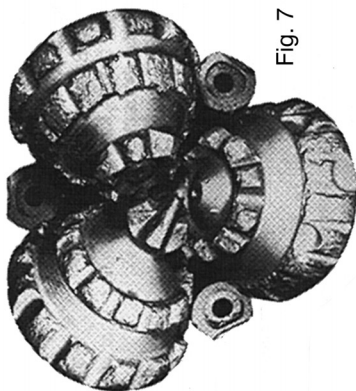


Fig. 7

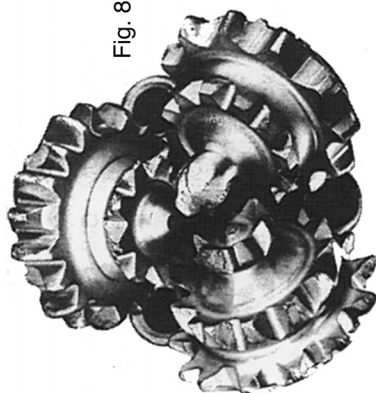


Fig. 8

EVALUATION OF USED TRI-CONE BITS

MILLED TOOTH BITS - UNDESIRABLE CONDITIONS (2)

Condition of bit	Comments/possible causes	Possible remedies
Unbalanced tooth wear	Improper bit type Tooth breakage (appears as wear when bit is pulled)	Use bit with deleted gauge row teeth (if inner row teeth are duller and bit shows no gauge wear) Prevent tooth breakage (see previously)
Off-centre wear Fig. 8	Bit 'walking' off centre When heavy mud weight is necessary	Use softer formation bit. Increase R.P.M. Use reamer and stabilisers on drill string before problem formation is drilled.
Excessive tooth wear	Excessive R.P.M. Improper bit type Use of non-hardfaced type bit	Decrease R.P.M. Use harder formation bit Use bit with hardfacing on teeth
Excessive cupping of tooth crests	Double hardfaced teeth Insufficient W.O.B. Improper formation for double hardfaced teeth	Use bit with single-face hardfacing or single-face and tipped hardfacing Increase W.O.B. -
Fluid cut teeth and cone	Excessive circulating rate Excessive sand content in mud	Decrease circulating rate Remove sand Use jet circulation bit

EVALUATION OF USED BITS

INSERT BITS - UNDESIRABLE CONDITIONS

Condition of bit	Comments/possible causes	Possible remedies
Damage from foreign material Fig. 9	Running on broken/lost inserts from previous bit Other Junk	More washing and pumping on bottom Use junk basket
Formation wear on cone shell around inserts Fig. 10	Inserts too short Insufficient cleaning under bit	Use bit with longer insert extension and more offset Review bit hydraulic horsepower
Excessive insert breakage Fig. 11	Improper break-in Improper bit type Improper drilling practices	Drill a few feet before applying desired W.O.B. If chisel inserts - use bit with less insert extension If formation is part lime and drilling fluid is light weight use projectile shaped inserts Decrease W.O.B. and/or RPM Use shock absorber
Gauge and outer rows broken Fig. 12	Excessive offset in bit Excessive RPM Insufficient stabilising	Use bit with less offset (which may also have less gauge insert extension) Decrease R.R.M. Stabilise drill collars
Excessive gauge wear	Excessive offset in bit Excessive RPM in abrasive formations High solids content in mud Insufficient stabilising	Use bit with less offset Decrease RPM Reduce solids content (if possible) Use stabilisers
Off-centre wear	Bit "walking" off centre When heavy mud weight is necessary	Use softer formation bit Increase R.R.M. Use reamer and stabilisers on drillstring before problem formation is drilled

USED BIT CONDITIONS
UNDESIRABLE

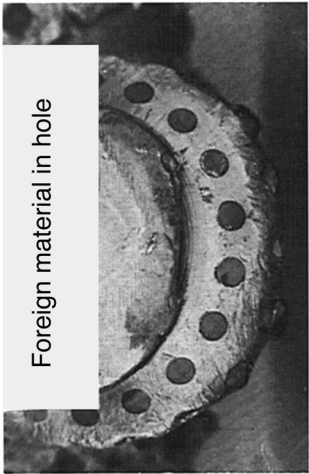


Fig. 9

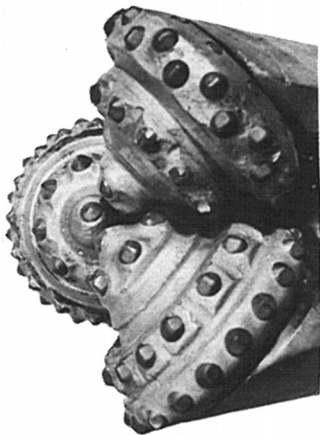


Fig. 10

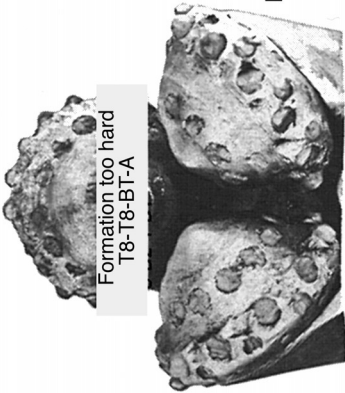


Fig. 11



Fig. 12

EVALUATION OF USED BITS

DIAMOND BITS

PDC bits

1. Generally, bit should not be re-run if cutting element is over 50% worn
2. Cutter wear
3. Cutter loss – caused by
 - excessive weight on bit
 - junk down hole
 - broken formation
 - bit bouncing
 - drastic formation change
 - braze failure
4. Cutter delamination
5. Cutter chipping, flaking, or spalling – caused by:
 - abrasion
 - overheating
 - excessive weight on bit
 - excessive side rake
6. Cutter erosion – caused by:
 - excessive fluid velocity across face
7. Bit badly worn – caused by:
 - fluid cutting on body – reasons :
 - excessive fluid velocity for body material
 - excessive solids/abrasive content in drilling fluid
 - formation wear on body – reasons:
 - too small a cutter standoff (exposure)
 - formation plastic
 - junk damage – reasons:
 - formation junk/broken formation
 - extraneous metal in hole
 - eccentric wear

Natural/thermally stable diamond bits

1. Diamond degradation – caused by:
 - abrasion or microscopic chipping
 - gross breakage – reasons:
 - excessive weight on bit
 - diamond size too large for formation
 - highly fractured formation/high impact loading
 - oxidation (occurs as low as 900° F in presence of oxygen)
 - matrix wear – reasons:
 - fluid erosion;
 - excessive fluid velocity
 - excessive abrasive content in drilling fluid
 - eccentric wear
 - heat damage (insufficient fluid to keep bit cool, thereby causing heat checking in matrix)
 - junk damage

Causes of bit failure

1. Misapplication
 - wrong bit for formation (bit too soft or too hard)
 - wrong diamond size for formation
2. Incorrect or inappropriate operating conditions
 - formation impaction (fluid volume/velocity insufficient to keep channels or bit face clean)
3. Loss of gauge
 - reaming to bottom too quickly or with excessive weight
 - insufficient gauge protection
4. Plugged bit
 - insufficient fluid to keep bit clean
 - washout in string
 - junk pumped down string causing plugging
 - LCM or swabbing to bottom too quickly
5. Ring out
 - junk in hole
 - insufficient cutter coverage
6. Bad Design

PDC CUTTER WEAR (13.3 MM DIAMETER)

PDC cutters are self sharpening and continue to drill until they are worn to approximately semi-circular shape.

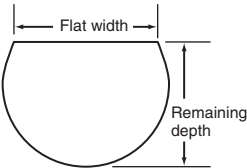
The percentage cutter wear figures given are based on the area of cutter that has been used divided by the area of a semi-circular cutter. (Because of this it is possible to obtain wear greater than 100%.)

It should be noted that, although the life of the individual PDC cutter is related to the given percentage wear figure, the life of the bit may vary widely depending on other conditions. Therefore, the decision of whether or not to re-run the bit should be based not only on cutter wear but on other factors such as:

- Number of missing or damaged cutters and their positions
- Wear on cutters adjacent to missing cutters
- Maximum cutter wear
- Gauge condition (PDC cutters and diamonds)
- Erosion
- Overall bit condition
- Section left to drill
- Formation

Up to 23% wear use the “Flat Width” and over 23% use “Remaining Cutter Depth”

Flat Width mm	Remaining Depth mm	% Worn
4		1
5		2
6		4
7		7
6		10
9		16
10	11	23
	10.5	30
	10	39
	9.5	47
	9	56
	8.5	65
	8	75
	7.5	84
	7	93
	6.5	103
	6	112
	5.5	122



AVAILABILITY OF ROCK BIT TYPES

The IADC classification of rock bits, or at least the first three digits, is a generic classification based on the bit construction and the type of formation for which it is suitable. The fourth character, a letter, indicates one or more special features of the bit. Each manufacturer, however, has his own nomenclature which is used in parallel with the IADC system, and which often allows a much more detailed specification of the special features.

The types of bit produced by each of the four major bit manufacturers (as of late 1997) are tabulated separately in the following eight pages, using their own nomenclature but arranged by IADC code. On the page facing each tabulation is a list of the codes used in that manufacturer's type names. Note that care is needed when requesting a bit with special features because some of the manufacturers base their codes on the IADC fourth character, but some are different, with the same letter used but corresponding to a different feature.

Not included in these tabulations are the sizes that are available for each bit type. All the manufacturers produce bits for the standard hole sizes, but you will have to refer to the manufacturers documentation to check the availability for non-standard sizes.

BIT SELECTION CHART

HUGHES CHRISTENSEN

IADC Classification									
Series	Form- ation	Third digit - bearings & features							
		1	2	4	5	6	7		
1	1	R1		ATX-1 GTX-1	ATX-G1 GTX-G1	MAX-G1 MAX-GT1	ATJ-1 ATJ-1S GT-1 STR-1	GT-G1 GT-G1H	ATM-G1
	3				ATX-G3 GTX-G3	MAX-G3 MAX-GT3			ATM-GT3
2	1						ATJ-4	ATJ-G4	
	2	DR-5							
3	2	R-7							
	4							ATJ-G8	
4	1				ATX-05 GTX-00 GTX-03	MAX-05 MAXGT-00 MAXGT-03		ATJ-05 GT-00 GT-03	ATM-05 ATMGT-03
	2							ATJ-05C GT-09C	ATMGT-09C
	3				ATX-11H GTX-09	MAX-11H MAX-11HG MAXGT-09		ATJ-11 ATJ-11H GT-09 STR-09	ATM-11H ATM-11HG ATMGT-09
	4					MAX-11CG MAXGT-18		ATJ-11C ATJ-18 GT-18 GT-18C	ATM-11CG ATMGT-18
5	1				ATX-22	MAX-22 MAX-22G		ATJ-22 ATJ-22S ATJ-22G GT-20 GT-20S STR-20	ATM-22 ATM-22G ATMGT-20
	2							ATJ-22C ATJ-28 ATJ-28C GT-20C GT-28 GT-28C	ATM-22C
	3							ATJ-33 ATJ-33S ATJ-33A ATJ-33H ATJ-35 STR-30	ATM-33 ATM-33G
	4				ATX-33C			ATJ-33C ATJ-35C	ATM-33C ATM-35CG
6	1		G44					ATJ-44 ATJ-44A ATJ-44G	
	2				ATX-44C	MAX-44C		ATJ-44C ATJ-44CA	
	3		G55			MAX-55		ATJ-55R ATJ-55RG ATJ-55 ATJ-55A	
	4							ATJ-66	
7	3		G77					ATJ-77	
	4							ATJ-88	
8	3		G99					ATJ-99	

Note : Only the series/formation combinations are shown for which bits are available from Hughes.
Similarly, empty "third digit" columns are not shown.

BIT TYPE DESIGNATIONS

HUGHES CHRISTENSEN

Product lines

MAXGT	Ball & roller bearing, metal seal, GT performance package
MAX	Ball & roller bearing, metal seal
ATMGT	Journal bearing, metal seal, GT performance package
ATM	Journal bearing, metal seal
GT	Journal bearing, elastomer seal, GT performance package
STR	Journal bearing, elastomer seal, STR (slim hole*) performance package
ATJ	Journal bearing, elastomer seal
GTX	Ball & roller bearing, elastomer seal, GT performance package
ATX	Ball & roller bearing, elastomer seal
R	Ball & roller bearing, non-sealed
G	Ball & roller bearing, non-sealed air

* 3⁷/₈" - 6³/₄"

Product features

		<u>Examples</u>
A	Air journal bearing, air nozzles	ATJ-33 A
C (prefix)	Centre jet	GT- C 18
C (suffix)	Conical shape inserts	GT-18 C
D	Diamond gauge compacts	ATM-22 D
G	Enhanced gauge	ATJ-33 G
M	Motor hardfacing	GT- M 1
P	Leg stabilisation wear pad	ATMGT- P 18
S	Shirt-tail compacts	GT- S 20
T	High flow extended nozzles	MAXGT- T 03

BIT SELECTION CHART

SMITH INTERNATIONAL

IADC Classification								
Series	Form- ation	Third digit - bearings & features						
		1	2	4	5	6	7	
1	1	DSJ		SDS	MSDSH MSDSSH MSDSHOD	FDS FDSS FDSS+	MFDSH MFDSSH MFDSHOD	
	2	DTJ				FDT		
	3	DGJ			SDGH MSDGH MSDGHOD	FDG	FDGH	MFDGH MFDGHOD
2	1	V2J			SVH MSVH	FV	FVH	
4	1				M01S M01SOD M02S M02SOD 02M		MF02 02MF	
	2				M05S M05MD 05M		F05 MF05 F07 05MF 05MFD	
	3				M1S M1SOD 10M 10MD 12M 12MD 12MY		F1 MF1 10MF 10MFD 12MF 12MFD 12MFY	
	4				15JS M15SD M15S M15SOD 15M 15MD		F15 MA15 F15D MF15 F15OD M15D 15MF M15OD 15MFD	
5	1				A1JSL MA1SL 2JS M2S 20M M2SD 20MD		A1 F15H F17 F25 F2 F25A F2H F2D MF2 MF2D 20MF 20MFD	
	2				M27S M27SD		F27 MF27 F27I MF27D	
	3				3JS M3S M3SOD		F3 MF3 F3D MF3D F3H MF3H MF3OD	
	4						F35 F35A F37A F37 MF37 F37D MF37D	
6	1		4GA		4JS		F4 F4A F4H F45H F45A F47 F47A	
	2		5GA 47JA		5JS 47JS		F47H MF5 F5 MF5D F5OD	
	3						F57 F57A F57D F57OD F57DD	
	4						F67OD	
7	3		7GA				F7 MF7 F7OD	
8	1						F8OD F8DD	
	3		9JA				F9	

Note : Only the series/formation combinations are shown for which bits are available from Smith.
Similarly, empty "third digit" columns are not shown.

BIT TYPE DESIGNATIONS

SMITH INTERNATIONAL

Prefixes

- F – Journal (pfinodal) bearing
- M – Steerable-motor bit bearing
- S – Sealed roller bearing

Suffixes

- A – Designed for air applications
- C – Centre jet
- D – Diamond enhanced gage inserts
- DD – Fully diamond enhanced cutting structure
- E – Full-extended nozzles
- G – Super D-Gun coating
- H – Heel inserts on milled tooth bits. Different, high wear-resistant grade of carbide on TCI bits for abrasive formations
- L – Lug pads
- N – Nominal gage diameter
- OD – Diamond enhanced heel row inserts
- P – Carbide compact in the leg back
- PD – Diamond SRT in the back of the leg
- Q – "Flow Plus" extended nozzles
- R – SRT inserts pressed in leg for stabilisation
- S – Sealed roller bearing

"Magnum" series suffixes

- M – Roller bearing, Trucut gauge
- MD – Roller bearing, diamond chisel gauge
- MF – Journal bearing, Trucut gauge
- MFD – Journal bearing, diamond chisel gauge
- Y – Conical cutting structure

Milled tooth cutting structure designations

- DS – Very soft formation cutting structure
- DT – Soft formation cutting structure
- DG – Medium formation cutting structure
- V – Medium-hard formation cutting structure

Tungsten carbide insert cutting structure designations

- 01 – Very soft formation chisel crest cutting structure
- 02 – Very soft formation chisel crest cutting structure
- 05 – Very soft formation chisel crest cutting structure
- 07 – Soft formation conical cutting structure
- 1 – Soft formation chisel crest cutting structure
- 15 – Soft-medium formation chisel crest cutting structure
- 17 – Soft-medium formation conical cutting structure
- 2 – Soft-medium formation chisel crest cutting structure
- 25 – Medium formation chisel crest cutting structure
- 27 – Medium formation conical cutting structure
- 3 – Medium formation chisel crest cutting structure
- 35 – Medium formation chisel crest cutting structure
- 37 – Medium formation conical cutting structure
- 4 – Medium formation chisel crest cutting structure
- 45 – Medium-hard formation chisel crest cutting structure
- 47 – Medium-hard formation conical cutting structure
- 5 – Medium-hard formation chisel crest cutting structure
- 57 – Medium-hard formation conical cutting structure
- 67 – Hard formation conical cutting structure
- 7 – Hard formation conical cutting structure
- 8 – Hard formation conical cutting structure
- 9 – Hard formation conical cutting structure

BIT SELECTION CHART

SECURITY DBS

IADC Classification							
Series	Form- ation	Third digit - bearings & features					
		1	3	5	6	7	
1	1	S3SJ		S33SG SS33SG	S33SF PSF	S33SGF MPSF	ERA MPSF
	2			SS33G	S33F	S33GF	S33TGF
	3	S4TJ	S4TGJ	SS44G		S44GF	
2	1	M4NJ		M44NG MM44NG	M44NF	M44NGF	
3	1	H7J					
	3			H77SG			
4	1			SS80		S80F ERA 03	ERA 03D
	2			SS81		S81F ERA 07	ERA 07C
	3			SS82		S82F S82CF ERA 13 ERA 13D	SS82F HZS82F ERA 13C ERA 14C
	4			SS83		S83F ERA 17	SS83F ERA 17D
5	1			SS84		S84F S84CF ERA 18C ERA 22C	SS84F HZS84F ERA 22 ERA 22D
	2					S85F ERA 25	S85CF ERA 25C
	3			S86 SS86		S86F S86CF ERA 33	SS86F ERA 33C
	4			SS88C		S88F S88CFH	S88CF S88FA
6	1			M84		M84F	MAF
	2			MM88 M89T		M84CF M86CF	M85F M89TF
	3					M89F	
7	1					H83F	
	3					H87F	
8	1					H89F	
	3			H100		H100F	

Note : Only the series/formation combinations are shown for which bits are available from Security DBS.
Similarly, empty "third digit" columns are not shown.

BIT TYPE DESIGNATIONS

SECURITY DBS

Feature	IADC fourth character	Security DBS nomenclature
Air application	A	A
Special bearing seal	B	Standard**
Centre jet	C	J4
Deviation control	D	D*
Extended jets	E	E
Extra gauge/body protection	G	G*, M*, D, SS*
Horizontal/steering	H	HZ*, SS*, MM*
Jet deflection	J	JD
Lug pads	L	L
Motor application	M	SS*, MM*, M*
Two cone	T	2*
Enhanced cutting structure	W	D
Chisel inserts	X	
Conical inserts	Y	CF

* - Prefix - others are suffixes

** - Special HDS seal available as standard feature

BIT SELECTION CHART

REED TOOL COMPANY

IADC Classification						
Series	Form- ation	Third digit - bearings & features				
		1	4	5	6	7
1	1	Y11	S11	S11G MS11G EMS11G MS11GD	HP11 EHT11	MHP11G
	2	Y12			HP12 EHP12	
	3	Y13		EMS13G MS13G	HP13	HP13G MHP13G MHP13GD
2	1			MS21G	HP21	HP21G
3	3					HP31G
4	1			EMS41H EMS41HD		HP41A EHP41 EHP41AD EHP41H
	2			EMS42H EMS42HD		
	3			S43A MS43A MS43AD-M MS43A-M	HP43 HP43A HP43A-M HP43-M	EHP43 EHP43A EHP43H EHP43AD EHP43HD
	4			MS44A EMS44A MS44AD EMS44AD EMS44H EMS44HD	HP44-M	EHP44H EHP44HD
5	1			S51A EMS51AD MS51A MS51A-M MS51AD-M		HP51 EHP51 HP51A EHP51A HP51A-M HP51AD HP51H EHP51H HP51H-M EHP51HD HP51X EMS51A HP51X-M
	2				HP52 HP52X	HP52A HP52-M
	3			S53A MS53 MS53D	HP53 HP53A HP53A-M HP53D HP53AD HP53JA	EHP53 EHP53A EHP53D EHP53AD
	4				HP54	
6	1				HP61 HP61A HP61AD	EHP61 EHP61A EHP61D EHP61AD
	2			S62A MS62 MS62D	HP62 HP62A HP62D HP62JAK	EHP62 EHP62A HP62AD
	3				HP63 HP63D HP64	EHP63
	4					
7	3				HP73 HP73D	EHP73
	4				HP74	
8	3				HP83 HP83D	EHP83 EHP83D

Note : Only the series/formation combinations are shown for which bits are available from Reed.
Similarly, empty "third digit" columns are not shown.

BIT TYPE DESIGNATIONS

REED TOOL COMPANY

Prefixes

- EHP – Enhanced performance : threaded ring journal bearing
- HP – Premium journal bearing bit
- S – Sealed roller bearing bit
- Y – Non-sealed roller bearing bit
- MHP – Premium journal bearing bit with high speed seal
- MS – Sealed roller bearing bit with high speed seal

Suffixes

- A – Chisel shaped inserts
- C – Centre jet
- D – Diamond heel pacs
- G – Tungsten carbide heel pacs on steel tooth bits
- H – Chisel shaped inserts in 417-517 designs with 3° skew.
- JA – Jet bit for air circulation
- K – Tungsten carbide inserts added to the shirt-tail to reduce wear and protect the seal
- L – Steel pads with tungsten carbide inserts which are welded to the bit body
- M – Mudpick II hydraulics
- X – Special cutting structure variations that may differ by bit type

D – BITS

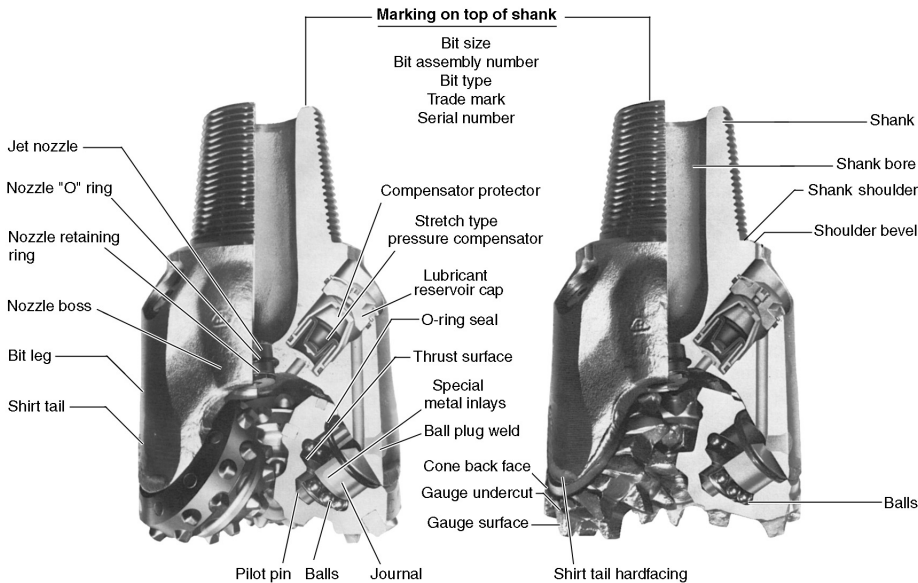
Clickable list

(Use the expanded list under "Bookmarks" to access individual tables)

Rock bit nomenclature	D-1
Rock bit classification schemes	D-5
Correlations of formations to IADC codes for tri-cone bits	D-8
General data	D-9
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Drilling practices	D-15
Drill off tests	D-19
Evaluation & comparison of bits	D-20
Evaluation of used bits	D-22
PDC cutter wear	D-30
Availability of rock bit types	D-31

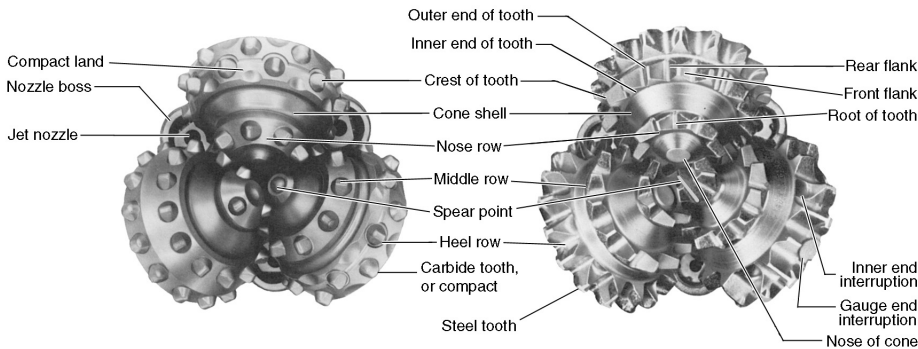
ROCK BIT NOMENCLATURE

TRI-CONE ROLLER BITS



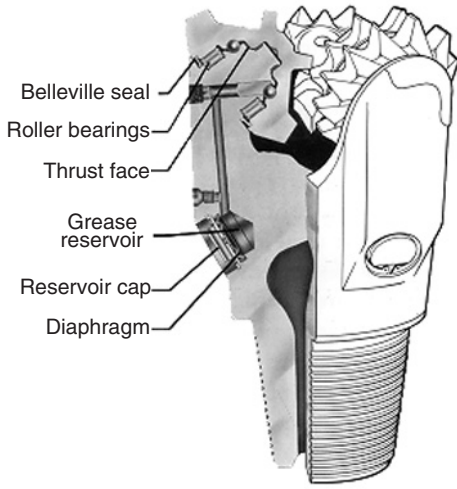
TUNGSTEN CARBIDE BIT
with sealed journal bearings

STEEL TOOTH BIT
with sealed journal bearings

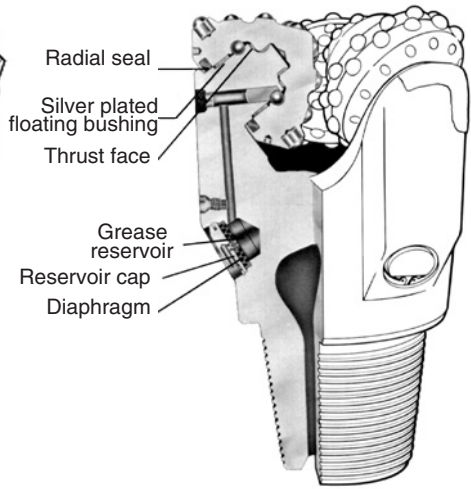


ROCK BIT NOMENCLATURE

DETAILS OF SEALED BEARINGS



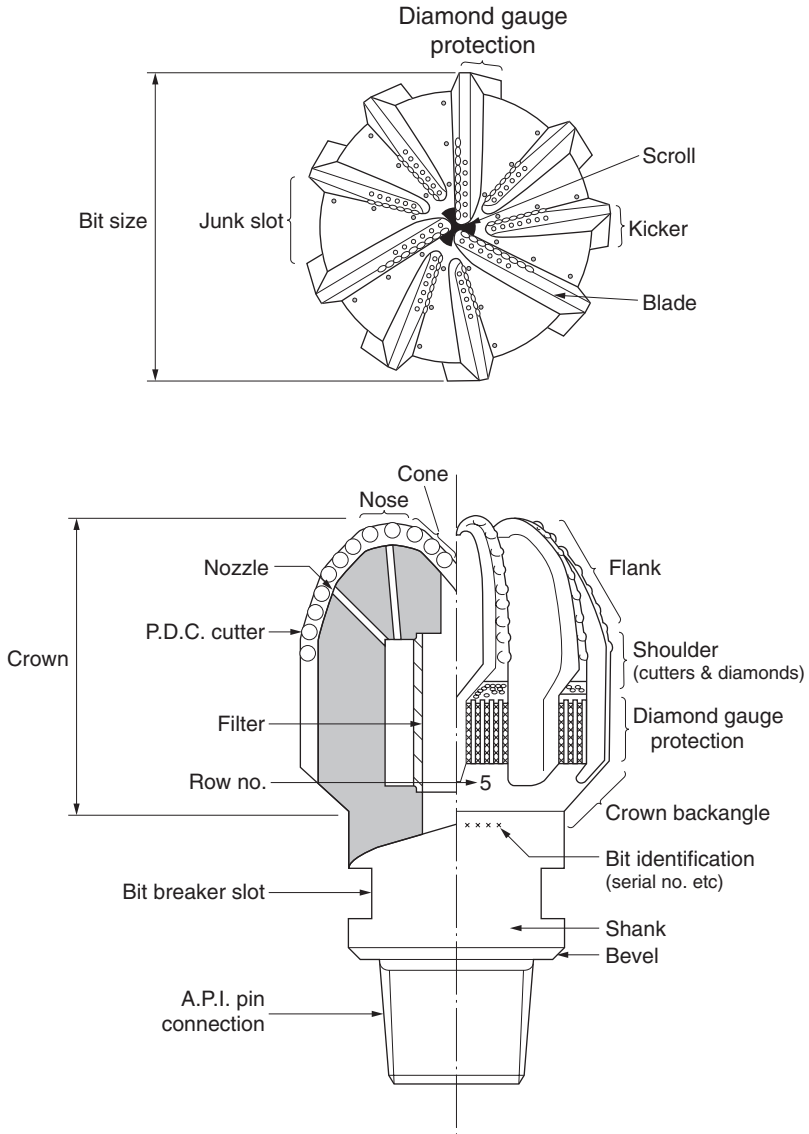
Roller bearings



Journal bearings

ROCK BIT NOMENCLATURE

PDC (Polycrystalline Diamond Compact) BITS

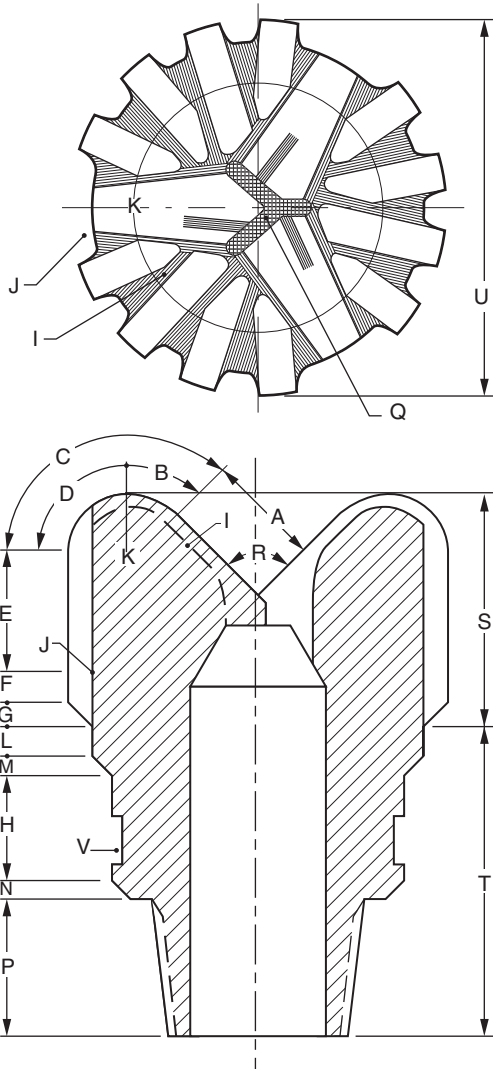


Typical part section through centre of bit

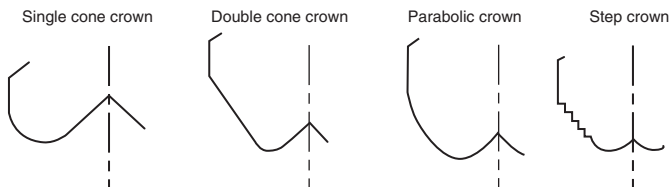
ROCK BIT NOMENCLATURE

DIAMOND BITS

- A - Throat
- B - I.D. Radius
- C - Nose
- D - O.D. Radius
- E - O.D. Gage
- F - O.D. Above Diamonds
- G - O.D. Angle
- H - Steel Shank
- I - Fluid Courses
- J - Junk Slots/Slabs
- K - Contact Point/Diameter
- L - Shoulder
- M - Shoulder Angle
- N - Shank Angle
- P - Thread Connection
- Q - Fluid Entrance
- R - Cone Angle
- S - Crown
- T - Pin Shank
- V - Breaker Slot



Diamond bit profiles



ROCK BIT CLASSIFICATION SCHEMES

INTRODUCTION

There are two classification schemes, one for roller cone bits and one for fixed cutter (diamond) bits. The current versions of each were introduced in 1987 jointly by the SPE (Society of Petroleum Engineers) and the IADC (International Association of Drilling Contractors).

The classifications schemes both make use of four characters. For roller cone bits this consists of three numbers and a letter, whereas diamond bits use a letter and three numbers (diamond bits may also use a letter as the third character). The basis for the classification is however slightly different as explained below, even though the end result is the same.

Roller cone bits

The system is based primarily on the formation characteristics with the first two characters indicating the hardness of the formation for which the bit is designed/suited, and also indicating whether it has milled teeth or tungsten carbide inserts. The second character is used to sub-divide the hardness classes defined by the first character.

The third and fourth characters indicate the general features of the bit itself, such as the type of bearing, whether there is gauge protection or not (which also reflect the type of formation for which it is intended) and whether the bit has any special features or whether it is intended for any special applications, such as air drilling.

The significance of these four characters is summarised in the boxes on [page D-6](#).

As an example a bit classified 6.3.5.Y would be a tungsten carbide insert bit with sealed roller bearings and gauge protection. It would have conical inserts intended for hard formations.

Diamond bits

The classification system of diamond bits is based much more on the construction and geometry of the bit than on the explicit formation type. For this reason the manufacturers sometimes quote not only the classification code for the diamond bit itself, but also the code for the tri-cone bit which would be appropriate for the same formations.

The first character indicates the cutter type and the body material. The second character indicates the profile of the cutting face of the bit. The third character indicates the design of the bit with regard to the flow of drilling fluid across its face. The fourth and last character indicates the size and density of the cutters.

The meanings of the four characters are shown in the boxes on [page D-7](#).

Charts

Each bit manufacturer produces a classification chart for tri-cone bits showing how their own and their competitors' bits fit into the system. The roller cone bits of four major manufacturers and one smaller one have been listed in the tabulations on pages D-32 to D-39 giving the manufacturers own type codes with the bits arranged according to the IADC classification

Equivalent classification charts for diamond bits do not exist, probably because the designs, and thus the type names, are changing much more rapidly than tri-cone bits, and any comparative chart would become out of date as soon as it was printed. The choice of diamond bits is made from the individual manufacturers catalogue and often in discussion with his representative.

ROCK BIT CLASSIFICATION SCHEMES

TRICONE BITS

First digit :

Tooth material and length

The numbers 1, 2 and 3 designate steel tooth bits and correspond to increasing formation hardness.

The numbers 4, 5, 6, 7 and 8 designate bits with tungsten carbide inserts and also correspond to increasing formation hardness.

Second digit :

Formation hardness (finer grading)

The numbers 1, 2, 3 and 4 denote a sub-classification of the formation hardness in each of the eight classes determined by the first digit.

Third digit :

Bearings and gauge protection

The numbers 1 to 7 define the type of bearing and specify the presence or absence of gage protection by tungsten carbide inserts, on the leading flanks of the bit cones:

- 1 = standard roller bearing
- 2 = roller bearing, air-cooled
- 3 = roller bearing, gage protected
- 4 = sealed roller bearing
- 5 = sealed roller bearing, gage protected
- 6 = sealed friction bearing
- 7 = sealed friction bearing, gage protected

The numbers 8 and 9 are reserved for future use. However, some bit manufacturers use this space to show their directional bits (8) and special application bits (9).

Additional letter :

Miscellaneous characteristics

- A = air application : journal bearing bits with air circulation nozzles.
- C = centre jet
- D = deviation control
- E = extended jets
- G = extra gauge/body protection
- J = jet deflection
- R = reinforced welds (for percussion applications)
- S = standard steel tooth model
- X = chisel insert
- Y = conical insert
- Z = other insert shape

ROCK BIT CLASSIFICATION SCHEMES

DIAMOND BITS

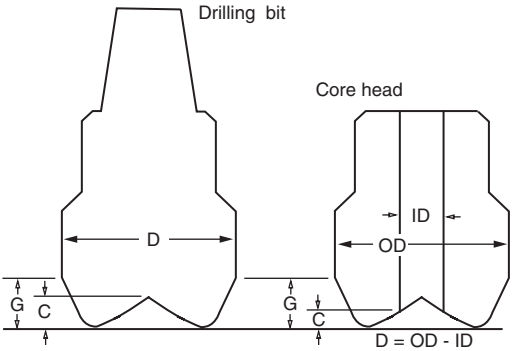
First character :

Cutter type & body material

D: Matrix body / Natural diamonds
M: Matrix body / PDC cutters
S: Steel body / PDC cutters
T: Matrix body / TSP cutters
O: Other

Second character:

Bit Profile Codes



G: Gage height	C : Cone height		
	High ($C > 1/4 D$)	Medium ($1/8 D \leq C \leq 1/4 D$)	Low ($C < 1/8 D$)
High G ($> 3/8 D$)	1	2	3
Medium G ($1/8 D \leq G \leq 3/8 D$)	4	5	6
Low G ($< 1/8 D$)	7	8	9

Third character :

Hydraulic design

Cutter distribution	Fluid exit		
	Changeable jets	Fixed ports	Open throat
Bladed(1)	1	2	3
Ribbed(2)	4	5	6
Open-faced	7	8	9
Alternate codes: R = Radial flow X = Feeder/collector flow O = Other			
These letters are used in preference to Numbers 6 & 9 for most natural diamond and TSP bits.			
(1) Bladed refers to raised, continuous flow restrictors with a standoff distance from the bit body of more than 1" (25.4 mm).			
(2) Ribbed refers to raised, continuous flow restrictors with a stand-off distance from the bit body of 1" (25.4 mm) or less.			

Fourth character :

Cutter size and density

Size	Density		
	Light	Medium	Heavy
Large	1	2	3
Medium	4	5	6
Small	7	8	9
0 = impregnated			
Cutter size range	Natural diamonds: stones/carat	Synthetic diamonds: usable cutter height	
Large	< 3	> 5/8"	
Medium	3 - 7	3/8" - 5/8"	
Small	> 7	< 3/8"	

CORRELATIONS OF FORMATIONS TO IADC CODES FOR TRI-CONE BITS

	Series	Type
Milled tooth	1. Soft formations having low compressive strength and high drillability	1. Very soft shale 2. Soft shales 3. Medium soft shale/lime 4. Medium lime shale
	2. Medium to medium hard formations with high compressive strengths	1. & 2. Medium lime/shale 3. Medium hard lime/sand/slate
	3. Hard semi-abrasive or abrasive formations	1. Hard lime 2. Hard lime/dolomite 3. Hard dolomite
Insert	4. Soft formations having low compressive strength and high drillability	1. Very soft shale 2. Soft shales 3. Medium soft shale/lime 4. Medium lime shale
	5. Soft to medium formations of high compressive strength	1. Very soft shale/sand 2. Soft shale/sand 3. Medium soft shale/lime
	6. Medium hard formations of high compressive strength	1. Medium lime/shale 2. Medium hard lime/sand 3. Medium hard lime/sand/slate
	7. Hard semi-abrasive and abrasive formations	1. Hard lime/dolomite 2. Hard sand/dolomite 3. Hard dolomite
	8. Extremely hard and abrasive formations	1. Hard chert 2. Very hard chert 3. Very hard granite

GENERAL DATA

ROCK BITS

IADC code*	Bearing type	Formation				WOB (lbs/inch bit diam).	RPM
		Soft	Med.soft				
			Med.hard Hard				
437	Friction	x				1,500-3,500	120-60
517	Friction	x				2,000-4,500	100-50
527	Friction	x				2,000-5,000	110-60
537	Friction		x			2,500-5,000	75-45
547	Friction		x			2,500-5,500	0-50
617	Friction		x			2,500-5,500	65-45
617	Friction			x		2,000-6,000	70-40
627	Friction			x		2,000-6,000	65-40
627	Friction			x		3,000-6,000	65-40
637	Friction			x		3,000-6,500	55-40
727	Friction				x	2,500-6,000	60-40
737	Friction				x	3,000-6,500	55-35
837	Friction				x	3,000-7,000	50-30
519	Friction	x				1,500-2,500	100-55
515	Sealed*	x				2,000-4,500	100-50
535	Sealed*		x			2,500-5,000	75-45
615	Sealed*		x			2,500-5,500	65-45
612	Air		x			2,000-5,000	70-45
622	Air			x		2,000-5,500	70-45
732	Air				x	2,500-6,000	65-45
832	Air				x	2,500-6,500	60-40
116	Friction	x				2,000-5,000	140-70
126	Friction	x				2,000-6,000	120-60
136	Friction		x			3,000-7,000	110-60
216	Friction			x		3,000-8,000	80-50
114	Sealed*	x				2,000-6,000	250-75
124	Sealed*	x				2,000-6,000	250-75
134	Sealed*		x			3,000-7,000	175-60
214	Sealed*			x		3,000-8,000	120-50
314	Sealed*				x	4,000-9,000	70-45
111	Open	x				2,000-6,000	250-75
121	Open	x				2,000-6,000	250-75
131	Open		x			3,000-7,000	175-60
211	Open			x		3,000-8,000	90-50
311	Open				x	4,000-9,000	70-45
231	Open			x		3,000-8,000	80-45
118	Open	x				2,000-6,000	250-75
128	Open	x				1,000-4,000	250-75

Pin connections - rock bits

Size range in inches	API Connection
55/8-63/4	31/2 Reg.
75/8-83/4	41/2 Reg.
91/2-141/2	65/8 Reg.
143/4-20	75/8 Reg.
22	65/8 Reg. or 75/8 Reg.
24 - 26	75/8 Reg. or 85/8 Reg.
28	75/8 Reg.

Note:

Certain sizes can be supplied with 75/8" or 85/8" API Reg. connections on special order

Recommended make-up torque - rock bits

	Minimum	Don't Exceed
	lbs-ft	
31/2 Reg.	7,000	9,000
41/2 Reg.	12,000	16,000
65/8 Reg.	28,000	32,000
75/8 Reg.	34,000	40,000
85/8 Reg.	40,000	60,000
	N-m	
31/2 Reg.	9,500	12,200
41/2 Reg.	16,300	21,700
65/8 Reg.	38,000	43,400
75/8 Reg.	46,100	54,200
85/8 Reg.	54,200	81,400

GENERAL DATA

DIAMOND BITS

Diamond bit drilling parameters

Bit size		RPM	Weight on bit		Flow Rate	
inches	mm		pounds	kdaN	gpm	dm ³ /sec
5 ⁷ / ₈	149.2	100-140	6-10,000	2.67-4.45	160-220	10-14
6	152.4	100-140	6-10,000	2.67-4.45	160-220	10-14
6 ¹ / ₂	165.1	100-140	6-12,000	2.67-5.34	160-220	10-14
6 ³ / ₄	171.5	100-140	6-14,000	2.67-6.23	160-220	10-14
7 ⁷ / ₈	200.0	80-120	8-16,000	3.56-7.12	200-300	13-19
8 ¹ / ₂	215.9	80-120	8-18,000	3.56-8.00	300-400	19-25
8 ³ / ₄	222.3	80-120	8-18,000	3.56-8.00	300-400	19-25
9 ⁷ / ₈	225.4	60-120	10-23,000	4.45-10.2	350-500	22-32
10 ⁵ / ₈	269.9	60-120	12-28,000	5.34-12.5	500-600	32-38
12 ¹ / ₄	311.2	60-100	15-39,000	6.67-17.4	550-700	34-44

Recommended make-up torque - PDC & diamond bits

Bit size		API pin size		Recommended torque	
inches	mm	inches	mm	lbs-ft	N-m
3 ³ / ₄ - 4 ¹ / ₂	95.2 - 114.3	2 ³ / ₈ Reg.	60.3	3,000 - 3,500	4,000 - 4,800
4 ⁵ / ₈ - 5	117.5 - 127.0	2 ⁷ / ₈ Reg.	73.0	6,000 - 7,000	8,000 - 9,500
5 ¹ / ₈ - 7 ³ / ₈	136.5 - 187.3	3 ¹ / ₂ Reg.	88.9	7,000 - 9,000	9,500 - 12,200
7 ⁵ / ₈ - 9	193.7 - 228.6	4 ¹ / ₂ Reg.	114.3	12,000 - 16,000	16,300 - 21,700
9 ¹ / ₂ - 26	241.3 - 660.4	6 ⁵ / ₈ Reg.	168.3	28,000 - 32,000	38,000 - 43,400
14 ³ / ₄ - 26	374.6 - 660.4	7 ⁵ / ₈ Reg.	193.7	34,000 - 40,000	46,100 - 54,200

GENERAL DATA

API TOLERANCES

on new rock bits

Size		Tolerance	
ins	mm	ins	mm
3 ³ / ₈ - 13 ³ / ₄	85.7 - 349.3	+ ¹ / ₃₂ ,0	+0.79 ,0
14 - 17 ¹ / ₂	355.6 - 444.5	+ ¹ / ₁₆ ,0	+1.59 ,0
17 ⁵ / ₈ or more	447.7 or more	+ ³ / ₃₂ ,0	+2.38 ,0

on new diamond and PDC bits

Size		Tolerance	
ins	mm	ins	mm
6 ³ / ₄ or less	171.5 or less	0, -0.015	0, -0.38
6 ²⁵ / ₃₂ - 9	172.2 - 228.6	0, -0.020	0, -0.51
9 ¹ / ₃₂ - 13 ³ / ₄	229.4 - 349.3	0, -0.030	0, -0.76
13 ²⁵ / ₃₂ - 17 ¹ / ₂	350.0 - 444.5	0, -0.045	0, -1 14
17 ¹⁷ / ₃₂ or more	445.3 or more	0, -0.063	0, -1.60

on casing drift mandrels

Size ins	Weight lbs/ft	Length		Tolerance	
		ins	mm	ins	mm
7	23.0	6	152	6.250	159
7	32.0	6	152	6.000	152
8 ⁵ / ₈	32.0	6	152	7.875	200
8 ⁵ / ₈	40.0	6	152	7.625	194
9 ⁵ / ₈	40.0	6	152	8.625	220
9 ⁵ / ₈	53.5	6	152	8.375	213
10 ³ / ₄	45.5	12	305	9.750	248
10 ³ / ₄	55.5	12	305	9.625	244
11 ³ / ₄	42.0	12	305	10.625	270
11 ³ / ₄	60.0	12	305	10.875	276
13 ³ / ₈	72.0	12	305	12.000	305

THE DULLNESS GRADING SYSTEM FOR USED BITS

The following information is recorded:

- Distance drilled
- Time taken
- Averaged drilling parameters (WOB, RPM, Pump speed)
- Average drilling fluid properties (type, density, viscosity, fluid loss)
- The condition of the bit when pulled.

The first four of these are objective measurements which can be obtained by reference to the standard daily reports. The condition however is a very subjective assessment made by the driller. In order to provide a measure of consistency between bit condition reports made by all drillers, world wide, a grading system has been introduced. This is the IADC system which applies to roller cone bits, diamond bits and core heads. It uses code characters for describing six categories of wear, grouped into the three sections cutters, bearings and gauge, and adds two codes for remarks.

If a standard bit report form is being completed there are eight boxes in which the individual codes are entered. If the bit condition is being discussed, or described in "free-format" text the three sections containing the description of the wear are each identified by a letter, or in one case a phrase.

Cutting Structure				Bearing or seal (B)	Gauge mm or 16ths (G)	Remarks	
Inner rows (I)	Outer rows (O)	Dullness character (D)	Location (L)			Other character (O)	Reason pulled (R)

The cutting structure

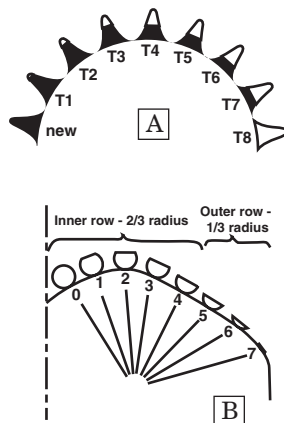
Four codes are used to describe the cutting structure - the teeth/inserts on a roller cone bit, or the cutting elements of a diamond bit. These are entered into the first four boxes of a standard report, otherwise they are identified by the letter "T" for roller cone bits or "cutting structure" for diamond bits.

The first two codes define the wear on the cutters using a scale of 0 to 8, where 0 represents no wear and 8 indicates that no usable cutting structure is left. The first code represents the average wear of the cutters in the inner two thirds of the bit radius, the second refers to the average wear of those in the outer third. Note that in the case of core bits the "radius" is to be interpreted as the distance from the ID to the OD of the core head, i.e. in Figure B the centre line shown would be the ID of the core head/OD of the core. For a roller cone bit the worst cone is taken for the grading.

The wear of milled teeth and PDC cutters is graded in eighths of the original tooth height - see Figures A & B.

For a bit worn as shown in figure B the first two codes would be $(0+1+2+3+4)/5=2$ and $(5+6+7)/3=6$.

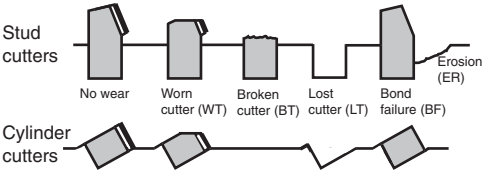
For roller cone insert bits and for cutting element wear on natural diamond and TSP bits the number of inserts or diamonds broken or missing is more relevant than the



actual wear on the individual inserts or cutting elements. It is a combination of wear and broken or missing inserts/diamonds which determines the amount of wear to be reported. The same scale of 1 to 8 is used with T1 representing 1/8 of the cutting elements lost or broken and T8 representing all the cutting elements lost or broken. Some experience is required to do this correctly.

The third box is for the code describing the primary wear characteristic of the cutting structure, chosen from the list in Table 1. The figure below shows how these “tooth” wear terms are applied to PDC cutters.

Wear characteristics - PDC cutters



The fourth part of the cutting structure code defines the basic location of the primary wear characteristic described in the third box of the eight. This can range from a specific part of the bit face to the entire bit. The codes are chosen from the list given in the second figure, opposite; in the case of a tri-cone bit the number(s) of the affected cone(s) is/are included.

The bearing/seals

The letter B is used to identify the code used for bearing/seal reporting. If a standard report form is being used the code is entered into the fifth box.

If no seals are used then the bearing wear is reported on a scale of 1 to 8, as for the teeth, with 0 representing “as new” condition, and 8 indicating that all bearing life has been used up (i.e. the bearings have failed). The condition of the worst bearing is reported.

For sealed bearings, only the condition of the seals is reported with either E indicating that the seals are still effective or F indicating that the seals have failed.

For fixed cutter bits without bearings or seals the code “X” is entered into standard report forms.

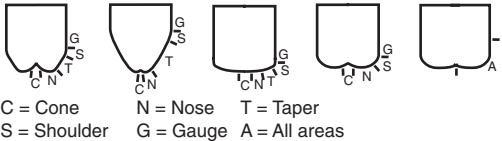
Table 1

*BC:	Broken Cone
BF:	Bond Failure
BT:	Broken Teeth/Cutters
BU:	Balled Up
*CC:	Cracked Cone
*CD:	Cone Dragged
CI:	Cone Interference
CR:	Cored
CT:	Chipped Teeth/Cutters
ER:	Erosion
FC:	Flat Crested Wear
HC:	Heat Checking (and spalling)
JD:	Junk Damage
*LC:	Lost Cone
LN:	Lost Nozzle
LT:	Lost Teeth/Cutters
NR:	Not Re-runnable
OC:	Off-CentreWear
PB:	Pinched Bit
PN:	Plugged Nozzle/Flow Passage
RG:	Rounded Gauge
RO:	Ring Out
RR:	Re-runnable
SD:	Shirt-tail Damage
SS:	Self Sharpening Wear
TR:	Tracking
WO:	Washed Out Bit
WT:	Worn Teeth/Cutters
NO:	No Major/Other Dull Characteristics
* Show cone number(s) under "Location".	

Roller cone location codes

N = Nose row, M = Middle row, G = Gauge row, A = All rows (followed by the number of the cone)

Fixed cutter location codes



The gauge

The letter G is used to identify the code used for reporting the condition of the bit gauge. If a standard report form is being used the code is entered into the sixth box.

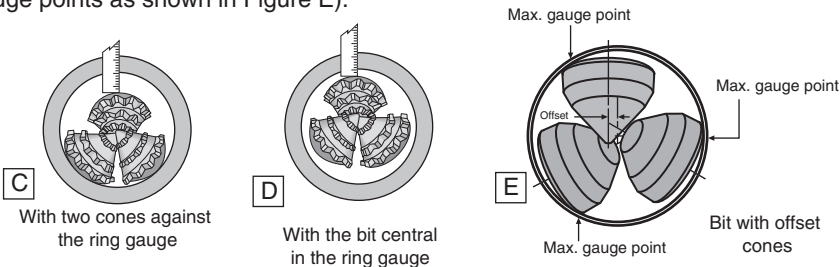
The reduction in diameter is measured in millimetres or in sixteenths of an inch. If the bit is full gauge an “1” indicates that it is in gauge. Working in SI units a “1” indicates that it is 0 to 1 mm under gauge. A “2” indicates that it is 1-2 mm under gauge. A “3” indicates that it is 2-3 mm under gauge, and so on.

In oilfield units the wear would be reported as “1/16”, indicating that the bit is zero to 1/16" under gauge. “2/16” would indicate that it is 1/16" to 1/8" under gauge, etc.

Gauge wear can be determined by using a ring gauge and ruler. This should be done with the bit standing on its cones so that they take up the position they had when cutting the hole.

There are two methods used to measure the wear. In the first, most common, method the ring gauge is pulled against the gauge points of two cones, and the space between the ring and third cone is measured (Figure C). Usually, this measurement is used for the amount of wear; however, to be exact, the measurement should be multiplied by $2/3$.

In the second method, the bit is centred in the gauge ring and the ruler is used to measure the distance from the ring to the outermost cutting surface (gauge surface) (Figure D). This measurement must be multiplied by 2 to give the loss in diameter and thus the total amount of wear. Offset bits should be measured at one of the maximum gauge points as shown in Figure E).



Remarks

In the first of the two places available for remarks at the end of the wear code more information is given on the state of the cutting structure and flow passage(s), chosen from the same list as for the primary characteristic of tooth/cutter wear.

In the last place the reason for pulling the bit is given. This is taken from the following list.

BHA: To change Bottom Hole Assembly	HR: Hours on Bit (estimated maximum usable hours reached)
DMF: Down-hole Motor Failure	LIH: Left in hole
DSF: Drill String Failure	PP: Unexpected variation in Pump Pressure
DST: To perform a Drill Stem Test	PR: Insufficient Penetration Rate
DTF: Down-hole Tool Failure	RIG: To carry out Rig Repairs
LOG: To run Logs	TD: Total Depth/Casing Depth reached
CM: To condition Mud	TQ: Unexpected variation in Torque
CP: Core Point reached	TW: Twist Off
DP: To Drill Plug	WC: Weather Conditions
FM: Formation Change	WO: Washout in drill string
HP: Hole Problems	

DRILLING PRACTICES

ROCK BITS

Rules of thumb for bit selection

- Shale has a better drilling response to RPM.
- Limestone has a better drilling response to bit weight.
- Bits with roller bearings can be run at a higher RPM than bits with journal bearings.
- Bits with sealed bearings can give longer life than bits with open bearings.
- Milled tooth bits with journal bearings can be run at higher weights than milled tooth bits with roller bearings.
- Diamond bits can run at higher RPM than tri-cone bits.
- Bits with high offset may wear more on gauge.
- Bits with high offset may cause more hole deviation.
- Cost per foot analysis can help you decide which bit to use.
- Examination of used bits can help you decide which bit to use.

Running in

- Make the bit up to proper torque.
- Hoist and lower the bit slowly through ledges and dog legs.
- Hoist and lower the bit slowly at liner tops.
- Rock bits are not designed for reaming. If you do ream, do it with light weight and low RPM.
- Protect nozzles from plugging with Smith Tool jet plugs.

Establish a bottom hole pattern

- Rotate the bit and circulate when approaching bottom. This will prevent plugged nozzles and clear out fill.
- Lightly tag the bottom with low RPM.
- Gradually increase the RPM.
- Gradually increase the weight.

If the expected penetration rate is not achieved

- Drilling fluid density may be too high with respect to formation pressure.
- Drilling fluid solids may need to be controlled.
- Pump pressure or pump volume may be too low.
- The bit used may be too hard for the formation.
- Formation hardness may have increased.
- RPM and weight may not be the best for bit type and formation — carry out a drill off test.
- Ensure that the bit is stabilised.

Before re-running green bits

- Make sure the bit is in gauge.
- Check any bit for complete cutting structure.
- Check any sealed bearing bit for effective seals .
- Soak any sealed bearing bit in water or diesel to loosen formation packed in the reservoir cap equalisation ports.
- Regrease 1 3/4" diameter and larger open bearing bits.

DRILLING PRACTICES

DIAMOND BITS - APPLICATIONS

Economics and Application

Regardless of how well designed or manufactured a bit, the situation in which it is used, and the applied practices, ultimately determine the success or failure of the run.

When to use a diamond bit

- When economics dictate
- Generally, the rate of penetration ultimately determines the economics of the bit run
- When on-bottom times are important
- When oil-phase mud systems are used (preferred)
- When water-phase systems are used in non-hydrating formations (preferred)
- When rotating at high speeds (turbine or PDM)
- When high bottom hole temperatures are encountered (approx. 300° F and higher)
- When drilling in deviated hole section requiring light bit weight -When drilling significantly overbalanced

Why use a diamond bit

- Economics
- To reduce the number of trips in order to :
 - minimise running through dangerous hole sections
 - minimise rig wear
- To avoid tripping in bad weather

Where to use a diamond bit (Formations in which diamond bits are normally beneficial)

- Polycrystalline Diamond Compact (PDC) Bits
 - Very weak, poorly consolidated, brittle, shallow sediments (e.g. Miocene sands, silts, clays)
 - Low strength, poorly compacted, brittle, non-abrasive, relatively shallow sediments, precipitates and evaporites (e.g. salt, anhydrite, marls, chalk – Devonian/Muschelkalk)
 - Moderately strong, somewhat abrasive and ductile, indurated medium-depth sediments, precipitates and evaporites (e.g. silty claystone, siliceous shales, porous carbonates, anhydrite — Eocene)
- Natural/Thermally Stable Diamond Bits
 - Moderately strong, somewhat abrasive and ductile, indurated medium-depth sediments, precipitates and evaporites (e.g. siliceous shales, porous carbonates, anhydrite, silty claystone— deep Miocenes)
 - Strong and abrasive indurated, very ductile deep sediments, precipitates and evaporites (e.g. sandy shales, calcareous sandstones, dolomites, limestone — Pennsylvanian/Mississippian)
 - Very strong and abrasive, indurated ductile and non-ductile sediments, precipitates and evaporites (e.g. Bunter sandstone, bromides, etc.)

Formations detrimental to diamond bits

- Polycrystalline Diamond Compact (PDC) Bits
 - Hard, cemented abrasive sandstone (e.g. sedimentary quartzite)
 - Hard dolomites (sedimentary or metamorphic)
 - Iron (e.g. pyrite — metamorphic or igneous)
 - Chert (metamorphic or sedimentary)
 - Granite and basalt (igneous)
- Natural/Thermally Stable Diamond Bits
 - Hard, cemented quartzitic sands that are highly fractured and abrasive

How to determine the application

- Use geological information to determine which offset wells in the area are likely to be representative for the well to be drilled.
- Review the bit records from the offset wells, in particular their condition when pulled and the calculated economics.
- Review the wireline logging data from the offset wells

DRILLING PRACTICES

DIAMOND BITS - FIELD OPERATIONS

Prior to running the bit

- Reach an agreement with the operator/contractor on what is expected of the bit including, if appropriate, its suitability for drilling float equipment
- Reach an agreement on what mechanical and hydraulic requirements are available and/or necessary to achieve optimum or expected performance
- Before running the diamond bit into the hole, have a junk basket run on the previous bit
- After the previous bit is pulled, inspect it for junk damage and other wear, then gauge it
- If the previous bit appears OK, the bit may be readied to run into the hole
- Check O-ring and install nozzles, if appropriate
- Check for cutter damage
- Check that the bit is within tolerance on diameter and that there is no foreign material inside it
- Recommend the use of drill pipe screens

Running the bit (rotary assembly)

- Handle the diamond bit with care. DO NOT set the bit down without placing wood or a rubber pad beneath the diamond cutters
- A correct bit breaker should be used and the bit should be made up to the correct torque as determined by the pin connection size
- Care should be taken in running the bit through the rotary table and through any known tight spots. Hitting ledges or running through tight spots carelessly may damage the bit gauge
- Reaming is not recommended, however, if necessary, pick up the kelly and run the maximum fluid possible. Rotate at about 60 RPM. Advance bit through tight spot with no more than 4000 pounds weight on bit (WOB) at any time
- As hole bottom is approached, the last three joints should be washed down slowly at full flow and with 40 to 60 RPM to avoid plugging the bit with fill
- The bottom is found by observing the rotary torque indicator as well as the weight indicator. The first on bottom indication is usually an increase in rotary torque
- Once the bottom is located, the bit should be lifted just off bottom (0 to 1 foot if possible) and full volume circulated while slowly rotating for about 5 to 10 minutes
- After circulating, ease back to bottom and be patient in establishing the bottom hole pattern
- When ready to start drilling, increase the rotary speed to about 100 RPM and start cutting a new bottom hole pattern with approx. 1000 to 4000 pounds WOB
- Cut at least one foot in this manner before determining optimum bit weight and RPM for drilling
- Determine optimum ROP through a drill-off test

Running the bit (PDM & turbine)

- Start the pumps and increase to the desired flow rate when approaching bottom
- After a short cleaning period, lower the bit to bottom and increase WOB slowly
- After establishing a bottom hole pattern, additional weight may be slowly added
- As weight is increased, pump pressure will increase, so the differential pressure and WOB must be kept within the recommended downhole motor specifications
- Drill pipe should be slowly rotated to prevent differential sticking
- All other operating practices are as per standard practices

Pull the bit when

- The bit stops drilling
- The bit ceases to be economical as shown by cost/foot calculations
- When very high on-bottom torque with little WOB and a decrease in ROP occurs - the bit may be undergauge in a tough formation
- There is a dramatic decrease in ROP and on-bottom torque
- If there are changes in stand pipe pressure
 - if it goes up there is probably a cutter structure failure
 - if it goes down there is probably washout or a nozzle has been lost

DRILLING PRACTICES

DIAMOND BITS - COMMON PROBLEMS

Problem	Probable cause	Preferred action
Difficulty going to bottom	Previous bit undergauge New bottom hole assembly Collapsed casing Out of drift Bit oversized Stabiliser oversized	Ream with roller cone bit When reaming to bottom, pick up and ream section again. If difficulty remains, check stabilisers. Roll casing with smaller bit Use bi centre bit or reduce bit size Gauge bit with API gauge; if not in tolerance, replace bit Replace with correct size stabiliser
Low pressure differential across nozzles or bit face	Flow area too large Flow area too small Different drilling parameters than designed for Washout in drill string	Increase flow rate and correct on next bit Increase flow rate/strokes Change liners Attempt to optimise flow area on next bit change Check bit pressure drop, drop softline, trip to check pipe and collars
High pressure differential across nozzles or bit face	Flow area too small Excessive flow rate Diamonds too small for formation Bit partially plugged (formation impaction) Formation change Ring out Downhole motor stalled	Reduce flow rate, change flow area on next bit Reduce flow rate If ROP acceptable, change on next bit If ROP unacceptable, pull bit and use bit with correct diamond size Check off bottom standpipe pressure Let bit drill off, circulate full volume for 10 minutes while rotating. Check off bottom pressure again Pick up, circulate, resume drilling at higher RPM, reset, drill off test On and off bottom pressure test, pull bit Refer to manufacturer's handbook
Fluctuating standpipe pressure	Drilling through fractured formation Formation breaking up beneath bit Stabilisers hanging up Equipment failure	If ROP acceptable, continue If ROP acceptable, continue Check equipment Try combination of lighter weight and higher RPM Check overpull Check stabilisers on next trip Repair equipment
Bit won't drill	Bottom not reached Stabilisers hanging up or too large Formation too plastic Establishing bottom hole pattern Core stump left Bit balled	Check tally Check torque, overpull Check pressure – increase flow rate, decrease/increase bit weight, RPM Can take up to an hour Attempt to carefully drill ahead with low bit weight Back off and increase flow rate, then slug with detergent or oil
Slow rate of penetration	Not enough weight on bit: hydraulic lift RPM too low/high Plastic formation Change in formation Overbalanced Diamonds flattened off Cutters flattened Pressure drop too low Wrong bit selection	Increase weight on bit Increase/decrease RPM Reset drill off Reset weight Reset drill off Accept ROP Pull bit Compare beginning and present pressure drops – new bit may be required Increase weight Pull bit Increase flow rate – new bit may be required Pull bit
Excessive torque	Excessive weight on bit Slow rotary speed Stabilisers too large Collars packing off Bit undergauge	Reduce weight and RPM Increase RPM Decrease weight Check bottom hole assembly, stabilisers should be 1/32" to 1/16" under hole size Increase flow rate and work up and down Pull bit
Bit bouncing	Slip-stick action Broken formation Pump off force	Change RPM/weight combination Reduce RPM and weight Increase weight Decrease volume

DRILL OFF TESTS

Drill off tests are performed in order to ascertain the optimum combination of weight on bit and rotary speed to maximize penetration rate. They should be done:

- At the start of a bit run
- On encountering a new formation
- If a reduction in ROP occurs

Procedure

1. Maintain a constant RPM. Select a WOB (near maximum allowable).
2. Record the time to drill off a weight increment, ie 5,000 lbs.
3. Re-apply the starting weight and record the length of pipe drilled during step 2.
4. From steps 2 and 3 the penetration rate may be found.
5. Repeat steps 2 and 3 at least four times. The last test should be at the same value as the first. This repeat test will determine if the formation has changed or not.
6. Plot seconds to drill off versus bit weight.
7. Plot penetration rate versus bit weight.
8. Select the bit weight which produced the fastest ROP. Maintain this WOB constant and repeat the above but varying the RPM.
9. Plot ROP versus RPM and select the RPM which resulted in the fastest ROP. This is the optimum rotary speed.
10. These values for WOB and RPM obtained will result in optimum progress for the particular formation and bit type.

ROCK BITS

EVALUATION AND COMPARISON

Rock bit evaluation

The Cost per Unit Length method can be used:

- To determine when to pull the bit, and
- To compare different types of bit

$$\text{Cost per Unit length} = \frac{\text{Bit cost} + \{\text{Rig cost/hour} \times (\text{Trip time} + \text{Drilling time})\}}{\text{Length of drilled interval}}$$

When to pull bit

$$\text{Ratio } \alpha = \frac{F}{D + T + B}$$

Where :

F = Distance drilled during present bit run

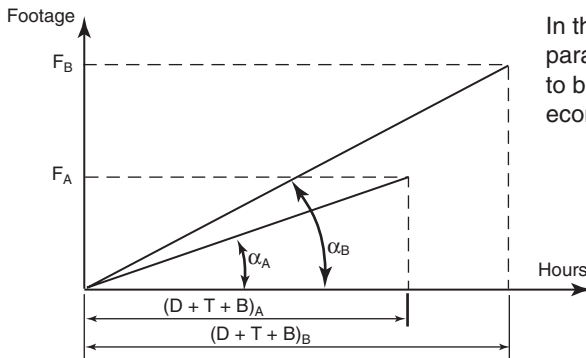
D = Drilling time during present bit run in hours

T = Trip time in hours

B = Cost of bit/rig cost per hour

The bit should be pulled when the ratio α starts to decrease after reaching a maximum

Bit comparison



In this example, assuming all other parameters including the formation to be constant, bit B was more economical than bit A.

Breakeven calculation

By analysis of good offset bit runs, the average cost per foot for a particular section can be calculated. This value can be used to find the breakeven line to which a more expensive bit must reach before it can be considered a good economical run.

1. Calculate cost per foot of comparison bit.
2. Draw breakeven line

$$F_0 = \frac{\text{cost of proposed bit} + (\text{trip time} \times \text{rig cost/hr})}{\text{cost per foot of comparison bit}}$$

$$F_{100} = \frac{\text{rig cost/hr} \times 100}{\text{cost per foot of comparison bit}} + F(0)$$

Notes :

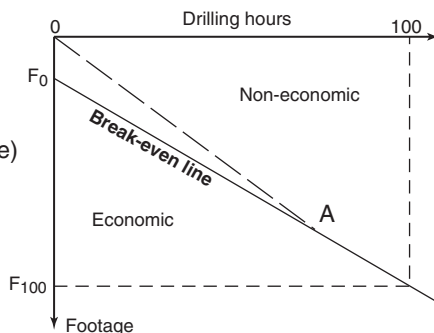
F_0 and F_{100} are breakeven footage at 0 hours and 100 hours.

The line drawn from $(0, F_0)$ to $(100, F_{100})$ represents the target cost per foot to breakeven.

As drilling progresses with the more expensive bit :

Plot footage drilled v. drilling hours (broken line)

Bit breaks even at point A.



Other factors which should be taken into account when doing a breakeven analysis include:

- turbine and PDM rentals
- clean up trips prior to running a stratapax bit
- wiper trips on long bit runs
- wear and tear of rig

EVALUATION OF USED TRI-CONE BITS

DESIRABLE CONDITIONS

Condition of bit	Comments/possible causes	Possible remedies
Soft and medium formation milled tooth bits Hardfacing has held cutting edge on tooth flanks (self-sharpening wear) Figs. 1 & 2	Good bit selection Good drilling practices	NA
Hard formation milled tooth bits Ragged teeth (not blunt) Fig. 3	As above	
Insert bits Balanced wear on bearings and teeth	As above	

UNDESIRABLE CONDITIONS

Condition of bit	Comments/possible causes	Possible remedies
All bits Excessive bearing wear	Excessive RPM Excessive rotating time Excessive W.O.B. Excessive sand in mud Unstabilised drillcollars Improper bit type	Decrease RPM Decrease rotating hours Decrease W.O.B. Decrease sand Stabilise drillcollars Use harder formation bit having larger bearing structure

USED BIT CONDITIONS
DESIRABLE

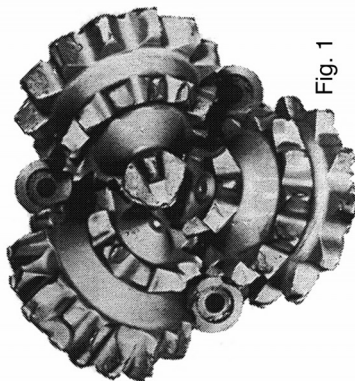


Fig. 1

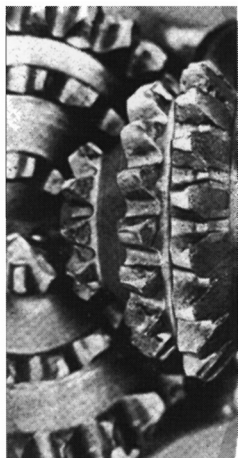


Fig. 2

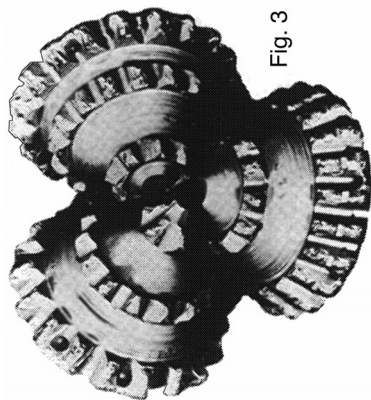


Fig. 3

EVALUATION OF USED TRI-CONE BITS

MILLED TOOTH BITS - UNDESIRABLE CONDITIONS (1)

Condition of bit	Comments/possible causes	Possible remedies
Rounded-off/blunt teeth (hard formation bit)	Formation too soft Insufficient W.O.B.	Use softer-formation bit Increase W.O.B.
Brinell marks Fig. 4	Seen under rollers and/or balls Impact loads	Caution while running-in, making connection etc.
Cones skidded, bearings good Fig. 5	Bit balling up Improper bit type Excessive W.O.B. Cones locking while drilling float shoe	Increase circulating rate (esp. in 'sticky' formations) Evaluation of hydraulics Use bit with wider spaced and longer teeth Decrease W.O.B. Decrease RPM (see torque more easily)
Excessive tooth breakage	Improper break-in Junk in hole Improper bit type Excessive W.O.B. and/or RPM	Drill a few feet before applying desired W.O.B. Wash on bottom before drilling. Run junk basket Use bit with shorter teeth Decrease W.O.B. and/or RPM
Excessive shirt-tail wear Fig. 6	Cuttings 'milling' around bit	Increase circulating rate Review bit hydraulic horsepower
Bradded teeth Fig. 7	Occurs more in soft-formation bits Excessive W.O.B. Failing to pull dull bit Formation too hard for bit type	- Decrease W.O.B. Replace bit when R.O.P becomes excessively slow Use harder formation bit
Heavy gauge wear and inner bearing loose	Improper bit type Excessive rotary speed and/or time Unstabilised drill collars	Use bit with less offset Use bit with more gauge protection Reduce rotary speed and/or time Stabilise drill collars

USED BIT CONDITIONS UNDESIRABLE



Fig. 4



Fig. 5



Inadequate cleaning

Fig. 6

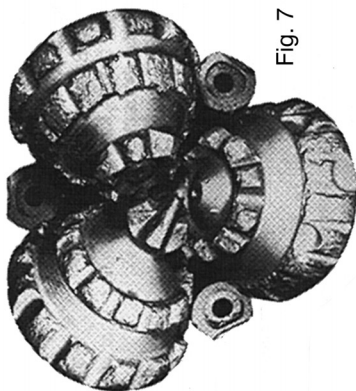


Fig. 7

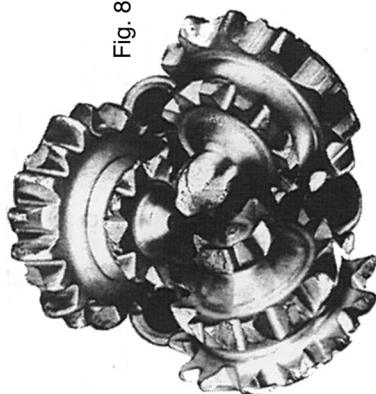


Fig. 8

EVALUATION OF USED TRI-CONE BITS

MILLED TOOTH BITS - UNDESIRABLE CONDITIONS (2)

Condition of bit	Comments/possible causes	Possible remedies
Unbalanced tooth wear	Improper bit type Tooth breakage (appears as wear when bit is pulled)	Use bit with deleted gauge row teeth (if inner row teeth are duller and bit shows no gauge wear) Prevent tooth breakage (see previously)
Off-centre wear Fig. 8	Bit 'walking' off centre When heavy mud weight is necessary	Use softer formation bit. Increase R.P.M. Use reamer and stabilisers on drill string before problem formation is drilled.
Excessive tooth wear	Excessive R.P.M. Improper bit type Use of non-hardfaced type bit	Decrease R.P.M. Use harder formation bit Use bit with hardfacing on teeth
Excessive cupping of tooth crests	Double hardfaced teeth Insufficient W.O.B. Improper formation for double hardfaced teeth	Use bit with single-face hardfacing or single-face and tipped hardfacing Increase W.O.B. -
Fluid cut teeth and cone	Excessive circulating rate Excessive sand content in mud	Decrease circulating rate Remove sand Use jet circulation bit

EVALUATION OF USED BITS

INSERT BITS - UNDESIRABLE CONDITIONS

Condition of bit	Comments/possible causes	Possible remedies
Damage from foreign material Fig. 9	Running on broken/lost inserts from previous bit Other Junk	More washing and pumping on bottom Use junk basket
Formation wear on cone shell around inserts Fig. 10	Inserts too short Insufficient cleaning under bit	Use bit with longer insert extension and more offset Review bit hydraulic horsepower
Excessive insert breakage Fig. 11	Improper break-in Improper bit type Improper drilling practices	Drill a few feet before applying desired W.O.B. If chisel inserts - use bit with less insert extension If formation is part lime and drilling fluid is light weight use projectile shaped inserts Decrease W.O.B. and/or RPM Use shock absorber
Gauge and outer rows broken Fig. 12	Excessive offset in bit Excessive RPM Insufficient stabilising	Use bit with less offset (which may also have less gauge insert extension) Decrease R.R.M. Stabilise drill collars
Excessive gauge wear	Excessive offset in bit Excessive RPM in abrasive formations High solids content in mud Insufficient stabilising	Use bit with less offset Decrease RPM Reduce solids content (if possible) Use stabilisers
Off-centre wear	Bit "walking" off centre When heavy mud weight is necessary	Use softer formation bit Increase R.R.M. Use reamer and stabilisers on drillstring before problem formation is drilled

USED BIT CONDITIONS
UNDESIRABLE

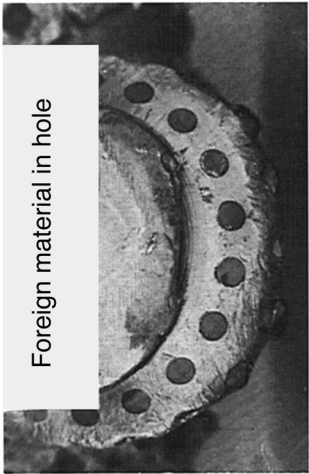


Fig. 9

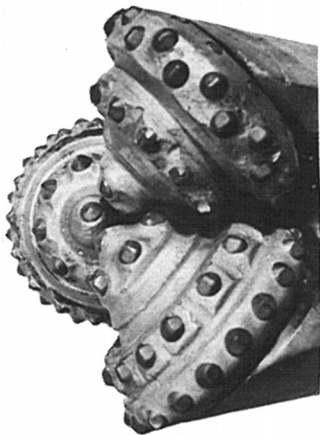


Fig. 10

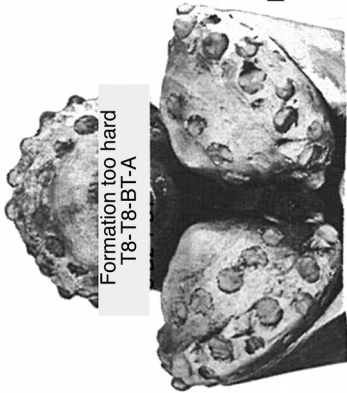


Fig. 11



Fig. 12

EVALUATION OF USED BITS

DIAMOND BITS

PDC bits

1. Generally, bit should not be re-run if cutting element is over 50% worn
2. Cutter wear
3. Cutter loss – caused by
 - excessive weight on bit
 - junk down hole
 - broken formation
 - bit bouncing
 - drastic formation change
 - braze failure
4. Cutter delamination
5. Cutter chipping, flaking, or spalling – caused by:
 - abrasion
 - overheating
 - excessive weight on bit
 - excessive side rake
6. Cutter erosion – caused by:
 - excessive fluid velocity across face
7. Bit badly worn – caused by:
 - fluid cutting on body – reasons :
 - excessive fluid velocity for body material
 - excessive solids/abrasive content in drilling fluid
 - formation wear on body – reasons:
 - too small a cutter standoff (exposure)
 - formation plastic
 - junk damage – reasons:
 - formation junk/broken formation
 - extraneous metal in hole
 - eccentric wear

Natural/thermally stable diamond bits

1. Diamond degradation – caused by:
 - abrasion or microscopic chipping
 - gross breakage – reasons:
 - excessive weight on bit
 - diamond size too large for formation
 - highly fractured formation/high impact loading
 - oxidation (occurs as low as 900° F in presence of oxygen)
 - matrix wear – reasons:
 - fluid erosion;
 - excessive fluid velocity
 - excessive abrasive content in drilling fluid
 - eccentric wear
 - heat damage (insufficient fluid to keep bit cool, thereby causing heat checking in matrix)
 - junk damage

Causes of bit failure

1. Misapplication
 - wrong bit for formation (bit too soft or too hard)
 - wrong diamond size for formation
2. Incorrect or inappropriate operating conditions
 - formation impaction (fluid volume/velocity insufficient to keep channels or bit face clean)
3. Loss of gauge
 - reaming to bottom too quickly or with excessive weight
 - insufficient gauge protection
4. Plugged bit
 - insufficient fluid to keep bit clean
 - washout in string
 - junk pumped down string causing plugging
 - LCM or swabbing to bottom too quickly
5. Ring out
 - junk in hole
 - insufficient cutter coverage
6. Bad Design

PDC CUTTER WEAR (13.3 MM DIAMETER)

PDC cutters are self sharpening and continue to drill until they are worn to approximately semi-circular shape.

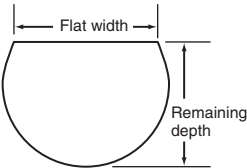
The percentage cutter wear figures given are based on the area of cutter that has been used divided by the area of a semi-circular cutter. (Because of this it is possible to obtain wear greater than 100%.)

It should be noted that, although the life of the individual PDC cutter is related to the given percentage wear figure, the life of the bit may vary widely depending on other conditions. Therefore, the decision of whether or not to re-run the bit should be based not only on cutter wear but on other factors such as:

- Number of missing or damaged cutters and their positions
- Wear on cutters adjacent to missing cutters
- Maximum cutter wear
- Gauge condition (PDC cutters and diamonds)
- Erosion
- Overall bit condition
- Section left to drill
- Formation

Up to 23% wear use the “Flat Width” and over 23% use “Remaining Cutter Depth”

Flat Width mm	Remaining Depth mm	% Worn
4		1
5		2
6		4
7		7
6		10
9		16
10	11	23
	10.5	30
	10	39
	9.5	47
	9	56
	8.5	65
	8	75
	7.5	84
	7	93
	6.5	103
	6	112
	5.5	122



AVAILABILITY OF ROCK BIT TYPES

The IADC classification of rock bits, or at least the first three digits, is a generic classification based on the bit construction and the type of formation for which it is suitable. The fourth character, a letter, indicates one or more special features of the bit. Each manufacturer, however, has his own nomenclature which is used in parallel with the IADC system, and which often allows a much more detailed specification of the special features.

The types of bit produced by each of the four major bit manufacturers (as of late 1997) are tabulated separately in the following eight pages, using their own nomenclature but arranged by IADC code. On the page facing each tabulation is a list of the codes used in that manufacturer's type names. Note that care is needed when requesting a bit with special features because some of the manufacturers base their codes on the IADC fourth character, but some are different, with the same letter used but corresponding to a different feature.

Not included in these tabulations are the sizes that are available for each bit type. All the manufacturers produce bits for the standard hole sizes, but you will have to refer to the manufacturers documentation to check the availability for non-standard sizes.

BIT SELECTION CHART

HUGHES CHRISTENSEN

IADC Classification									
Series	Form- ation	Third digit - bearings & features							
		1	2	4	5	6	7		
1	1	R1		ATX-1 GTX-1	ATX-G1 GTX-G1	MAX-G1 MAX-GT1	ATJ-1 ATJ-1S GT-1 STR-1	GT-G1 GT-G1H	ATM-G1
	3				ATX-G3 GTX-G3	MAX-G3 MAX-GT3			ATM-GT3
2	1						ATJ-4	ATJ-G4	
	2	DR-5							
3	2	R-7							
	4							ATJ-G8	
4	1				ATX-05 GTX-00 GTX-03	MAX-05 MAXGT-00 MAXGT-03		ATJ-05 GT-00 GT-03	ATM-05 ATMGT-03
	2							ATJ-05C GT-09C	ATMGT-09C
	3				ATX-11H GTX-09	MAX-11H MAX-11HG MAXGT-09		ATJ-11 ATJ-11H GT-09 STR-09	ATM-11H ATM-11HG ATMGT-09
	4					MAX-11CG MAXGT-18		ATJ-11C ATJ-18 GT-18 GT-18C	ATM-11CG ATMGT-18
5	1				ATX-22	MAX-22 MAX-22G		ATJ-22 ATJ-22S ATJ-22G GT-20 GT-20S STR-20	ATM-22 ATM-22G ATMGT-20
	2							ATJ-22C ATJ-28 ATJ-28C GT-20C GT-28 GT-28C	ATM-22C
	3							ATJ-33 ATJ-33S ATJ-33A ATJ-33H ATJ-35 STR-30	ATM-33 ATM-33G
	4				ATX-33C			ATJ-33C ATJ-35C	ATM-33C ATM-35CG
6	1		G44					ATJ-44 ATJ-44A ATJ-44G	
	2				ATX-44C	MAX-44C		ATJ-44C ATJ-44CA	
	3		G55			MAX-55		ATJ-55R ATJ-55RG ATJ-55 ATJ-55A	
	4							ATJ-66	
7	3		G77					ATJ-77	
	4							ATJ-88	
8	3		G99					ATJ-99	

Note : Only the series/formation combinations are shown for which bits are available from Hughes.
Similarly, empty "third digit" columns are not shown.

BIT TYPE DESIGNATIONS

HUGHES CHRISTENSEN

Product lines

MAXGT	Ball & roller bearing, metal seal, GT performance package
MAX	Ball & roller bearing, metal seal
ATMGT	Journal bearing, metal seal, GT performance package
ATM	Journal bearing, metal seal
GT	Journal bearing, elastomer seal, GT performance package
STR	Journal bearing, elastomer seal, STR (slim hole*) performance package
ATJ	Journal bearing, elastomer seal
GTX	Ball & roller bearing, elastomer seal, GT performance package
ATX	Ball & roller bearing, elastomer seal
R	Ball & roller bearing, non-sealed
G	Ball & roller bearing, non-sealed air

* 3⁷/₈" - 6³/₄"

Product features

		<u>Examples</u>
A	Air journal bearing, air nozzles	ATJ-33 A
C (prefix)	Centre jet	GT- C 18
C (suffix)	Conical shape inserts	GT-18 C
D	Diamond gauge compacts	ATM-22 D
G	Enhanced gauge	ATJ-33 G
M	Motor hardfacing	GT- M 1
P	Leg stabilisation wear pad	ATMGT- P 18
S	Shirt-tail compacts	GT- S 20
T	High flow extended nozzles	MAXGT- T 03

BIT SELECTION CHART

SMITH INTERNATIONAL

IADC Classification								
Series	Form- ation	Third digit - bearings & features						
		1	2	4	5	6	7	
1	1	DSJ		SDS	MSDSH MSDSSH MSDSHOD	FDS FDSS FDSS+	MFDSH MFDSSH MFDSHOD	
	2	DTJ				FDT		
	3	DGJ			SDGH MSDGH MSDGHOD	FDG	FDGH	MFDGH MFDGHOD
2	1	V2J			SVH MSVH	FV	FVH	
4	1				M01S M01SOD M02S M02SOD 02M		MF02 02MF	
	2				M05S M05MD 05M		F05 MF05 F07 05MF 05MFD	
	3				M1S M1SOD 10M 10MD 12M 12MD 12MY		F1 MF1 10MF 10MFD 12MF 12MFD 12MFY	
	4				15JS M15SD M15S M15SOD 15M 15MD		F15 MA15 F15D MF15 F15OD M15D 15MF M15OD 15MFD	
5	1				A1JSL MA1SL 2JS M2S 20M M2SD 20MD		A1 F15H F17 F25 F2 F25A F2H F2D MF2 MF2D 20MF 20MFD	
	2				M27S M27SD		F27 MF27 F27I MF27D	
	3				3JS M3S M3SOD		F3 MF3 F3D MF3D F3H MF3H MF3OD	
	4						F35 F35A F37A F37 MF37 F37D MF37D	
6	1		4GA		4JS		F4 F4A F4H F45H F45A F47 F47A	
	2		5GA 47JA		5JS 47JS		F47H MF5 F5 MF5D F5OD	
	3						F57 F57A F57D F57OD F57DD	
	4						F67OD	
7	3		7GA				F7 MF7 F7OD	
8	1						F8OD F8DD	
	3		9JA				F9	

Note : Only the series/formation combinations are shown for which bits are available from Smith.
Similarly, empty "third digit" columns are not shown.

BIT TYPE DESIGNATIONS

SMITH INTERNATIONAL

Prefixes

- F – Journal (pfinodal) bearing
- M – Steerable-motor bit bearing
- S – Sealed roller bearing

Suffixes

- A – Designed for air applications
- C – Centre jet
- D – Diamond enhanced gage inserts
- DD – Fully diamond enhanced cutting structure
- E – Full-extended nozzles
- G – Super D-Gun coating
- H – Heel inserts on milled tooth bits. Different, high wear-resistant grade of carbide on TCI bits for abrasive formations
- L – Lug pads
- N – Nominal gage diameter
- OD – Diamond enhanced heel row inserts
- P – Carbide compact in the leg back
- PD – Diamond SRT in the back of the leg
- Q – "Flow Plus" extended nozzles
- R – SRT inserts pressed in leg for stabilisation
- S – Sealed roller bearing

"Magnum" series suffixes

- M – Roller bearing, Trucut gauge
- MD – Roller bearing, diamond chisel gauge
- MF – Journal bearing, Trucut gauge
- MFD – Journal bearing, diamond chisel gauge
- Y – Conical cutting structure

Milled tooth cutting structure designations

- DS – Very soft formation cutting structure
- DT – Soft formation cutting structure
- DG – Medium formation cutting structure
- V – Medium-hard formation cutting structure

Tungsten carbide insert cutting structure designations

- 01 – Very soft formation chisel crest cutting structure
- 02 – Very soft formation chisel crest cutting structure
- 05 – Very soft formation chisel crest cutting structure
- 07 – Soft formation conical cutting structure
- 1 – Soft formation chisel crest cutting structure
- 15 – Soft-medium formation chisel crest cutting structure
- 17 – Soft-medium formation conical cutting structure
- 2 – Soft-medium formation chisel crest cutting structure
- 25 – Medium formation chisel crest cutting structure
- 27 – Medium formation conical cutting structure
- 3 – Medium formation chisel crest cutting structure
- 35 – Medium formation chisel crest cutting structure
- 37 – Medium formation conical cutting structure
- 4 – Medium formation chisel crest cutting structure
- 45 – Medium-hard formation chisel crest cutting structure
- 47 – Medium-hard formation conical cutting structure
- 5 – Medium-hard formation chisel crest cutting structure
- 57 – Medium-hard formation conical cutting structure
- 67 – Hard formation conical cutting structure
- 7 – Hard formation conical cutting structure
- 8 – Hard formation conical cutting structure
- 9 – Hard formation conical cutting structure

BIT SELECTION CHART

SECURITY DBS

IADC Classification							
Series	Form- ation	Third digit - bearings & features					
		1	3	5	6	7	
1	1	S3SJ		S33SG SS33SG	S33SF PSF	S33SGF MPSF	ERA MPSF
	2			SS33G	S33F	S33GF	S33TGF
	3	S4TJ	S4TGJ	SS44G		S44GF	
2	1	M4NJ		M44NG MM44NG	M44NF	M44NGF	
3	1	H7J					
	3			H77SG			
4	1			SS80		S80F ERA 03	ERA 03D
	2			SS81		S81F ERA 07	ERA 07C
	3			SS82		S82F S82CF ERA 13 ERA 13D	SS82F HZS82F ERA 13C ERA 14C
	4			SS83		S83F ERA 17	SS83F ERA 17D
5	1			SS84		S84F S84CF ERA 18C ERA 22C	SS84F HZS84F ERA 22 ERA 22D
	2					S85F ERA 25	S85CF ERA 25C
	3			S86 SS86		S86F S86CF ERA 33	SS86F ERA 33C
	4			SS88C		S88F S88CFH	S88CF S88FA
6	1			M84		M84F	MAF
	2			MM88 M89T		M84CF M86CF	M85F M89TF
	3					M89F	
7	1					H83F	
	3					H87F	
8	1					H89F	
	3			H100		H100F	

Note : Only the series/formation combinations are shown for which bits are available from Security DBS.
Similarly, empty "third digit" columns are not shown.

BIT TYPE DESIGNATIONS

SECURITY DBS

Feature	IADC fourth character	Security DBS nomenclature
Air application	A	A
Special bearing seal	B	Standard**
Centre jet	C	J4
Deviation control	D	D*
Extended jets	E	E
Extra gauge/body protection	G	G*, M*, D, SS*
Horizontal/steering	H	HZ*, SS*, MM*
Jet deflection	J	JD
Lug pads	L	L
Motor application	M	SS*, MM*, M*
Two cone	T	2*
Enhanced cutting structure	W	D
Chisel inserts	X	
Conical inserts	Y	CF

* - Prefix - others are suffixes

** - Special HDS seal available as standard feature

BIT SELECTION CHART

REED TOOL COMPANY

IADC Classification						
Series	Form- ation	Third digit - bearings & features				
		1	4	5	6	7
1	1	Y11	S11	S11G MS11G EMS11G MS11GD	HP11 EHT11	MHP11G
	2	Y12			HP12 EHP12	
	3	Y13		EMS13G MS13G	HP13	HP13G MHP13G MHP13GD
2	1			MS21G	HP21	HP21G
3	3					HP31G
4	1			EMS41H EMS41HD		HP41A EHP41 EHP41AD EHP41H
	2			EMS42H EMS42HD		
	3			S43A MS43A MS43AD-M MS43A-M	HP43 HP43A HP43A-M HP43-M	EHP43 EHP43A EHP43H EHP43AD EHP43HD
	4			MS44A EMS44A MS44AD EMS44AD EMS44H EMS44HD	HP44-M	EHP44H EHP44HD
5	1			S51A EMS51AD MS51A MS51A-M MS51AD-M		HP51 EHP51 HP51A EHP51A HP51A-M HP51AD HP51H EHP51H HP51H-M EHP51HD HP51X EMS51A HP51X-M
	2				HP52 HP52X	HP52A HP52-M
	3			S53A MS53 MS53D	HP53 HP53A HP53A-M HP53D HP53AD HP53JA	EHP53 EHP53A EHP53D EHP53AD
	4				HP54	
6	1				HP61 HP61A HP61AD	EHP61 EHP61A EHP61D EHP61AD
	2			S62A MS62 MS62D	HP62 HP62A HP62D HP62JAK	EHP62 EHP62A HP62AD
	3				HP63 HP63D HP64	EHP63
	4					
7	3				HP73 HP73D	EHP73
	4				HP74	
8	3				HP83 HP83D	EHP83 EHP83D

Note : Only the series/formation combinations are shown for which bits are available from Reed.
Similarly, empty "third digit" columns are not shown.

BIT TYPE DESIGNATIONS

REED TOOL COMPANY

Prefixes

- EHP – Enhanced performance : threaded ring journal bearing
- HP – Premium journal bearing bit
- S – Sealed roller bearing bit
- Y – Non-sealed roller bearing bit
- MHP – Premium journal bearing bit with high speed seal
- MS – Sealed roller bearing bit with high speed seal

Suffixes

- A – Chisel shaped inserts
- C – Centre jet
- D – Diamond heel pacs
- G – Tungsten carbide heel pacs on steel tooth bits
- H – Chisel shaped inserts in 417-517 designs with 3° skew.
- JA – Jet bit for air circulation
- K – Tungsten carbide inserts added to the shirt-tail to reduce wear and protect the seal
- L – Steel pads with tungsten carbide inserts which are welded to the bit body
- M – Mudpick II hydraulics
- X – Special cutting structure variations that may differ by bit type

E – HYDRAULICS

Clickable list

(Use the hierarchical list under "Bookmarks" to access individual tables and/or sub-topics)

Nomenclature	E-1
Pump outputs	E-2
Optimum bit hydraulics	E-3
Nozzle sizes & flow areas	E-4
Hole cleaning	E-5
Slip velocities	E-6
Miscellaneous equations (1)	E-7
Miscellaneous equations (2)	E-8

NOMENCLATURE

The following are the symbols and units used in this Section, except where otherwise noted :

		Units	
		Field	SI
A_n	= total area of bit nozzles	inches ²	mm ²
C	= coefficient of Q^N in equation for ΔP_s	-	-
D_h	= diameter of hole	inches	mm
D_p	= diameter of drillpipe	inches	mm
D_1	= small diameter	inches	mm
D_2	= large diameter	inches	mm
D	= inside diameter (conduit or pump liner)	inches	mm
d_c	= chip diameter or greatest dimension	inches	mm
HHP_t	= total hydraulic horsepower	hp	kW
HHP_s	= hydraulic horsepower expended in system	hp	kW
HHP_b	= hydraulic horsepower expended at bit	hp	kW
IF	= jet impact force	lbs	N
J	= nozzle size (e.g. 12, 14 etc.)	$J/_{32}$ "	mm
L	= length of conduit	ft	m
	or length of pump stroke	inches	mm
N	= exponent of Q in equation giving ΔP_s	-	-
P_1	= surface pressure losses	psi	kPa
ΔP	= pressure drop	psi	kPa
ΔP_t	= total pressure drop	psi	kPa
	(or pump discharge pressure)		
ΔP_s	= pressure drop in system	psi	kPa
ΔP_b	= pressure drop across bit nozzles	psi	kPa
$\overline{PV}$	= plastic viscosity	cp	cp
Q	= flow rate	galls/min	dm ³ /min
R_n	= Reynold's number	-	-
V_a	= annular velocity	ft/min	m/min
V_p	= velocity of fluid inside circular pipe	ft/min	m/min
V_s	= slip velocity	ft/min	m/min
V_c	= critical velocity	ft/min	m/min
V_n	= jet velocity	ft/sec	m/sec
$\overline{YP}$	= yield point	lbs/100ft ²	lbs/100ft ²
μ	= effective viscosity	cp	cp
ρ_{df}	= pressure gradient of drilling fluid	psi/ft	kPa/m
ρ_c	= pressure gradient of cuttings	psi/ft	kPa/m
	(usually sg = 2.51)		

Note: nozzle sizes are given as numbers (e.g. 12,14, etc.) meaning in fact $12/_{32}$ " , $14/_{32}$ " , $x/_{32}$ " , etc.

PUMP OUTPUTS

Double Acting Duplex Pump

$$\begin{array}{llll} \text{gal/min} & = & 0.00679 & \times \\ \text{bbl/min} & = & 0.000162 & \times \\ \text{ft}^3/\text{min} & = & 0.000909 & \times \\ \text{L/min} & = & 0.0257 & \times \end{array} \quad L \times (2D^2 - d^2) \times \text{SPM} \times \text{fractional volumetric efficiency}$$

Single Acting Triplex Pump

$$\begin{array}{llll} \text{gal/min} & = & 0.01020 & \times \\ \text{bbl/min} & = & 0.000243 & \times \\ \text{ft}^3/\text{min} & = & 0.001364 & \times \\ \text{L/min} & = & 0.0386 & \times \end{array} \quad L \times D^2 \times \text{SPM} \times \text{fractional volumetric efficiency}$$

All of the above equations are valid when the pump sizes are specified in **inches**, as is normally the case even on rigs where SI units are standard.

When dimensions are quoted in **millimetres** the applicable equations to obtain the output in SI units are :

Double Acting Duplex Pump

$$\text{L/min} = 1.568 \times 10^{-6} \times L \times (2D^2 - d^2) \times \text{SPM} \times \text{fractional volumetric efficiency}$$

Single Acting Triplex Pump

$$\text{L/min} = 2.355 \times 10^{-6} \times L \times D^2 \times \text{SPM} \times \text{fractional volumetric efficiency}$$

Note :

In pump calculations a “stroke” actually means one rotation of the pump crank. Thus a triplex pump for example will deliver three piston pumping actions per stroke.

OPTIMUM BIT HYDRAULICS

Two approaches can be used - either to optimise the bit hydraulic horsepower, which will occur when ΔP_b is approximately equal to $^{2/3} \Delta P_t$, or to optimise the jet impact force, which will occur when ΔP_b is approximately equal to $^{1/2} \Delta P_t$. Nozzles can then be chosen to achieve the required result.

Note that the relationships quoted above are approximate and are presented to give a feeling for the order of magnitude of the values required. For a more accurate estimate of the required P_b the properties of the drilling fluid need to be taken into account - in this case the parameters C and N in the equation $\Delta P_s = C.Q^N$. The procedure below shows how to determine these, and how to apply them to calculate the optimum nozzle sizes corresponding to each approach.

Note also, however, that the optimisation of bit hydraulics is often compromised by other hydraulic requirements such as hole cleaning requirements and the pressure drops / flowrate restrictions associated with certain pieces of downhole equipment.

In the approximations given above N has been taken as 2 for simplicity; however until a better value has been determined, as follows, it is recommended to use 1.82 in calculations.

Prior to pulling out of hole to change bit, determine the following:

1. Total Pressure Drop (ΔP_t)

Observe ΔP_t for two or three different pump outputs (Q), preferably close to the rate used when drilling.

2. N & C values

a) Find the bit pressure drop (ΔP_b) for different values of Q

Field units

$$\Delta P_b = \frac{\rho_{df} \times Q^2}{564 \times A_n^2}$$

SI units

$$\Delta P_b = \frac{15.7 \times \rho_{df} \times Q^2}{A_n^2}$$

b) Find the system pressure drop (ΔP_s) for different values of Q

$$\Delta P_s = \Delta P_t - \Delta P_b$$

$$N = \frac{\text{Log } \frac{\Delta P_{s1}}{\Delta P_{s2}}}{\text{Log } \frac{Q_1}{Q_2}} \quad \text{and} \quad C = \frac{\Delta P_{s1}}{Q_1^N} = \frac{\Delta P_{s2}}{Q_2^N}$$

$$\Delta P_s = C.Q^N$$

3. Nozzle area (A_n) for optimum use of available power

a) P_t (max) should be known

b) Find system pressure drop (ΔP_s)

To optimise Bit Hydraulic Horsepower - $\Delta P_s = \frac{P_t}{N+1}$

To optimise Jet Impact Force - $\Delta P_s = \frac{2P_t}{N+2}$

c) Find pump output to give ΔP_s - $Q_{opt} = \left(\frac{\Delta P_s}{C}\right)^{1/N}$

d) Find available bit pressure drop - $\Delta P_b = P_t - \Delta P_s$

so,

Field units

$$A_n = \frac{Q_{opt}}{23.75} \sqrt{\frac{\rho_{df}}{\Delta P_b}}$$

SI units

$$A_n = 3.962 \cdot Q_{opt} \sqrt{\frac{\rho_{df}}{\Delta P_b}}$$

Nozzle sizes and flow areas

Nozzle size		Nozzle number	Flow area of 1 nozzle		Flow area of 2 nozzles		Flow area of 3 nozzles	
In	mm		Inch ²	mm ²	inch ²	mm ²	inch ²	mm ²
7/32	5.5	7	0.0376	24.3	0.0752	48.5	0.1127	72.7
1/4	6.4	8	0.0491	31.7	0.0982	63.4	0.1473	95.0
9/32	7.1	9	0.0621	40.1	0.1242	80.1	0.1864	120.2
5/16	7.9	10	0.0767	49.5	0.1534	99.0	0.2301	148.4
11/32	8.7	11	0.0928	59.9	0.1856	119.7	0.2784	179.6
3/8	9.5	12	0.1104	71.2	0.2209	142.5	0.3313	213.7
13/32	10.2	13	0.1296	83.6	0.2592	167.2	0.3889	250.9
7/16	11.1	14	0.1503	97.0	0.3007	194.0	0.4510	291.0
15/32	11.9	15	0.1726	111.4	0.3451	222.6	0.5177	334.0
1/2	12.7	16	0.1963	126.6	0.3927	253.4	0.5890	380.0
9/16	14.3	18	0.2485	160.3	0.4970	320.6	0.7455	481.0
5/8	15.9	20	0.3068	197.9	0.6136	395.9	0.9204	593.8
11/16	17.5	22	0.3712	239.5	0.7424	479.0	1.1137	718.5
3/4	19.0	24	0.4418	285.0	0.8836	570.1	1.3254	855.2
7/8	22.3	28	0.6013	287.9	1.2026	775.9	1.8040	1163.9

HOLE CLEANING

Insufficient hole cleaning is a common cause of stuck pipe especially in deviated wells. Good hole cleaning depends on the properties of the drilling fluid, the flow rate of the drilling fluid and the procedures used. The ABC of Stuck Pipe, Supplement 2, Hole Cleaning (EP94-1908) is a useful reference on this subject.

Remember ;

- Holes between 30 and 60 degrees inclination are the hardest to clean due to the formation of unstable solids beds on the low side of the hole. These unstable beds can “avalanche” down the hole.
- At greater than 60 degrees inclination stable beds form which are very difficult to remove without some mechanical action. Pipe rotation is ideal for this even when only slow rotation is possible.
- If no other guide to the minimum annular velocity is available, then a rule of thumb is to try to maintain a minimum of 45 m/minute. The best guide to hole cleaning is what you see on the weight indicator and shakers.
- Balanced combination pills are very effective for sweeping the hole. A low viscosity low density pill is followed by a high viscosity high density (ca 2 kPa/m over the existing fluid gradient) pill. Pumping must be continuous while the pills are in the hole. Make sure the low density pill does not underbalance the well at any stage (eg when opposite the BHA).
- If possible rotate and reciprocate the string while circulating clean. The reciprocation stroke should be greater than the length of a single to avoid building ridges on the low side of the hole.
- Solids beds move up the hole far slower than the fluid velocity in the middle of the largest part of the annular space, perhaps 3 to 5 times slower. This means that extra circulating time is needed when cleaning the hole.
- Initial overpull when tripping should be limited to 10 to 15 kdaN (no rotation or circulation) if a hole cleaning problem is suspected. If this limit is reached more hole cleaning should be considered prior to pulling out of hole.
- Good drag charts are essential for spotting problems early.

A set of equations are presented on the [next page](#) that can be used to estimate the slip velocity of cuttings under different circumstances. It is recommended that slip velocity should be less than half the annular velocity (averaged across the cross-section).

SLIP VELOCITIES

Turbulent flow - spherical chips

$$V_s = K \sqrt{\frac{d_c(\rho_c - \rho_{df})}{\rho_{df}}}$$

Where K = 9.41 for SI units and 156 for field units

Turbulent flow - flat chips

$$V_s = K \sqrt{\frac{d_c(\rho_c - \rho_{df})}{\rho_{df}}}$$

Where K = 3.66 for SI units and 60.6 for field units

Laminar flow - spherical chips

$$V_s = \frac{K d_c^2 (\rho_c - \rho_{df})}{\mu}$$

Where K = 75.0 for SI units and 160,000 for field units
(see below for the value of μ)

Laminar flow - flat chips

$$V_s = \frac{K d_c^2 (\rho_c - \rho_{df})}{\mu}$$

Where K = 29.1 for SI units and 62,100 for field units
(see below for the value of μ)

The value of μ in the laminar flow equations is given by :

$$\mu = \overline{P}V + \frac{K \cdot \overline{Y}P(D_h - D_p)}{V_a}$$

Where K = 4.79 for SI units and 399 for field units

MISCELLANEOUS EQUATIONS (1)

Pump output to give annular velocity

$$Q = \frac{V_a(D_h^2 - D_p^2)}{K} \quad \text{Where } K = 1,270 \text{ for SI units and } 24.5 \text{ for field units}$$

Fluid velocity inside pipe

$$V_p = \frac{KQ}{D^2} \quad \text{Where } K = 1,270 \text{ for SI units and } 24.5 \text{ for field units}$$

Fluid velocity in annulus

$$V_a = \frac{KQ}{(D_2^2 - D_1^2)} \quad \text{Where } K = 1,270 \text{ for SI units and } 24.5 \text{ for field units}$$

Critical velocity inside pipe ($R_N = 2,000$)

$$V_c = \frac{K_1(\overline{PV} + \sqrt{\overline{PV}^2 + (K_2 \times D^2 \times \overline{YP} \times \rho_{df})})}{\rho_{df} \times D} \quad \begin{array}{l} \text{Where } K_1 = 588 \text{ for SI units and } 3.36 \text{ for field units} \\ \text{and } K_2 = 0.0163 \text{ for SI units and } 238 \text{ for field units} \end{array}$$

Critical velocity in annulus ($R_N = 2,000$)

$$V_c = \frac{K_1(\overline{PV} + \sqrt{\overline{PV}^2 + (K_2 \times (D_2 - D_1)^2 \times \overline{YP} \times \rho_{df})})}{\rho_{df} \times (D_2 - D_1)} \quad \begin{array}{l} \text{Where } K_1 = 588 \text{ for SI units and } 3.36 \text{ for field units} \\ \text{and } K_2 = 0.0122 \text{ for SI units and } 179 \text{ for field units} \end{array}$$

Compare V_c with V_a or V_p for each section of the annulus, drill string and surface equipment, thus determining whether flow is laminar or turbulent.

Pressure losses in system (Equations are those used for Hydraulic Slide Rules)

Turbulent flow in a circular pipe :

$$\Delta P = \frac{K \times Q^{1.82} \times \rho_{df}^{0.82} \times \overline{PV}^{0.18} \times L}{D^{4.82}} \quad \text{Where } K = 794 \text{ for SI units and } 8.65 \times 10^{-4} \text{ for field units}$$

Turbulent flow in an annulus :

$$\Delta P = \frac{K \times Q^{1.82} \times \rho_{df}^{0.82} \times \overline{PV}^{0.18} \times L}{(D_2 - D_1)^3 \times (D_2 + D_1)^{1.82}} \quad \text{Where } K = 794 \text{ for SI units and } 8.65 \times 10^{-4} \text{ for field units}$$

Laminar flow in a circular pipe :

$$\Delta P = \frac{L \cdot \overline{YP}}{K_1 \cdot D} + \frac{L \cdot \overline{PV} \cdot V_p}{K_2 \cdot D^2} \quad \begin{array}{l} \text{Where } K_1 = 0.392 \text{ for SI units and } 225 \text{ for field units} \\ \text{and } K_2 = 1.88 \text{ for SI units and } 90,000 \text{ for field units} \end{array}$$

Laminar flow in an annulus :

$$\Delta P = \frac{L \cdot \overline{YP}}{K_1(D_2 - D_1)} + \frac{L \cdot \overline{PV} \cdot V_a}{K_2(D_2 - D_1)^2} \quad \text{Where } K_1 \text{ \& } K_2 \text{ are as for the previous equation}$$

Repeat for all sections of annulus, drill string and surface equipment

MISCELLANEOUS EQUATIONS (2)

Bit pressure drop available

$$\Delta P_b = \Delta P_t - \Delta P_s \quad \text{or} \quad \text{HHP}_b = \text{HHP}_t - \text{HHP}_s$$

where HHP_t is the input horsepower x the mechanical efficiency of the pump

$$\text{HHP} = \frac{\Delta P \times Q}{K}$$

Where K = 60,000 for SI units and 1,714 for field units

Note: ΔP_t could be limited (2,000-2,500 kPa/3,000-3,500 psi)

Nozzle area to produce ΔP_b

$$A_n = K \cdot Q \sqrt{\frac{\rho_{df}}{\Delta P_b}}$$

Where K = 3.96 for SI units and 0.0421 for field units

$$\text{therefore : } \Delta P_b = \frac{K^2 \cdot Q^2 \cdot \rho_{df}}{A_n^2}$$

Jet velocity

$$V_n = \frac{K \cdot Q}{A_n}$$

Where K = 16.7 for SI units and 0.32 for field units

Jet impact force

$$\text{IF} = \frac{K \cdot Q^2 \cdot \rho_{df}}{A_n}$$

Where K = 0.0283 for SI units and 0.0032 for field units

Surface connection losses (P_1)

$$P_1 = E \rho^{0.8} Q^{1.8} (\overline{PV})^{0.2}$$

where E is a constant depending on the type of surface equipment and units used

Surface equipment type	Value of E	
	Field units	SI units
1	2.7×10^{-3}	1.4×10^{-4}
2	1.0×10^{-3}	5.3×10^{-5}
3	5.6×10^{-4}	2.9×10^{-5}
4	4.5×10^{-4}	2.3×10^{-5}

Surface equipment type	Standpipe				Rotary hose				Swivel				Kelly			
	Length		ID		Length		ID		Length		ID		Length		ID	
	ft	m	ins	mm	ft	m	ins	mm	ft	m	ins	mm	ft	m	ins	mm
1	40	12.19	3.0	76.2	45	13.72	2.0	50.8	4	1.22	2.0	50.8	40	12.19	2.25	57.2
2	40	12.19	3.5	88.9	55	16.76	2.5	63.5	5	1.52	2.5	63.5	40	12.19	3.25	82.6
3	45	13.72	4.0	101.6	55	16.76	3.0	76.2	5	1.52	2.5	63.5	40	12.19	3.25	82.6
4	45	13.72	4.0	101.6	55	16.76	3.0	76.2	6	1.83	3.0	76.2	40	12.19	4.00	101.6

F – PRESSURE CONTROL

Clickable list

(Use the hierarchical list under "Bookmarks" to access individual tables and/or sub-topics)

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PRIMARY WELL CONTROL

OVERBALANCE

Primary Well Control relies on the use of hydrostatic pressure to control the pore pressure in exposed formations.

Trip margin

In order to counteract any minor pressure reductions on bottom due to swabbing effects when tripping it is customary to maintain a hydrostatic column with a pressure slightly exceeding the pore pressure of the exposed formation. This overbalance, safety margin, or trip margin, P_{tm} , varies depending on circumstances but is generally of the order of 1,500 kPa (200 psi) for conventional operations. The required density

ρ_f to control a formation at a depth, D , with a pressure, P_0 , is given by :
$$\rho_f = \frac{P_0 + P_{tm}}{D}$$

Riser Margin

Whilst drilling from a semi-submersible rig or drillship a riser margin is also commonly added to the drilling fluid gradient so that, in case of an accidental disconnect or a failure of the marine riser close to the BOP stack, control is maintained by the remaining drilling fluid column plus sea-water. The riser margin is given by:

$$\rho_{rm} = \frac{(L \times \rho_{df}) - (D_w \times \rho_{sw})}{D - L}$$

Where : ρ_{rm} = Riser margin
 ρ_{df} = Drilling fluid gradient needed to control the formation pressure with the riser installed
 ρ_{sw} = Seawater gradient
 L = Riser length
 D = depth of hole (TV BDF)
 D_w = water depth

It might not be possible to add the riser margin; in that case add the trip margin. Do not add both. If the trip margin is higher than the riser margin use the trip margin.

Loss of overbalance

Four reasons for loss of overbalance are:

- insufficient drilling fluid density
- losses
- swabbing
- failure to keep the hole full

Loss of head due to gas-cut drilling fluid

From the Strong-White equation
$$\Delta P = P_{wh} \left\{ \frac{\rho_1}{\rho_2} - 1 \right\} \times \ln \left\{ \frac{P}{P_{wh}} \right\}$$

Where : ρ_1 = original drilling fluid gradient
 ρ_2 = gas-cut drilling fluid gradient
 P = original BHP with ρ_1 drilling fluid (absolute)
 P_{wh} = atmospheric pressure (absolute)
 ΔP = reduction in BHP

PRIMARY WELL CONTROL

WARNING SIGNS WHILE DRILLING

Gas cut or contaminated mud

This could be an indication of decreasing overbalance especially if increasing connection gas is seen (swabbed shows).

Alternatively it could be due to drilled shows (gas released from drilled cuttings). This does not require an increase in mud weight and is characterised by :

- a constant elevated gas content in the drilling fluid, correlated with the lithology.
- no peak in gas cutting or contamination of the drilling fluid at bottoms up.

Flow checks should be made when the gas cutting first appears and, if the fluid density is not increased, at regular intervals thereafter. In the case of drilled shows a flow check should be made if the amount of gas increases with no associated increase in penetration rate.

A drilling break

A significant change in penetration rate, particularly an increase, can indicate that a new formation has been penetrated. Generally the driller should make a flow check. In a known area where formation tops, pressures and top hydrocarbons are known with some confidence, this may not be necessary.

PRIMARY WELL CONTROL

DRILLING FLUID LOSSES

As soon as it is noticed that drilling fluid is being lost into the formation, drilling must be discontinued while the severity of the problem, in particular with respect to well control but also with respect to borehole stability, is assessed. The rate at which fluid is being lost will give an indication of the cause of losses and will determine the reaction to the problem. In all cases endeavour to **KEEP THE HOLE FULL** under static conditions. This may mean pumping water into the annulus.

The causes of drilling fluid losses, and the methods available for restoring full circulation, are discussed on [pages I-11 to I-13](#).

Once the hole can be kept full under static conditions, the possibility of restoring a situation of full (or close to full) circulation can be evaluated as per the tabulation on page I-13.

If this can be done the well control situation must be evaluated, taking into account:

- the current scale of the losses
- the possible future scale of the losses
- the formations and pressures already exposed in the open-hole section
- the formations and pressures that may be encountered as drilling proceeds

Based on this it is necessary to make an updated estimate of how much further the current open-hole section may be deepened without creating a situation in which primary control may be lost and an uncontrollable flow occur, either to surface or between two formations. In the worst case the additional depth may be zero.

Note that once fluid losses have occurred, even if they have apparently been cured, the minimum formation strength in the open hole will probably have been reduced. The implications of this have to be taken into account when assessing the maximum safe drilling depth.

Over time, loss zones sometimes “heal” to a certain extent giving an apparent increase in formation strength. This should NOT be used as a justification for drilling ahead if there is any potential for taking a kick.

In a situation when full (or close to full) circulation cannot be restored well control after a kick becomes much more difficult, and, unless there is a very high degree of confidence that no influx will occur, casing will have to be set and cemented. Even so, it will usually be necessary to drill ahead with little or no returns until the end of the loss zone has been reached, otherwise a string of casing will have been set only to lose circulation again immediately after drilling out of the shoe.

Note that in these circumstances it will also be difficult to achieve a successful primary cement job.

PRIMARY WELL CONTROL

CONDUCTOR DESIGN to prevent formation breakdown while drilling

We need to calculate the minimum conductor setting depth based on the following data:

1. Formation strength gradient - ρ_{fsg}
2. Sea water gradient - ρ_{sw}
3. Return drilling fluid gradient - ρ_{df}
4. Water depth - D_w
5. Flow line height above seabed - F_h

All expressed in consistent units.

If X = the casing setting depth below seabed, then $X \cdot \rho_{fsg} + D_w \cdot \rho_{sw} = (F_h + X) \cdot \rho_{df}$

Example: $\rho_{fsg} = 0.67$ psi/ft
 $D_w = 175$ ft
 $\rho_{sw} = 0.445$ psi/ft
 $F_h = 240$ ft
 $\rho_{df}^* = 0.48$ psi/ft
Flowline outlet is 10 ft below derrick floor

$$0.67X + (175 \times 0.445) = (240 + X)0.48$$

$$(0.67 - 0.48)X = 115.2 - 77.87$$

$$0.19X = 37.33$$

$$X = 196.5 \text{ (say 200 ft)}$$

Depth of conductor BDF = $10 + 240 + 200 = 450$ ft

* Density of drilling fluid in annulus based on calculated maximum drilling rate to avoid overloading (see next page).

PRIMARY WELL CONTROL

MAXIMUM DRILLING RATE to avoid overloading of annulus

The maximum drilling rate can be calculated based on the following data:

1. Pump rate - Q
2. Volume of cuttings produced per unit time - V
3. Maximum density of fluid in annulus - D_a
4. Density of drilled formation - D_{fm}
5. Density of drilling fluid in use - D_{df}

All expressed in consistent units. Note that in this case the units of density have to be expressed as mass per unit volume rather than as a gradient

$$Q.D_{df} + V.D_{fm} = (Q + V).D_a$$

$$Q(D_a - D_{df}) = V(D_{fm} - D_a)$$

$$V = \frac{Q(D_a - D_{df})}{(D_{fm} - D_a)}$$

Example: $Q = 2.83 \text{ m}^3/\text{min}$
 $D_a = 1,114 \text{ kg/m}^3$
 $D_{df} = 1,033 \text{ kg/m}^3$
 $D_{fm} = 2,268 \text{ kg/m}^3$

$$V = \frac{2.83(1,114 - 1,033)}{(2,268 - 1,114)}$$
$$= 0.199 \text{ m}^3/\text{min}$$

In a 26" hole the volume of formation cut per metre drilled = 0.3425 m^3 (assuming a gauge hole)

The maximum drilling rate then becomes $0.199/0.3425 = 0.581 \text{ m/min}$, equivalent to approximately 15 mins/joint.

PRIMARY WELL CONTROL

SWABBING

If high volume swabbing is likely to be a problem, consider running a float sub and check that the pipe is full when tripping out.

Avoid low volume swabbing by controlling pulling speed and drilling fluid properties - water loss and yield point.

Use the trip tank at all times when tripping (in fact whenever the well will not be circulated, even if for only a short length of time). Keep a trip sheet and monitor volumes accurately.

Do not pump a trip pill unless hole conditions are known to be good. This may mean waiting until the bit is at the casing shoe. After pumping a pill be aware that it may take a short while for the fluid levels to stabilise which can disguise swabbing.

Avoid pulling wet pipe (subject to the point above). If it is unavoidable, use a mud box to minimise drilling fluid losses on the drill floor and allow accurate measurement of the volume taken by the hole.

SECONDARY WELL CONTROL

WELL PLANNING & CONSTRUCTION

During well planning

Design the well so that it can contain a kick, the size and pressure of which is based on engineering assumptions about :

- the formation strengths & pressures
- possible scenarios under which a kick might occur, and
- the ability / preparation of the rig and its crew.

During well construction

- Continuously check that the assumptions made in the well design (shoe strength, etc) are still valid.
- Carry out casing seat tests according to the guidelines and procedures described on pages F-26 and F-27.
- As soon as a casing seat test has been carried out, calculate the Maximum Allowable Annular Surface Pressure, using the drilling fluid density used for drilling below the shoe.
- Ensure that the “pre-kick” section of the appropriate IWCF kill sheet is kept up to date, especially after changes in drilling fluid parameters. (Examples of these sheets are shown at the end of this Section)

SECONDARY WELL CONTROL

SHUT-IN PROCEDURES – DRILLING OR TRIPPING PHASES

Operators and Contractors differ in their operating practices regarding the open and closed position of valves on the choke manifold during the drilling and/or tripping phase. The choke to be used during well control (normally a remotely operated adjustable choke) should be kept in a closed position during normal operations. The valve immediately upstream of this choke should be open during normal operations and closed following a shut-in. The outer remotely operated choke line valve adjacent to the BOPs (or all fail-safe choke line valves on sub-sea BOPs) are to be kept in a closed position.

SECONDARY WELL CONTROL

IMMEDIATE ACTIONS IN CASE OF A KICK WHILE DRILLING

With a surface BOP stack

- Stop drilling (stop rotation when kelly is used)
- Raise the string to shut-in position (lower kelly cock or Top-drive internal BOP above rotary)
- Stop the pump(s)
- Close the annular preventer and open the remotely operated choke line valve(s)
- Inform Supervisor, Toolpusher and crew members
- Check space out and close the pipe rams; bleed off any pressure between the pipe rams and the annular preventer (if so required) and open the annular preventer
- Record the casing and drill pipe pressures and the pit gain

In high pressure wells:

- Close the lower kelly cock or top drive internal BOP, install and test the kill assembly, pressure up to the closed-in drill pipe pressure and open the lower kelly cock or top drive internal BOP.

With a sub-sea BOP stack

- Stop drilling (stop rotation when kelly is used)
- Raise the string to the hang-off position (lower kelly cock or Top-drive internal BOP above rotary)
- Stop the pump(s)
- Close the annular preventer and open the upper fail-safe choke line valves
- Inform Supervisor, Toolpusher and crew members
- Check the space out, close the (middle) pipe rams to be used for hanging off and close the ram locks (if so required).
- Hang off the drill string and adjust the heave compensator at mid-stroke
- Bleed off any pressure between the pipe rams and the annular preventer (if so required) and open the annular preventer. Check that the pipe rams are not leaking.
- Record the casing and drill pipe pressures and the pit gain

In high pressure wells :

- Close the lower kelly cock or top drive internal BOP, install and test the kill assembly, pressure up to the closed-in drill pipe pressure and open the lower kelly cock or top drive internal BOP

Notes

1. Raising the string is only done if time permits; in extreme or critical circumstances this operation should be carried out in as short a period as practically possible.
2. Closing the ram locks is optional on surface stacks. Certain ram locks on sub-sea stacks activate in any position along the piston stroke and are an integral part of the operating system.
3. Ensure that the valve upstream of the choke is closed when monitoring the casing pressure.
4. The drill pipe pressure might be zero if a float valve is present in the string.

SECONDARY WELL CONTROL

IMMEDIATE ACTIONS IN CASE OF A KICK WHILE TRIPPING

With a surface BOP stack

- Set slips below a drill pipe tool joint (run in stand if pulling out)
- Install a Full Opening Safety Valve and close it
- Close the annular preventer and open the remotely operated choke line valve(s)
- Inform Supervisor, Toolpusher and crew members
- Check the space out and close the pipe rams; bleed off any pressure between the pipe rams and the annular preventer (if so required) and open the annular preventer
- Make up the kelly or top drive and open the Full Opening Safety Valve
- Record the casing and drill pipe pressures and the trip gain

In high pressure wells:

- Close the lower kelly cock or top drive internal BOP, install and test the kill assembly, pressure up to the closed-in drill pipe pressure and open the lower kelly cock or top drive internal BOP

With a sub-sea BOP stack

- Set the slips below a drill pipe tool joint (run in stand if pulling out)
- Install the Full Opening Safety Valve and close it
- Close the annular preventer and open the upper fail-safe choke line valves
- Inform Supervisor, Toolpusher and crew members
- Check the space out, close the (middle) pipe rams to be used for hanging off and close the ram locks (if so required)
- Hang off the drill string and adjust the heave compensator at mid-stroke
- Bleed off any pressure between the pipe rams and the annular preventer (if so required) and open the annular preventer. Check that the pipe rams are not leaking.
- Record the casing and drill pipe pressures and the pit gain

In high pressure wells:

- Close the lower kelly cock or top drive internal BOP, install and test the kill assembly, pressure up to the closed-in drill pipe pressure and open the lower kelly cock or top drive internal BOP

Notes

1. If unable to install the Full Opening Safety Valve due to strong flow:
 - with no top drive: consider dropping the string or closing the shear rams (provided there is no tool joint opposite the shear rams)
 - with top drive: lower, stab and make up the top drive
2. Closing the ram locks is optional on surface stacks. Certain ram locks on sub-sea stacks activate in any position along the piston stroke and are an integral part of the operating system
3. Ensure that the valve upstream of choke is closed when monitoring the casing pressure
4. The drill pipe pressure might be zero if a float valve is present in the string

REMOVAL OF INFLUX

GENERAL

Fill in the kick data and complete the relevant IWCF kill sheet (see the examples at the end of this section).

The basic principles are to :

- keep the bottom hole pressure equal to or just above formation pressure at all times.
- minimise the pressure exerted on the open formations (1st priority), the casing and the BOP.

Given these principles, determine which of the following methods will be used to remove the influx from the well :

- Wait and weight method (see below and [page F-15](#))
- Driller's method (see below and [page F-17](#))
- Combined volumetric and stripping method (see [page F-18](#))
- Bullheading (see [page F-24](#))

The two circulating methods

The following table gives the advantages and disadvantages of the “wait & weight” method compared with the “driller's method”.

Advantages of “wait & weight” method	Disadvantages of “wait & weight” method
Surface pressures are lower during later stages of the kill due to the presence of heavy fluid in the annulus. The maximum pressure exerted on the casing shoe is sometimes lower. This occurs if kill fluid enters the annulus before the top of the influx reaches the shoe. The well is killed more quickly since only one circulation via the BPM is required. This reduces the amount of time that the surface equipment is under pressure.	The method is more complex to manage during execution. In the driller's method, circulation can be started without making calculations (except those required for calculating the circulating pressure and bottoms up volume). Ensuring that the drilling fluid pumped into the well is weighted up correctly and that a constant density is maintained can be problematic. With the drillers method, pumping can begin as soon as stabilised drillpipe pressure is established. This could be important in case of a low MAASP in combination with a gas influx.

REMOVAL OF INFLUX

CONSTRUCTION OF KILL GRAPH IN A DEVIATED HOLE

General

During Phase I the pressure will drop due to higher density drilling fluid entering the drill pipe string and the pressure will rise due to higher friction loss of the ρ_2 drilling fluid. Therefore at a point of interest 'x' :

$$P_{st_x} = P_{st_1} + (P_{c_2} - P_{c_1}) \frac{V_x}{V_t} - D_{x_{tvd}} (\rho_2 - \rho_1)$$

Where : P_{st_x} = standpipe pressure at the point of interest

P_{st_1} = standpipe pressure observed at start of kill (kPa or psi)

P_{c_1} = circulating pressure at start of kill (kPa or psi)

P_{c_2} = circulating pressure at end of kill (kPa or psi)

V_x = string volume down to point of interest (m^3 or ft^3)

V_t = total string volume (m^3 or ft^3)

$D_{x_{tvd}}$ = true vertical depth at point of interest (m or ft)

To construct the kill graph four points of interest are calculated:

- P_{st_1} and P_{c_1} at start of kill, in the usual manner
- P_{c_2} at end Phase 1 in the usual manner
- P_{st} at K.O.P. with the above equation
- P_{st} at end of build up section with the above equation

In an "S" bend hole two more points are calculated (start and end of drop off section).

To construct the P_{dp} graph (static conditions) the above four points of interest are calculated using the following equation :

$$P_{dp_x} = P_{dp_1} - D_{x_{tvd}} (\rho_2 - \rho_1)$$

Where : P_{dp_x} = drill pipe pressure at the point of interest

P_{dp_1} = drill pipe pressure at start of kill

Note:

For non-tapered strings the uppermost equation above may be simplified by using string lengths instead of string volumes (ignoring in this case the drill collars and the HWDP).

REMOVAL OF INFLUX

HORIZONTAL WELLS

In principle, well control calculations for deviated wells also apply for horizontal wells. However, a bottom hole angle of 90° for the horizontal section cannot be used in the calculations, because of practical arithmetical reasons. An assumed bottom hole angle of 89° should be used instead. For hydrostatic pressure-related calculations the TVD of the “deepest” part of the horizontal section should be used. Most well control methods are also applicable to horizontal wells. However, when the string is off bottom, or when circulation at bottom is not possible, well control options become limited, because the volumetric method or bullheading are unlikely to be successful or very effective in the horizontal section.

Other kick control considerations for horizontal wells are:

- When a kick is encountered, the influx may take place over the entire exposed horizontal reservoir section at once.
- The overbalance at the “beginning” of a truly horizontal hole section through a reservoir is the same as the “end” of that hole section, provided it does not penetrate any sealing faults or other permeability barrier.
- There may be a dispersion effect in the horizontal section, depending on hole and flow conditions. This can result in long circulating times to get the fluid in the well gas free and with a homogeneous density.
- Lower than expected annular pressures will occur due to the dispersion effect.
- A proper standpipe kill graph for a deviated well should be used to ensure that the correct bottom hole pressure is applied during the well killing process.
- As formation pressures are often known accurately by the time a horizontal section is drilled, the majority of well control problems are related to swabbing.
- In horizontal wells the previous casing is often set just above the producing zone. This means that formation strength should not be a limiting factor and is a strong driver to use the Driller’s Method.

REMOVAL OF INFLUX

AMOUNT OF WEIGHTING MATERIAL TO WEIGHT UP THE DRILLING FLUID

Calculating the amount of weighting material needed:

The amount of weighting material that is needed to weight up the drilling fluid is calculated using the following equation :

$$N_1 = \frac{1000 \times d_w \times (\rho_2 - \rho_1)}{g \times (d_w - \rho_2)} \text{ kg/m}^3 = \frac{808.5 \times d_w \times (\rho_2 - \rho_1)}{(d_w - \rho_2)} \text{ lbs/bbl}$$

Where : d_w = the gradient of weighting material (kPa/m or psi/ft)

g = 9.807 m/s²

N_1 = the amount of weighting material used to weight up 1 m³ (1 bbl) of drilling fluid

Note :

This method uses the gradient of the weighting material in kPa/m or psi/ft and not the density (min. field gradient of barytes = 1.82 psi/ft or 41.2 kPa/m).

Calculating the increase in volume

The increase in volume is given by the following :

$$\Delta V = \frac{N_1 \times g}{1000 d_w} \text{ m}^3 = \frac{N_1}{808.5 d_w} \text{ bbls}$$

Where : ΔV = The increase in volume (m³ or bbl)

N_t = The amount of weighting material to weight up the total volume of drilling fluid (kg or lbs)

REMOVAL OF INFLUX

WAIT AND WEIGHT METHOD

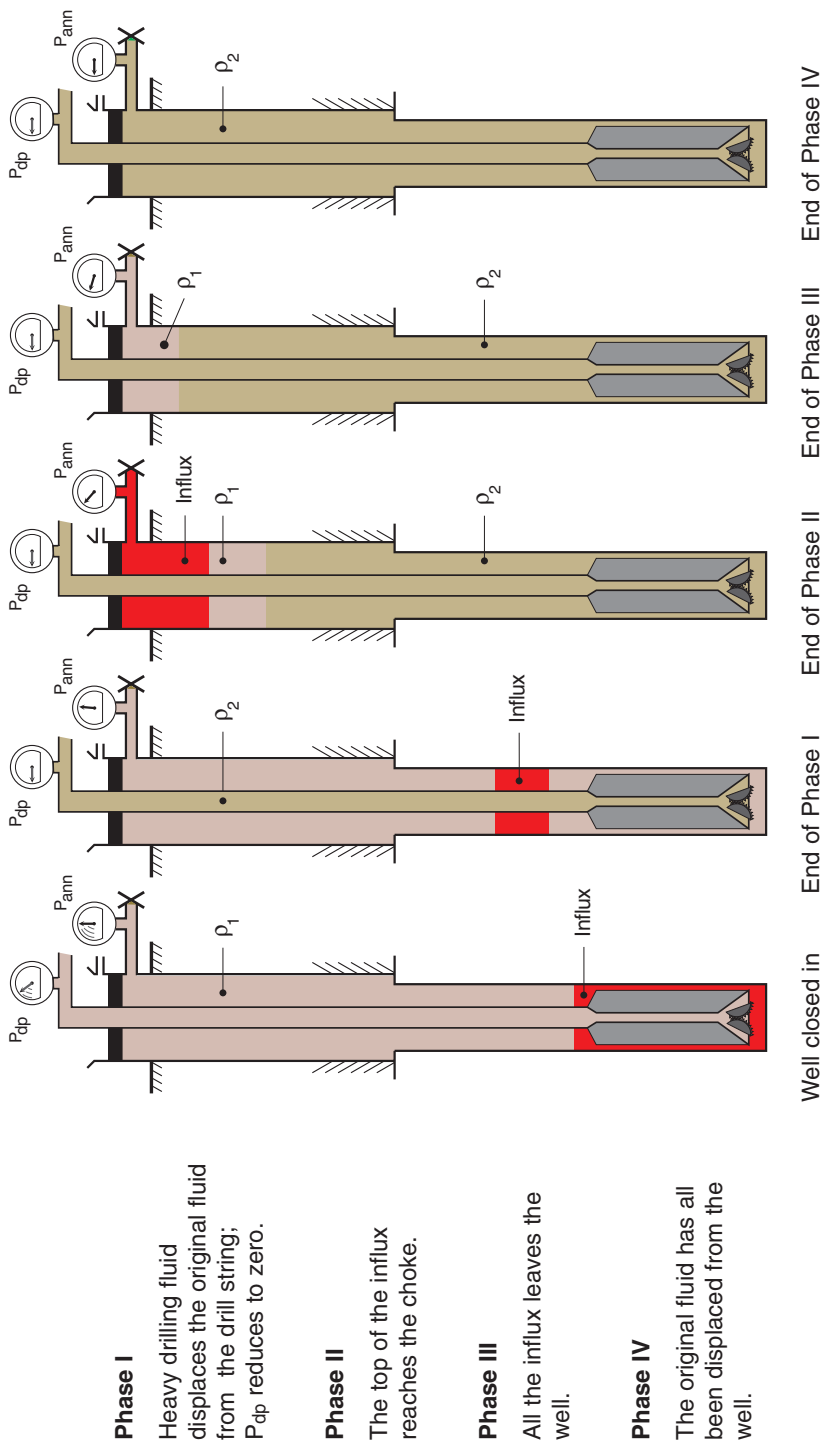
After selecting the well killing parameters the procedure for the first circulation is:

- Before starting the killing operations, bleed off at the choke any increase in P_{dp} resulting from a second build up to, or near, the initial stabilised closed-in pressure. (i.e. the closed-in pressure after the first build up):
- Open the choke and simultaneously start pumping the ρ_2 drilling fluid, bringing the pump rate to the selected killing pump speed.
- While reaching and maintaining the pre-selected pump speed adjust the choke opening until the choke pressure P_{ch} equals the closed-in annulus pressure P_{ann} immediately before starting the pump. (Record the choke pressure throughout the first circulation.)
- Read the standpipe pressure P_{st1} . It should agree with the calculated value, i.e. the normal pre-kick circulating pressure drop P_{c1} at the selected pump speed plus the closed-in drill pipe pressure P_{dp} . If the observed standpipe pressure does not agree with the calculated value, consider the observed pressure to be “correct” and calculate the actual P_{c1} . The revised P_{c1} should then be used to calculate a corrected P_{c2} and the standpipe kill graph then redrawn.
- Continue pumping at the pre-selected killing speed and keep the standpipe pressure in line with the calculated pressure (the sloping line of phase I on the kill graph and thereafter the horizontal line of the ρ_2 drilling fluid circulating pressure drop P_{c2}) by opening and closing the choke as required.
- Continue pumping with P_{st} equal to P_{c2} until the influx is circulated out. Be prepared to cope with a substantial increase in total surface volume of drilling fluid due to a gas influx expanding.
- When all the influx and the original ρ_1 drilling fluid from the string has been circulated out i.e. at the end of phase IV, stop the pump and close the choke. Check for pressures on the drill pipe and annulus gauges. They should both read zero. If not, and if $P_{dp} = P_{ann}$, the choke may have been closed a little too quickly trapping pressure from the pump. Bleed off a little pressure from the annulus, checking that both pressures drop and do not rise again once the choke is closed.
- Open the BOP and make a flow check. Some imbalance between the annulus and drill string is likely, but a definitive flow can usually be confirmed. If a positive flow is still apparent, close the BOP and continue circulation under controlled conditions, i.e. via the choke.
- If there is no flow, start the pump and check whether the string is free. If free the string should be moved at regular intervals until ready to pull out.
- Make a second circulation raising the drilling fluid gradient again to include a trip margin and conditioning the fluid to remove the effects of any contamination from the influx.

While circulating during the well kill, the following actions should also be carried out :

- Maintain and record the density of the drilling fluid pumped into the drill string. Ensure that it has the correct value
- Measure and record the properties of the drilling fluid returns
- De-gas, treat, or separate for disposal, any contaminated drilling fluid returns

THE FOUR PHASES IN WELL KILLING, WAIT & WEIGHT METHOD



REMOVAL OF INFLUX

DRILLER'S METHOD

First circulation

The procedure for the first circulation is :

- Open the choke and simultaneously start pumping the ρ_1 drilling fluid at the selected pump rate.
- Whilst reaching and maintaining the selected pump rate adjust the choke opening until the choke pressure equals the closed-in annulus pressure. Record the choke pressure throughout the first circulation.
- Read standpipe pressure. It should equal the normal pre-kick pump test circulation pressure at the selected pump speed plus the closed-in drill pipe pressure. If the observed standpipe pressure does not agree with the calculated value, consider the observed pressure to be correct.
- Maintain constant standpipe pressure and pump rate whilst the influx is circulated out.
- When all influx has been circulated out, stop the pump and close in the well to check the closed-in drill pipe and annulus pressures. At the end of the first circulation the closed-in pressures of the annulus and drill pipe should be the same and equal to the initial closed-in drill pipe pressure. The well is controlled but not killed.
- Weight up spare drilling fluid to the required density.

Second circulation

Once the drilling fluid has been weighed up to the correct density the second circulation can begin. This is carried out in exactly the same way, including construction of the standpipe kill graph, as the first circulation of the "Wait & Weight" method. However, since the influx has been displaced with ρ_1 drilling fluid during the first circulation, large fluctuations in fluid returns and, therefore, choke position are not expected. Thus, if possible, the density of the drilling fluid in the well is raised directly to that required to resume normal operations including a trip margin.

REMOVAL OF INFLUX

STATIC VOLUMETRIC AND STRIPPING METHOD (1)

Introduction

Should there be a fluid gain while tripping, close-in the well immediately. The example decision tree shown on [page F-29](#) illustrates how the most appropriate course of action may then be identified.

If pipe stripping is an alternative, the following must be considered.

- Accurate measurement of any mud volume bled off must be possible. (pay close attention to the stripping tank (Fig. 1) which is an addition to standard rig equipment)
- Accurate pressure gauges should be available.
- Pressure regulators on blowout preventers must be in good working order.
- The equipment and procedure to be used should be known and practiced.
(Implementation of “strip drills” in addition to the usual “pit drills” are required to evaluate the stripping characteristics of the preventer in use.)

If stripping cannot for some reason be considered, it is essential that well control is maintained whilst decisions are made concerning the most effective well killing method to be employed. The rig must be suitably rigged-up (Fig.1) to implement the volumetric method immediately.

In general, the annular preventer (Fig. 2) is used for stripping pipe. To minimize wear of the element, the pipe should be well lubricated with grease and the closing pressure applied to the annular kept to a minimum whilst avoiding leakage. As a rule drillpipe/casing protectors are no longer used, however if present, they should be removed. Sharp edges and tong marks on the pipe body and tool joints should be removed or in extreme cases the pipe laid down.

In the notes which follow it is assumed that the influx is gas, as this is the case that gives the complication of an expanding influx. In the case of an oil or water influx there is little difficulty in controlling the pressures while stripping in order to circulate it out.

Accurate fluid measurement

Measure all fluid that comes out of the well bore. Formation fluid that has entered the well may be gas and during the stripping operation migration may take place.

If there is no migration of fluid, the volume of mud released from the well bore, as pipe is stripped, will equal the closed-end pipe displacement and the choke pressure should remain the same.

Effect of running into the influx

When the bit and DCs enter the influx, a dramatic loss of hydrostatic head will take place. The loss of hydrostatic head can be anticipated and a corresponding additional back pressure (Ps) should be added at the very start of the stripping operation to prevent a second influx.

Effect of expansion

When a gas influx migrates, the surface pressure will increase even though the volume of drilling fluid released at the surface is exactly equal to the closed-end pipe displacement stripped in. In this case it is necessary to bleed off additional fluid to let the gas expand.

Figure 1 : Rig layout for combined stripping and static volumetric method

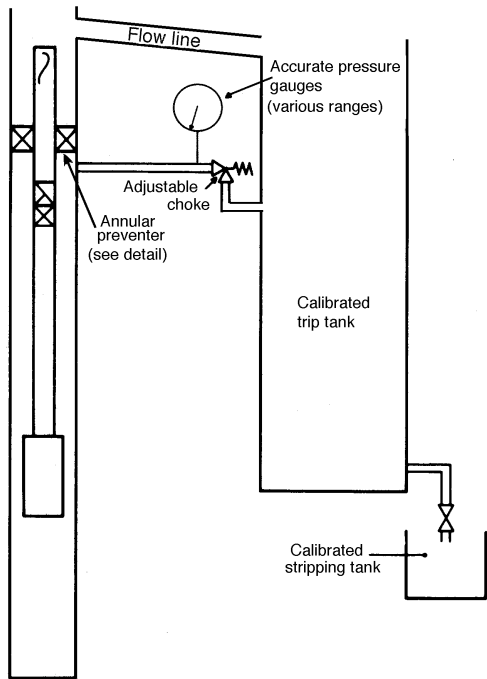
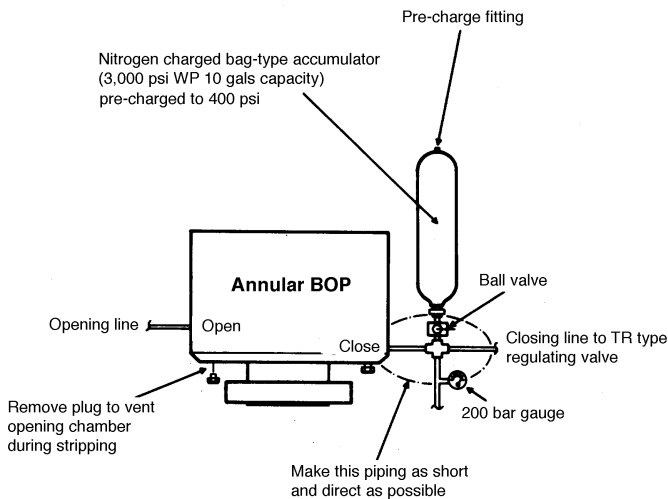


Figure 2 : Equipment requirements for stripping with an annular preventer



REMOVAL OF INFLUX

STATIC VOLUMETRIC AND STRIPPING METHOD (2)

Operational considerations

To simplify the method and to allow easy operational application, the following assumptions, which give “worst case” situations, have been made.

- The influx remains as a continuous slug, occupying the entire annular cross section.
- For all calculations relating to loss of hydrostatic head (gas expansion) the influx is assumed to be opposite the smallest annular cross section. It should be remembered that by the time the bit has been returned to bottom, an extra backpressure will have been created in the well due to the position of the influx which will now be situated opposite the drillpipe/ openhole annulus.

As a further simplification to aid well-site use the method is divided into three separate operations:

- 1) A minimum closing pressure is applied to the annular preventer sufficient to eliminate any leakage when the tool joint passes through. The driller then lowers pipe at a constant rate, avoiding pressure surges in the well.
The approximate closing pressure required can be determined from strip drills or manufacturers data.
- 2) The man at the choke maintains a constant predetermined choke pressure, increasing pressure only at the instruction of the Supervisor.
- 3) The man at the trip tank/stripping tank continuously monitors net changes in the level of the trip tank and bleeds off the closed-end displacement volume of each stand into the stripping tank.

To make it easier to read tank levels it is suggested to add defoamer, as required, to the fluid in the trip tank.

DO NOT CONFUSE THE TRIP TANK WITH THE STRIPPING TANK

REMOVAL OF INFLUX

STATIC VOLUMETRIC AND STRIPPING METHOD (3)

Operational procedure (stripping)

- 1) After closing in the well, determine the influx volume and record pressures at two minute intervals. After closed-in pressures have stabilised complete the upper left box in the form shown on page F-30 and further record pressures at five minute intervals or after stripping in each stand.
- 2) Determine the volume of drilling fluid in the OH/DC annulus equivalent to one psi (one kPa) of hydrostatic head (i.e. the volume of fluid equivalent to a change of hydrostatic head of one psi or kPa).
$$\text{Volume (V) per unit pressure increment} = \frac{\text{annular volume per unit length}}{\text{drilling fluid gradient}}$$
- 3) Determine a convenient working pressure increment P_w bearing in mind the scale divisions of available pressure gauges (see step 8).
- 4) Multiply P_w by V to obtain an equivalent working volume ΔV in the OH-DC annulus (the volume of fluid to be used for volumetric control steps).
- 5) Determine the height that this volume will occupy in the calibrated trip tank and complete the upper right box in the form shown on page F-30.
- 6) Determine the extra back pressure P_s to compensate for the loss of hydrostatic pressure as the bit and drill collars are run into the influx. If the influx is assumed to be in the open hole beneath the bit, an increase in surface pressure will be required to maintain BHP above P_0 when this event occurs. It is unknown when the extra back pressure will be required since the exact position of the influx is unknown. It is therefore advisable to adopt a suitable safety factor from the very start of the stripping operation. Since an overbalance (trip margin) will exist in nearly all wells which kick during round tripping, it is not possible to use closed-in annulus pressure P_{ann} to make an accurate estimate of the magnitude of the influx and thus the additional back pressure required to compensate for the aforementioned loss of hydrostatic head. It is therefore essential accurately to measure the influx volume gained at surface, and by application of a factor based on the ratio open hole to OH-DC annulus, calculate the expected loss of hydrostatic head as the DCs enter the influx.
- 7) Adjust the closing pressure on the annular preventer to a minimum, but avoid leakage. Whilst reducing closing pressure check continuously for flow (leakage past the annular rubber).
- 8) Allow annulus pressure to build up to P_{choke} .

$$P_{choke} = P_{ann} + P_s + P_w$$

where : P_{ann} = Initial closed-in annulus pressure before second build-up

P_s = Allowance for loss of hydrostatic head as the DCs enter the influx

P_w = Working pressure increment

During the pressure build-up to calculated P_{choke} value, commence with stripping in the first stand(s), whilst bleeding off the respective closed-end displacement volume from trip tank to strip tank. To facilitate this operation the trip tank needs to have a starting volume.

- 9) Maintain P_{choke} constant whilst further stripping pipe. The volume increase due to closed-end displacement of drillpipe is purged into the trip tank and, after stripping the entire stand, bled off into the stripping tank thus ensuring that any increase in the trip tank volume is due to the gas influx only, and reflects the loss of hydrostatic head in the well. (See notes on [page F-18](#))

REMOVAL OF INFLUX

STATIC VOLUMETRIC AND STRIPPING METHOD (4)

- 10) Avoid excessive surge pressures by adjusting the pipe lowering rate to allow the chokeman to maintain P_{ch} constant.
- 11) Maintain P_{ch} constant at the initial value until a volume of mud ΔV has accumulated in the trip tank. Simultaneously strip pipe in the hole.
- 12) When the additional mud volume ΔV has accumulated in the trip tank (at constant choke pressure), P_{ch} is allowed to increase again by the value P_w and now becomes P_{ch1} (where $P_{ch1} = P_{ch} + P_w$).
- 13) Fill each stand run and file off any sharp edges or tong marks from the pipe body and tool joints.
- 14) By repeating this cycle as often as necessary gas is able to percolate upwards and expand while a nearly constant BHP is maintained.
- 15) Values of pressure and volume should be recorded throughout the stripping exercise.
- 16) On bottom the well can be killed using the "driller's method" but first ensure that the entire string is full of drilling fluid. Pump at a slow rate the volume from the bit to the Gray valve (some gas may have entered the string) then stop pumping, check for trapped pressures and continue with the circulation.
- 17) To avoid differential sticking consider moving the string through the preventer.

Note

Should, during the stripping operation, bottom hole pressure inadvertently drop below formation pressure ($BHP < P_0$), a second influx will take place. The method makes allowance for this eventuality and re-establishes the required P_{ch} by overcompensating for the loss of hydrostatic head caused by the new influx. This is achieved automatically due to the manner in which P_w has been calculated. Well pressure (P_w) compensates for loss of hydrostatic head assumed opposite the DCs. A second influx will enter in the open hole section resulting in a volume gain at surface, where it will be interpreted as a volumetric step. The well will be closed in and P_{ch} allowed to increase P_w . The effect, of course, will be overcompensation of the underbalance that existed in the well. In other words it is impossible to lose hydrostatic control of the well since the method is self-correcting.

Calculation of P_s

In the equation $P_{choke} = P_{ann} + P_s + P_w$

P_s = the loss of hydrostatic head as the DCs enter the influx

= $F \times$ influx volume, where :

$$F = \frac{1}{OH \text{ capacity}} \times (\text{drilling fluid gradient} - \text{gas gradient}) \times \left\{ \frac{OH \text{ capacity}}{OH-DC \text{ capacity}} - 1 \right\}$$

An alternative method of calculation is :

Let H_1 be the height of the influx in open hole = $\frac{\text{influx volume}}{OH \text{ capacity}}$

Let H_2 be the height of the influx in the open hole/drill collar annulus = $\frac{\text{influx volume}}{OH-DC \text{ capacity}}$

The reduction in hydrostatic head as the influx is displaced behind the drill collars, and thus the value of P_s , is given by $P_s = (H_2 - H_1) \times (\text{drilling fluid gradient} - \text{gas gradient})$

REMOVAL OF INFLUX

CIRCULATING VOLUMETRIC CONTROL METHOD

The Volumetric (and combined Stripping) Method can be safely applied on surface and sub-sea stack operations. However when the influx (we assume gas) enters the long choke line above the sub-sea stack it could lead to more complex situations and unacceptably high bottom hole pressures could result. These higher pressures will occur when the relatively low capacity of the choke line has not been taken into account when calculating the working volume increment dV i.e. dV is derived solely from the capacities of the OH/DC and OH/DP annuli. Over and above these higher bottom hole pressures, as a result of capacity changes, the long choke line will fill up with the influx over a substantial length, thus creating high annular surface pressures. One method used to reduce these high surface and subsequent bottom hole pressures is that of the Circulating Volumetric Control. With this method circulation is maintained across the BOP stack (kill line in, choke line out), while choke pressure and pit gain is being controlled by using the kill line pressure. In this manner the influx is diluted continuously whilst being routed to the choke manifold. The active pit used for circulating should be almost as small as a trip tank to record noticeable changes in level.

Application

- Ensure that the kill line is full of original drilling fluid with the correct density
- Identify and/or calculate when the influx is about to arrive at the choke line (this might not be simple but a prudent (conservative) approach should be considered)
- Connect up to small active pit and make sure that all surface lines including the return line to the mud/gas separator are filled with drilling fluid; it is important to account for the circulating volume which usually draws a known volume from the active pit
- Bring the pump on the kill line up to speed, increasing the pressure by an amount equal to the kill line pressure loss.
- When the influx is rising and thus expanding in the casing annulus, a pit gain will be seen which needs to be compensated for by a higher kill line pump pressure; when the influx is entering the choke line, a more rapid increase in choke pressure will be noticed.
- The amount by which the kill line pumping pressure should be increased depends on the equivalent pit level gain, mud density and annular line capacity (influx still inside casing) or choke nominal capacity (influx in part or complete inside choke line). The slope of the kill line pump pressure (dP versus dV) should reflect this.
- When the influx arrives at the choke and is bled from the well, the situation goes into reverse and the pit level will drop accordingly. In order to maintain a constant bottom hole pressure (or as low an overbalance as practically possible), we should now decrease the kill line pumping pressure. The amount of decrease depends on the equivalent pit level loss, mud density and choke nominal capacity. The slope of the kill line pump pressure (dP versus dV) should once again reflect this.
- The well is considered killed once influx returns has ceased and the kill line pumping pressure is equal to all pressure losses combined (kill line, choke line and choke manifold circulated through open choke).

Note

Although the method is rarely applied, simulation tests have proven this alternative volumetric control application to be a most useful tool when drilling in deep water. Careful preparation and a true understanding of volumetric well control in general is a definite prerequisite in order to be successful in avoiding fracturing relative weak formations.

REMOVAL OF INFLUX

BULLHEADING

Bullheading involves applying pressure at surface to re-inject an influx either into the formation from which it came or another more permeable/weaker formation.

Application of this technique can help in circumstances which do not lend themselves to normal methods. Examples are ;

- The influx contains H₂S which will cause a safety hazard if brought to surface.
- A combined kick and losses situation has arisen.
- The MAASP is likely to be exceeded by a large margin before the influx enters the shoe. This can be the case in high pressure high temperature wells where there is only a small margin between the drilling fluid gradient and the formation strength and where a high degree of expansion (and thus high surface pressures) is required to bring a gas influx to surface.

Unfortunately, bullheading is not a routine method during drilling operations (although it is often used when killing a well for workover, using brine). A considerable amount of whole drilling fluid may have to be squeezed away in order to remove a migrating gas influx from the well. This might well result in considerable formation damage and a permanent loss situation, jeopardising the hole section and the objectives of the well. Bullheading can only be used if hole conditions permit and each case must be judged on its own merits.

Factors which will affect the success of the operation are :

- formation permeability
- the type of influx
- contamination of the influx with drilling fluid
- the position of the influx relative to the weakest formation exposed
- the burst strength of the casing and
- the pressure rating of the BOP

FORMATION STRENGTH TESTS

GUIDELINES

Exploration/Appraisal wells

It is recommended (EP89-1500) that formation strength tests are carried out for all casing shoes below which drilling will be carried out. This includes the conductor string if a BOP is planned to be used.

Development wells

Formation strength tests are justified in the majority of cases below all casing shoes. Tests may be omitted, however, if no hydrocarbon bearing or over-pressured formations are to be penetrated in the following hole section.

Important notes

- 1) All formation strength tests should be carried out with the lowest drilling fluid density necessary for primary well control of the formations exposed during the test. The drilling fluid density should only be increased for the rest of the section after the test is complete.
- 2) In situations when good zonal isolation behind the casing is critical to the well's success in both short and long term it is recommended to carry out the formation strength test using a retrievable packer, to avoid the creation of micro-annuli.
- 3) When testing below intermediate casing strings, the annulus outside the casing being tested should be left open and observed for returns. Do not neglect to close the side outlet valves following the test.
- 4) Information obtained from formation strength tests is dependant on the inclination of the hole. Data from vertical holes is not generally applicable to deviated ones.
- 5) Breakdown tests (e.g. minifrac tests) can be carried out when abandoning a well to gain valuable information on breakdown and fracture propagation pressures

FORMATION STRENGTH TESTS

PROCEDURES

The cement filled pocket is drilled out along with a minimum of about 20 ft (6m) of new formation. The test is conducted using a low volume high pressure pump (i.e. the cementing pump) and calibrated pressure gauges over a variety of ranges. **The drilling fluid system pumps and gauges are not sufficiently accurate enough to perform the operation.**

The procedure is:

- Circulate and condition the drilling fluid to a consistent density. If high gel strengths are present it may be necessary to reduce them to ensure good pressure transmission. Conditioning can be done with the bit at the shoe to avoid wash outs.
- Pull back the bit into the casing shoe.
- Ensure the well is full. Close in the well using the BOP pipe rams around the drill pipe.
- Open the annulus between the current and the previous casing string and monitor for flow.
- Slowly pump down the drill string until surface pressure approaches ca. 100 psi (700 kPa).
- Carefully measure tank levels etc.
- Pump uniform increments of volume – 0.1 to 0.25 bbl (0.016 to 0.4 m³) then stop and wait two minutes for pressures to stabilise. For each increment the following are noted:
 - cumulative volume pumped
 - pressure immediately after pumping ceases (final pumping pressure)
 - static pressure after two minutes (final static pressure)

Some operators prefer to apply the continuous pumping method whereby pumping is performed at a selected slow rate with simultaneous monitoring and plotting of pressures and volumes. This method is used when testing consolidated formations, during which the two minute pressure stabilisation period does not markedly influence the final test result.

- Plot the cumulative volume pumped against both the dynamic and static pressures on a graph.
- Continue to pump incremental volumes until one of the following occurs:
 - a pre-determined limit pressure has been reached
 - the static pressure line deviates from a straight line (i.e. a linear relationship between pressure and volume pumped).

Note that the difference in elevation between the derrick floor and the cementing unit should be taken into account when determining the limit pressure and using the result of the test.

If the pump pressure suddenly drops, stop pumping but leave the well closed in. This indicates a leak in the system, cement failure or formation breakdown. Record the pressures every minute until they stabilise. If the drop in pressure is related to formation breakdown, this data can be used to derive the minimum in-situ stress.

- Bleed off pressure at surface and monitor the returns. Determine how much fluid has been lost to the formation.
- Top up and close the annulus

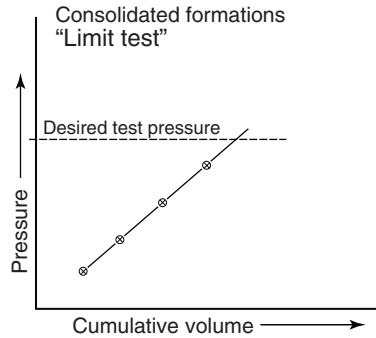
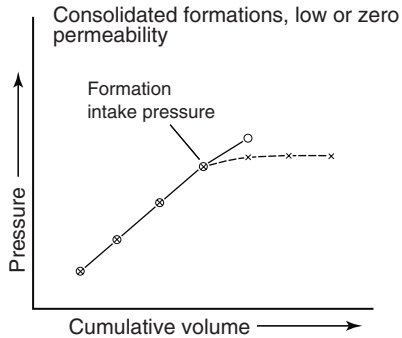
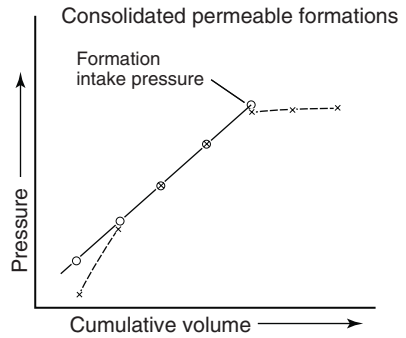
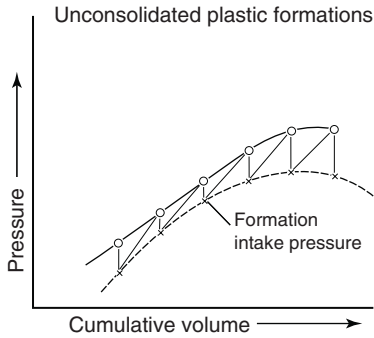
It should be noted that the objective of a Formation Strength Test is not to break down the formation and generate/propagate a fracture. The point at which the pressure/ volume plot deviates from a linear relationship is called the leak off point. It should be taken as the last measured point on the straight line; no extrapolation should normally be performed that would yield an increased formation strength.

The leak off point is sometimes also called the formation intake point.

A formation strength test that is terminated when a leak off point is identified is called a Leak Off Test. A formation strength test that is terminated when a pre-determined pressure is reached is called a Limit Test.

FORMATION STRENGTH TESTS

ILLUSTRATIONS



- Final pump pressure after each increment
- × Final pump pressure after waiting period

CAPACITIES and HEIGHT OF INFLUX

$$\text{Open Hole Capacity} = \frac{D_h^2}{1,029} \text{ bbls/ft} = \frac{D_h^2}{1.273 \times 10^6} \text{ m}^3/\text{m}$$

$$\text{Annular capacity} = \frac{D_h^2 - D_p^2}{1,029} \text{ bbls/ft} = \frac{D_h^2 - D_p^2}{1.273 \times 10^6} \text{ m}^3/\text{m}$$

Where : D_h = Hole diameter (ins/mm)
 D_p = Pipe/casing size (ins/mm)

Calculation of annular capacity between casing and pipe, when ID of casing is unknown (for steel pipe only)

$$\begin{aligned} \text{Capacity} &= \frac{(\text{Casing OD})^2 - \frac{\text{Casing lbs/ft}}{2.665} - (\text{Pipe OD})^2}{1,029} \text{ bbls/ft} \\ &= \frac{(\text{Casing OD})^2 - (163.9 \times \text{Casing kg/m}) - (\text{Pipe OD})^2}{1.273 \times 10^6} \text{ m}^3/\text{m} \end{aligned}$$

Calculation of I.D. of casing (for steel casing only)

$$\text{ID} = \sqrt{\text{OD}^2 - \frac{\text{lbs/ft}}{2.665}} \text{ inches} = \sqrt{\text{OD}^2 - (163.9 \times \text{kg/m})} \text{ mm}$$

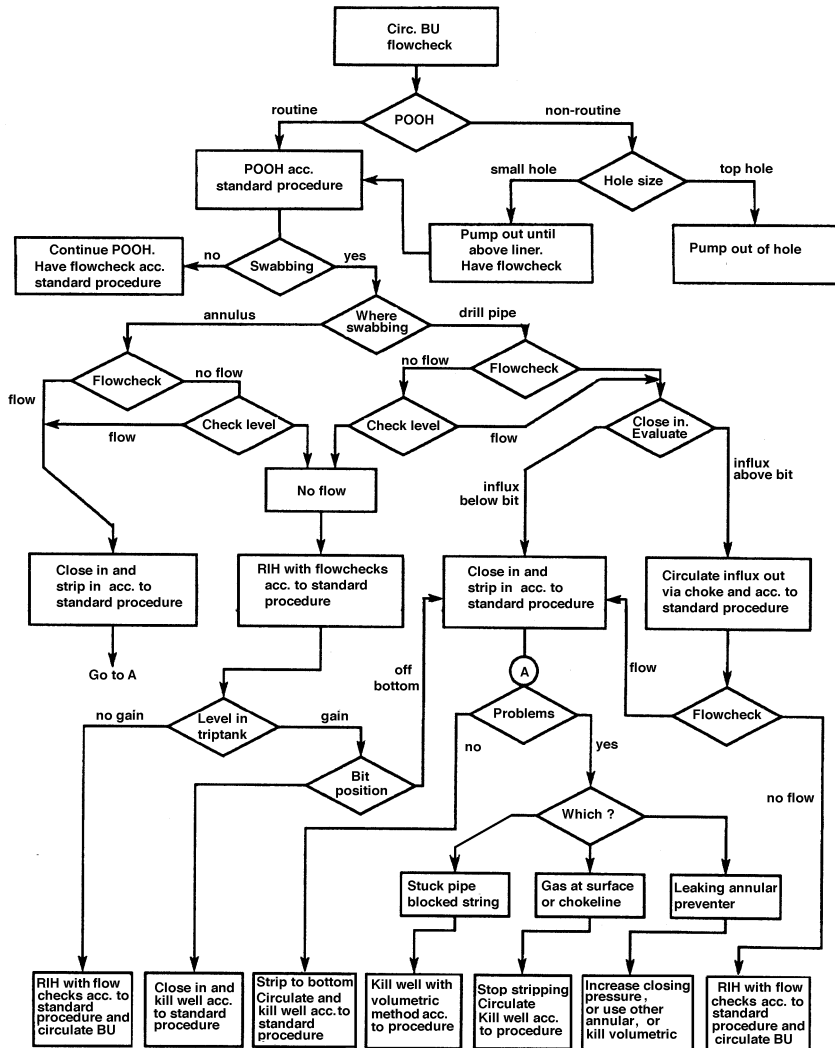
Height of a gas influx at any point in the annulus

$$h = \frac{P_0 Z T h_b}{P Z_1 T_1}$$

Where : h = The height of the gas column at any given point
 P_0 = Formation pore pressure
 P = The pressure at the bottom of the gas column at the point of interest
 Z_1 = Initial compressibility factor of gas
 Z = Compressibility factor of the gas at the point of interest
 T_1 = Initial temperature of the gas
 T = Temperature of the gas at the point of interest
 h_b = HEIGHT OF GAS COLUMN AT THE BOTTOM OF THE HOLE OR, IF THE AREA OF THE ANNULUS CHANGES, THE EQUIVALENT HEIGHT AT THE BOTTOM BASED ON THE ANNULAR AREA AT THE POINT OF INTEREST

TRIPPING OUT OF HOLE

FLOW CHART AND DECISION TREE



F-30

Form for pressure and volume records during stripping operations

Trip tank level with required P_{choke}

$P_{choke} = P_{ann} + P_s + P_w =$	$= \dots\dots$
P_{ann}	$= \dots\dots$
F - factor	$= \dots\dots$
Volume influx	$= \dots\dots$
$P_s = F \times V_i$	$= \dots\dots$
P_w selected	$= \dots\dots$
ΔV	$= \dots\dots$
	divisions in trip tank

[illegible][illegible]

KICK CONTROL WORKSHEET

SURFACE BOP - FIELD UNITS

PAGE 1 OF 2

IWCF CERTIFICATION TEST SURFACE BOP KILL SHEET

DATE : _____

NAME : _____

UNITS : FIELD

FORMATION STRENGTH DATA :

SURFACE LEAK-OFF PRESSURE FROM
FORMATION STRENGTH TEST

(A) _____ psi

DRILLING FLUID DENS. AT TEST (B) _____ ppg

MAX. ALLOWABLE DRILLING FLUID DENSITY =
(A)

(B) + 0.052 x SHOE T.V. DEPTH = (C) _____ ppg

INITIAL MAASP =

((C) - CURR. DENS.) x SHOE T.V. DEPTH x 0.052
= _____ psi

CURRENT WELL DATA :

DRILLING FLUID DATA

DENSITY _____ ppg

GRADIENT _____ psi/ft

CASING & SHOE DATA

SIZE _____ in

M. DEPTH _____ ft

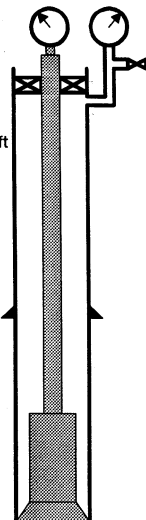
T.V. DEPTH _____ ft

HOLE DATA

SIZE _____ in

M. DEPTH _____ ft

T.V. DEPTH _____ ft



PUMP NO. 1 DISPL.	PUMP NO. 2 DISPL.
bbl / stk	bbl / stk

SLOW PUMP RATE DATA	DYNAMIC PRESSURE LOSS	
	PUMP NO. 1	PUMP NO. 2
SPM	psi	psi
SPM	psi	psi

PRE - RECORDED VOLUME DATA :	LENGTH ft	CAPACITY bbl / ft	VOLUME bbl	PUMP STROKES stks	TIME minutes
DRILL PIPE	x	=			
HEVI WATE	x	=	+		
DRILL COLLAR	x	=	+		
DRILL STRING VOLUME			(D) bbl	(E) stks	min
DC x OPEN HOLE	x	=			
DP/ HWDP x OPEN HOLE	x	=	+		
OPEN HOLE VOLUME			(F) bbl	stks	min
DP x CASING	x	= (G)	+	stks	min
TOTAL ANNULUS VOLUME		(F+G) = (H)	bbl	stks	min
TOTAL WELL SYSTEM VOLUME		(D+H) = (I)	bbl	stks	min
ACTIVE SURFACE VOLUME		(J)	bbl	stks	
TOTAL ACTIVE FLUID SYSTEM		(I+J)	bbl	stks	

CALCULATIONS CAN BE MADE USING EITHER DRILLING FLUID DENSITY OR DRILLING FLUID GRADIENT

SUKILL1.DRW

REV.12.03.1995

SURFACE BOP - FIELD UNITS

IWCF CERTIFICATION TEST SURFACE BOP KILL SHEET

NAME : _____

UNITS : FIELD

SHUT IN DRILL PIPE PRESSURE --- SIDPP psiSHUT IN CASING PRESSURE ----- SICP psi

PIT GAIN bbl

1 bbl = 42 US gall

$$\text{CURRENT DRILLING FLUID DENSITY} + \frac{\text{SIDPP}}{\text{TVD} \times 0.052}$$
$$\frac{\text{.....}}{\text{.....} \times 0.052} = \text{..... ppg}$$
$$\text{CURRENT DRILLING FLUID GRADIENT} + \frac{\text{SIDPP}}{\text{TVD}}$$
$$\text{.....} + \frac{\text{.....}}{\text{.....}} = \text{..... psi / ft}$$

DYNAMIC PRESSURE LOSS + SIDPP

..... + = psi

$$\frac{\text{KILL FLUID DENSITY}}{\text{CURRENT DRILLING FLUID DENSITY}} \times \text{DYNAMIC PRESSURE LOSS}$$
$$\frac{\text{.....}}{\text{.....}} \times \text{.....} = \text{.....}$$
$$\frac{\text{KILL FLUID GRADIENT}}{\text{CURRENT DRILLING FLUID GRADIENT}} \times \text{DYNAMIC PRESSURE LOSS}$$
$$\frac{\text{.....}}{\text{.....}} \times \text{.....} = \text{.....}$$
$$(K) = ICP - FCP = \dots - \dots = \dots \text{ ps}$$
$$\frac{(K) \times 100}{(E)} = \frac{\text{.....} \times 100}{\text{.....}} = \text{..... psi / 100 strokes}$$
[illegible]

↑
D R I L L P I P E P R E S S U R E
S T A T I C & D Y N A M I C

0

STROKES →

KICK CONTROL WORKSHEET

SURFACE BOP - S.I. UNITS

PAGE 1 OF 2

IWCF CERTIFICATION TEST SURFACE BOP KILL SHEET

DATE : _____
NAME : _____
UNITS : SI

FORMATION STRENGTH DATA :

SURFACE LEAK-OFF PRESSURE FROM
FORMATION STRENGTH TEST

(A) kPa

DRILLING FLUID DENS. AT TEST (B) kg/m³

MAX. ALLOWABLE DRILLING FLUID DENSITY =
(A) x 102

(B) + SHOE T.V. DEPTH = (C) kg/m³

INITIAL MAASP =

((C) - CURR. DENS.) x SHOE T.V. DEPTH
102

kPa

CURRENT WELL DATA :

DRILLING FLUID DATA

DENSITY kg/m³

GRADIENT kPa/m

CASING & SHOE DATA

SIZE mm

M. DEPTH m

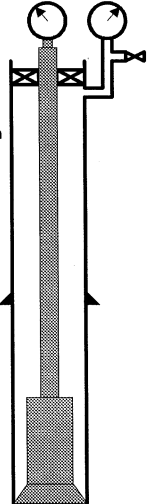
T.V. DEPTH m

HOLE DATA

SIZE mm

M. DEPTH m

T.V. DEPTH m



PUMP NO. 1 DISPL.	PUMP NO. 2 DISPL.
m ³ / stk	m ³ / stk

SLOW PUMP RATE DATA	DYNAMIC PRESSURE LOSS	
	PUMP NO. 1	PUMP NO. 2
SPM	kPa	kPa
SPM	kPa	kPa

PRE - RECORDED VOLUME DATA :	LENGTH m	CAPACITY m ³ /m	VOLUME m ³	PUMP STROKES stks	TIME minutes
DRILL PIPE	x	=		VOLUME	PUMP STROKES
HEVI WATE	x	=	+	PUMP DISPLACEMENT	SLOW PUMP RATE
DRILL COLLAR	x	=	+		
DRILL STRING VOLUME			(D) m ³	(E) stks	min
DC x OPEN HOLE	x	=			
DP/ HWDP x OPEN HOLE	x	=	+		
OPEN HOLE VOLUME			(F) m ³	stks	min
DP x CASING	x	= (G)	+	stks	min
TOTAL ANNULUS VOLUME		(F+G) = (H)	m ³	stks	min
TOTAL WELL SYSTEM VOLUME		(D+H) = (I)	m ³	stks	min
ACTIVE SURFACE VOLUME		(J)	m ³	stks	
TOTAL ACTIVE FLUID SYSTEM		(I+J)	m ³	stks	

CALCULATIONS CAN BE MADE USING EITHER DRILLING FLUID DENSITY OR DRILLING FLUID GRADIENT

SUKILL3.DRW

REV.01.03.1994

SURFACE BOP - S.I. UNITS

IWCF CERTIFICATION TEST SURFACE BOP KILL SHEET

NAME : _____

UNITS · SI

SHUT IN DRILL PIPE PRESSURE — SIDPP kPaSHUT IN CASING PRESSURE ----- SICP kPa

PIT GAIN m³

$kN = 102 \text{ kgf}$

$$\text{CURRENT DRILLING FLUID DENSITY} + \frac{102 \times \text{SIDPP}}{\text{TVD}} \\ \text{.....} + \frac{102 \times \text{.....}}{\text{.....}} = \text{..... kg / m}^3$$

CURRENT DRILLING FLUID GRADIENT + $\frac{\text{SIDPP}}{\text{TVD}}$

..... + $\frac{\text{.....}}{\text{.....}}$ = kPa / m

DYNAMIC PRESSURE LOSS + SIDPP

..... + = kPa

SICP kPa

$$\frac{\text{KILL FLUID DENSITY}}{\text{CURRENT DRILLING FLUID DENSITY}} \times \text{DYNAMIC PRESSURE LOSS} = \text{..... kPa}$$

$$\frac{\text{KILL FLUID GRADIENT}}{\text{CURRENT DRILLING FLUID GRADIENT}} \times \text{DYNAMIC PRESSURE LOSS}$$

$$\text{.....} \times \text{.....} = \text{..... kPa}$$
$$(K) = ICP - FCP = \dots\dots\dots - \dots\dots\dots = \dots\dots\dots \text{ kPa} \quad \frac{(K) \times 100}{(E)} = \frac{\dots\dots\dots \times 100}{\dots\dots\dots} = \dots\dots\dots \text{ kPa} / 100 \text{ strokes}$$

STROKES	PRESSURE
---------	----------

DRILL PIPE PRESSURE STATIC & DYNAMIC

9

STROKES →

REV.01.03.1994

KICK CONTROL WORKSHEET

SUB-SURFACE BOP - FIELD UNITS

PAGE 1 OF 2

IWCF CERTIFICATION TEST SUBSEA BOP KILL SHEET								DATE : _____ NAME : _____ UNITS : FIELD																																																																																					
FORMATION STRENGTH DATA : SURFACE LEAK-OFF PRESSURE FROM FORMATION STRENGTH TEST (A) psi DRILLING FLUID DENS. AT TEST (B) ppg MAX. ALLOWABLE DRILLING FLUID DENSITY = $\frac{(B) + 0.052 \times \text{SHOE T.V. DEPTH}}{(A)} = (C) \text{ ppg}$ INITIAL MAASP = $((C) - \text{CURR. DENS.}) \times \text{SHOE T.V. DEPTH} \times 0.052 = \text{ } \text{psi}$					CURRENT WELL DATA : MARINE RISER LENGTH ft CHOKELINE LENGTH ft DRILLING FLUID DATA DENSITY ppg GRADIENT psi/ft CASING & SHOE DATA SIZE in M. DEPTH ft T.V. DEPTH ft																																																																																								
<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th style="width: 50%;">PUMP NO. 1 DISPL.</th> <th style="width: 50%;">PUMP NO. 2 DISPL.</th> </tr> <tr> <td style="text-align: center;">bbl / stk</td> <td style="text-align: center;">bbl / stk</td> </tr> </table>		PUMP NO. 1 DISPL.	PUMP NO. 2 DISPL.	bbl / stk	bbl / stk	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th colspan="7">DYNAMIC PRESSURE LOSS psi</th> </tr> <tr> <th colspan="3">PUMP NO. 1</th> <th colspan="4">PUMP NO. 2</th> </tr> <tr> <th>SLOW PUMP RATE DATA</th> <th>RISER</th> <th>CHOKELINE</th> <th>CL. FRIC.</th> <th>RISER</th> <th>CHOKELINE</th> <th>CL. FRIC.</th> </tr> <tr> <td>SPM</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>SPM</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table>				DYNAMIC PRESSURE LOSS psi							PUMP NO. 1			PUMP NO. 2				SLOW PUMP RATE DATA	RISER	CHOKELINE	CL. FRIC.	RISER	CHOKELINE	CL. FRIC.	SPM							SPM							HOLE DATA SIZE in M. DEPTH ft T.V. DEPTH ft																																																
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PRE - RECORDED VOLUME DATA :	LENGTH ft	CAPACITY bbl / ft	VOLUME bbl	PUMP STROKES stks	TIME minutes																																																																																								
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MARINE RISER x DP		x	=	bbl	stks																																																																																								

SSKILL1.DRW

REV.01.03.1994

KICK CONTROL WORKSHEET

SUB-SURFACE BOP - FIELD UNITS

PAGE 2 OF 2

IWCF CERTIFICATION TEST SUBSEA BOP KILL SHEET

DATE : _____
NAME : _____
UNITS : FIELD

KICK DATA :

SHUT IN DRILL PIPE PRESSURE ---- SIDPP psi
SHUT IN CASING PRESSURE ----- SICP psi
PIT GAIN bbl

1 bbl = 42 US gal

KILL FLUID DENSITY	CURRENT DRILLING FLUID DENSITY + $\frac{\text{SIDPP}}{\text{TVD} \times 0.052}$ + = ppg
--------------------	--

KILL FLUID GRADIENT	CURRENT DRILLING FLUID GRADIENT + $\frac{\text{SIDPP}}{\text{TVD}}$ + = psi / ft
---------------------	---

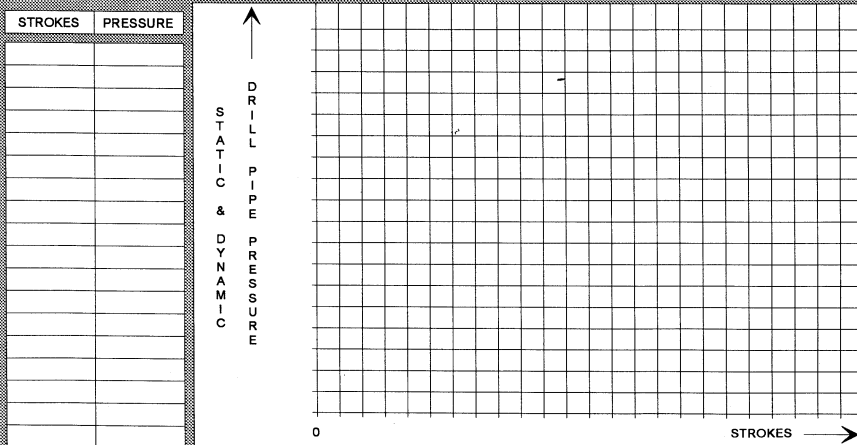
INITIAL CIRCULATING PRESS. ICP	DYNAMIC PRESSURE LOSS THROUGH RISER CIRCUIT + SIDPP + = psi
--------------------------------	--

INITIAL DYNAMIC CASING PRESS AT KILL PUMP RATE	SICP - CHOKE LINE FRICTION - = psi
--	---

FINAL CIRCULATING PRESS. FCP	$\frac{\text{KILL FLUID DENSITY}}{\text{CURRENT DRILLING FLUID DENSITY}} \times \text{DYN. PRESS. LOSS RISER CIRC.}$ x = psi
------------------------------	---

FINAL CIRCULATING PRESS. FCP	$\frac{\text{KILL FLUID GRADIENT}}{\text{CURRENT DRILLING FLUID GRADIENT}} \times \text{DYN. PRESS. LOSS RISER CIRC.}$ x = psi
------------------------------	---

(L) = ICP - FCP = - = psi
 $\frac{(L) \times 100}{(E)} = \frac{\text{.....} \times 100}{\text{.....}} = \text{.....} \text{ psi / 100 strokes}$



SSKILL2.DRW

REV.01.03.1994

KICK CONTROL WORKSHEET

SUB-SURFACE BOP - S.I. UNITS

PAGE 1 OF 2

IWCF CERTIFICATION TEST SUBSEA BOP KILL SHEET

DATE : _____
NAME : _____
UNITS : SI

FORMATION STRENGTH DATA :

SURFACE LEAK-OFF PRESSURE FROM
FORMATION STRENGTH TEST (A) kPa
DRILLING FLUID DENS. AT TEST (B) kg/m³
MAX. ALLOWABLE DRILLING FLUID DENSITY =
(A) x 102
(B) + SHOE T.V. DEPTH = (C) kg/m³
INITIAL MAASP =
((C) - CURR. DENS.) x SHOE T.V. DEPTH =
102 _____ kPa

CURRENT WELL DATA :

MARINE RISER
LENGTH _____ m
CHOKELINE
LENGTH _____ m

DRILLING FLUID DATA

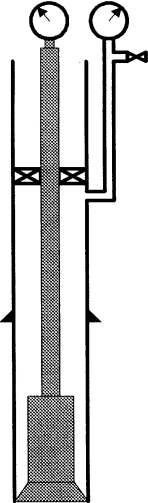
DENSITY _____ kg/m³
GRADIENT _____ kPa/m

CASING & SHOE DATA

SIZE _____ mm
M. DEPTH _____ m
T.V. DEPTH _____ m

HOLE DATA

SIZE _____ mm
M. DEPTH _____ m
T.V. DEPTH _____ m



PUMP NO. 1 DISPL.	PUMP NO. 2 DISPL.
m ³ / stk	m ³ / stk

SLOW PUMP RATE DATA	DYNAMIC PRESSURE LOSS kPa					
	PUMP NO. 1			PUMP NO. 2		
	RISER	CHOKELINE	CL. FRIC.	RISER	CHOKELINE	CL. FRIC.
SPM						
SPM						

PRE - RECORDED VOLUME DATA :	LENGTH m	CAPACITY m ³ / m	VOLUME m ³	PUMP STROKES stks	TIME minutes
DRILL PIPE	x	=		VOLUME	PUMP STROKES
HEVI WATE	x	=	+	PUMP DISPLACEMENT	SLOW PUMP RATE
DRILL COLLAR	x	=	+		
DRILL STRING VOLUME			(D) m ³	(E) stks	min
DC x OPEN HOLE	x	=			
DP/ HWDP x OPEN HOLE	x	=	+		
OPEN HOLE VOLUME			(F) m ³	stks	min
DP x CASING	x	= (G)	+	stks	min
CHOKELINE	x	= (H)	+	stks	min
TOTAL ANNULUS / CHOKELINE VOL.		(F+G+H) = (I)	m ³	stks	min
TOTAL WELL SYSTEM VOLUME		(D+I) = (J)	m ³	stks	min
ACTIVE SURFACE VOLUME		(K)	m ³	stks	
TOTAL ACTIVE FLUID SYSTEM		(J+K)	m ³	stks	
MARINE RISER x DP	x	=	m ³	stks	

CALCULATIONS CAN BE MADE USING EITHER DRILLING FLUID DENSITY OR DRILLING FLUID GRADIENT

SSKILL3.DRW

REV.01.03.1994

SUB-SURFACE BOP - S.I. UNITS

PAGE 2 OF 2

IWCF CERTIFICATION TEST

SUBSEA BOP KILL SHEET

DATE : _____

NAME : _____

UNITS : SI

KICK DATA :

SHUT IN DRILL PIPE PRESSURE --- SIDPP kPa

SHUT IN CASING PRESSURE ----- SICP kPa

PIT GAIN	m³
----------	--

1 kN = 102 kgf

KILL FLUID DENSITY

$$\text{CURRENT DRILLING FLUID DENSITY} + \frac{102 \times \text{SIDPP}}{\text{TVD}}$$
$$\dots\dots\dots + \frac{102 \times \dots\dots\dots}{\dots\dots\dots} = \dots\dots\dots \text{ kg / m}^3$$

KILL FLUID GRADIENT

$$\text{CURRENT DRILLING FLUID GRADIENT} + \frac{\text{SIDPP}}{\text{TVD}}$$
$$\frac{\text{.....}}{\text{.....}} + \frac{\text{.....}}{\text{.....}} = \text{..... kPa / m}$$

INITIAL CIRCULATING PRESS.
ICP

DYNAMIC PRESSURE LOSS THROUGH RISER CIRCUIT + SIDPP

..... + = kPa

INITIAL DYNAMIC CASING
PRESS AT KILL PUMP RATE

SICP - CHOKE LINE FRICTION

..... - = kPa

FINAL CIRCULATING PRESS.
FCP

$$\frac{\text{KILL FLUID DENSITY}}{\text{CURRENT DRILLING FLUID DENSITY}} \times \text{DYN. PRESS. LOSS RISER CIRC.}$$

$$\text{*****} \times \text{*****} =$$

..... kPa

FINAL CIRCULATING PRESS.
FCP

$$\frac{\text{KILL FLUID GRADIENT}}{\text{CURRENT DRILLING FLUID GRADIENT}} \times \text{DYN. PRESS. LOSS RISER CIRC.}$$

..... kPa

$$(L) = ICP - FCP = \dots\dots\dots - \dots\dots\dots = \dots\dots\dots \text{ kPa} \quad \frac{(L) \times 100}{(E)} = \frac{\dots\dots\dots \times 100}{\dots\dots\dots} = \dots\dots\dots \text{ kPa} / 100 \text{ strokes}$$

STROKES	PRESSURE
---------	----------

DRILL PIPE PRESSURE
↑
STATIC & DYNAMIC

0

STROKES

SSKILL4.DRW

REV.01.03.1994

KICK CONTROL WORKSHEET

DEVIATED WELL - SURFACE BOP - FIELD UNITS

PAGE 1 OF 3

IWCF CERTIFICATION TEST SURFACE BOP KILL SHEET DEVIATED WELL

DATE : _____

NAME : _____

UNITS : FIELD

FORMATION STRENGTH DATA:

SURFACE LEAK -OFF PRESSURE FROM
FORMATION STRENGTH TEST (A) psi

DRILLING FLUID DENS. AT TEST (B) ppg

MAX. ALLOWABLE DRILLING FLUID DENSITY =

(B) + $\frac{(A)}{\text{SHOE T.V. DEPTH} \times 0.052}$ = (C) ppg

INITIAL MAASP =

((C) - CURR. DENS.) x SHOE T.V. DEPTH x 0.052

= _____ psi

CURRENT WELL DATA:

DRILLING FLUID DATA:

DENSITY _____ ppg

GRADIENT _____ psi/ft

DEVIATION DATA:

KOP M.D. _____ ft

KOP T.V.D. _____ ft

EOB M.D. _____ ft

EOB T.V.D. _____ ft

CASING SHOE DATA:

SIZE _____ in

M. DEPTH _____ ft

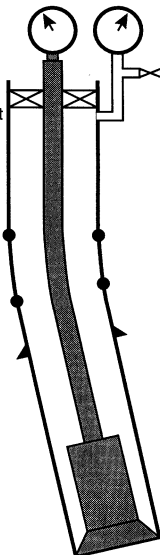
T.V. DEPTH _____ ft

HOLE DATA:

SIZE _____ in

M. DEPTH _____ ft

T.V. DEPTH _____ ft



PUMP NO. 1 DISPL.	PUMP NO. 2 DISPL.
bbl / stroke	bbl / stroke

SLOW PUMP RATE DATA:	(PL) DYNAMIC PRESSURE LOSS	
	PUMP NO. 1	PUMP NO. 2
SPM	psi	psi
SPM	psi	psi

PRE-RECORDED VOLUME DATA:	LENGTH ft	CAPACITY bbl / ft	VOLUME bbl	PUMP STROKES stks	TIME minutes
DP - SURFACE TO KOP	x	=		(L) stks	
DP - KOP TO EOB	x	=	+	(M) stks	
DP - EOB TO BHA	x	=	+	(N ₁) stks	
HEVI WALL DRILL PIPE	x	=	+	(N ₂) stks	
DRILL COLLAR	x	=	+	(N ₃) stks	
DRILL STRING VOLUME			(D) bbl	stks	min
DC x OPEN HOLE	x	=			
DP / HWDP x OPEN HOLE	x	=	+		
OPEN HOLE VOLUME			(F) bbl	stks	min
DP x CASING	x	=	(G) +	stks	min
TOTAL ANNULUS VOLUME			(F+G) = (H) bbl	stks	min
TOTAL WELL SYSTEM VOLUME			(D+H) = (I) bbl	stks	min
ACTIVE SURFACE VOLUME			(J) bbl		
TOTAL ACTIVE FLUID SYSTEM			(I+J) bbl		

10-01-96
Dr No TCM 7036 (FI)

KICK CONTROL WORKSHEET

DEVIATED WELL - SURFACE BOP - FIELD UNITS

PAGE 2 OF 3

IWCF CERTIFICATION TEST SURFACE BOP KILL SHEET DEVIATED WELL

DATE : _____

NAME : _____

UNITS : FIELD

KICK DATA :

SIDPP psi

SICP psi

PIT GAIN bbl

KILL FLUID DENSITY

$$\text{CURRENT DRILLING FLUID DENSITY} + \frac{\text{SIDPP}}{\text{TVD} \times 0.052}$$

KMD

$$\text{.....} + \frac{\text{.....}}{\text{.....} \times 0.052} = \text{..... ppg}$$

INITIAL CIRC. PRESS.

ICP

DYNAMIC PRESSURE LOSS + SIDPP

$$\text{.....} + \text{.....} = \text{..... psi}$$

FINAL CIRCULATING
PRESSURE

FCP

$$\frac{\text{KILL FLUID DENSITY}}{\text{CURRENT DRILLING FLUID DENSITY}} \times \text{DYNAMIC PRESSURE LOSS}$$

$$\text{.....} \times \text{.....} = \text{..... psi}$$

DYNAMIC PRESSURE
LOSS AT KOP (O)

$$\text{PL} + \left[(\text{FCP} - \text{PL}) \times \frac{\text{KOP}_{\text{MD}}}{\text{TD}_{\text{MD}}} \right] = \text{.....} + \left[(\text{.....}) \times \frac{\text{.....}}{\text{.....}} \right] = \text{..... psi}$$

REMAINING SIDPP
AT KOP (P)

$$\text{SIDPP} - \left[(\text{KMD} - \text{OMD}) \times 0.052 \times \text{KOP}_{\text{TVD}} \right]$$

$$= \text{.....} - \left[(\text{.....}) \times 0.052 \times \text{.....} \right] = \text{..... psi}$$

CIRCULATING PRESS.
AT KOP (KOP CP)

$$(O) + (P) = \text{.....} + \text{.....} = \text{..... psi}$$

DYNAMIC PRESS. LOSS
AT EOB (R)

$$\text{PL} + \left[(\text{FCP} - \text{PL}) \times \frac{\text{EOB}_{\text{MD}}}{\text{TD}_{\text{MD}}} \right] = \text{.....} + \left[(\text{.....}) \times \frac{\text{.....}}{\text{.....}} \right] = \text{..... psi}$$

REMAINING SIDPP
AT EOB (S)

$$\text{SIDPP} - \left[(\text{KMD} - \text{OMD}) \times 0.052 \times \text{EOB}_{\text{TVD}} \right]$$

$$= \text{.....} - \left[(\text{.....}) \times 0.052 \times \text{.....} \right] = \text{..... psi}$$

CIRCULATING PRESS.
AT EOB (EOB CP)

$$(R) + (S) = \text{.....} + \text{.....} = \text{..... psi}$$

$$(T) = \text{ICP} - \text{KOP CP} = \text{.....} - \text{.....} = \text{..... psi}$$

$$\frac{(T) \times 100}{(L)} = \frac{\text{.....} \times 100}{\text{.....}} = \text{.....} \frac{\text{psi}}{100 \text{ strokes}}$$

$$(U) = \text{KOP CP} - \text{EOB CP} = \text{.....} - \text{.....} = \text{..... psi}$$

$$\frac{(U) \times 100}{(M)} = \frac{\text{.....} \times 100}{\text{.....}} = \text{.....} \frac{\text{psi}}{100 \text{ strokes}}$$

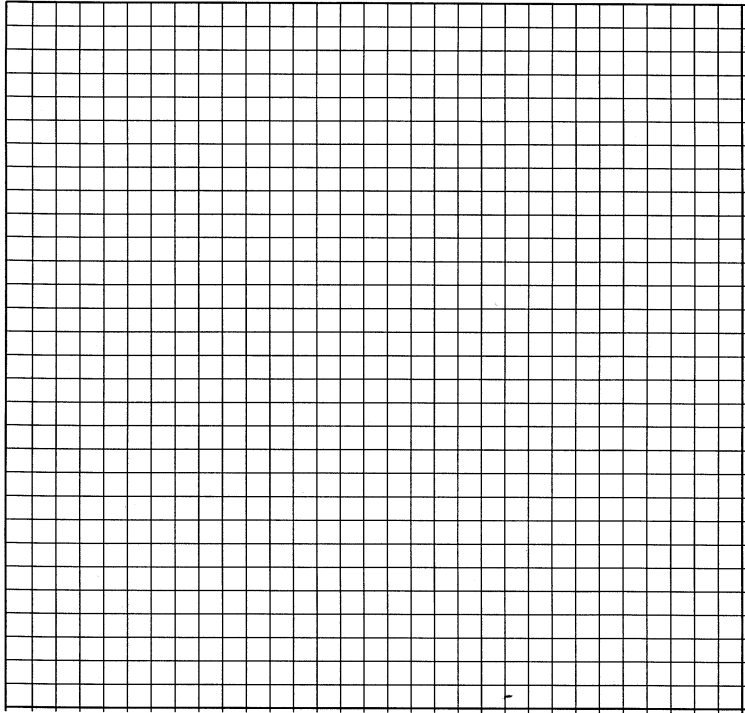
$$(W) = \text{EOB CP} - \text{FCP} = \text{.....} - \text{.....} = \text{..... psi}$$

$$\frac{(W) \times 100}{(N_1 + N_2 + N_3)} = \frac{\text{.....} \times 100}{\text{.....}} = \text{.....} \frac{\text{psi}}{100 \text{ strokes}}$$

DEVIATED WELL - SURFACE BOP - FIELD UNITS

IWCF CERTIFICATION TEST
SURFACE BOP KILL SHEET
DEVIATED WELL

UNITS : FIELD



STROKES — 

← **STATIC & DYNAMIC DRILL PIPE PRESSURE [psi]**

[illegible]

KICK CONTROL WORKSHEET

DEVIATED WELL - SURFACE BOP - S.I. UNITS

PAGE 1 OF 3

IWCF CERTIFICATION TEST SURFACE BOP KILL SHEET DEVIATED WELL

DATE : _____

NAME : _____

UNITS : S.I.

FORMATION STRENGTH DATA:

SURFACE LEAK -OFF PRESSURE FROM
FORMATION STRENGTH TEST (A) kPa

DRILLING FLUID DENS. AT TEST (B) kg/m³

MAX. ALLOWABLE DRILLING FLUID DENSITY =

(B) + $\frac{(A) \times 102}{\text{SHOE T.V. DEPTH}}$ = (C) kg/m³

INITIAL MAASP =

$\frac{((C) - \text{CURR. DENS.}) \times \text{SHOE T.V. DEPTH}}{102}$ = kPa

CURRENT WELL DATA:

DRILLING FLUID DATA:

DENSITY kg/m³

DEVIATION DATA:

KOP M.D. m

KOP T.V.D. m

EOB M.D. m

EOB T.V.D. m

CASING SHOE DATA:

SIZE mm

M. DEPTH m

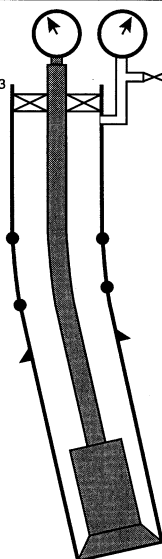
T.V. DEPTH m

HOLE DATA:

SIZE mm

M. DEPTH m

T.V. DEPTH m



PUMP NO. 1 DISPL.	PUMP NO. 2 DISPL.
m ³ / stroke	m ³ / stroke

SLOW PUMP RATE DATA:	(PL) DYNAMIC PRESSURE LOSS	
	PUMP NO. 1	PUMP NO. 2
SPM	kPa	kPa
SPM	kPa	kPa

PRE-RECORDED VOLUME DATA:	LENGTH m	CAPACITY m ³ / m	VOLUME m ³	PUMP STROKES stks	TIME minutes
DP - SURFACE TO KOP	x	=		(L) stks	
DP - KOP TO EOB	x	=	+	(M) stks	
DP - EOB TO BHA	x	=	+	(N ₁) stks	
HEVI WALL DRILL PIPE	x	=	+	(N ₂) stks	
DRILL COLLAR	x	=	+	(N ₃) stks	
DRILL STRING VOLUME			(D) m ³	stks	min
DC x OPEN HOLE	x	=			
DP / HWDP x OPEN HOLE	x	=	+		
OPEN HOLE VOLUME			(F) m ³	stks	min
DP x CASING	x	=	(G) +	stks	min
TOTAL ANNULUS VOLUME		(F+G) = (H)	m ³	stks	min
TOTAL WELL SYSTEM VOLUME		(D+H) = (I)	m ³	stks	min
ACTIVE SURFACE VOLUME		(J)	m ³		
TOTAL ACTIVE FLUID SYSTEM		(I+J)	m ³		

15-01-96
Dr No TCM 7036 (S.I.)

KICK CONTROL WORKSHEET

DEVIATED WELL - SURFACE BOP - S.I. UNITS

PAGE 2 OF 3

IWCF CERTIFICATION TEST SURFACE BOP KILL SHEET DEVIATED WELL

DATE : _____

NAME : _____

UNITS : S.I.

KICK DATA :

SIDPP kPa

SICP kPa

PIT GAIN m³

KILL FLUID DENSITY

KMD

$$\text{CURRENT DRILLING FLUID DENSITY} + \frac{102 \times \text{SIDPP}}{\text{TVD}}$$

$$\dots\dots\dots + \frac{102 \times \dots\dots\dots}{\dots\dots\dots} = \dots\dots\dots \text{ kg / m}^3$$

INITIAL CIRC. PRESS.

ICP

$$\text{DYNAMIC PRESSURE LOSS} + \text{SIDPP}$$

$$\dots\dots\dots + \dots\dots\dots = \dots\dots\dots \text{ kPa}$$

FINAL CIRCULATING PRESSURE

FCP

$$\frac{\text{KILL FLUID DENSITY}}{\text{CURRENT DRILLING FLUID DENSITY}} \times \text{DYNAMIC PRESSURE LOSS}$$

$$\frac{\dots\dots\dots}{\dots\dots\dots} \times \dots\dots\dots = \dots\dots\dots \text{ kPa}$$

DYNAMIC PRESSURE LOSS AT KOP (O)

$$\text{PL} + \left[(\text{FCP} - \text{PL}) \times \frac{\text{KOP}_{\text{MD}}}{\text{TD}_{\text{MD}}} \right] = \dots\dots\dots + \left[(\dots\dots\dots) \times \frac{\dots\dots\dots}{\dots\dots\dots} \right] = \dots\dots\dots \text{ kPa}$$

REMAINING SIDPP AT KOP (P)

$$\text{SIDPP} - \left[\frac{(\text{KMD} - \text{OMD}) \times \text{KOP}_{\text{TVD}}}{102} \right]$$

$$= \dots\dots\dots - \left[\frac{(\dots\dots\dots) \times \dots\dots\dots}{102} \right] = \dots\dots\dots \text{ kPa}$$

CIRCULATING PRESS. AT KOP (KOP CP)

$$(O) + (P) = \dots\dots\dots + \dots\dots\dots = \dots\dots\dots \text{ kPa}$$

DYNAMIC PRESS. LOSS AT EOB (R)

$$\text{PL} + \left[(\text{FCP} - \text{PL}) \times \frac{\text{EOB}_{\text{MD}}}{\text{TD}_{\text{MD}}} \right] = \dots\dots\dots + \left[(\dots\dots\dots) \times \frac{\dots\dots\dots}{\dots\dots\dots} \right] = \dots\dots\dots \text{ kPa}$$

REMAINING SIDPP AT EOB (S)

$$\text{SIDPP} - \left[\frac{(\text{KMD} - \text{OMD}) \times \text{EOB}_{\text{TVD}}}{102} \right]$$

$$= \dots\dots\dots - \left[\frac{(\dots\dots\dots) \times \dots\dots\dots}{102} \right] = \dots\dots\dots \text{ kPa}$$

CIRCULATING PRESS. AT EOB (EOB CP)

$$(R) + (S) = \dots\dots\dots + \dots\dots\dots = \dots\dots\dots \text{ kPa}$$

$$(T) = \text{ICP} - \text{KOP CP} = \dots\dots\dots - \dots\dots\dots = \dots\dots\dots \text{ kPa}$$

$$\frac{(T) \times 100}{(L)} = \frac{\dots\dots\dots \times 100}{\dots\dots\dots} = \dots\dots\dots \frac{\text{kPa}}{100 \text{ strokes}}$$

$$(U) = \text{KOP CP} - \text{EOB CP} = \dots\dots\dots - \dots\dots\dots = \dots\dots\dots \text{ kPa}$$

$$\frac{(U) \times 100}{(M)} = \frac{\dots\dots\dots \times 100}{\dots\dots\dots} = \dots\dots\dots \frac{\text{kPa}}{100 \text{ strokes}}$$

$$(W) = \text{EOB CP} - \text{FCP} = \dots\dots\dots - \dots\dots\dots = \dots\dots\dots \text{ kPa}$$

$$\frac{(W) \times 100}{(N_1 + N_2 + N_3)} = \frac{\dots\dots\dots \times 100}{\dots\dots\dots} = \dots\dots\dots \frac{\text{kPa}}{100 \text{ strokes}}$$

15-01-96
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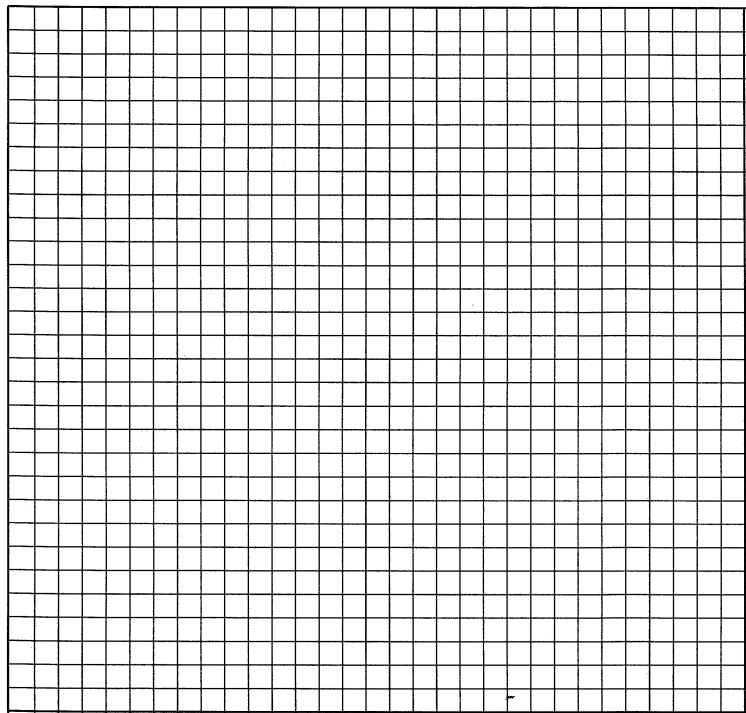
DEVIATED WELL - SURFACE BOP - S.I. UNITS

IWCF CERTIFICATION TEST

SURFACE BOP KILL SHEET

DEVIATED WELL

UNITS : S.I.



STROKES —

← STATIC & DYNAMIC DRILL PIPE PRESSURE [kPa]

[illegible]

KICK CONTROL WORKSHEET

DEVIATED WELL - SUB-SURFACE BOP - FIELD UNITS

PAGE 1 OF 3

IWCF CERTIFICATION TEST SUBSEA BOP KILL SHEET DEVIATED WELL

DATE : _____

NAME : _____

UNITS : FIELD

FORMATION STRENGTH DATA:

SURFACE LEAK -OFF PRESSURE FROM
FORMATION STRENGTH TEST (A) _____ psi

DRILLING FLUID DENS. AT TEST (B) _____ ppg

MAX. ALLOWABLE DRILLING FLUID DENSITY =
(B) + $\frac{(A)}{\text{SHOE T.V. DEPTH} \times 0.052}$ = (C) _____ ppg

INITIAL MAASP =
((C) - CURR. DENS.) x SHOE T.V. DEPTH x 0.052
= _____ psi

CURRENT DRILLING FLUID:

DENSITY _____ ppg

SUBSEA BOP DATA:

MARINE RISER
LENGTH _____ ft

CHOKELINE
LENGTH _____ ft

DEVIATION DATA:

KOP M.D. _____ ft

KOP T.V.D. _____ ft

EOB M.D. _____ ft

EOB T.V.D. _____ ft

CASING SHOE DATA:

SIZE _____ in

M. DEPTH _____ ft

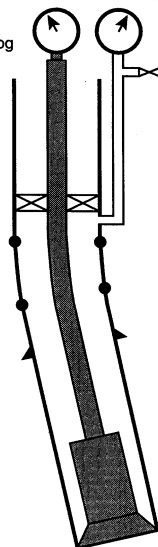
T.V. DEPTH _____ ft

HOLE DATA:

SIZE _____ in

M. DEPTH _____ ft

T.V. DEPTH _____ ft



PUMP NO. 1 DISPL.	PUMP NO. 2 DISPL.
bbl / stroke	bbl / stroke

(PL) DYNAMIC PRESSURE LOSS [psi]						
SLOW PUMP RATE DATA:	PUMP NO. 1			PUMP NO. 2		
	Riser	Choke Line	Friction Choke Line	Riser	Choke Line	Friction Choke Line
SPM						
SPM						

PRE-RECORDED VOLUME DATA:	LENGTH ft	CAPACITY bbl / ft	VOLUME bbl	PUMP STROKES stks	TIME minutes
DP - SURFACE TO KOP	x	=		(L) _____ stks	
DP - KOP TO EOB	x	=	+	(M) _____ stks	
DP - EOB TO BHA	x	=	+	(N ₁) _____ stks	
HEVI WALL DRILL PIPE	x	=	+	(N ₂) _____ stks	
DRILL COLLAR	x	=	+	(N ₃) _____ stks	
DRILL STRING VOLUME			(D) _____ bbl	_____ stks	_____ min
DC x OPEN HOLE	x	=			
DP / HWDP x OPEN HOLE	x	=	+		
OPEN HOLE VOLUME			(F) _____ bbl	_____ stks	_____ min
DP x CASING	x	=	(G) _____ bbl	_____ stks	_____ min
CHOKELINE	x	=	(H) _____ bbl	_____ stks	_____ min
TOTAL ANNULUS/CHOKELINE VOLUME		(F+G+H) = (I)	_____ bbl	_____ stks	_____ min
TOTAL WELL SYSTEM VOLUME		(D+I) = (J)	_____ bbl	_____ stks	_____ min
ACTIVE SURFACE VOLUME		(K)	_____ bbl	_____ stks	
TOTAL ACTIVE FLUID SYSTEM		(J+K)	_____ bbl	_____ stks	
MARINE RISER x DP	x	=	_____ bbl	_____ stks	

19-01-96
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KICK CONTROL WORKSHEET

DEVIATED WELL - SUB-SURFACE BOP - FIELD UNITS

PAGE 2 OF 3

IWCF CERTIFICATION TEST SUBSEA BOP KILL SHEET DEVIATED WELL

DATE : _____

NAME : _____

UNITS : FIELD

KICK DATA : SIDPP psi SICP psi PIT GAIN bbl

KILL FLUID DENSITY
KMD CURRENT DRILLING FLUID DENSITY + $\frac{\text{SIDPP}}{\text{TVD} \times 0.052}$
..... + x 0.052 = ppg

INITIAL CIRC. PRESS.
ICP DYNAMIC PRESSURE LOSS + SIDPP
..... + = psi

INITIAL DYNAMIC
CASING PRESS AT
KILL PUMP RATE **SICP - CHOKE LINE FRICTION**
= - = psi

FINAL CIRCULATING
PRESSURE
FCP $\frac{\text{KILL FLUID DENSITY}}{\text{CURRENT DRILLING FLUID DENSITY}} \times \text{DYNAMIC PRESSURE LOSS}$
..... x = psi

DYNAMIC PRESSURE
LOSS AT KOP (**O**) $PL + \left[(\text{FCP} - PL) \times \frac{\text{KOP}_{MD}}{\text{TD}_{MD}} \right] = \dots + \left[(\dots - \dots) \times \frac{\dots}{\dots} \right] = \dots$ psi

REMAINING SIDPP
AT KOP (**P**) $\text{SIDPP} - \left[(\text{KMD} - \text{OMD}) \times \text{KOP}_{TVD} \times 0.052 \right]$
= - $\left[(\dots - \dots) \times 0.052 \times \dots \right] = \dots$ psi

CIRCULATING PRESS.
AT KOP (**KOP CP**) $(O) + (P) = \dots + \dots = \dots$ psi

DYNAMIC PRESS. LOSS
AT EOB (**R**) $PL + \left[(\text{FCP} - PL) \times \frac{\text{EOB}_{MD}}{\text{TD}_{MD}} \right] = \dots + \left[(\dots - \dots) \times \frac{\dots}{\dots} \right] = \dots$ psi

REMAINING SIDPP
AT EOB (**S**) $\text{SIDPP} - \left[(\text{KMD} - \text{OMD}) \times \text{EOB}_{TVD} \times 0.052 \right]$
= - $\left[(\dots - \dots) \times 0.052 \times \dots \right] = \dots$ psi

CIRCULATING PRESS.
AT EOB (**EOB CP**) $(R) + (S) = \dots + \dots = \dots$ psi

$(T) = \text{ICP} - \text{KOP CP} = \dots - \dots = \dots$ psi $\frac{(T) \times 100}{(L)} = \frac{\dots \times 100}{\dots} = \dots$ psi
100 strokes

$(U) = \text{KOP CP} - \text{EOB CP} = \dots - \dots = \dots$ psi $\frac{(U) \times 100}{(M)} = \frac{\dots \times 100}{\dots} = \dots$ psi
100 strokes

$(W) = \text{EOB CP} - \text{FCP} = \dots - \dots = \dots$ psi $\frac{(W) \times 100}{(N_1 + N_2 + N_3)} = \frac{\dots \times 100}{\dots} = \dots$ psi
100 strokes

DEVIATED WELL - SUB-SURFACE BOP - FIELD UNITS

IWCF CERTIFICATION TEST
SUBSEA BOP KILL SHEET
DEVIATED WELL

UNITS : FIELD

A full-page sheet of white graph paper with a uniform black grid pattern. The grid consists of small squares covering the entire area. There are no margins, text, or other markings on the page.

STROKES —

← **STATIC & DYNAMIC DRILL PIPE PRESSURE [psi]**

[illegible]

KICK CONTROL WORKSHEET

DEVIATED WELL - SUB-SURFACE BOP - SI UNITS

PAGE 1 OF 3

IWCF CERTIFICATION TEST SUBSEA BOP KILL SHEET DEVIATED WELL

DATE : _____

NAME : _____

UNITS : S.I.

FORMATION STRENGTH DATA:

SURFACE LEAK -OFF PRESSURE FROM

FORMATION STRENGTH TEST

(A) kPa

DRILLING FLUID DENS. AT TEST

(B) kg/m³

MAX. ALLOWABLE DRILLING FLUID DENSITY =

(B) + $\frac{(A) \times 102}{\text{SHOE T.V. DEPTH}}$ = (C) kg/m³

INITIAL MAASP =

((C) - CURR. DENS.) x SHOE T.V. DEPTH
102 = _____ kPa

CURRENT DRILLING FLUID:

DENSITY _____ kg/m³

SUBSEA BOP DATA:

MARINE RISER

LENGTH _____ m

CHOKELINE

LENGTH _____ m

DEVIATION DATA:

KOP M.D. _____ m

KOP T.V.D. _____ m

EOB M.D. _____ m

EOB T.V.D. _____ m

CASING SHOE DATA:

SIZE _____ mm

M. DEPTH _____ m

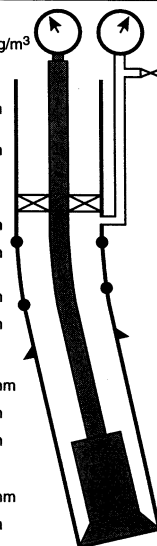
T.V. DEPTH _____ m

HOLE DATA:

SIZE _____ mm

M. DEPTH _____ m

T.V. DEPTH _____ m



PUMP NO. 1 DISPL.	PUMP NO. 2 DISPL.
m ³ / stroke	m ³ / stroke

(PL) DYNAMIC PRESSURE LOSS [kPa]

SLOW PUMP RATE DATA:	PUMP NO. 1			PUMP NO. 2		
	Riser	Choke Line	Friction Choke Line	Riser	Choke Line	Friction Choke Line
SPM						
SPM						

PRE-RECORDED VOLUME DATA:	LENGTH m	CAPACITY m ³ / m	VOLUME m ³	PUMP STROKES stks	TIME minutes
DP - SURFACE TO KOP	x	=		(L) _____ stks	
DP - KOP TO EOB	x	=	+	(M) _____ stks	
DP - EOB TO BHA	x	=	+	(N ₁) _____ stks	
HEVI WALL DRILL PIPE	x	=	+	(N ₂) _____ stks	
DRILL COLLAR	x	=	+	(N ₃) _____ stks	
DRILL STRING VOLUME			(D) _____ m ³	_____ stks	_____ min
DC x OPEN HOLE	x	=			
DP / HWDP x OPEN HOLE	x	=	+		
OPEN HOLE VOLUME			(F) _____ m ³	_____ stks	_____ min
DP x CASING	x	=	(G) _____ +	_____ stks	_____ min
CHOKELINE	x	=	(H) _____ +	_____ stks	_____ min
TOTAL ANNULUS/CHOKELINE VOLUME			(F+G+H) = (I) _____ m ³	_____ stks	_____ min
TOTAL WELL SYSTEM VOLUME			(D+I) = (J) _____ m ³	_____ stks	_____ min
ACTIVE SURFACE VOLUME			(K) _____ m ³	_____ stks	
TOTAL ACTIVE FLUID SYSTEM			(J+K) _____ m ³	_____ stks	
MARINE RISER x DP	x	=	_____ m ³	_____ stks	

19-01-96

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KICK CONTROL WORKSHEET

DEVIATED WELL - SUB-SURFACE BOP - SI UNITS

PAGE 2 OF 3

IWCF CERTIFICATION TEST SUBSEA BOP KILL SHEET DEVIATED WELL

DATE : _____

NAME : _____

UNITS : S.I.

KICK DATA :

SIDPP kPa

SICP kPa

PIT GAIN m³

KILL FLUID DENSITY

CURRENT DRILLING FLUID DENSITY + $\frac{102 \times \text{SIDPP}}{\text{TVD}}$

KMD

..... + $\frac{102 \times \text{.....}}{\text{.....}} = \text{..... kg/m}^3$

INITIAL CIRC. PRESS.

DYNAMIC PRESSURE LOSS + SIDPP

ICP

..... + = kPa

INITIAL DYNAMIC
CASING PRESS AT
KILL PUMP RATE

SICP - CHOKE LINE FRICTION

= - = kPa

FINAL CIRCULATING
PRESSURE
FCP

$\frac{\text{KILL FLUID DENSITY}}{\text{CURRENT DRILLING FLUID DENSITY}} \times \text{DYNAMIC PRESSURE LOSS}$

..... x = kPa

DYNAMIC PRESSURE
LOSS AT KOP (O)

$\text{PL} + \left[(\text{FCP} - \text{PL}) \times \frac{\text{KOP}_{\text{MD}}}{\text{TD}_{\text{MD}}} \right] = \text{.....} + \left[(\text{.....} - \text{.....}) \times \frac{\text{.....}}{\text{.....}} \right] = \text{..... kPa}$

REMAINING SIDPP
AT KOP (P)

$\text{SIDPP} - \left[\frac{(\text{KMD} - \text{OMD}) \times \text{KOP}_{\text{TVD}}}{102} \right]$
= - $\left[\frac{(\text{.....} - \text{.....}) \times \text{.....}}{102} \right] = \text{..... kPa}$

CIRCULATING PRESS.
AT KOP (KOP CP)

(O) + (P) = + = kPa

DYNAMIC PRESS. LOSS
AT EOB (R)

$\text{PL} + \left[(\text{FCP} - \text{PL}) \times \frac{\text{EOB}_{\text{MD}}}{\text{TD}_{\text{MD}}} \right] = \text{.....} + \left[(\text{.....} - \text{.....}) \times \frac{\text{.....}}{\text{.....}} \right] = \text{..... kPa}$

REMAINING SIDPP
AT EOB (S)

$\text{SIDPP} - \left[\frac{(\text{KMD} - \text{OMD}) \times \text{EOB}_{\text{TVD}}}{102} \right]$
= - $\left[\frac{(\text{.....} - \text{.....}) \times \text{.....}}{102} \right] = \text{..... kPa}$

CIRCULATING PRESS.
AT EOB (EOB CP)

(R) + (S) = + = kPa

(T) = ICP - KOP CP = - = kPa

$\frac{(T) \times 100}{(L)} = \frac{\text{.....} \times 100}{\text{.....}} = \text{..... kPa}$
100 strokes

(U) = KOP CP - EOB CP = - = kPa

$\frac{(U) \times 100}{(M)} = \frac{\text{.....} \times 100}{\text{.....}} = \text{..... kPa}$
100 strokes

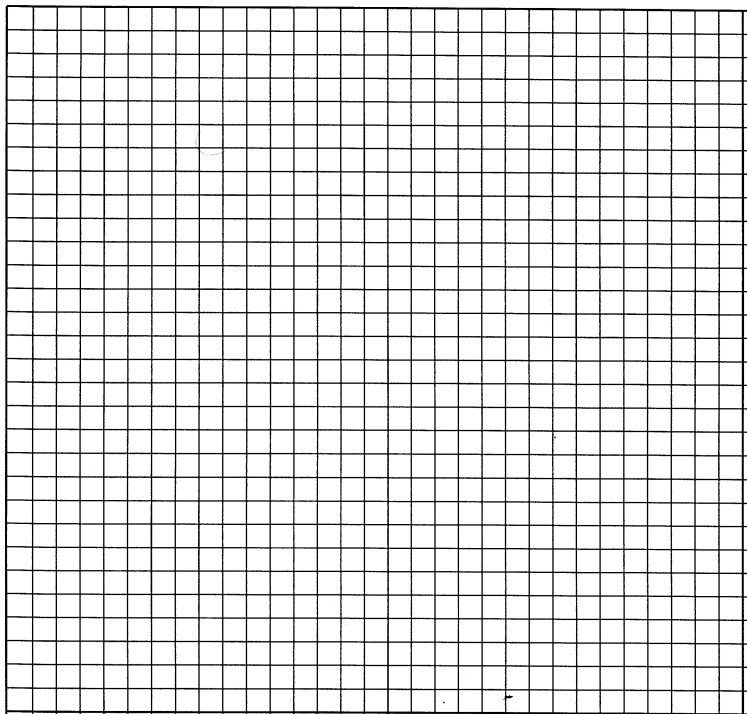
(W) = EOB CP - FCP = - = kPa

$\frac{(W) \times 100}{(N_1 + N_2 + N_3)} = \frac{\text{.....} \times 100}{\text{.....}} = \text{..... kPa}$
100 strokes

DEVIATED WELL - SUB-SURFACE BOP - SI UNITS

PAGE 3 OF 3

DATE : _____
NAME : _____
UNITS : S.I.



STROKES →

← [kPa] STATIC & DYNAMIC DRILL PIPE PRESSURE

[illegible]

G – STUCK PIPE AND FISHING

Clickable list

(Use the hierarchical list under "Bookmarks" to access individual tables and/or sub-topics)

Avoid stuck pipe	G-1
Sticking mechanisms	G-2
Free point location	G-3
Backing off	G-5
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Recovery of tubular fish	G-11
Recovery of a wireline fish	G-12
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Freeing stuck pipe with hydrochloric acid	G-20

STUCK PIPE & FISHING

The first rule is

AVOID STUCK PIPE !

Stuck pipe is a major cause of non-productive time and costs. Well Engineering personnel are strongly recommended to obtain and read the "ABC of Stuck Pipe" series of reports (numbers EP91-1908, EP93-1908 & EP94-1908). Some general points which have been culled from those reports are given below (see also the advice given at the beginning of Sections C and E)

- Design your drill string to allow a minimum of 50 kdaN overpull, taking drag fully into account.
- Develop and update a drag chart for all deviated wells.
- Ensure that drillers and assistant drillers are conversant with the different sticking mechanisms that could be encountered in your well and their first actions if the pipe does become stuck.
- Ensure that key personnel are fully conversant with the operating procedures of the jars you are using.
- Use BHAs with well stabilised lightweight drill collar sections, using HWDP in compression providing it remains within its critical buckling load (hole inclination dependant).
- Use barrel shaped stabilisers and back reaming tools where appropriate.

STUCK PIPE & FISHING

STICKING MECHANISMS

Sticking mechanisms can be grouped into three categories.

Geometry

- Types : Undergauge hole, keyseat, assembly too stiff, ledges, mobile formations.
- Symptoms : Problem occurs when moving the string, affects motion in one direction only and does not affect circulation.
- First Action : Attempt to work free in the opposite direction to the direction of movement when the string became stuck. Gradually increase the force used (setdown, overpull, jarring, torque).

Solids

- Types : Settled cavings and cuttings, hole collapse, reactive formations, geopressured formations, fractured and faulted formations, junk, cement blocks, soft cement.
- Symptoms : Problem mainly occurs when pulling out, affects motion in one direction, is often associated with inadequate hole cleaning and often results in restriction of circulation.
- First Actions : Attempt to work free in the opposite direction to the direction of movement when the string became stuck. Gradually increase the force used (set-down, over-pull, jarring, torque). Break circulation as soon as possible (be aware of FBG, pump out forces opposing attempts to go down, effect of pump open forces on jar operation).

Differential Sticking – refer also to [page I-14](#)

- Conditions required : Permeable zone covered with mud filter cake, static overbalance, wall contact, stationary string.
- Promoted by : Inadequate stabilisation, long drill collar sections.
- Symptoms : String becomes stuck while stationary, sometimes after a very brief time. Circulation is unaffected.
- First Actions : Work pipe with MAXIMUM FORCE as soon as possible (the sticking force will increase rapidly with time) up or down. If possible, reduce the overbalance.

FREE POINT LOCATION (1)

There are two methods for estimating the depth at which a string is stuck.

- by measuring the pipe stretch under tension
- by locating the free point with a free point indicating tool

Measuring the pipe stretch under tension

The method is based upon Hooke's Law. Knowing the stretch under a particular tensile load enables the unstretched length to be calculated. This equals the length of pipe between the stuck point and surface. In practice the length of free pipe remaining in a straight hole is estimated by applying two different tensions to the string and measuring the difference in the resulting stretches. This is done in order to ensure that the stretch measured is actual stretch and is not due to straightening buckled pipe.

The string should be pulled until the weight reading is at least equal to the pre-stuck situation. When this weight is pulled the string is marked at a point level with the rotary table. Then a known amount of additional pull is applied and the string marked again. The amount of overpull is obviously limited by the maximum allowable pull on the pipe.

The applicable equation is :

$$L = \frac{K \cdot W_{dp} \cdot e}{P} \quad \text{Where :}$$

L	= Length of free pipe	SI units	field units
		metres	feet
W_{dp}	= Plain end pipe weight (see page C-2)	kg/m	lbs/ft
e	= Differential stretch	mm	inches
P	= Differential pull	kN	lbs
K	=	26.37	735,294

Reasonable estimates of the depth of a stuck point in near-vertical holes can be obtained in this way. The values obtained are less reliable as the deviation increases due to a) down hole friction and b) the support provided by the bore hole wall. Another minor inaccuracy is introduced by neglecting the changing cross-section of the string at the upsets and tool joints.

Related to the stretch of stuck pipe is the stretch of a length of pipe suspended in a liquid due to its own weight.

The applicable equation is :

$$e = \frac{L^2}{K_2} (K_1 - 1.44 \rho_{df}) \quad \text{Where :}$$

e	= Differential stretch	SI units	field units
		mm	inches
L	= Length of suspended pipe	metres	feet
ρ_{df}	= Drilling fluid gradient	kPa/m	psi/ft
K_1	=	77.0	3.40
K_2	=	4.12×10^5	5.00×10^6

FREE POINT LOCATION (2)

Utilisation of a free-point indicating tool

A stuck- or free-point indicator service is offered by the wireline logging companies. A sensitive electronic strain gauge is run on the logging cable within the stuck string and anchored to the inner surface of the pipe. Tension and torque are then applied to the string at the surface and the strain gauge readings are transmitted to surface, indicating whether the pipe reacts at that depth to the applied tension and the applied torque. By repeating this procedure the deepest point to which tension can be transmitted can be identified, and similarly the deepest point to which torque can be transmitted. These are the points below which the pipe cannot be moved up or rotated respectively. The effective stuck point is the lower of these.

Note that pipe which appears to be free in tension does not always react to applied torque, and vice versa. A back-off can only succeed if the pipe is free in both senses.

Separate slim acoustic logs are available that are designed to indicate intervals of stuck, partially stuck or free pipe which may exist below the upper stuck point.

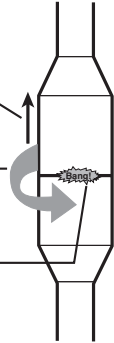
BACKING OFF

PROCEDURE

Drillpipe or collars can be unscrewed downhole by exploding a charge known as a string-shot (prima-cord folded up inside a piece of tubular plastic) inside a selected tool-joint connection, just above the stuck point. A connection should be selected which has been broken during the round trip prior to the pipe becoming stuck.

A successful back-off depends upon having the following :

- zero or slightly positive tension at the joint
- sufficient left-hand, or reverse torque at the joint - 50% to 75% of make-up torque is suggested
- a sufficiently large explosive charge, accurately located at the joint



For a safe operation carry out the following checks :

- ensure that tong and slips dies are clean, sharp and the proper size for the string above the rotary
- check that tong, snub and jerk lines are in excellent condition
- ensure that slip handles are tied together with strong line, to prevent the slips being kicked out of the table and thrown clear when the pipe breaks out
- ensure that elevators are latched around the pipe and slackened off under a tool joint with the hook locked when torque is being applied to the string
- ensure that no torque remains in the string when it is picked out of the slips, unless the pipe is properly held with a back-up tong

Particular care should always be taken when applying torque or releasing it from the string. Keep the forces involved fully under control and keep men out of the potentially dangerous area.

The following two pages give information about the tension and torque to be applied.

Note: Torque should be worked down the string before the string shot is fired, this may take some time. If the string fails to back off after firing the charge, continue to work the torque down the string before trying another string shot.

BACKING OFF

MAINTAINING THE APPROPRIATE TENSION

The ideal tensile load is zero, i.e. with the threads subject to neither compression nor tension. However, since a zero tensile load is difficult to achieve, pull is applied which will develop a slight tension rather than compression. Over the years there has been some debate regarding the surface pull required to achieve this condition. Since the pipe is held down then it can be assumed that buoyancy does not affect the pipe above the stuck point. However, as soon as the joint is cracked buoyancy will act on the freed pipe.

If buoyancy does not apply then the pull required to maintain the drillpipe in tension will be the total weight of pipe above the stuck point plus the weight of other equipment such as blocks.

An alternative method for finding the required pull is to use the actual hook load observed by the Driller just before getting stuck :

$$\text{Required Pull} = \frac{\text{Hook load} - \text{weight of blocks} - \text{weight of fish in mud}}{\text{Buoyancy Factor}} + \text{weight of blocks}$$

In deviated wells with excessive drag and pull it will be difficult to develop the correct tension at the joint, and more than one attempt may be necessary before a successful back-off is achieved. In a highly deviated well the pipe weight may be partially supported.

If the hook load while moving the string slowly up has been observed prior to becoming stuck, the following method can be used to estimate the required pull:

- Calculate the theoretical weight of the whole string in air (using approximate weight for drillpipe)
- Subtract from this the observed weight of the string (hook load – blocks)
- This gives the weight loss due to buoyancy, friction and wall support which can be expressed as a percentage.
- Calculate the theoretical weight of pipe in air down to the stuck point (using approximate weights - see [page C-2](#)) then subtract the percentage weight loss due to buoyancy and wall support etc.
- Add the weight of the blocks etc. and this will be the tension prior to back-off.

BACKING OFF TORQUE

TORQUE VERSUS NUMBER OF TURNS (PIPE BODY)

Torque in N-m(lbs-ft) = K x turns/100m (turns/1000 ft)

where K is given in the following table:

Outside Diameter		Inside Diameter		Nominal Weight		K-factor new-pipe		K-factor premium-pipe	
inch	mm	inch	mm	lbs/ft	kg/m	field units	S.I. units	field units	S.I. units
3 1/2	88.9	2.764	70.2	13.30	19.79	4,600	19,500	3,410	14,400
3 1/2	88.9	2.602	66.1	15.50	23.07	5,230	22,100	3,800	16,100
4 1/2	114.3	3.958	100.5	13.75	20.46	8,260	35,000	6,350	26,900
4 1/2	114.3	3.826	97.2	16.60	24.70	9,820	41,500	7,460	31,600
4 1/2	114.3	3.640	92.5	20.00	29.76	11,800	49,700	8,790	37,200
5	127.0	4.408	112.0	16.25	24.18	12,400	52,400	9,550	40,300
5	127.0	4.276	108.6	19.50	29.02	14,600	61,700	11,100	47,000
5	127.0	4.000	101.6	25.60	38.10	18,500	78,300	13,800	58,300

Note :

in S.I. units : $K = 0.00051 (D^4 - d^4)$ [D and d in mm]

in field units : $K = 50.16 (D^4 - d^4)$ [D and d in inches]

These factors are based on a shear modulus of 8.274×10^{10} N/m² (11.71×10^6 psi)

Example

S.I. units :

127 mm IEU 29.02 kg/m, grade E, premium class drill pipe with NC50 tool joints.

Stuck at 3,630 m

The approximate weight (see [page C-9](#)) of the DP is 28.9 kg/m

The weight of free pipe in air is

$$3,630 \times 28.9 \times 9.81/10 = 102,900 \text{ daN}$$

Using a design factor of 1.15 the allowable torque is 1,850 daN-m ([page C-43](#))

$$\text{Turns per 100 m} = (1,850 \times 10) / 47,000 = 0.394$$

$$\text{Number of turns is } 0.391 \times 36.30 = 14.3$$

Field units :

5" IEU 19.5 lbs/ft, grade E, premium class drill pipe with NC50 tool joints.

Stuck at 11,900 ft.

The approximate weight (see [page C-7](#)) of the DP is 19.4 lbs/ft

The weight of free pipe in air is

$$11,900 \times 19.4 = 230,900 \text{ lbs.}$$

Using a design factor of 1.15 the allowable torque is 13,300 lbs-ft ([page C-42](#))

$$\text{Turns per 1000 ft} = 13,300 / 11,100 = 1.20$$

$$\text{Number of turns is } 1.20 \times 11.9 = 14.3$$

Note:

Remember that if the tool joint make-up torque is less than the allowable pipe body torque then when applying left hand torque the pipe may back off before the allowable pipe body torque has been reached. If this is not desired the upper torque limit is determined by the lowest actually used tool joint make up torque, reduced by a safety factor.

FISHING TOOLS

GENERAL

Type of fishing job	Type of fishing tool	Names of tools
Recovery of tubular Fish	Connecting tools	
	External catch	Overshot Die collar
	Internal catch	Taper tap (poor class of tool: overshot always preferable if available) Spear (provides very good connection,
	Accessories	Bent drillpipe single Hydraulic knuckle joint Hydraulic wall hook Wall hook
	Washover tools	Washover safety joint Washover pipe Washover shoe
	Force multiplier tools	Jar, hydraulic or mechanical Bumper sub Surface bumper-jar Accelerator Hydraulic pulling tool
Recovery of fish	Disengagement tools	Safety joint Bumper safety joint External tubing/drillpipe cutter Internal tubing/drillpipe cutter Flash cutter (Schlumberger, etc.) Jet cutter (Halliburton, etc.) Chemical cutter (Baroid, etc.) Electrical cable back-off (Schlumberger, etc.)
	Information tools	Impression block Free-point indicator
Recovery of non-tubular fish		Junk basket Circulating junk basket Reverse circulating globe-type basket Magnet Wireline spear Junk sub
Fish destruction		Milling shoe Packer retriever Section mill Jet bottom-hole cutter

FISHING TOOLS

SPECIFIC

Listed below are fishing tools often kept on the rig site for various hole sizes drilled.

Fishing Tools for 26" - 17¹/₂" - 12¹/₄" Holes

- 8" Hydraulic jar 6⁵/₈" Reg. pin x box
- 8" Accelerator 6⁵/₈" Reg. pin x box
- 8" Fishing bumper sub 6⁵/₈" Reg. pin x box
- 7" Surface jar 4¹/₂" IF pin x box
- 11³/₄" Overshot, c/w extension subs and 15" & 22" guides, to catch 9¹/₂" & 8¹/₄" DCs, 5" DP & 6⁵/₈" tool joints.
- 11¹/₄" Reverse circulating basket 6⁵/₈" Reg. box
- 12" Magnet 6⁵/₈" Reg. pin (optional)
- 9¹/₂" Junk sub 6⁵/₈" Reg. box x box
- 8¹/₈" Overshot, c/w extension sub and 11" guides to catch 5" DP+ 6³/₈" tool joints.
- 11¹/₄" Globe basket (or equivalent)
- 8" circulating sub 6⁵/₈" Reg. pin x box

Fishing Tools for 8¹/₂" hole

- 6¹/₄" Hydraulic jar 4" IF pin x box
- 6¹/₄" Accelerator 4" IF pin x box
- 6¹/₄" Fishing bumper sub 4" IF pin x box
- 7" Surface jar 4¹/₂" IF pin x box
- 8¹/₈"/7⁷/₈" Overshots, c/w extension subs to catch 5" DP, 6¹/₄" DCs & 6³/₈" tool joints
- 7⁷/₈" Reverse circulating basket 4" IF box
- 8" Magnet 4¹/₂" Reg. pin
- 6⁵/₈" Junk sub 4¹/₂" Reg. box x 4" IF box up
- 7⁷/₈" Globe basket (or equivalent)
- 6¹/₄" circulation sub 4" IF pin x box

Fishing tools for 5⁷/₈" or 6" holes

- 4³/₄" Hydraulic jar 3¹/₂" IF pin x box
- 4³/₄" Accelerator 3¹/₂" pin x box
- 4³/₄" Fishing bumper sub 3¹/₂" pin x box
- 7" Surface jar 4¹/₂" IF pin x box
- Sub 3¹/₂" IF pin x 4¹/₂" IF box
- Sub 4¹/₂" IF pin x 3¹/₂" IF box
- 5⁵/₈" Overshot, c/w extension subs to catch 3¹/₂" DP, 4³/₄" DCs & tool joints
- 5⁵/₈" Reverse Circulating basket 3¹/₂" IF box
- 5" Magnet 3¹/₂" Reg. pin (optional)
- 5¹/₂" Junk sub 3¹/₂" Reg. box x 3¹/₂" IF box
- 5⁷/₈" Junk mill 3¹/₂" Reg. pin up
- 4³/₄" circulation sub 3¹/₂" IF pin x box

FISHING ASSEMBLIES

The choice of fishing tools to use in a fishing assembly is directly related to the prospective efficiency of the operation. In short it is better to fish for a longer time with a high chance of success rather than do a quick fishing operation with low chances of success. Experience has narrowed the choice of commonly used fishing tools and assemblies to a few practical combinations (see the previous page). A typical standard fishing assembly would consist of the following:

OVERSHOT	or, if back off achieved before fishing, a screw in connection is preferred. Data on a common type of overshot can be found on page G-14
BUMPER SUB HYDRAULIC JAR	Data on a common type of jar can be found on page G-16 .
DRILL COLLARS (JAR INTENSIFIER OR ACCELERATOR)	equal to weight of fish in hole. If an accelerator is used a lower weight is required. Data on a common type of accelerator, including the reduced DC weight requirement, can be found on page G-19 .
HWDP	optional
DP	
KELLY	should always be used if heavy jarring or high over-pulls are necessary for the operation.

Where losses are expected the use of a circulation sub in the fishing assembly should be considered.

RECOVERY OF TUBULAR FISH

GENERAL POINTS ON RECOVERY OF TUBULAR FISH USING CONNECTING TOOLS

Standard Assembly

A typical fishing assembly when using connecting tools will consist of the catching tool plus fishing bumper sub, jar, drill collars and accelerator. When a non-releasing tool such as a tap or die collar is being employed as the catching tool, the assembly should also include a safety joint between the catching tool and jar. However, since the safety tool will not transmit reverse torque, it would not be possible to back off below it using a string shot. The bore of the tools run above the overshot should be large enough to allow the passage of a cutting tool or back-off shot that can operate within the fish.

Circulation

If the string parts while drilling, the annulus may be loaded with cuttings. It may be useful to circulate the hole clean above the fish before pulling out. This will prevent sand and cuttings settling around the top of the fish. However if you circulate at only one place close above the fish there is a risk of enlarging the hole, thus the circulation should be done in several stages at different levels above the fish during the trip out of the hole.

A good pack-off or seal in the connecting tool is a valuable asset because once a fish is engaged it is good practice to circulate through it if possible, particularly if potential reservoirs are exposed. If possible, you should circulate bottoms-up before pulling out with a fish to ensure that the hole is gas-free. Well control is particularly important when tripping out because overshot and fish together make a good swabbing assembly.

Size of guide shoe and grapple.

A guide-shoe should be used with the overshot having an outside diameter approximately 25 mm/1 inch less than the hole size. This prevents bypassing the fish.

The recovered part of the string will give a good indication of the dimensions of the top of the fish remaining in the hole. If an overshot grapple can be pushed over it by hand it is too large and a size smaller should be run. Where possible use the stronger spiral grapple in preference to the basket type. (Refer to the Bowen Instruction Manual No 5/1150).

Make sure that overshots and suitable grapples are on-site for all relevant combinations of hole size and component OD.

WIRELINE FISHING (overstripping)

Logging tools may become stuck downhole, for different reasons :

- Hole collapsing or loose formation
- Hole bridging
- Torpedo or cable head caught in a key seat
- Cable or tool differentially stuck
- Tool stopped in a split casing shoe.

Once the tool is stuck, **pulling on the cable does not help**; on the contrary it will definitely trap the tool for good!

When the wireline is still intact it is best to use a cable guide technique: the wireline will hold the fish in a centralised position and serve as a guide for the overshot.

The “cut and thread technique”

This method has a potential of 100% recovery if the proper procedures are followed.

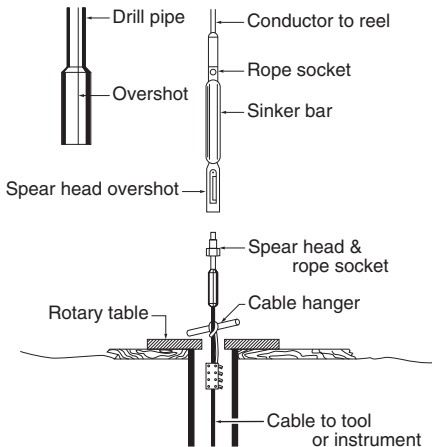


Figure 1 : The cable guide fishing assembly

1. Preparing the line

The cable is set under tension to remove any slack and the cable hanger, which will rest on the rotary table, is clamped on the cable. The cable is then cut 2-3 m (6-10 ft) above the hanger, and a spearhead rope socket is made on the end of the cable remaining in the well. Allow for sufficient excess line ! A rope socket, sinker bar and spear head overshot are made up on the end of cable hanging in the derrick (Figure 1). With the overshot engaged to the spearhead, the wireline can be put under tension again. When the cable hanger is removed a C-plate is used to hang the cable in the rotary table.

2. Threading the cable through the drillpipe

The spearhead overshot is released and drawn up to the monkey board. The stand of drillpipe with an overshot dressed to fish the logging tool is picked up and held over the rotary table. The derrick man guides and sends the spear head overshot down the stand of drillpipe. The spear head overshot is attached to the spear head in the rotary. A little strain is pulled on the cable and the C-plate is removed. The drillpipe is then lowered through the rotary table and set in the slips. The C-plate is placed on top of the drillpipe tool joint sticking up in the rotary table. The spear head overshot is released, pulled up to the monkey board and fed into the next stand of drill pipe. This procedure is repeated until the overshot is within a short distance of the fish (Figure 2).

3. Approaching the fish

A special circulating head is installed on the last stand and circulation is started to clean the end of the pipe, the overshot and the top of the fish. The fish is then engaged; a record of pump strokes per minute versus pressure will indicate if the fish is caught in the overshot.

4. Breaking the weak point

Once established that the fish is caught the cable hanger is clamped on the cable below the rope sockets, the rope sockets removed and the hanger is set in the elevators. The weak point is broken by pulling on the cable with the elevators. The cable is pulled out of the drill pipe. The string is then pulled out of the hole with the fish attached.

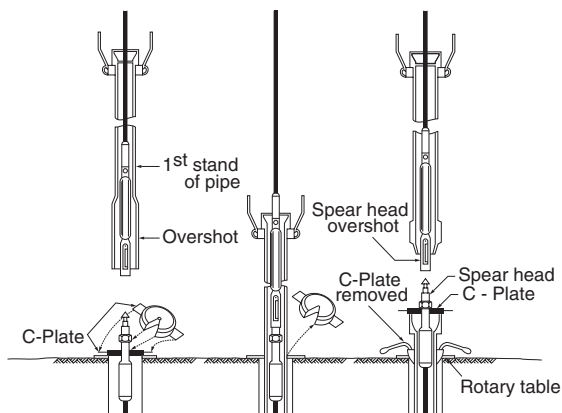


Figure 2 : Cable guide fishing method

Note : Never try to break the weak point in a wire line by pulling with the winch. The greatest tension in a wireline is at the surface and if the line parts there rather than at depth the recoil will be violent.

SERIES 150 BOWEN RELEASING AND CIRCULATING OVERSHOT

The Series 150 Bowen releasing and circulating overshot has a simple and rugged construction that has made it one of the more popular tools available to externally engage, pack off and pull a fish.

It has three body parts; the top sub, the bowl, and the guide. The basic overshot may be dressed with either of two sets of internal parts, depending on whether the fish to be caught is near maximum size for the particular overshot.

If the fish diameter is near the maximum catch of the overshot, a spiral grapple, spiral grapple control and type "A" packer are used. If it is considerably below maximum catch size (usually $1/2$ "), a basket grapple and a mill control packer are used.

For a list of the available overshot sizes, and details of the accessories, you should refer to the current Bowen Tools Inc. catalogue.

Gripping and releasing mechanism

The bowl of the overshot is designed with helically tapered spiral section in its inside diameter. The gripping member (spiral grapple or basket grapple), is fitted into this section. When an upward pull is exerted against a fish, the expansion and compression forces are spread evenly over long sections of the bowl and fish respectively, minimising damage to, and distortion of, both overshot and fish.

A spiral grapple is formed as a left-hand helix, whereas a basket grapple is an expandable cylinder. Both have a tapered exterior, to conform to the helically tapered section in the bowl, and a wickered interior for engagement with the fish.

Three types of basket grapple are available to meet the need for catching various types of fish:

- The plain basket grapple (as shown) is wickered for its entire interior length. It is used to catch any plain single diameter fish.
- The basket grapple with long catch stop has an internal shoulder located at the upper end to stop the fish in the best catch position. It is designed to stop and catch collars and tool joints, with sufficient length left below the grapple to allow the joint to be packed-off with a basket control packer.
- The basket grapple with short catch stop has a double set of wickers, of two different internal diameters. It is used to stop and catch a coupling with a ruptured piece of pipe engaged in its upper end. The upper set of wickers will catch the ruptured pipe, and act as a stop against the coupling, while the lower set of wickers will catch the coupling.

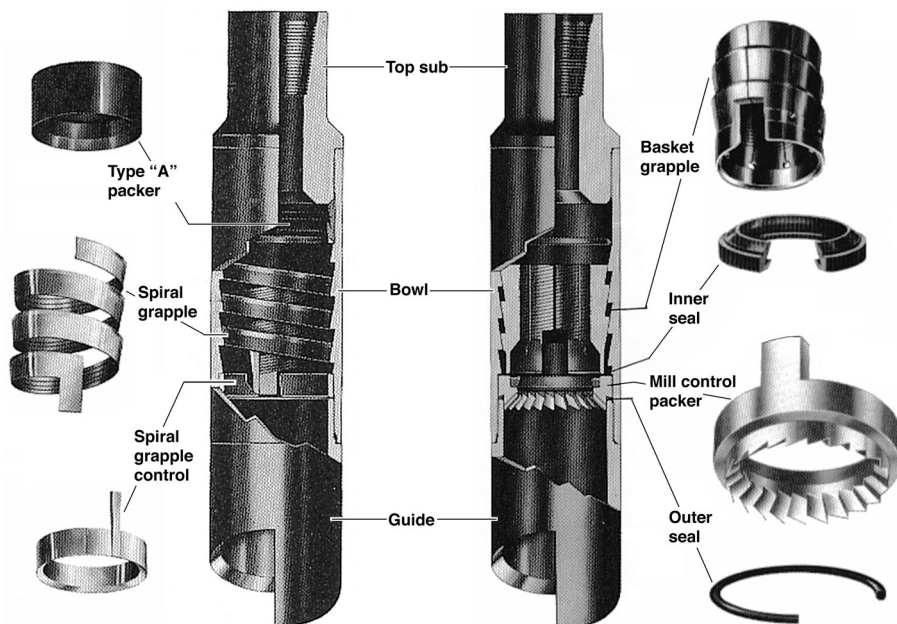
Grapple controls are of two types corresponding to the type of grapple used. They are used as a special key, to allow the grapple to move up and down during operation while simultaneously transmitting full torque from the grapple to the bowl. Spiral grapple controls are always plain; basket grapple controls may be either plain or include a pack-off. In addition to the pack-off, they may include mill teeth, as shown in the figure opposite - see "Pack-off mechanism" below.

In operation, the overshot functions in the same manner whether dressed with spiral grapple parts or basket grapple parts.

Pack-off mechanism

The type of pack-off used depends on how the overshot is dressed.

- A type "A" packer is used when the overshot is dressed with a spiral grapple. This is a sleeve type sealing at its O.D. against the inside of the bowl. It has an internal lip which seals around the fish.
- Control packers are used when the overshot is dressed with a basket grapple. A plain control packer is used when the milling operation has already been performed prior to the fishing operation. A mill control packer is used when light dressing is required prior to engagement of the fish.
- Plain controls are used when no pack-off is required. They are installed in the same location as the control packer.



Operating procedures

During the engaging operation, as the overshoot is rotated to the right and lowered, the grapple will expand when the fish is engaged, allowing the fish to enter the grapple. Thereafter, with rotation ceased and upward pull exerted, the grapple is contacted by the tapers in the bowl and its deep wickers grip the fish firmly.

During the releasing operation, a sharp downward bump places the larger portion of the bowl tapers opposite the grapple, breaking the hold. Thereafter, when the overshoot is rotated to the right, and slowly elevated, the wickers will screw the grapple off the fish, effecting release.

The fact that these overshoots require right hand rotation only, during both engaging and releasing operations, is an important feature that eliminates the risk of backing off the string.

- To engage and pull the fish:

Connect the overshoot to the fishing string and run it in the hole. As the top of the fish is reached, slowly rotate the fishing string to the right and gradually lower the overshoot over the fish. Allow the right-hand torque to slack out of the fishing string and pull on the fish by elevating the fishing string. If the fish does not come, start the circulating pumps and maintain a heavy upward strain while fluid is forced through the fish.

- To release from the fish:

Drop the weight of the fishing string heavily against the overshoot, then simultaneously rotate to the right and slowly elevate the fishing string until the overshoot is clear of the fish.

To release from a recovered fish, follow the same procedure while holding the fish below the overshoot.

HOUSTON ENGINEERS “HYDRA-JAR”

OPERATING PROCEDURES

The Houston Engineers “Hydra-Jar” is a hydraulic, double acting drilling jar that can also be used for fishing operations. The following are the operational procedures for its use.

To jar up

- Establish the jarring-up force, which should not exceed the maximum detent working load (given in the accompanying specifications table).
- Reduce the weight down by 15,000-20,000 lbs at the jar to set the jarring-up cycle.
- Pick up again immediately to the up-weight of the total string minus the weight below the jar plus the specified jarring-up force.
- Set the brake, and wait for the Hydra-Jar to fire (30 to 60 seconds). There will be a small loss of indicator weight due to jar travel.
- Once the jar has fired, additional pull can be applied up to the limits of the drill string.

To jar down

- Establish the jarring-down force, which should not exceed the maximum detent working load (given in the accompanying specifications table) or the weight of the drill collars and heavy wall drill pipe above the Hydra-Jar.
- Set down to the down-weight of the total string minus the weight below the jar minus the specified jarring-down force minus the “pump open” effect (see below).
- Wait for the jar to fire.

To jar down again

- Pull up 15,000 to 20,000 lbs on the jar to set the down cycle. Set weight down as described above. Wait for the jar to fire.

To jar faster (or slower)

- Use less (or more) weight to set the Hydra-Jar.

Pump-open force.

The design of the jar is such that a differential pressure between the inside and outside of the jar will create an upwards thrust on it, known as the “pump-open” force. This reduces the jarring-down force and has to be compensated for by increasing the weight set down on the jar. The amount of this “pump-open” force for the various sized tools is shown in the graph on [page G-18](#).

Note:

The specifications of the “Hydra-Jar”, and the above procedures, have been taken from Houston Engineers documentation.

The procedures may be different for other types of jar - you should always check the specifications of, and procedures for, the particular jar that you have in the hole.

HOUSTON ENGINEERS "HYDRA-JAR"

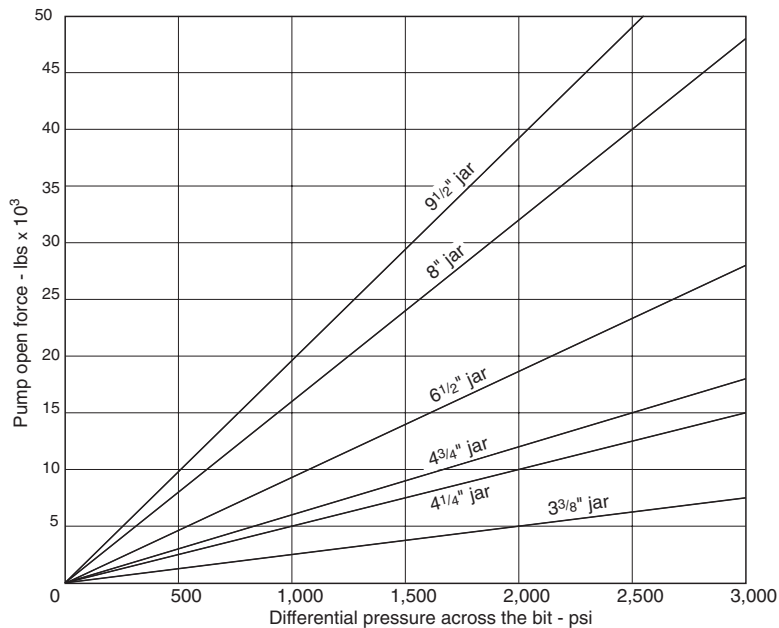
SPECIFICATIONS

Tool OD	inches mm	31/8 79.4	33/8 85.7	41/4 108.0	43/4 120.7	61/4 158.8	61/2 165.1	7 177.8	71/4 184.2	73/4 196.9	8 203.2	81/4 209.6	81/2 215.9	91/2 241.3
Tool ID	inches mm	11/4 31.8	11/2 38.1	2 50.8	21/4 57.2	23/4 69.9	23/4 69.9	23/4 69.9	23/4 69.9	3 76.2	3 76.2	3 76.2	3 76.2	3 76.2
Tool joint connections		23/8" API Reg	23/8" QPI IF	27/8" API IF	31/2" API IF	41/2" XH	41/2" API IF	5" H90	51/2" H90	65/8" API Reg	65/8" API Reg	65/8" API Reg	65/8" API Reg	75/8" API Reg
Overall length "extended"	ft-ins mm	25' 1" 7,645	24' 5" 7,442	29' 10" 9,093	29' 10" 9,093	31' 2" 9,500	31' 2" 9,500	31' 6" 9,601	31' 6" 9,601	32' 9,754	32' 9,754	32' 9,754	32' 9,754	32' 6" 9,906
Max. detent working load	lbs x 10 ³ N x 10 ³	45 200	44 196	70 311	80 356	150 667	175 778	230 1,023	240 1,068	260 1,156	300 1,334	350 1,557	350 1,557	500 2,224
Tensile yield strength	lbs x 10 ³ N x 10 ³	210 934	233 1,034	310 1,379	460 2,046	730 3,247	900 4,003	1100 4,893	1200 5,338	1300 5,782	1600 7,117	1700 7,562	1700 7,562	2000 8,896
Torsion yield strength	lbs-ft x 10 ³ N-m x 10 ³	8.5 11.5	6.1 8.3	16.0 21.7	21.0 28.5	50.0 67.8	61.0 82.7	80.0 108	97.0 132	118 160	118 160	118 160	118 160	200 271
Up stroke	inches mm	6 152	7 178	8 203	8 203	8 203	8 203	8 203	8 203	8 203	8 203	8 203	8 203	8 203
Down stroke	inches mm	6 152	7 178	7 178	7 178	7 178	7 178	8 203	8 203	7 178	7 178	8 203	8 203	8 203
Total stroke	inches mm	18 457	21 533	25 635	25 635	25 635	25 635	25 635	25 635	25 635	25 635	25 635	25 635	25 635
Tool weight	lbs Kg	350 159	500 227	800 362	1,050 476	1,600 725	1,850 839	2,600 1,180	3,000 1,360	3,200 1,450	3,550 1,610	4,000 1,810	4,500 2,040	5,600 2,540

The torsion yield strength is based on the tool joint connection. The tensile yield, torsion yield and maximum overpull values are calculated per API RP7G, utilising the published yield strength of the material. In critical cases the service company (Houston Engineering Inc.) should be consulted.

HOUSTON ENGINEERS “HYDRA-JAR”

“PUMP OPEN” FORCES



BOWEN JAR INTENSIFIERS

GENERAL DATA

Intensifier assembly	O.D. inches	I.D. inches	Connection	Stroke (inches)	Recommended DC weight range (lbs)	Pull load to open fully (lbs)	Minimum pull required (above weight of string and collars) to obtain effective blow (Lbs)	Calculated strength data			Fluid capacity (gals)	Used with jar no.	Used with Super fishing jar no.
								Tensile load at yield in lbs	Torque in lbs-ft				
									Recom-mended	at yield			
70957	1 ⁵ / ₈	1/4	Per order	6	1,100-1,400	14,000	8,400	43,200 46,300	200	420	0.13	70822	
64460	1 ¹³ / ₁₆	5/16	1 ¹³ / ₁₆ " Wilson FJ	6	1,360-1,800	18,100	10,800	59,400	370	640	0.195	74223 21150 78074	
50640	2 1/4	3/8	1 1/4" API Reg	8	1,560-2,100	20,700	13,800	118,500	1,700	2,200	0.112	18775 54020	
68262	2 ²⁹ / ₃₂	1	2 ³ / ₈ " PH-6	12 ³ / ₄	2,200-3,000	37,000	24,600	194,800	1,600	5,200	0.692	68010	
55867	3 1/8	1	2 ³ / ₈ " API Reg	8 ³ / ₄	2,400-3,300	30,000	21,000	229,200	3,500	7,600	0.375	42736 52504	72888
55895	3 ³ / ₄	1 1/4	2 ⁷ / ₈ " API Reg	8 1/4	4,200-5,700	52,000	36,000	345,000	3,800	13,500	0.82	38040 13255 52506	145737
55747	3 ³ / ₄	1 1/2	2 ³ / ₈ " API IF	7 ⁷ / ₈	3,400-4,600	43,500	30,000	299,700	3,800	13,000	0.63	37406 52528	
50660	3 ³ / ₄	1 7/8	2 ³ / ₈ " E.U.E	7 ⁵ / ₈	3,500-4,700	43,000	30,000	179,500	2,500	8,200	0.613	41355 20150 52497	
55664	4 1/4	1 ¹⁵ / ₁₆	2 ⁷ / ₈ " API IF	8 ⁵ / ₈	3,500-4,700	43,000	30,000	430,300	6,600	24,500	0.92	44483 13640 52502	80468
50708	4 1/2	2 ³ / ₈	2 ⁷ / ₈ " E.U.E.	10 ³ / ₈	3,600-4,900	49,000	32,000	375,000	4,000	25,900	1.15	35849 52653	
50700	4 ³ / ₄	1 1/2	3 1/2" API FH	8 ⁷ / ₈	6,300-8,500	78,000	54,000	591,900	9,500	27,600	1.0	25960 52530	
50700	4 ³ / ₄	1 1/2	3 1/2" API FH	8 ⁷ / ₈	6,300-8,500	78,000	54,000	591,900	9,500	27,600	1.0	25960 52530	
55812	4 ³ / ₄	2	3 1/2" API FH,IF	10 1/8	5,600-7,500	63,000	43,000	468,800	9,500	27,100	1.35	38110 52500	79789
55860	6	2	4 1/2" API FH	8 ⁵ / ₈	10,200-13,800	128,500	77,000	937,000	17,000	52,600	1.57	14710 52496	145484
55905	6 1/4	2 1/4	4 1/2" API IF	13	11,800-16,000	147,000	102,000	917,400	21,000	56,900	4.24	12370 52544	79691
50720	6 ³ / ₄	2 ³ / ₈	5 1/2" API Reg	13	13,000-17,500	172,900	102,000	1,013,800	24,000	74,200	3.45	11130 52680	145440
55910	7 ³ / ₄	3 1/16	6 ⁵ / ₈ " API Reg	13	11,000-15,000	126,000	88,000	1,587,900	45,000	145,300	4.65	15160 52711	
78964	7 ³ / ₄	3 1/16	6 ⁵ / ₈ " API Reg	12	12,100-20,500	220,000	123,000	1,600,000	45,500	130,000	...	...	72978
66372	9	3 ³ / ₄	7 ⁵ / ₈ " API Reg	13	12,000-16,000	200,000	100,000	1,621,000	70,000	224,700	3.2	66346	

Notes:

- The strengths shown are theoretical calculations based on the yield strength of the material used in each case. The strengths shown are therefore accurate to plus or minus 20% of the figure shown only. The manufacturers (Bowen Tools Inc. in this case) state that the strengths are not guaranteed, and that they are meant to serve as a guide only and that appropriate safety factors should be used.
- All jarring and pulling loads shown assume that the force is acting alone and is essentially along the major axis of the tool. If torque and tension or bending and tension are used together, the resulting combined stresses may lead to failure at substantially less than rated loads. Rotation and bending together can lead to fatigue.
- Users of jars and bumper subs should be aware that milling or drilling operations may develop stresses in these tools that are more complex than the simple torsional and tension values listed. If unstabilised, the weight necessary for milling can induce bending forces that combine with torsional forces to generate very high stresses in some areas of the tool. Rotating in a deviated hole or with the tool at a neutral point may have the same effect. It is not the intention to advise against the use of such tools in these operations, but merely to caution the user of possible dangers when rotating under the conditions described.
- Weight consisting of DCs, sinker bars, HWDP, etc, should not be run above a jar intensifier for at least 1,000 feet.

FREEING STUCK PIPE WITH HYDROCHLORIC ACID

A very successful technique for freeing stuck pipe in carbonate formations, including chalk, is to spot hydrochloric acid (HCl) around the contact zone and allow it to soak in. The HCl reaction with these formations will degrade/dissolve the formation and thus reduce the pipe contact area.

The procedure is applied as follows:

- Pump a pre-determined volume (e.g. 6 m³ or 40 bbls for a 8 1/2" hole section) of a spacer liquid (water or otherwise). Ensure that the spacer is buffered with soda ash if it is water based.
- Pump the HCl pill (15% concentration only) in volumes of 3 to 4 m³ (20 to 30 bbls) and displace with the spacer liquid (1.5 to 3 m³ or 10 to 20 bbls). Spot the acid pill directly across the contact zone.
- It is important to allow the acid pill to soak into the formation for a minimum of 1 hour, but no longer than 2 hours, before working or jarring on the drill string in order to prevent burying the drill string into a soft well bore wall.

Repeat the soaking period with the remainder of the acid pill, as required.

- When the pills are displaced from the hole they can be allowed to mix into the drilling fluid system, adjusting the pH with caustic soda or lime. They should be circulated out through the choke at a low pump rate to vent the carbon dioxide reaction product which could behave much like a gas influx.
- It is not advisable to use HCl when the opportunity for hydrocarbon contact exists, including contact with any diesel based freeing pills that may have been used prior to the acid pills. HCl can crack the hydrocarbon structure at high temperatures and pressures, creating extremely volatile and flammable gases when vented to the atmosphere.

H – CASING AND CEMENTING

Clickable list

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CHECK LISTS FOR CASING RUNNING/CEMENTING OPERATIONS

RUNNING THE CASING

Order all tools & materials in good time !

When ordering casing specify

- Size & weight per unit length
- Grade
- Range
- Length of each casing type (plus 5-10% excess)

Order/check the availability of casing accessories

- Float shoe & collar
- Stage cementing collar (if required)
- Centralisers
- Scratchers
- Stop rings
- Drift mandrels
- Thread lock compound
- Casing thread compound
- Landing joint

Prepare the casing

- Identify casing as ordered
- Number joints
- Check casing condition. Drift casing
- Measure casing (cross-check tally)
- Prepare casing running list (have an independent check made)
- Mark ALL joints to be left out
- Mark off landing joint
- Check cement. Mix a sample

Calculate

- Casing ton-miles. Slip & cut if necessary
- Total weight of string before & after cementation
- Maximum hook load & safety factor (if safety factor is too low, consider stringing more lines)
- Estimated crown load; compare to allowable derrick design load
- Check substructure loading with drill string in the derrick & casing landed in table
- Check actual height of casing shoe above Section TD

Check running equipment

- Elevators; single joint/side door/slip types
- Slips
- Spider
- Spare dies for slips & spider
- Casing tongs
- Power tongs
- Torque gauges
- "Klamp-on" thread protectors
- Casing rams
- Drift mandrel
- Casing circulating head & swings
- Casing spear
- Stabbing board

CHECK LISTS FOR CASING RUNNING/CEMENTING OPERATIONS

RUNNING THE CASING (2)

Check the wellhead equipment

- Casing head housing (weld-on or threaded)
- Casing head spool
- Slip & seal assembly or BRX type casing hanger
- Wear bushing
- X-bushing, plastic sticks & injection tool
- Ring joint gaskets
- Studs & nuts/clamp assemblies
- Steel gate valves/companion flanges
- Plug type tester
- Cup type tester

Operational preparations

- Only run casing if hole is in good condition after wiper trip
- Lay down drill collars & drill pipe if required
- Rig up casing tongs
- Install rotary table insert or casing spider
- Change top pipe rams to casing rams
- Test stack

While running-in

- Ensure correct torque for make-up
- Apply thread lock to **all** connections of the shoe track
- Run in at controlled speed (approximately 30 sec/joint) to minimise pressure surges
- Check returns constantly
- Fill up each joint
- Change to slip elevator at shoe of last casing

After reaching planned depth

- Check casing left on rack with pipe tally.
- Check string weight, upward and downward strokes.
- Install cementation head/plug housing.
- Break circulation slowly! Observe returns.
- Check for losses
- Circulate at least the casing contents (preferably circulate completely).
- Record circulating rates & pressures.

CHECK LISTS FOR CASING RUNNING/CEMENTING OPERATIONS

CEMENTING

Ordering cement

- Type & quantity as specified in programme
- Specify additives required

Order/check availability of cementing accessories

- Cementing head / plug housing
- Top & bottom plugs
- Metal petal basket (if required)
- Additives;
 - accelerator or retarder (if required)
 - water loss control agent
 - friction reducer
 - lost circulation material
 - slurry density reducer
 - slurry density enhancer

Slurry design

- Send samples to laboratory for checking in time to allow several re-runs of the test
- Send representative samples of :
 - cement
 - bentonite (if to be used)
 - mix water
 - any other additives that the laboratory could not be expected already to have representative samples of

Calculate

- Volume of cement slurry required (study caliper log)
- Volume of mix water for required density
- Total time for job (compare to thickening time)
- Downward & upward forces (will casing float?)
- Volume to displace top plug to collar (convert to pump strokes & time)
- Annular velocity during displacement
- Total drilling fluid returns (casing + cement + displacement volume)
- Pressure differential prior to bumping plug
- Expected top of cement in annulus

CASING JOINT LENGTHS & COLOUR CODING

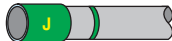
Casing range

Casing joints are not manufactured in exactly equal lengths. To facilitate handling arrangements, the API specify three ranges into which pipe lengths must fall. These are :

Range	Length (m)	Average Length (m)	Length (ft)	Average Length (ft)
1	4.9 - 7.6	6.7	16.1 - 25	22.0
2	7.6 - 10.4	9.4	25.0 - 34.1	30.8
3	>10.4	12.8	>34.1	42.0

Casing colour coding

The API has defined a colour code for the different grades of steel used for making casing and collars. These are shown below.

Grade	Color Coating on Coupling	Color Band on Coupling	Color Band on Pipe Body	
H-40	Black	-	-	
J-55	Green	Letter "J" (Yellow)	One Green band	
K-55	Green	-	Two Green bands	
N-80	Red	-	One Red band	
L-80	Red	One Brown band	One Red band and One Brown band	
L80-CR9	Red	Two Yellow bands	One Red band, One Brown band and Two Yellow bands	
L80-CR13	Red	One Yellow band	One Red band, One Brown band and One Yellow bands	
C-90	Purple	-	One Purple band	
C-95	Brown	-	One Brown band	
T-95	Silver	-	One Silver band	
P-110	White	-	One White band	
Q-125	Orange	-	One Orange band	

19-07-1995
Dr. No. TCM 6626

CASING TEST PRESSURES

The purpose of any casing pressure test is to verify that the casing string integrity is sufficient to contain the maximum anticipated burst loads i.e. the design load case. Integrity for collapse loads is generally only tested, indirectly, when inflow testing liner laps. Exerting a suitable differential test pressure at any point in the casing string is complicated by the fact that the fluids inside and outside the casing during the test are unlikely to be those expected to be present for the design load case. This means that the application of a given pressure at surface for a single test may result in insufficient or excessive differential pressures deeper in the well.

Ideally, casing pressure tests should thus be designed so that the differential pressure exerted at any point is equal to or exceeds the maximum expected load but remains less than 91% of the rated internal yield pressure. For new casings the latter value can be ascertained from data handbooks. Where wear has occurred the following equation may be used :

$$P = \frac{0.875 \times 2 \times t \times Y}{D}$$

Where : P = the internal yield pressure, without safety factors

Y = the specified minimum yield strength for the given casing grade.

t = the actual wall thickness

D = the actual OD

all in consistent units

Even when more than one weight and/or grade of casing is not present, it will often require the use of a retrievable packer to test a casing string adequately.

The preferred time to test the casing is immediately following cementation prior to the cement setting, a so called "green cement test". This avoids the possibility of creating a micro-annulus, but such a test may not be sufficient as it is further limited to ensure that:

- the differential pressure at the casing shoe does not exceed the pressure rating of the float equipment. This is commonly 21 MPa but equipment can be supplied with ratings of 34.5 MPa and above.
- the resultant tensile load does not exceed 77% of the rated pipe body yield strength at the critical point of the string.

The pipe body yield strength is given by $T = \frac{\pi Y}{4} (D^2 - d^2)$

Where Y and D are as above and d = the actual ID

Notes :

- EP 89-1500 also **recommends** that a green cement pressure test is restricted to 75% of the casing internal yield pressure.
- When testing with a retrievable packer, it should preferably be set above the top of cement. In any case EP 89-1500 states that it **shall** not be placed within 80 m of the shoe or within 80 m of a hydrocarbon bearing zone.
- Casing pressure tests should be carried out for 10 minutes (EP89-1500).
- Problems may be experienced when trying to set packers in high grade casings due to the problem of getting the slips to bite.

BASIC CEMENTING PROCEDURES

Single stage cementation

- Circulate (at least casing contents)
- Pump pre-flush
- Release bottom plug
- Pump slurry (monitor weight continuously and take samples)
- Release top plug
- Pump spacer
- Chase with drilling fluid using rig pumps in turbulent flow (create turbulent flow, count strokes, reciprocation of string may be necessary)
- Reduce pump speed as top plug approaches float collar (note pressure differential)
- Bump plug gently.

Two-stage cementation

- Circulate (at least casing contents)
- Pump water spacer
- Drop by-pass plug
- Pump first-stage slurry
- Drop first-stage top plug
- Displace with water spacer
- Chase with drilling fluid (create turbulent flow, count strokes, reciprocation of string may be necessary)
- Reduce pump speed as plug approaches float collar (note pressure differential)
- Bump plug gently
- Pressure test casing
- Check for back flow
- Drop opening-bomb and wait 15 minutes
- Pressurise the casing to the manufacturers-specified value to open stage collar. This is often approximately 5.17 MPa (750 psi)
- Circulate out drilling fluid/cement through stage collar
- Wait for cement to set
- Pump second-stage water spacer
- Pump second-stage slurry
- Drop second-stage top plug (to close stage collar)
- Displace with water
- Chase with drilling fluid using the rig pumps
- Pump at minimum of 1.2 m³/min or 7.5 bbl/min as plug approaches stage collar (note pressure differential)
- Bump plug (do not slow down the pump until the stage collar closing pressure is reached)

Note: The equipment manufacturer's standard procedures should be checked prior to any job.

BASIC CEMENTING PROCEDURES

Stinger cementation

- Circulate (annulus and string contents)
- Pump pre-flush or marker ahead
- Pump slurry until (i) returns are seen, or
(ii) marker is seen, or
(iii) returns are considered impossible
- Displace stinger contents using drilling fluid
- Watch casing/string annulus for returns during entire job
- Check for back flow
- Pull stinger. Do not rotate

Liner cementation

Having set liner and retracted running sleeve :

- Pump water spacer
- Pump slurry
- Release pump down plug
- Pump water spacer
- Displace cement at recommended rate/pressure
- Pump until pump down plug reaches liner wiper plug
- Shear liner wiper plug
- Displace combined wiper plugs to float collar
- Pull out stinger

Setting a balanced plug

- Ensure that the stinger is long enough to keep the drill pipe above the cement column
- Pump water spacer
- Pump slurry
- Pump water spacer
- Displace with drilling fluid, either (i) underdisplaced, or
(ii) balanced)
- Pull stinger slowly without rotating string (will disturb cement)

CEMENT SLURRY GRADIENT & YIELD

CLASS 'G' OIL WELL CEMENT

This tabulation is based on :
Mix water density = 1.00 kg/l
Portland cement grain density = 3.10, 3.14 and 3.20 kg/l

Gradient (kPa/m)	Yield in m³/tonne		Water requirements in m³/tonne		Gradient (psi/ft)	Yield in ft³/sack		Water requirements in gals/sack	
	gd=3.10	gd=3.14	gd=3.10	gd=3.14		gd=3.10	gd=3.14	gd=3.10	gd=3.20
17.00	0.924	0.930	0.602	0.611	0.750	1.39	1.40	6.77	6.88
17.10	0.912	0.917	0.589	0.599	0.756	1.37	1.38	6.63	6.73
17.20	0.899	0.905	0.577	0.586	0.759	1.35	1.35	6.49	6.60
17.30	0.887	0.893	0.565	0.574	0.764	1.33	1.34	6.35	6.46
17.40	0.876	0.881	0.553	0.562	0.768	1.32	1.32	6.22	6.33
17.50	0.864	0.869	0.542	0.551	0.772	1.30	1.31	6.09	6.20
17.60	0.853	0.858	0.530	0.540	0.777	1.28	1.29	5.97	6.07
17.70	0.842	0.847	0.520	0.529	0.781	1.27	1.27	5.85	5.95
17.80	0.832	0.837	0.509	0.518	0.786	1.25	1.25	5.73	5.83
17.90	0.821	0.826	0.499	0.508	0.790	1.24	1.24	5.61	5.71
18.00	0.811	0.816	0.489	0.498	0.794	1.22	1.23	5.50	5.60
18.10	0.802	0.806	0.479	0.488	0.799	1.21	1.21	5.39	5.49
18.20	0.792	0.797	0.469	0.478	0.803	1.19	1.20	5.28	5.38
18.30	0.783	0.787	0.460	0.469	0.808	1.18	1.18	5.18	5.28
18.40	0.774	0.778	0.451	0.460	0.812	1.16	1.17	5.07	5.17
18.50	0.765	0.769	0.442	0.451	0.817	1.15	1.15	4.97	5.07
18.60	0.756	0.761	0.433	0.442	0.821	1.14	1.14	4.88	4.97
18.70	0.748	0.752	0.425	0.434	0.825	1.12	1.13	4.78	4.88
18.80	0.739	0.744	0.417	0.425	0.830	1.11	1.12	4.69	4.78
18.90	0.731	0.736	0.408	0.417	0.834	1.10	1.11	4.60	4.69
19.00	0.723	0.728	0.401	0.409	0.839	1.09	1.09	4.51	4.60
19.10	0.715	0.720	0.393	0.401	0.843	1.08	1.08	4.42	4.51
19.20	0.708	0.712	0.385	0.394	0.847	1.06	1.07	4.33	4.43
19.30	0.700	0.705	0.378	0.386	0.852	1.05	1.06	4.25	4.34
19.40	0.693	0.697	0.370	0.379	0.856	1.04	1.05	4.17	4.26
19.50	0.686	0.690	0.363	0.371	0.861	1.03	1.04	4.09	4.18
19.60	0.679	0.683	0.356	0.364	0.865	1.02	1.03	4.01	4.10
19.70	0.672	0.676	0.349	0.358	0.870	1.01	1.02	3.93	4.02
19.80	0.665	0.669	0.343	0.351	0.874	1.00	1.01	3.85	3.95
19.90	0.659	0.663	0.336	0.344	0.878	0.99	1.00	3.78	3.87
20.00	0.652	0.656	0.330	0.338	0.883	0.98	0.99	3.71	3.80

POZZOLAN CEMENT

Average grain density in kg/l = 2.88 (60/40 mix), 2.82 (50/50 mix) or 2.76 (40/60 mix)

Average grain density in kg/l = 2.88 (60/40 mix), 2.82 (50/50 mix) or 2.76 (40/60 mix)

I – DRILLING FLUIDS

Clickable list

(Use the hierarchical list under "Bookmarks" to access individual tables and/or sub-topics)

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DRILLING FLUID FUNCTIONS

Although originally designed to bring the drilled cuttings from the bottom of the hole to surface, drilling fluid now serves at least twelve important functions in modern drilling operations.

Drilling fluid assists in making hole by :

1. Removing the cuttings
2. Cooling and lubricating the bit and drillstring
3. Transmitting power to bit nozzles or turbines

It assists in hole preservation by :

4. Supporting and stabilising the borehole wall
5. Minimising hole wash outs due to turbulence or dissolution

It also :

6. Produces sufficient pressure within the borehole to prevent the inflow of formation fluids
7. Supports the weight of pipe and casing
8. Serves as a medium for formation logging
9. Must be compatible with drilled formations and encountered formation fluids.

It must not :

10. Corrode the bit, the drillstring, the casing or surface facilities
11. Impair the productivity of the productive intervals
12. Pollute the environment

DRILLING FLUID PROPERTIES

Density

A sufficiently high drilling fluid density, or specific gravity, is required for the control of bottom hole pressures and is a key factor in hole stability. However the density must also be kept as low as possible consistent with these requirements because an increase of drilling fluid density causes a considerable reduction in penetration rate and a significant increase in friction losses. The drilling fluid density is generally expressed as a pressure gradient such as kPa/m or psi/ft. A table giving the conversion factors into other units is presented in Section A.

Viscosity

The viscosity of the drilling fluid is very important for the optimisation of various different functions. Viscosity is measured with two different instruments: The Marsh Funnel and the Fann Viscometer.

The Marsh Funnel is a very simple piece of equipment - as the name implies, it is a funnel. It has a standard size holding approximately a litre, and the MF viscosity is the time taken in seconds for 946 ml (= 1 U.S. quart) to run out after starting with the liquid surface at a defined level. It is used routinely in drilling operations to establish whether changes in the drilling fluid properties occur. Further conclusions cannot be drawn from the results produced.

The Fann Viscometer is a far more versatile instrument. It consists of a rotating cylinder and a bob (stator) which is connected to a spring. The cylinder is rotated at 600 rpm and then at 300 rpm, and readings of the bob rotation are taken at each speed. Subsequently gel values are determined by rotating at a low speed.

The results of the Fann Viscometer test can be used to define two different models for the rheological behaviour of the drilling fluid, these being the Bingham model and the Power Law model.

The Bingham model comprises a Plastic Viscosity and a Yield Point and is used to determine the treatment requirements for the drilling fluid. The Power Law model comprises a Power Index (n) and a Consistency Index (k) and is used for pressure drop calculations and to determine the carrying capacity of the fluid.

Bingham model

Plastic Viscosity (PV)

The PV is the difference between the readings at 600 and 300 rpm ($R_{600} - R_{300}$). The dimensions of the viscometer are such that the numerical result gives, approximately, the viscosity in cP.

The significance of the PV is that it is that part of the resistance to flow caused by mechanical friction. It is mainly dependent on the number of solid particles in the drilling fluid. The shape of the particles and the viscosity of the liquid phase have secondary effects on the PV.

PV is increased by :-

- an increase in solids content from the drilled formation due to increased penetration rate
- an increase in solids content due to inadequate solids removal in the surface system, especially clay solids when drilling through clay layers.
- the addition of weighting agents.
- the addition of polymers (CMC HV, starch HV) to the drilling fluid.

The lowest possible PV is essential for :-

- low frictional losses
- optimal hole cleaning.

The PV can be reduced by :-

- lowering the solids concentration (dilution)
- removal of solids (centrifuge, desander, desilter)

Yield Point (YP)

The Yield Point is calculated by subtracting the PV from the Fann reading at 300 rpm ($R_{300} - PV$), giving a result in lbs/100 ft². The significance of the Yield Point is that it is that part of the resistance to flow caused by attractive forces between particles. The YP is a function of :-

- the type of solids and surface charge associated with them
- the solids concentration
- the ionic concentration in the liquid phase.

Clays suspended in water generally develop negative charges on the faces of the individual platelets and positive charges on the edges. Attraction between these charges leads to the build-up of a card house type of structure which results in a high YP.

The YP is increased by :-

- the addition of bentonite clay particles to the drilling fluid
- the addition of biopolymers to the drilling fluid
- contamination of drilling fluid with, for example, salts, cement or gypsum.

The YP is decreased by :-

- shielding of the positive clay charges with thinners (e.g. lignosulfonates)
- the reduction of solids content (solids removal, watering back)
- the chemical neutralisation of contaminants.

An optimal YP is essential for :-

- carrying capacity of the drilling fluid
(a rule of thumb is that $YP = \pm 0.75 \times \text{hole size in inches}$)
- stability of drilling fluid towards settling of solids
(a rule of thumb is that $YP = \pm 9 \times \text{specific gravity}$)
- the hole cleaning capacity of the drilling fluid

Gels

Gel values are a measure of the build-up of gel structures in the drilling fluid under static conditions. Gels originate from the same forces and parameters as the YP. Two gel values are measured - one when the fluid has been stationary for 10 seconds, and one when it has been stationary for 10 minutes.

A reasonable 10 second gel is essential to prevent immediate settling of solids when circulation is stopped.

A large difference between 10 second gel and 10 minute gel indicates a slow but ongoing build-up of structure. This may result in the development of very high gel strengths during a round trip and hence in high swab/surge pressures, which can then cause hole failure. Optimally the 10' gel value is 1.5 x the 10" gel value.

Power law model

Power law viscosities can be obtained from Fann Viscometer results in various ways. In their simplest form n and k are calculated as follows

$$n = 0.5 \log \left\{ \frac{R_{600}}{R_{300}} \right\}$$
$$k = 5.11 \times \frac{R_{300}}{511^n} \text{ poise}$$

The " n " value is the most important for direct drilling fluid engineering applications. A low " n " value will generally improve the carrying capacity of drilling fluid. For displacement of drilling fluid under turbulent flow conditions during cementations the " n " value should be as close to 1 as possible.

Additional notes about rheology in general and the two fluid behaviour models can be found in the Well Engineering Distance Learning Package, Section 6, Part 1, Topic 1.3.

Fluid loss/Filter cake

As the hydrostatic head of the drilling fluid is generally higher than the pore pressure in the formation, liquid from the drilling fluid will be forced into the formation. Consequently a filter cake consisting of the solids in the drilling fluid is formed on the borehole wall. For various reasons it is essential that this drilling fluidcake is as thin and impermeable as possible. Dispersed and deflocculated clay particles are very small and flat and can form a thin and impermeable filter cake. If larger particles are present (clay aggregates, flocculated clay, sand, weighting material) the space between the particles in the filter cake is bigger and various chemicals are required to control the fluid loss

The fluid loss is determined according to an API method and expressed in ml/30 min. Local experience is required to establish the optimal fluid loss while drilling particular hole intervals. The condition and thickness of the filter cake obtained from the API test are very important for, for example, the sticking tendency of the hole.

DRILLING FLUID CHEMICALS

Bentonite

Bentonite (active clay) is the most important constituent of almost all water base drilling fluids. When suspended in fresh water bentonite provides viscosity (mainly YP/gel) and fluid loss properties. It generally takes some six to twelve hours after mixing before a bentonite suspension has developed its full properties. In saline solutions bentonite does not develop viscosity. In saline drilling fluids bentonite is therefore added in a prehydrated form, i.e. premixed in fresh water.

Lignosulphonates, lignites (thinners)

Lignosulphonates and lignites are available under numerous trade names. These chemicals are thinners and dispersants. They reduce the yield point (dynamic attractive forces) and the gels (static attractive forces), and hence the funnel viscosity. The yield point and gels are reduced because the thinner “coats” the bentonite particles and neutralises charged particles on the surface of the clay. With the surface charges neutralized, the clays can disperse properly and thus form a thin filter cake with a uniform overlapping texture. It is by this mechanism that dispersants allow bentonite to reduce fluid loss.

The use of thinners is discouraged as it detracts from the optimum performance of bentonite viscosifiers.

Caustic Soda/Lime

Caustic soda (sodium hydroxide NaOH) and lime (calcium hydroxide Ca(OH)_2) are both alkaline products which increase the pH of a drilling fluid. The amount of pH increase, however, is also dependent on the concentration of buffering agents in the drilling fluid. The combination of buffering agents and OH concentration is a measure of the alkalinity (Pf) of the drilling fluid. Alkaline fluids are required to guard against acid corrosion and to inhibit bacterial growth.

Calcium sulphate (gypsum)

Gypsum is sometimes added to the drilling fluid when large amounts of active clay are present or when active clay or anhydrite layers are to be drilled. The gypsum partly dissolves and the calcium ions now present in the drilling fluid prevent clay swelling or further dissolution of anhydrite.

Potassium chloride

Potassium chloride (KCl) is sometimes added to the drilling fluid when troublesome clay or shale layers must be drilled. Potassium ions adhere to clay/shale particles and prevent swelling. When operating a KCl drilling fluid it is essential closely to monitor the ratio of K^+ and Cl^- ion concentrations. This ratio should be higher than 0.5.

Sodium carbonate/sodium bicarbonate

Sodium carbonate (soda ash, Na_2CO_3) or sodium bicarbonate (NaHCO_3) can be added to a drilling fluid when the concentration of calcium ions is too high (due for example to overdosage of gypsum, drilling into chalk layers or cement contamination) .

Too high a concentration of Ca^{++} ions will spoil the viscosity and fluid loss properties of clay.

CMC/Starch

A large number of polymers is available worldwide under many different trade names. The base products are normally Cellulose (CMC, Carbocel etc.) or Starch (Flocgel, Stabilose). These polymers affect both the fluid loss and the viscosity of the drilling fluid. LV (low viscosity) versions are generally designed to lower the fluid loss without major effects on the viscosity. HV (high viscosity) versions have a major effect on the viscosity (PV) and generally reduce the fluid loss as well. HV polymers should be used only when high viscosity is required.

Biopolymers

Recently several biopolymers (X-C Polymer, Rhodopol, Enerflo, Drillam X/84) have found world wide application in drilling fluids. When used in concentrations as low as 2-3 kg/m³ (1 lb/bbl) they provide low 'n', pseudoplastic, fluids with good hole cleaning capability.

Salts

Various salts (sodium chloride, magnesium chloride and potassium chloride) are used when drilling through salt sections. The drilling fluid is saturated with respect to the salt that is expected in the hole and thus washouts due to the dissolution of the salt layers are minimised. Refer also to the section on salt saturated drilling fluids.

Weighting material

Barytes, Dolomite and Iron Oxide are used as weighting materials. Prior to adding weighting material the drilling fluid must have a YP sufficient to keep this dense material in suspension. Weighting materials add solids and proportionally reduce the amount of free water in the system.

COMMON DRILLING FLUID ADDITIVES, NAMES AND FORMULAE

Common name	Chemical name	Chemical formula
Lime (cement)	Calcium Hydroxide	$\text{Ca}(\text{OH})_2$
Caustic Soda	Sodium Hydroxide	NaOH
Gypsum	Calcium Sulphate	CaSO_4
Soda Ash	Sodium Carbonate	Na_2CO_3
Bicarb	Sodium Bicarbonate	NaHCO_3
Haematite	Iron Oxide	$\text{Fe}_2\text{O}_3 \cdot \text{FeO}$
Barytes	Barium Sulphate	BaSO_4

PREHYDRATED BENTONITE

Uses of Prehydrated Bentonite

Bentonite is added to water based drilling fluids to increase the viscosity and gel strength, increasing their ability to suspend solids and their carrying capacity. It forms a filter cake and, if properly dispersed, is the main agent for reducing water loss.

Bentonite will “yield” in fresh water (less than 50 g/l salt) but not in salty water. Hence, if bentonite is to be used in salty water, it must be prehydrated in fresh water to form a premix. Before adding bentonite (or any other chemical) to the water to make such a premix, test the water for salinity and hardness. As a guide, use soda ash at 1 kg/25m³ (1.4 lb/100 bbls) water for every one ppm total hardness, then add caustic soda to obtain the required pH. Note that in salt saturated drilling fluids even prehydrated bentonite may have limited use as osmotic forces will dehydrate the bentonite again.

Preparation of a premix

- Pre-treat the water with soda ash and caustic as required.
- Add 75 - 100 kg/m³ (25-35 ppb) bentonite.
- Allow 8-10 hours hydration time.
- Maintain the pH at 9 with caustic.

Advantages of using a premix

- Maximum viscosity is obtained with minimum solids.
- Filtration control is easier.
- Filter cake is thinner and stronger.

SPUD MUD

“Spud mud” is the name given to the drilling fluid used for top hole sections. When drilling top hole vast amounts of cuttings are generated due to high penetration rates and large hole sizes. In view of the limited pump capacity the carrying capacity of the drilling fluid is of prime importance, hence a low “n” value (i.e. a high YP/PV ratio) is required. Spud mud normally consists of some 40-60 kg/m³ bentonite in fresh water. The pH is maintained at 9-10 with caustic soda. Sometimes some CMC-HV polymer is required for extra viscosity.

General Properties

Density	1.05 - 1.15 kg/l
MF visc	80 - 100 secs
PV	± 20
YP	20 - 30
0' gel	5 - 15
Fluid loss	± 30 ml API
pH	9 - 10

When drilling top hole it often occurs that there are no returns. In such cases water is used as the drilling fluid. Slugs of prehydrated bentonite with MF viscosities between 100 and 120 secs are then circulated occasionally for hole cleaning.

BENTONITE/LIGNOSULPHONATE DRILLING FLUID

Fresh water bentonite drilling fluid is a relatively inexpensive drilling fluid which is used widely in drilling operations. The main drilling fluid parameters are maintained by the careful balancing of clay and lignosulphonate additions.

Addition of clay will increase the YP and gel values whereas lignosulphonate additions will lower the fluid loss and YP/gel values. When drilling fluid properties cannot be maintained additions of CMC-LV are required. The PV is kept low by optimal solids removal and dilution. Batchwise dilution (replacement of 20-30% of the total drilling fluid volume in one circulation) is more effective than continuous dilution.

General Properties

Density	< 1.20 kg/l
MF visc	45 secs
PV	15 - 20
YP	8 - 12
Gels	2/4
Fluid loss	as required
pH	9.5 - 10.5

GYP/LIGNOSULPHONATE DRILLING FLUID

Gypsum is sometimes added to the drilling fluid when large amounts of active clay must be drilled. The calcium ions from the gyp convert the clay particles into the relatively harmless calcium form whereby strong increases in viscosity are prevented. Additions of ligno sulphonate further reduce the viscosity. CMC or starches are required for fluid loss control as the clay particles are now aggregated and consequently no thin filter cake can be obtained. A bentonite fresh water lignosulphonate drilling fluid may also be converted to a gypsum type of fluid if large sections of anhydrite layers have to be drilled.

General Properties

Density	< 1.30 kg/l
PV	± 20
YP	10 - 15
Gels	8/12
Fluid loss	± 10 ml API
pH	9.5 - 10.5
Ca ⁺⁺	600-1200 ppm

SALT DRILLING FLUIDS

Salt saturated drilling fluids are used for drilling salt formations. The drilling fluid is saturated with respect to the salt (or salt mixture) that is to be drilled. Given that most salts are plastic, and tend to close the freshly drilled hole, high fluid densities are often required to maintain a stable borehole. In view of these density requirements no average properties of salt saturated drilling fluids can be given. The composition of salt drilling fluids is given below.

Salt to be drilled	Halite NaCl	Bischoffite NaCl/MgCl ₂	Carnallite NaCl/MgCl ₂ /KCl
Brine composition	300 kg/m ³ NaCl	200 kg/m ³ Cl ⁻ 40 kg/m ³ Mg ⁺⁺	300 kg/m ³ Cl ⁻ 80 kg/m ³ Mg ⁺⁺ 40 kg/m ³ K ⁺
YP gel	XC pol 3-5 kg/m ³	XC pol 3-5 kg/m ³	XC pol 3-5 kg/m ³
Fluid loss	Starch LV* 20-30 kg/m ³	Starch LV* 20-30 kg/m ³	Starch LV* 20-30 kg/m ³

* Starch may be replaced by synthetic polymers such as polyacrylates or a mixture of MgO/Mg(OH)₂ called Magnemagic.

LOST CIRCULATION

Lost circulation is one of the most expensive problems in drilling with the possibility of large quantities of drilling fluid being lost before the losses are cured or reduced to a reasonable level. Lost circulation does not necessarily imply that there are total losses to the formation, but can include partial or seepage losses.

Causes of lost circulation

Penetration of a coarsely permeable formation

Highly porous and permeable formations such as coarse sandstone or gravel can allow drilling fluid particles to penetrate the formation. The degree of losses will depend on the size of the formation openings and the composition of the fluid. Generally this only causes mild or seepage losses which should reduce if LCM is used.

Cavernous or vugular zones

These are often found in dolomitic limestone, are water filled and may cause sudden total or very severe losses accompanied, in the case of cavernous formations, by a sudden drop of the drill string. They are very difficult, if not impossible, to cure, and require the use of the most extreme measures if a reduction in the rate of loss is to be achieved, e.g. special cements and/or diesel oil/bentonite (DOB/Gunk) plugs. LCM is unlikely to have much effect but it is worth pumping some while you prepare for cementing. Several cement and/or DOB plugs may be required and for best effect attempts should be made to balance the fluid column with formation pressure once these are in place. With large vugs only a slight overbalance will cause huge losses so a reduced drilling fluid density may slow static losses — dynamic losses will be unaffected. The ability to bullhead the annular contents easily into the loss zone makes blind or floating mud cap drilling to the next logging or casing point viable options. This will depend on the pressure regimes in the open hole section and the absence of mobile hydrocarbons or H₂S.

Natural or induced fractures

Fractures may exist naturally in a formation due to tectonic movement or they may be induced while drilling by:

- using too high a drilling fluid density
- operating with a high equivalent circulating gradient
- creating pressure surges by running in pipe too fast
- pressure surges caused by bit or stabiliser balling
- attempting to place too high a column of cement in a casing annulus
- applying too high a pressure during a formation strength test

Fractures will take drilling fluid when the hydrostatic pressure exceeds the formation breakdown pressure (inducing fractures) or the fracture closure pressure (natural or previously induced fractures). Losses induced by pressure surges (moving the drillstring too rapidly or breaking circulation too quickly) are a problem when operating with a drilling fluid density which is close to or above the fracture closure gradient. This should only be done following a very thorough assessment of the risks - it has been the cause of at least one blowout in the Group. The most effective way to stop these losses is to reduce the drilling fluid gradient. Be aware that this may cause other formations to flow or lead to hole instability so risks will have to be balanced.

“Ballooning”

Some formations, especially shales, will “give back” lost fluid once well bore pressure is reduced. This returning fluid may bring hydrocarbons with it. However if the volume increase is interpreted and treated as a kick, considerable wasted time can result.

LOST CIRCULATION (2)

Analysis of the losses

Check surface equipment for leaks. If this fails to show any leaking, then reduce drilling fluid density if safe to do so.

The conditions at the time of losses can give an indication of the reasons for the lost circulation. Losses during tripping are probably due to running pipe too fast. During drilling, a change in drilling rate or change in lithology from cuttings would indicate a weaker or porous formation or a fault had been reached. The drilling fluid density and viscosity may have also increased. The hole should always be observed with the pump shut off since if the level then remains static, the drilling fluid density or viscosity only needs to be reduced slightly and/or a light treatment carried out with lost circulation material.

The depths where losses can be expected for each particular well are usually mentioned on the Drilling Programme if such information is available.

An estimate can be made on the maximum pressure the formation can withstand by filling up the hole with water until, if at all, circulation is regained and the annulus level can be seen. The new drilling fluid gradient can then be calculated.

Curing the losses

Basically, lost circulation occurs because the vugs and fractures in the formation are larger than the bridging particles in the drilling fluid. Sealing materials can be added to the drilling fluid to cure these problems.

A wide range of materials can be used depending on their availability and price. They can be classified as flakes, granules, fibres and mixtures. Examples of lost circulation material are: micas, cellophane, nut shells, wood shavings, feathers, textile fibres, etc. and the effectiveness of each will depend upon the size of the formation openings. Very often mixtures of these are used to give a wide range of sizes of bridging particles. After the lost circulation material has formed a bridge (often far into the formation), the drilling fluid cake can then form.

Recently a series of products consisting of calcium carbonate in various sizes is being applied for curing lost circulation. The sizing of the particles is chosen in such a way that a combination of three or four materials with consecutive sizes will give optimal blockage of the pores.

A 5-15 m³ (30-100 bbl) pill containing a mixture of lost circulation materials should have between 15-50 kg/m³ (5-15 lbs/bbl) of each type of LCM, with a total LCM concentration of 50-75 kg/m³ (15-25 lbs/bbl). The pill is preferably spotted where the losses are occurring and allowed to stand there for a short period. The remaining pill can be squeezed into the formation if required. If losses are not cured, a second pill can be used. The shale shakers will also tend to plug so that they will have to be cleaned continuously when lost circulation material is being used. If drilling ahead, do not by-pass shakers as this will cause the drilling fluid column to get heavier and possibly make losses worse.

When deciding which LCM will be used the size of the bit nozzles and drilling tool flow restrictions should not be overlooked. Prior to adding LCM to the drilling fluid the pump screens should be removed.

LOST CIRCULATION (3)

PROCEDURES

Type of losses	Definition	Possible actions
Seepage	0.2 to 2 m ³ /h	Measure loss rate and track trend. Can mud gradient be reduced ? Consider using LCM if no improvement, depending on economics of mud loss vs rig time to cure and possible effect on downhole equipment.
Mild/Partial	2 to 8 m ³ /h Lost mud can be replaced indefinitely (chemical supply, mixing rate).	As for seepage. However, increased loss rate means increased risk that the hole will not be kept full - evaluate the consequences.
Severe/Total	Mud supply will be exhausted within a given time.	Keep hole full. Pump water (base oil for OBM) if necessary. Identify likely cause of losses and identify minimum mud gradient that can be used for formations already open. Make preparations to pump cement/DOB. Consider whether blind or mud cap drilling is feasible / safe.

Lost circulation pills

For a typical lost circulation pill add the following to 10 m³ drilling fluid :-

Slight losses	400 kg 40 µm granular material	
	300 kg 100 µm	"
	300 kg 400 µm	"
Moderate losses	600 kg 40 µm	"
	450 kg 100 µm	"
	450 kg 400 µm	"
	150 kg 1 mm	"
Severe losses	800 kg 0.1 mm	"
	600 kg 0.4 mm	"
	600 kg 1.0 mm	"
	200 kg 2.5 mm	"

Add HEC or XCD Polymer for suspension in **completion/workover** fluids

A WAITING PERIOD IS SOMETIMES BENEFICIAL !

It is imperative that an LCM pill is lost in order to have a long term effect. A quick cure usually means that the problem will recur again soon.

DIFFERENTIAL STICKING

Differential sticking is caused by a difference between the hydrostatic head of the drilling fluid and pore pressure in a permeable formation. When the drill string is not moving it can be pressed into the filter cake by this pressure differential. Very high pulling forces are then required to overcome the frictional forces before the pipe can be moved again; often these are so great that the pipe is in practice immovable, i.e. stuck. A thin impermeable filtercake is essential to prevent differential sticking. Once the pipe is stuck rapid action is required as the filter cake at the stuck point dehydrates and sticking increases with time. The first priority is to reduce the pressure differential (i.e. the drilling fluid hydrostatic head) as far as safely possible. This can be achieved by pumping water or base oil into the drillpipe and/or annulus (do not allow it to enter the open hole section) and thereafter bleeding off the differential pressure in steps, thus allowing the columns to equalize gradually whilst working the string. Pills of lubricating oil (polylube, cebulube, etc.) are often used in an attempt to reduce frictional forces.

Furthermore, mixtures of lubricating oil and surfactants (pipe lax, b-free) are applied to “dissolve” the filter cake. Recipes for these “soak pills” are different for most Operating Units. When used at depths below 3,000 m the combination of vegetable lubricating oil and barytes may cause polymerisation resulting in even more stickyness. Recipes including these chemicals must have been tested before their application.

DRILLING FLUID CONTAMINANTS

Contamination

Contamination of the drilling fluid is a continuous process while drilling, as drilled solids enter the drilling fluid. While much of the drilled solids is inert, there are certain materials which can cause severe problems with the drilling fluid properties or with corrosion, and the effect on these will depend on how quickly the contaminant enters the drilling fluid. Any contamination should be treated immediately, so careful and frequent checks should be kept on the actual drilling fluid properties. Advance treatment to prevent contaminants affecting the drilling fluid can sometimes be made, for example on the basis of the geological prognosis or when drilling out cement. With the majority of water based drilling fluids, the contaminant will tend to affect the clay particles most.

Salt contamination

Salt contamination can come from drilling salt beds, or from a formation water influx. This can easily be detected by an increase in chlorides. The electrolyte effect tends to flocculate the clay with the sodium ion replacing the hydrogen ion in the clay. There may be a slight decrease in pH, and an increase in viscosities, gels and fluid loss. There is no chemical treatment for severe salt contamination and treatment depends on dilution and building new drilling fluid, or converting to another type of fluid such as salt water or a salt saturated drilling fluid.

Cement contamination

When drilling out cement, calcium hydroxide is formed and severe flocculation of the clays contained in the drilling fluid is observed. The contamination is easily detected due to the increased level of calcium (seen as an increase in hardness), increased viscosities and an increased pH. Cement contamination can be cured by the use of soda ash or sodium bicarbonate. Sodium bicarbonate is preferred to soda ash if a low pH is required.

For drilling out a hard cement plug or shoetrack, a pre-treatment of 5-6 sacks of sodium bicarbonate should be used to guard against contamination. If green (or very soft) cement has to be drilled or circulated out, the contamination may be too severe to cure, and the contaminated fluid should be discarded at surface.

Calcium contamination

Due to various reasons calcium ions may become sufficiently concentrated to flocculate the drilling fluid. As the calcium replaces the sodium ions in the clay, the clay particles tend to aggregate. The result of the contamination is an increase in calcium concentration, increase in fluid loss, and an increase in yield point and gels. The plastic viscosity will increase initially, then slowly decrease.

The contamination may be cured by the use of soda ash to precipitate the calcium, however the amount of soda ash should be comparable to the level of calcium present. The contamination is not usually a problem with gyp drilling fluids.

Carbonate-bicarbonate contamination

In alkaline drilling fluids, a CO₂ influx can form bicarbonate or carbonate ions. Over-treatment of other contaminants with soda ash or sodium bicarbonate can also contribute to the problem known as “carbonate alkalinity”. In practice, this results in an inability of the lignosulphonate to treat high yield point and gel strengths.

If it is suspected that carbonate-bicarbonate contamination is taking place in the drilling fluid, the system should always first be treated with caustic. If this treatment is not effective then soluble calcium ions may be introduced in the form of lime. However this should be done with the utmost care to avoid flocculating the drilling fluid clays. The lime should be added to the system stepwise, such that each addition produces no more than 10 ppm increase in filtrate hardness.

If possible a pilot test should be performed, before treating the system as a whole.

Active clays

Drilling active clays will tend to build up the colloidal solids in the drilling fluid and cause high viscosities and gels, and often give a thick filter cake. The clay content can be reduced by watering back.

Clay inhibitors such as KCl may be used to reduce the effect of swelling clays. Ideally the system should be converted to a polymer or invert oil system, but this is rarely practicable.

Foaming

Several drilling fluid additives such as lignosulphonates, polymers etc. can cause foaming, particularly when a large batch of drilling fluid is prepared. The foam can easily be reduced by adding aluminium tristearate to a bucket of diesel oil and mixing this into the drilling fluid. As an alternative, proprietary liquid defoamers such as NF-1 can be used.

Gas cutting

When a large quantity of gas enters the drilling fluid, the drilling fluid will have a reduced density on surface but can have a very high viscosity and often shows a degree of foaming. Generally the degasser can easily remove the gas but only gas-free drilling fluid should be pumped down the hole.

FUNCTIONS AND PROPERTIES OF OIL WELL DRILLING FLUIDS

Function	Relevant property	Effect of property on penetration rate	Recommended value	Chemicals for control of water based drilling fluids
Control of formation pressures	Drilling fluid density	Increased drilling fluid density decreases penetration rate	Has to be calculated from well depth and expected pressures Safety factor: 2-300 psi overpressure(1500-2000 KPa)	Raise by adding barytes Lower by adding water (check viscosity)
Removal of cuttings	Plastic Viscosity	Increased drilling fluid viscosity decreases penetration rate	Keep as low as is practically possible 35-50 secs M. F. AV 12-20 cp, PV 10-15 cp	Lower by adding water (check density) or thinner.
	Yield Point	Increased yield point and gel strength decreases penetration rate	YP $\pm$ 9 x drilling fluid density in kg/l	Raise by adding bentonite or XC-polymer. Reduce with dilution.
	Gel strength		0'gel (drilling fluid density-1) x 10 10'gel (drilling fluid density-1) x 15	Lower by adding thinner or dilution
Protection and support of bore hole wall by the formation of an impermeable filter cake which also minimises formation contamination	Fluid loss	Decreased fluid loss slightly decreases penetration rate	Spud fluid $\pm$ 20 ml Shallow, no producing zones : 10 ml Below 10,000 ft : 5 ml. Hole troubles or producing zones < 5 ml	Lower by adding CMC. or tStarch Raise by adding water
	Solids content	Increased solids content decreases penetration rate	In unweighted fluids < 10% vol	Keep as low as possible by continuous removal of unwanted clay, silt, sand and cuttings

TROUBLE SHOOTING IN FRESH WATER DRILLING FLUIDS

Symptoms	Possible causes	Recommended treatment
High water loss (normal viscosity)	Inadequate fluid loss control	Add starch or CMC/LV to system.
High water loss (high viscosity)	Inadequate fluid loss control and solids build up	Run solids removal equipment. Prepare batch of new drilling fluid with excess starch or CMC LV and add slowly to system.
High fluid loss Filter cake thick and spongy	Poor dispersion of bentonite	Treat with thinner and starch or CMC
High viscosity (high PV, YP, gels and solids)	Build up of drilled solids in drilling fluid	Run solids removal equipment.
High viscosity (high PV, solids. normal YP, Gels)	Build up of drilled solids in drilling fluid	Run solids removal equipment. Dilution may also be needed.
High viscosity (normal PV, solids high YP, gels)	Excess interaction of solid particles in drilling fluid.	Add thinner cautiously.
High viscosity (high PV, YP, normal gels, solids)	Possibly combination of excess drilled solids and excess particles	Run solids removal equipment. Dilution may also be needed.
Flocculation (high water loss, high YP, gels, increase in hardness and pH)	Grouping together of bentonite particles. Typically caused by cement or calcium contamination	Treat with soda ash and thinner.
Foaming/aeration (foam formed on top of drilling fluid tanks)	Foaming is usually caused by ligno-sulphonates or polymers	Alcohol defoamer.
Unstable drilling fluid, barytes settling out	Fluid viscosity unable to support barytes	Increase YP viscosity by addition of bentonite/biopolymer.

Symptoms	Possible causes	Recommended treatment
Starch fermentation (bubbling in drilling fluid, sometimes with unpleasant smell)	pH too low	Add caustic to raise pH above 10.5. The pH of any drilling fluid containing starch should be maintained at least as high as 11 to avoid fermentation .
Salt contamination (high viscosity, gels, water loss increased salinity)	Drilling salt formation	Usually it is necessary to convert to salt-saturated system. Very small contaminations may be treated with thinners,CMC and dilution.
Bit balled	Bit heavily packed with cuttings	Maintain viscosity and gels at lowest possible values. Add phosphate soap
Differential sticking	String against permeable formation, high solids content, high fluid loss	To prevent differential sticking keep fluid loss at a minimum and maintain a thin, slick filter cake by addition of starch or CMC. If stuck pipe occurs, spot pipe-freeing agent across zone where pipe is stuck.
Sloughing shale (excessive cuttings of splintered shale, tight connections)	Drilling fluid weight and/or hole cleaning inadequate	Increase drilling fluid weight, if possible. Maintain low fluid loss.

OIL BASE DRILLING FLUID

INTRODUCTION

Oil base drilling fluids are used world-wide for several different applications. They are especially suitable for drilling slim and deviated holes, depleted zones and water sensitive formations. Two different types of oil base drilling fluids can be distinguished, pure oil base fluids and invert oil emulsion fluids (IOEM). Pure oil base drilling fluid contains less than 3% vol water. This water is considered as unavoidable contaminant and the drilling fluid properties are created by adding chemicals. Invert oil emulsion fluids contains 5 - 40% vol water which is well emulsified. In this case water replaces expensive oil and provides part of the drilling fluid properties. It is generally recommended to use chemicals that belong to the same fluid system as most oil base systems contain chemicals that are adapted to each other. In the following section a brief description is given of the various base chemicals. Trade names are not mentioned as they vary between systems.

OIL BASE DRILLING FLUID

THE OIL PHASE

The following chemicals are used in the oil phase of IOEM:

Base oil

Diesel oil has been the widest used base oil for oil base drilling fluid. In view of its large concentration of aromatic compounds (toxicity!) diesel has been phased out and replaced by less toxic base oils. A wide range of low toxic (aromatic content less than 2%) base oils is available. The flashpoint and viscosity of base oils vary widely. A list of base oils with their various properties is presented on [page I-25](#).

Primary Emulsifier

The water that is present in the IOEM is sheared into very fine droplets. The primary emulsifier is a surfactant that puts itself on the oil/water interface. It forms a skin around the water droplets. Due to this skin the water droplets cannot approach each other and will therefore stay very small (typical size: a few microns). The presence of well emulsified water droplets causes an increase in Plastic Viscosity.

Secondary emulsifier

A secondary emulsifier is normally added to IOEM to improve the emulsion stability. Moreover, the secondary emulsifier should emulsify water entering the drilling fluid from e.g. the drilled formation.

Fluid loss agent

A fluid loss agent, e.g. blown asphalt, is added to form a thin impermeable filter cake.

Primary viscosifier

The primary viscosifier is often a surfactant that interacts with the emulsifiers. Due to this interaction a weak structure occurs. This leads to an increase in Yield Point and gel values.

Secondary viscosifier

The YP/gel values obtained with the primary viscosifiers are generally not sufficient for a good carrying capacity and suspension stability. Therefore, clays that are coated with organophilic material are used as secondary viscosifiers. These organophilic clays form structures in the oil phase that are similar to the normal clay structures in water base drilling fluids. It is often advertised to add a very little amount of water together with the secondary emulsifier. Although this is indeed beneficial extreme care should be taken as the slightest water overdose will do a lot more harm than good.

Oil wetting agent

All solids that enter an IOEM must remain in the oil phase in order to avoid the dropping out of water droplets that have become too heavy. Therefore sufficient oil wetting agent must be available to coat all solids with an oil wet layer. Extra additions of oil wetting agent are required particularly when adding weighting material or with increased penetration rates.

OIL BASE DRILLING FLUID

Weighting material

Normal weighting materials that are used in water base drilling fluids (barytes, iron oxide, dolomite) can also be used in IOEM. It must be emphasized that the total solids content (drilled solids + weighting material) plus the emulsified water content should not exceed 45% of the total volume. When weighting up an IOEM it may therefore be required to add extra oil (see also [page I-25](#)). Furthermore extra oil wetting agent must be added when adding weighting material (See [previous page](#)).

OIL BASE DRILLING FLUID

THE WATER PHASE

The composition of the water phase is a key-factor in the emulsion stability and drilling fluid/formation interactions. The effect of the water composition on the emulsion stability is often not appreciated and expensive oil phase chemicals are sometimes wasted where cheap water phase chemicals could have been used. Chemicals that are added to an IOEM in order to treat the water phase must find a way to the water droplets through the oil phase. It is therefore recommended to use fine powders rather than flakes in order to promote fast and homogeneous dissolution.

Lime

Lime is added to IOEM to provide alkalinity to the water phase, thus reducing corrosivity. Moreover, lime counteracts the effects of contaminants such as H_2S or CO_2 gas. Furthermore the performance of emulsifiers is optimised by a certain amount of lime in the water phase.

Calcium Chloride

Calcium Chloride is added to IOEM in order to match the salinity (activity) of the water phase with the salinity of formation water. This “balanced activity concept” should prevent the transfer of water from the drilling fluid into the formation and vice versa. Although the concept cannot be proven theoretically it is widely (and often successfully) applied in practice. Moreover, like lime, the presence of calcium chloride optimises the performance of emulsifiers. When the properties of an IOEM are not up to standard it is often easier - and cheaper - to treat the water phase first. i.e. prior to adding more emulsifier to the oil phase.

OIL BASE DRILLING FLUID

PROPERTIES

Apart from the standard properties that are measured for all drilling fluids, oil base drilling fluids require some special properties.

Electrical stability

The electrical stability of an IOEM is a measure of the voltage required to initiate the flow of an electric current through the drilling fluid. A change in electrical stability normally represents a change in the emulsion stability of the drilling fluid. Several parameters determine the absolute value of electrical stability:

- Total water content and emulsion droplet size
- Electrolyte concentration in the water phase
- Temperature of the drilling fluid
- The presence of water wet solids.

The electrical stability of field drilling fluids varies generally between 500 and 1,000 volts.

Activity

The activity of the water phase of IOEM is a measure of the salinity (CaCl_2 content) of the water. The relation between activity and CaCl_2 content is presented on [page I-26](#). The salinity is often expressed as activity to enable a comparison with the activity of formation liquids or cuttings.

Solids content - oil/water ratio

Both the water droplets and the solid particles (weighting material and drilled cuttings) are distributed over the continuous oil phase. The solids content is normally expressed as volume percentage in the total drilling fluid. The water content is expressed as oil/water ratio. i.e. not taking into account the solids. One litre of drilling fluid with 25% solids and an oil/water ratio of 80/20 will therefore contain 250 ml solids and 150 ml water.

OIL BASE DRILLING FLUID

MISCELLANEOUS

Weighting up IOEM

For economical reasons the oil/water ratio of IOEM is often adapted to the drilling fluid density. At low drilling fluid densities (i.e. low solids contents) relatively large amounts of water are often tolerated to save on expensive oil costs. Approximate ranges of solids, water and oil are presented on [page I-26](#). When such a drilling fluid must be weighted up (in a kick situation) it is not possible to add barytes and keep the oil/water ratio unchanged. This would lead to an unacceptably low oil content. The drilling fluid will become very unstable and extreme viscosities will occur.

When weighting up an IOEM it is essential that prior to addition of weighting material the oil/water ratio is adapted to the envisaged final density. During the addition of weighting material sufficient oil wetting agent must be added.

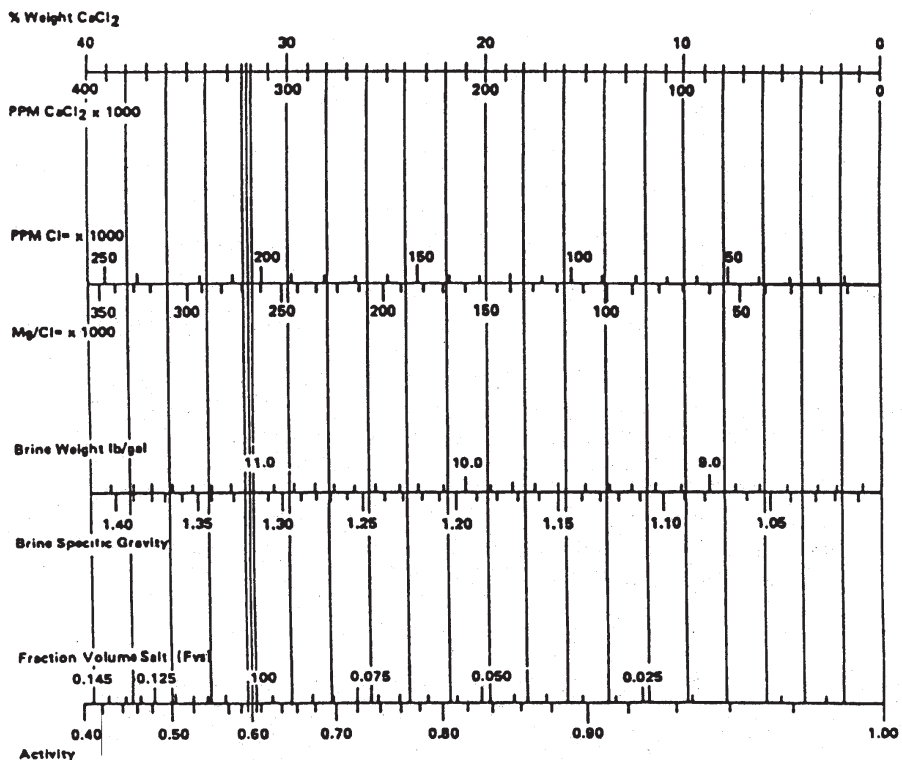
Gas solubility in oil base drilling fluid

When drilling into deep gas reservoirs the hydrostatic head of the drilling fluid is sufficiently high for the gas to dissolve in the drilling fluid. In such cases it is difficult to detect an influx; the detection depending on the size of the influx. When drilling fluid containing dissolved gas is circulated out, however, the gas will come out of solution at a given depth. It will then try to expand to the volume corresponding to the pressure at which this process takes place. This will cause a vast expansion and, therefore, high choke velocities and high well head pressures.

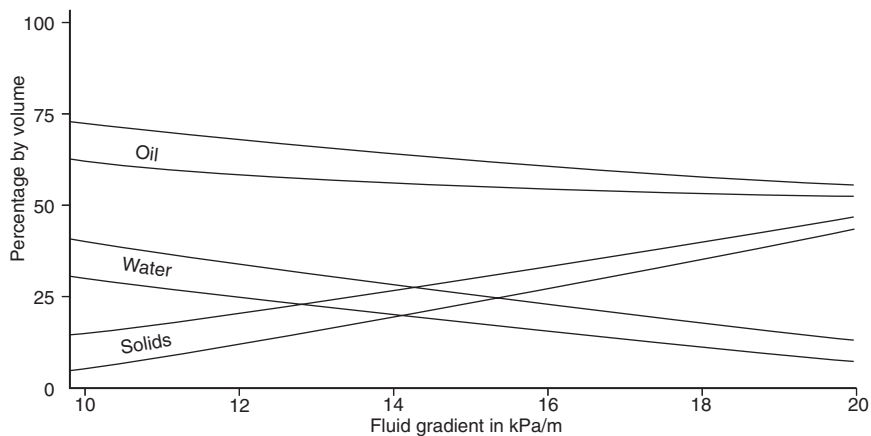
Properties of various low aromatic oils

	Shell		BP	Exxon	
	D70	DMA	HPO	SOM31	ENL 195
Naphthenics, %	38	40	42	32	33
Aromatics, %	0.5	0.5	2	0.5-1	4
Boiling, °C	195	225	190	205	210
Range, °C	250	265	260	245	355
Flashpoint, °C	72	97	66	80	91
Density, 15°, kg/l	0.792	0.807	0.785	0.80	0.82
Viscosity, cp	1.6	2.1	1.7	1.7	3.6

RELATION BETWEEN CALCIUM CHLORIDE CONCENTRATION AND ACTIVITY



AVERAGE COMPOSITION OF FIELD OIL BASED MUDS



WORKOVER/COMPLETION FLUIDS

INTRODUCTION

Workover/completion fluids are fluids which are used for all operations in a well bore after termination of the drilling phase. As these operations may vary between wireline manipulation of completion parts and plug back/sidetracking of a well the requirements for workover/completion fluids will vary widely for the different applications.

Some considerations, however, are a prerequisite for all applications of workover/completion fluids.

Formation Damage

In virtually all cases the workover/completion fluid will be in direct contact with producing or injection intervals. Prevention of formation damage is therefore always the No.1 priority in the choice of the fluid. Possible mechanisms of formation damage are described overleaf.

Fluid Stability

In workover operations it may occur that at least part of the fluid in the hole is not circulated for long periods. This implies that the rheological and fluid loss properties in particular of the chosen fluid must remain constant over long periods of exposure to high temperatures and pressures.

Corrosion

Certain frequently used workover/completion fluids can produce high corrosion rates. This may cause severe damage to well tubulars and, furthermore, corrosion products may cause formation impairment. It is therefore essential to treat the fluids in such a way that acceptably low corrosion rates are maintained throughout all operations.

Economics

The cost of workover/completion fluids may be as high as US\$ 3-4 per litre. This leads to substantial investments especially when the fluid is kept in the hole for long periods. Moreover, treatment costs may rise up to US\$150 per m³ (US\$25 per barrel) of circulation volume. Careful programming of the use, restoration and re-use of the fluids, and proper selection of treatment systems, will have considerable effect on the total economics of workover/completion operations.

Workover/completion fluids are usually divided in two categories, solids-free liquids and solids-laden liquids. Solids-free liquids are generally solutions of salts in water. The presence of solids is avoided as much as possible by careful handling and filtration. The solids present in solids-laden liquids are selected in order to ensure minimal permanent formation damage.

Detailed descriptions of solids-free liquids and solids-laden liquids are presented on the following pages.

Additional information on the subject of completion and workover fluids may be found in the Distance Learning Package Part 6.1, Topic 1.15 .

WORKOVER/COMPLETION FLUIDS

FORMATION DAMAGE

As pointed out on the previous page, the prevention of formation damage is an important factor in the selection of workover/completion fluids. The following are the various mechanisms of possible formation damage.

Solid particles

Solid particles suspended in the workover/completion fluid invading a formation may cause severe formation damage. If a solids-laden fluid is used the amount and size of the solids must be selected in such a way that rapid creation of a filter cake on the formation is ensured. It is generally accepted that a few percent solids with a minimum size of $\frac{1}{3}$ of the pore opening are required to ensure proper formation of filter cake.

Fluid/formation rock interactions

A workover/completion fluid entering a clay-containing formation may cause swelling and dispersion of clay, resulting in formation impairment. When a minimum of 3%w KCl or some other salt, e.g. NaCl, is present in the fluid, damage will generally be prevented. At high brine densities (i.e. high salinities) CaBr_2 brines (s.g. between 1.70 and 1.80) have shown formation damage due to clay shrinkage.

Furthermore, a pH value above 11 will most probably cause clay dispersion and must therefore be avoided although corrosion is effectively controlled. Adherence to the above mentioned criteria will generally be sufficient to avoid problems. In some cases core flooding studies may be required.

Fluid/formation water interactions

If the workover/completion fluid is not compatible with the formation water precipitates may form resulting in formation damage. A classic example is the use of calcium chloride brines in wells containing CO_2 gas or carbonate rich formation water leading CaCO_3 deposition. Other possible sources of deposits are sulphates, magnesium oxide-hydroxide and iron sulphate.

For these cases compatibility studies must be carried out under downhole conditions.

Fluid/hydrocarbon interactions

When a workover/completion fluid enters an oil bearing formation a viscous emulsion may be formed. This emulsion will be stabilised by surfactants which are present in the crude and/or the workover/completion fluid. These emulsions are generally only formed at rather high mixing rates. Hence, if the rate at which the fluid enters the formation can be kept low, emulsion forming is generally not a problem.

BRINES

INTRODUCTION

Brines - solutions of salt in water - are widely used as workover/completion fluids. Depending on the type of salt and its concentration densities between 1.0 kg/l and 2.3 kg/l can be obtained. Occasionally brines must be viscified to prevent losses to the formation or to ensure sufficient carrying capacity in well clean-outs. Filtration is required to avoid formation damage by solids.

The composition, properties and treatment of various brines are discussed in the following pages. Prices of the various brines vary strongly per area. A relative price as function of brine density is presented in Figure 1.

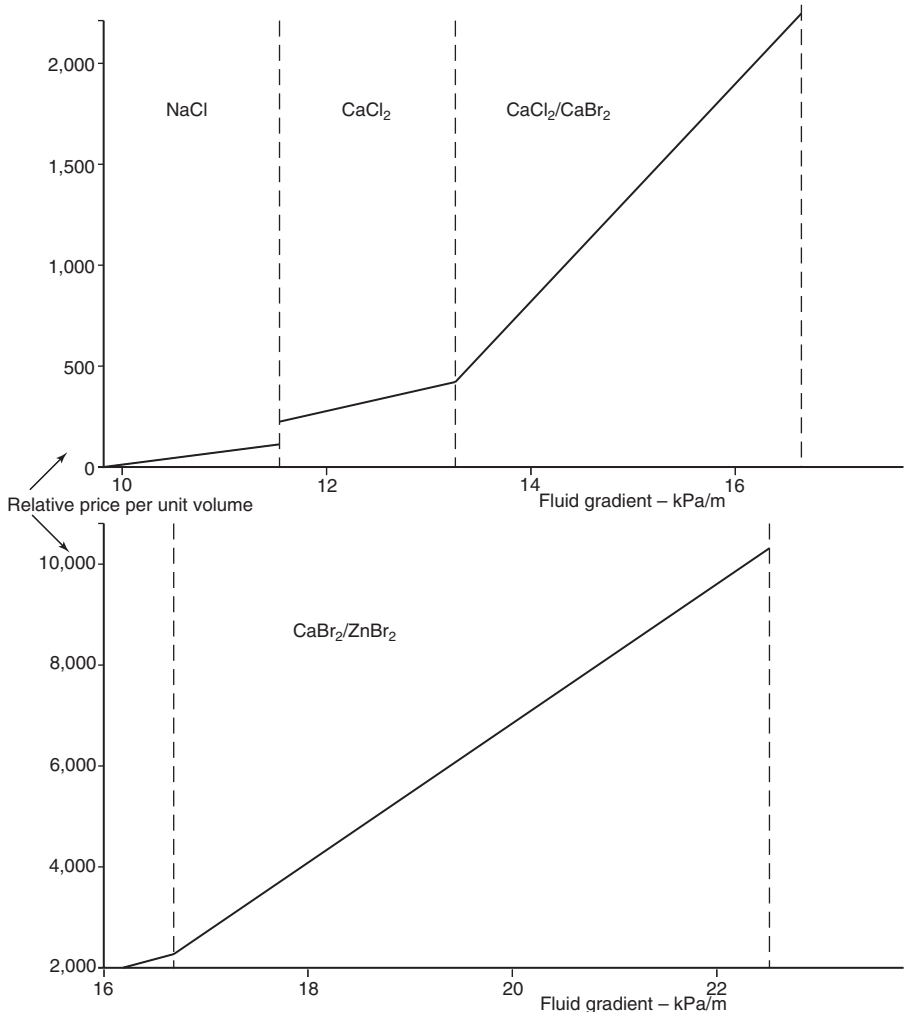


Figure 1 – Relative price of various brines vs. density

BRINES

SEA WATER, PRODUCED WATER AND SODIUM CHLORIDE BRINES

Sea water/produced water

Sea water and produced water are frequently used as cheap easily available completion brines. In view of their nature, however, complications may occur with, for example, the deposition of salts at higher temperatures or due to air in solution.

Sodium chloride (NaCl) brine

NaCl brine is made up from sacked salt and fresh water. The maximum obtainable density is 1.18 kg/l. NaCl brine is fairly cheap and easy to operate. The pH can be adjusted with caustic soda or lime and NaCl brine can be viscified with commercial viscosifiers.

The composition and density of NaCl brines is depicted below.

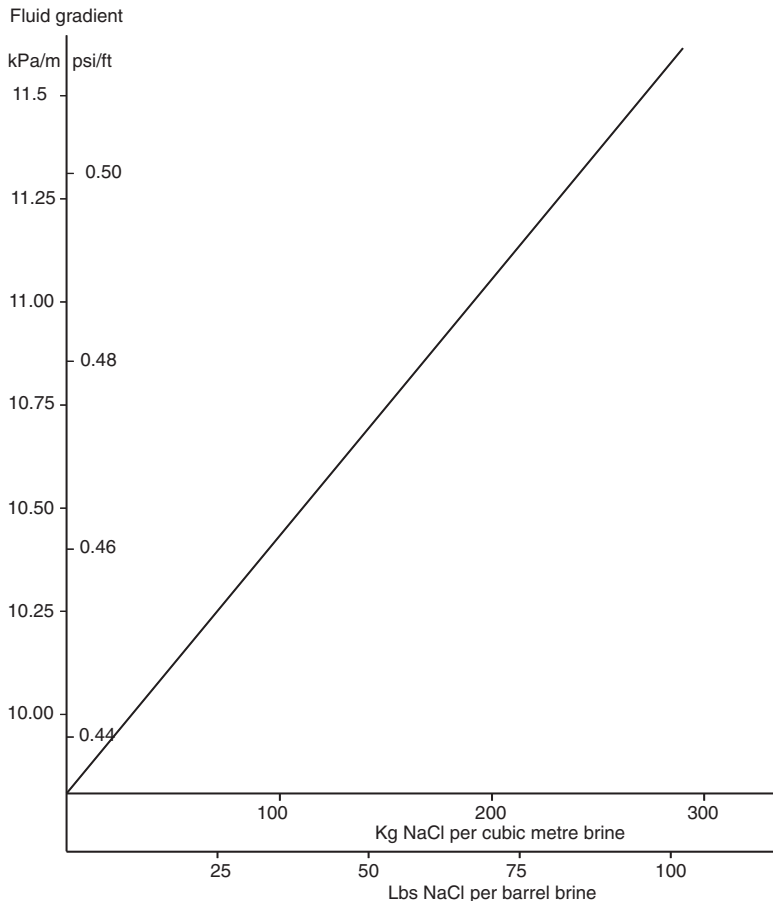


Figure 2 – Composition of NaCl brines

BRINES

CALCIUM CHLORIDE BRINES

Calcium chloride (CaCl_2) brine is a widely used workover/completion fluid. It is generally made by adding sacked CaCl_2 salt ($\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$ technical grade or 96-98% pure CaCl_2 powder) to fresh water. A potential problem is the precipitation of calcium phosphates which may take place even after filtration. Furthermore, precipitates of calcium carbonate will be formed when a fluid comes into contact with CO_2 or HCO_3/CO_3 containing waters.

The maximum density at which CaCl_2 brines can be used is determined by the crystallisation temperature.

The brines can be viscosified with commercial viscosifiers and the pH can be adjusted with lime. The composition and crystallisation temperatures of CaCl_2 brines are presented below.

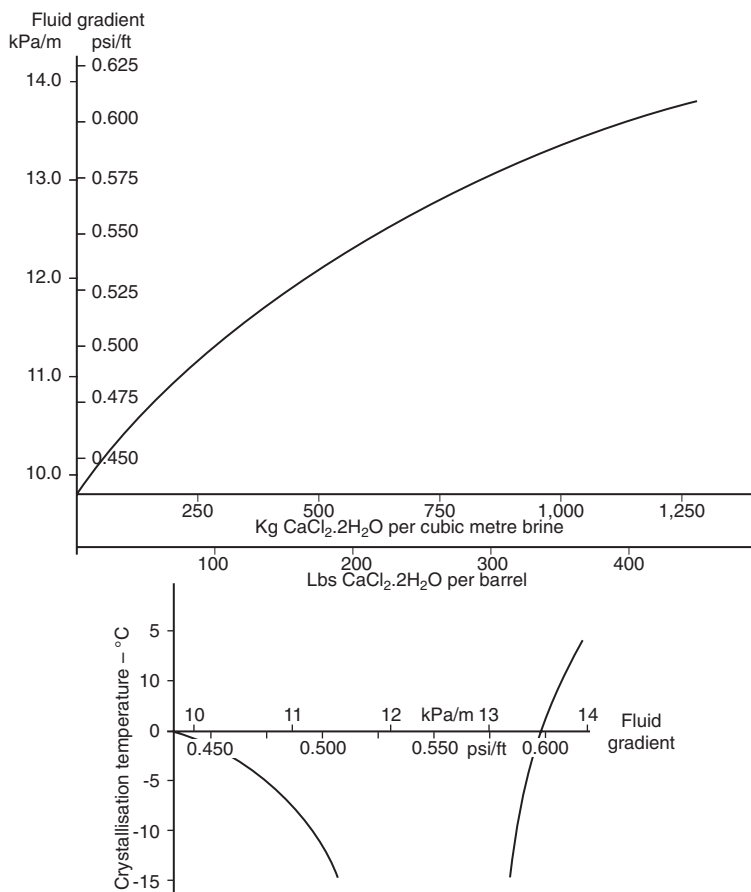


Figure 3 – Composition and crystallisation temperatures of CaCl_2 brines

BRINES

CALCIUM BROMIDE BRINES

Calcium bromide (CaBr_2) brines are used at densities up to 1.70 kg/l (14.2 ppg). A bulk liquid with this density is supplied by several manufacturers and is frequently used as a base liquid to obtain lower densities. Calcium bromide is also supplied as 95% pure powder. This powder, however, is highly hygroscopic and serious skin effects occur on longer exposures. To obtain densities below 1.70 kg/l calcium bromide brines are mixed with 1.35 kg/l calcium chloride brine. The mixing rate depends on the desired density and crystallisation temperature. Two composition/density graphs with different crystallisation temperatures are given below.

Calcium bromide brines can be viscosified with HEC or biopolymers. Polymer solution rates are however low. In view of the costs involved CaBr_2 brines should be handled with care. After use the brines are generally transported to base and conditioned for re-use.

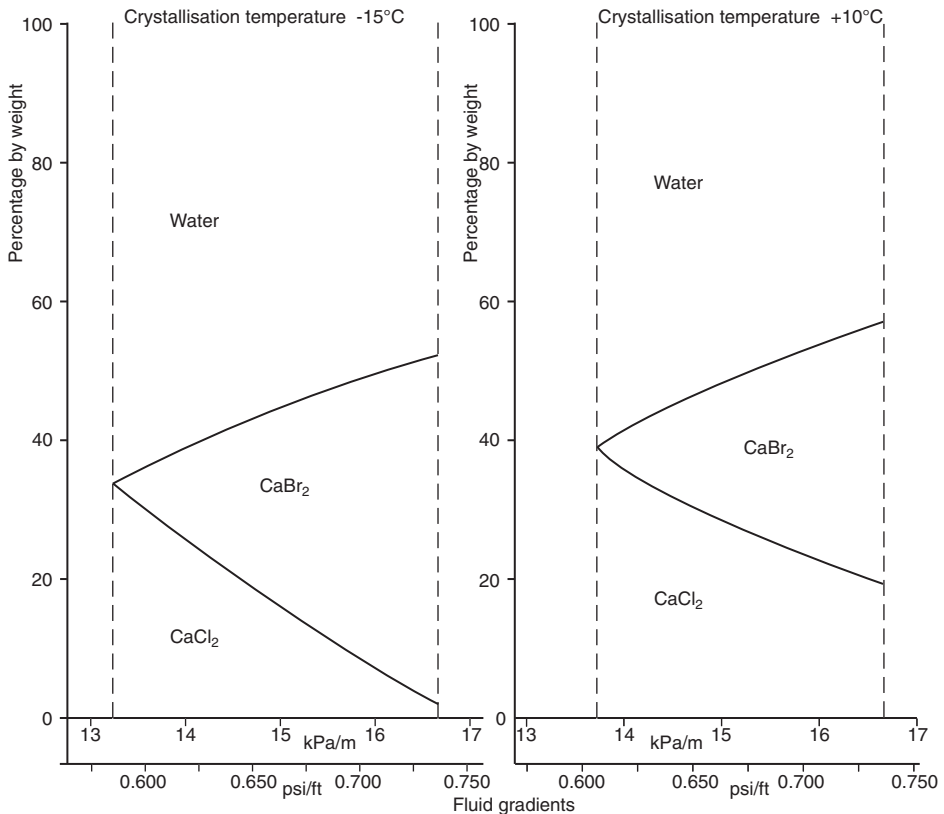


Figure 4 – Composition of $\text{CaCl}_2/\text{CaBr}_2$ brines at different crystallisation temperatures

BRINES

ZINC BROMIDE BRINES

Zinc bromide (ZnBr_2) brine must be used when densities higher than 1.70 kg/l are required. It is supplied in plastic lined drums as a 2.30 kg/l solution. The density is adjusted by mixing this solution with 1.70 kg/l $\text{CaBr}_2/\text{CaCl}_2$ brine.

Zinc bromide brines are toxic and may cause severe burns on the skin. They are also acidic and highly corrosive. Oxygen scavenger and corrosion inhibitors must be added continuously as the effectiveness of these materials decreases sharply with time.

Zinc bromide brines are always reconditioned for re-use.

BRINES

VISCOSITY

The viscosity of most brines can be adjusted by additions of HEC or biopolymers. The viscosity characteristics of these products vary strongly with temperature. In order to judge the downhole viscosity measurements must be carried out at static BHT. To prevent losses of brine into the formation viscosities must be increasing with decreasing shear rates. Hence, a high YP is a prerequisite to prevent brine losses.

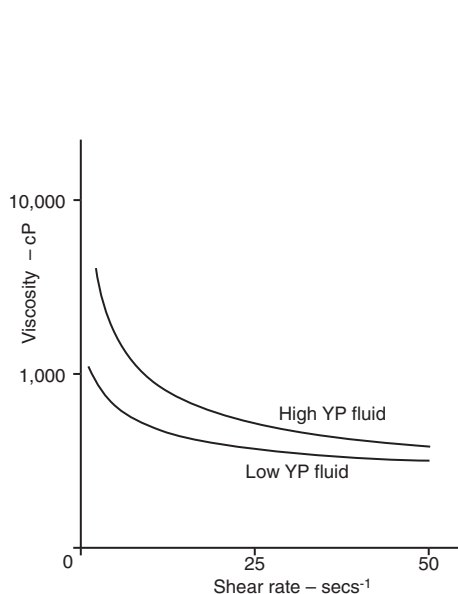


Figure 5a – Effect of YP on viscosity at low shear rates

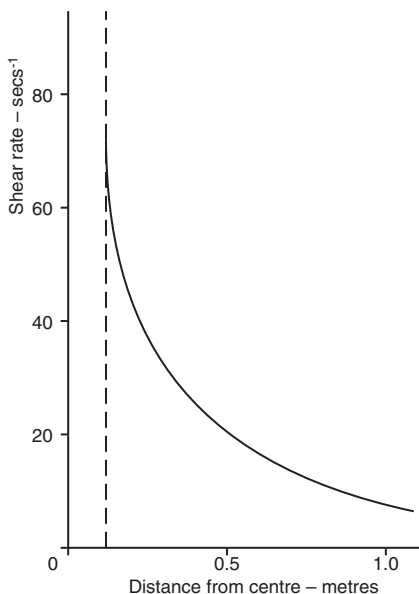


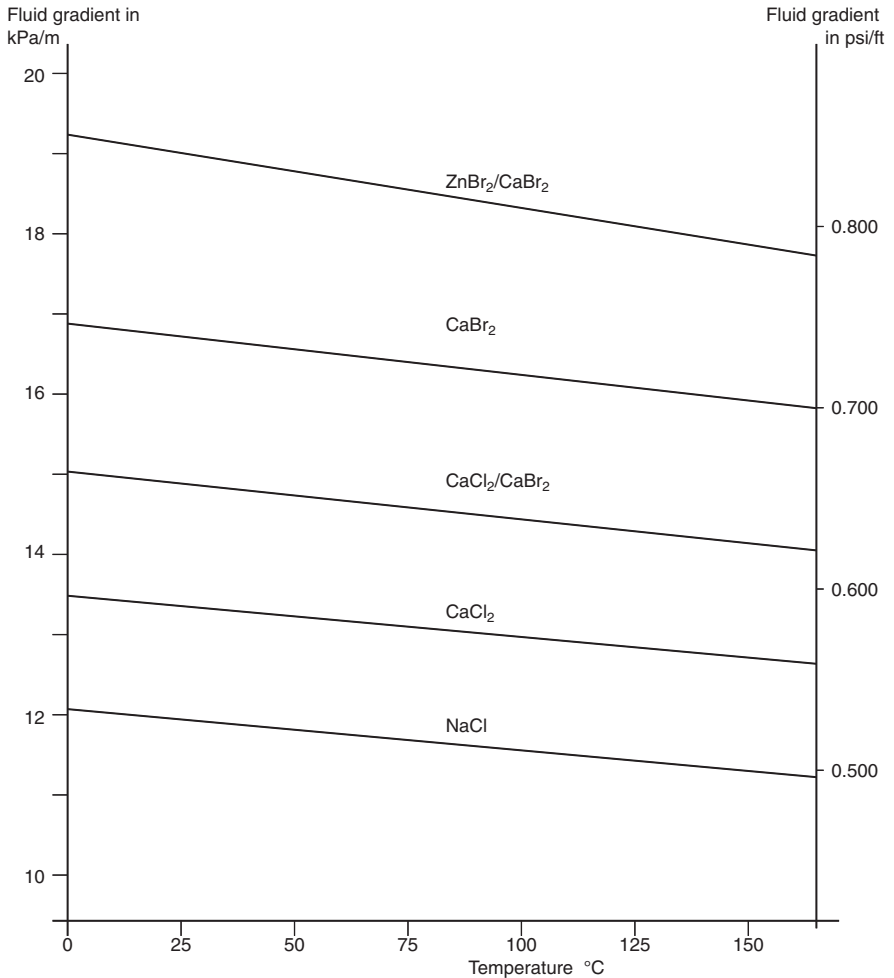
Figure 5b – Typical shear rates around a borehole

BRINES

DENSITY/TEMPERATURE EFFECTS

Temperature may have a dramatic effect on the density of brine. Due to thermal expansion brine densities will decrease as the temperature increases. This may well lead to under balance and thus to a flowing well.

The effect of temperature is illustrated in the chart shown below. If the effect is likely to be critical a chart for the exact brine mixture in use should be consulted.



BRINES

FILTRATION

Removal of all suspended solids from a brine is essential to prevent formation damage. Several innovative techniques such as centrifuges and flocculation/flotation are under development. Filtration, however, is still the most widely used method.

Two type of filters are commonly used.

Precoat filters

Precoat filters consist of a coarse back-up screen on which the actual filter is built by particles added to the liquid stream. Diatomaceous earth (DE) is the most commonly used filter medium (some OUs prefer alternative material such as perlite as DE is a suspected health hazard).

Precoat filters are relatively expensive on a day-to-day rental basis. Operating costs, however, are low.

Depending on the choice of filter material good to excellent brine qualities are obtained at high to moderate throughput rates.

Cartridge filters

Filter cartridges are available as wound fibre (non-absolute) or as cellulose or glass fibre (absolute) pleated membranes.

Non-absolute filters are easy to operate but need continuous supervision as wash-outs frequently occur.

Absolute filters can give perfect brine qualities but tend to get blocked by traces of oil in the brine. Cartridge filters are relatively cheap on a day-to-day rental basis. Operating costs (cartridges !), however, can be quite high.

U. S. MESH SIZES

US Mesh	D (mm)	D (ins)	OE/
2 ^{1/2}	8.0	0.315	-3.0
3	6.73	0.265	-2.75
3 ^{1/2}	5.66	0.223	-2.5
4	4.76	0.187	-2.25
5	4.0	0.157	-2.0
6	3.36	0.132	-1.75
7	2.83	0.111	-1.5
8	2.38	0.094	-1.25
10	2.0	0.079	-1.0
12	1.68	0.066	-0.75
14	1.41	0.056	-0.5
16	1.19	0.047	-0.25
18	1.0	0.039	0.0
20	0.841	0.033	0.25
25	0.707	0.028	0.5
30	0.595	0.023	0.75
35	0.500	0.02	1.0
40	0.420	0.017	1.25
45	0.354	0.014	1.5
50	0.297	0.012	1.75
60	0.25	0.0098	2.0
70	0.21	0.0083	2.25
80	0.177	0.007	2.5
100	0.149	0.0059	2.75
120	0.125	0.0049	3.0
140	0.105	0.0041	3.25
170	0.088	0.0035	3.5
200	0.074	0.0029	3.75
230	0.063	0.0025	4.0
270	0.053	0.0021	4.25
320	0.044	0.0017	4.5
400	0.037	0.0015	4.75

J – LOGGING

Clickable list

(Use the hierarchical list under "Bookmarks" to access individual tables and/or sub-topics)

Summary of tools, logs & samples	J-1
Quicklook evaluation	J-2
Logging tips for well site geologists	J-6
Common logging tools: output curves & mnemonics	J-10

SUMMARY OF TOOLS, LOGS & SAMPLES

Device/Sample	Qualitative use	Quantitative use
SP log	Detection of reservoirs Correlation	Formation water salinity Thickness determination
Conventional resistivity log	Correlation Bed delineation Locating lost pipe	Formation resistivity Water saturation
Laterolog	Correlation Locating lost pipe	Formation resistivity Water saturation
Induction log		Formation resistivity Water saturation
Microresistivity log	Correlation Locating lost pipe	Flushed zone resistivity Water saturation
Sonic log	Cement Bond inspection Lithology Correlation Detection of fractures Locating lost pipe Lateral prediction	Porosity Seismic velocity
Formation density log	Identification of minerals Lateral prediction	Porosity Density Seismic velocity
Neutron logs	Correlation	Porosity
Density and Neutron log in combination	Complex Lithology Shaliness Gas detection	Porosity in complex lithologies
Gamma ray log	Correlation Distinction between shale and non-shale Detection of radio-active minerals Estimate shale in 'dirty' sands	Shale content Depth control Net/Gross ratio in reservoir section.
Repeat Formation tester	Fluid samples	PVT Analysis Pressure Data
Gamma ray collar locator	Locating lost pipe Casing collar location	Depth measurement for perforation
4-arm dipmeter		Formation dip
Free point indicator		Depth of free point
Caliper survey		Hole diameter
Continuous directional survey		Inclination & azimuth
Cutting	Lithology Fluid type (gas or oil)	
Sidewall samples	Lithology Fluid type (gas or oil)	
Cores	Formation homogeneity Show fractures, fossils Deposition patterns Determine stimulation possibilities	Porosity Calibrate porosity logs Formation permeability

QUICKLOOK EVALUATION STEP BY STEP

Review the logs

1. Inspect the mud log for intervals with reservoir rock, hydrocarbon shows and mud gains.
2. Review the quality of the wireline logs checking headers, depths, scales calibrations and tool checks as required. *Read the remarks section, if present.*
3. Use logs from surrounding wells, if available, to identify any obvious anomalies in the data.

Identify Reservoir Rock

4. Discriminate potential reservoir rock from non-reservoir rock using the GR, SP, caliper (mud cake) and porosity logs. Prepare a sand count using 1:200 scale logs (preferably the density curve).
5. Square porosity and resistivity log readings in the reservoir sections.

Porosity

6. Calculate porosity using density and/or neutron logs depending on lithology:

In **sandstones** calculate the porosity from the density log using a matrix density of 2.65 g/ml (unless otherwise known):

$$\phi = \frac{\rho_{ma} - \rho_{LOG}}{\rho_{ma} - \rho_{fl}}$$

In **carbonates** use the FDC/CNL crossplot provided in chart books to establish the matrix density, ρ_{ma} , of any limestone/dolomite mixture before using the above formula.

Estimate the fluid density, ρ_{fl} , based on the salinity of the mud filtrate e.g. from R_{mf} and resistivity vs. salinity charts. In hydrocarbon bearing zones approximate the invaded zone fluid density using the mud filtrate density and an estimated hydrocarbon density :

$$\rho_{fl} = 0.7\rho_{mf} + 0.3\rho_{hc}$$

Hydrocarbon Saturation

7. Calculate the approximate true resistivity, R_t , from the deep laterolog, R_{LLD} , and the shallow laterolog, R_{LLS} , using the superdeep equation:

$$R_t = 1.7R_{LLD} - 0.7 R_{LLS}$$

In the absence of a laterolog, assume the deep induction log approximates R_t .

8. Identify an appropriate fully water bearing section of the logs and use this to evaluate the formation water resistivity, R_w :

$$R_w = R_0\phi^m$$

Where no water bearing reservoir rock is present define R_w using local well data, the SP (PHB Petrophysics section 2.2.4) or an R_w atlas.

9. Calculate hydrocarbon saturation, S_h , from the Archie equation using $m=n=1.8$ in sandstones and 2.0 in carbonates:

$$S_w = \left(\frac{R_w}{R_t\phi^m} \right)^{\frac{1}{n}}$$

$$S_h = 1 - S_w$$

Hydrocarbon Distribution

10. Determine, as far as possible, the presence of the various fluid contacts (GOC, OWC, GDT, OUT, ODT WUT) from the logs. Identify the presence of transition zones
11. Use SWS's, RFT pressure data and RFT fluid samples to confirm the presence of oil and gas and identify pressure regimes. Target RFT and SWS at areas of uncertainty from the log evaluation, particularly where calculated S_h values are between 50 % and 70 % pv.

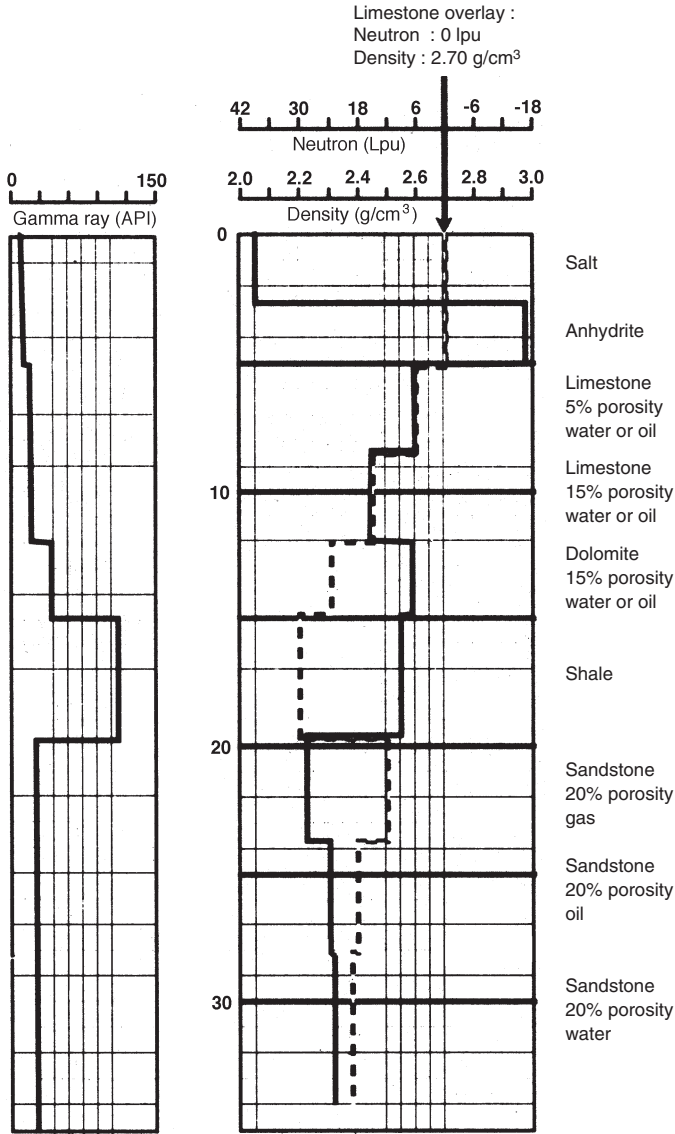
When selecting RFT and SWS depths take note of the following:

- Specify depths with respect to a named and dated log (e.g. GR of FDC/CNL/GR of 4/5/94)
- Use the caliper to identify smooth hole on-gauge hole sections
- For the RFT pick high porosity intervals where possible to avoid supercharging - identify these by high porosity and low GR (shale content)
- In picking RFT pressures, consider the requirement spacing and position required for gradient calculation and establishing communication between reservoir units. In long reservoir units take sufficient pressures to identify changes in fluid properties with depth.

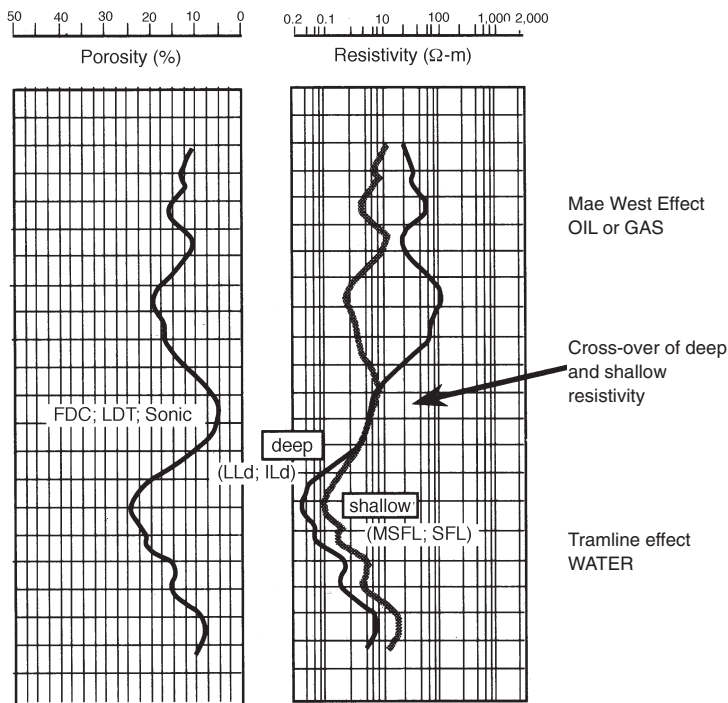
Reporting

12. Report the results of the quicklook evaluation summarising the following elements for each major reservoir and fluid type:
 - Total net hydrocarbon sand count (Net/Gross ratio if gross interval known/defined)
 - Average porosity
 - Average hydrocarbon saturation (transition zone separate)
 - Observed fluid contacts and source (RFT or logs)
 - Petrophysical parameters used (ρ_{ma} , R_w etc.)
 - Special considerations or peculiarities of the evaluation

GR - Density - Neutron : Lithology determination
Gas/Oil differentiation



Quick Look methods to distinguish hydrocarbons and water



LOGGING TIPS FOR WELLSITE GEOLOGISTS

Safety

- Safety meeting:

It should be a standard practice to hold a safety meeting before rigging up and also before any explosives are run into the hole. This should include the logging crew, geologist, company man, and most importantly the rig crew. Most important points as far as the rig crew are concerned are the location of any nuclear sources and where the crew should not be when the sources are being loaded into the tools. Similarly for explosives. The engineer should also explain any safety points regarding the equipment they are using, particularly on the rig floor. Often a good rig crew will help during the rig up. In these situations it is vital that they understand all the dangers associated with the equipment.

- Safety signs:

There should be signs posted around the area where the nuclear sources are kept warning people to keep away, similarly with explosives. When the sources are being loaded signs should warn the crew to keep off the rig floor. For explosive runs signs should also be posted, particularly on the radio/telephones and also on any welding sets.

Equipment

Obviously all the equipment the contractor brings to the site will be in good condition. A good paint job and clean tools etc. will generally indicate that the level of maintenance is high and you shouldn't need to look any further, however you may just like to have a quick look at the following:

- The cable:

A good cable is easy to spot. It will be clean and shiny. Excessive rust, worn or broken strands, or 'birdcaging' all indicate that it may not be as strong as its design strength and in a 'difficult' well may cause problems (break) especially if an overpull is required. A quick look inside the cab at the cable chart will show you how up to date the cable records are. If the cable looks poor and the records are out of up to date then you may be looking at problems. Ask the engineer.

- Depth system (IDW):

The depth system must be functional or else the logs will be off depth. Take a quick look at the two 'encoder' wheels. If they are pitted or corroded they will cause inaccuracies in the measurement and should be replaced.

Before logging

- Toolstrings:

It is very important to confirm the tools required for the job in advance, so that the engineer can prepare them.

The composition of the toolstring is very important. Whilst most tools can be positioned in a number of different locations in the toolstring there is generally a standard configuration for different places in the world. Sometimes tools will have been specially modified for the local conditions so it may be impossible to run them in different positions in the string. This becomes important when specific readings, or samples are needed right down at the bottom of the hole (first readings). In order to ensure that the required 'first readings' are obtained discuss the toolstring configuration with the engineer. These requirements will then determine the length of rathole required and this should be communicated to the driller.

For a typical 'Super Combo' toolstring the rathole required is around 150 ft, although this is shrinking dramatically with the introduction of new tools, such as Platform Express.

- Logging intervals:

Discuss all the logging intervals required before the job so that the engineer can plan his runs in the most efficient way, to reduce time spent in the hole. Sort out what information you really need first (generally the main log) and make this the first run. If time is really important stop the main log above the zone of interest and then go back down and do the high resolution passes. After these continue the main log out of the hole (these can be spliced together later on). This avoids the scenario of logging all the way out of the hole and then having to run all the way back in to perform the hi-res passes.

If the well is very hot, the likelihood of tool failure is increased. In these cases it is imperative that a downlog is performed as the tool is run into the hole (this should be performed anyway) and then the main log should be started as soon as the tool reaches the bottom of the well. This is where prior planning is vital as it reduces the time required in the hottest part of the well.

- Responsibility:

Clarify who on the rig is responsible for what. Generally the company man is in overall control, but during logging the geologist often has most of the responsibility. This will rarely cause problems for the engineer except in extreme circumstances such as stuck tools, fishing, etc.

A typical scenario would be where the tools have become stuck in the hole and the geologist assumes responsibility for the situation and tells the engineer to pull beyond the safe working limit of the cable in an effort to free the tool. If the cable breaks who was responsible? Normally the company man is required to make that decision. A fishing job resulting from a broken cable can be very expensive and you don't want to have the bill dropped in your lap.

- Mud sample:

There are occasionally problems caused during the mud sample measurements due to incorrect temperature measurements, particularly if the mud is hot, but the testing machine is cold. A good way to remove this problem is to ensure that the sample is obtained immediately after last circulation and then left a few hours to cool down to ambient temperature.

The engineer will need about 1 litre of homogenous mud, a good filter cake sample on a piece of filter paper, and about 10 cc's of filtrate. Try to ensure that the filter cake sample is left floating (still attached to the filter paper) in the mud sample as this will stop it drying out. It is also a good idea to ask the engineer to keep hold of the samples until after the logging job is complete in case there are any anomalies. This will allow a second test to be performed.

- **Deliverables/data transmission:**

Discuss the data you will need and when you will need it. During logging ensure the engineer produces a real time print. Although this may still require some depth adjustments you should be able to obtain all the data you need from it. Try not to burden the engineer with too many requests for print outs etc. during logging, unless you REALLY need them.

If data transmission is required, find out how urgently it is needed. The phone lines on a rig are at their busiest during the day and particularly in the early evening (drilling reports). The best time to transmit is often at night when phone lines are 'quiet'. Also, if it is 7pm on a Friday evening how likely is it that anyone will be looking at the data that evening, or even during the weekend?

In summary, try to minimise your requests for prints, data transmission, etc. during logging as much as possible as this is often quite a stressful time for the engineer, particularly if things aren't going to plan, tools failing, etc. Once the logging is finished he/she will be only too willing to provide whatever it is you need as the stress level will be back down to normal by then. Also, if the engineer is busy concentrating on playing back logs etc. when the tools are in the hole, he will not be able to monitor what is going on and this increases the risk of a major problem occurring. Basically, try to plan the logging job as much as possible before it starts.

Operation:

- **Calibrations:**

A few tools require master calibrations and many more require wellsite calibrations, both before they are run into the hole, and also after they come back out. It is a good idea to ask the engineer for a copy of the calibrations before logging. The calibration dates will be listed on the printout. Ensure that the master calibration dates are within the allotted time periods and that the wellsite calibrations have been done within the last couple of days, but more particularly on the wellsite you are on. The calibrations should also all be in tolerance.

- **Depth Control:**

During logging, if there appear to be any anomalies with the depth always look at the 'encoder' wheels first. Nothing should be allowed to be deposited on the wheels. If deposition occurs it leads to an increased diameter of the wheel and hence the depth system will read shallow. Normally there are scrapers on the wheels to keep them clean, but these can wear away or be bent. If the scrapers fail, deposition continues and the wheels may stop turning. This happens most regularly when the cable is brand new because the tar that is used to protect the cable after manufacture deposits itself on the wheels and eventually jams them.

A good indication of problems with the depth system is the 'E1-E2' reading on the depth system display. This is a measure of the difference in depth recorded by the two encoder wheels. This value should generally remain in single figures, indicating that both systems are reading the same. If it starts increasing rapidly this indicates that one of the wheels is stuck. If it is increasing slowly but steadily and reading high values this is an indication that one of the wheels is badly worn and should be replaced.

It is very important to list all the reference depths before logging begins, i.e. permanent datums, seabed, casing shoe, TD, depths of problem zones, etc. In vertical wells depth control is relatively easy, but in deviated wells it becomes much more complex and can be very confusing. Try to understand what depth control really is. Get the engineer to explain what he is doing when he is applying corrections.

While logging is going on just keep an eye on the depth system. If the engineer continuously makes adjustments to it start asking why.

- Tool testing:

There are often quick checks that can be performed in the hole to prove that the tools are working correctly.

Calipers - Do a quick uplog in the casing (or even at the casing shoe) opening all the calipers before running to TD. The diameter of the casing is known and the caliper should read this value (± 0.1 ").

Resistivity - In shales resistance tool readings should all overlay one another.

Density and lithology - In clean waterfilled formations (sands, limestones, etc) densities and PEF's should be known (look in the log interpretation chart book). Check these against the values obtained from the logs to prove that the tool is reading correctly..

RFT/MDT - Prove that the packer is sealing properly by setting the tool in casing before running down into the open hole. The pressure reading should drop rapidly to zero on drawdown and remain there.

Sonic - The velocity of sound in steel is a known value. During the caliper check in casing check that the sonic (DT, DTL) is reading $\pm 56 \mu\text{s/ft}$ ($\pm 1 \mu\text{s/ft}$). Beware; it is sometimes difficult to get a good reading for this in good cement as the sound waves are attenuated by the good cement/casing bond, and you may find that you are reading the formation value rather than the steel.

SP - A good check that the SP is working is to look for a kick in the reading at the casing shoe during the log down, otherwise look for steady values in clean sands and shales. Sometimes you may get a problem which is commonly called 'magnetised' SP. This is caused by the cable becoming magnetised due to DC currents used during perforating runs. This manifests itself as a sinusoidally oscillating SP value, caused by the voltage generated as the magnetised drum rotates. There is no way to get rid of this other than to demagnetise the cable which can only be done with the tools out of the hole. In reality it is not a major problem (as an average SP value can still be derived from the trace, it just doesn't look very nice). If the SP is magnetised you can ask the engineer to use the compensated SPARC reading instead.

COMMON LOGGING TOOLS/OUTPUT CURVES MNEMONICS (1)

Tool/Curves	Schlumberger	Western Atlas	Halliburton	BPB
Logging unit	Cyber service unit CSU MAXIS 500	Computerised logging service CLS ECLIPS		
Spontaneous pot.	SP	SP	SP	SP
Induction tools	Induction-Electrical Survey IES Dual Induction Tool DIT Phasor Induction tool DITE Array Induction Imager Tool AIT	Induction-Electrolog IEL Dual Induction Focussed log DIFL Dual Phase Induction log DPIL High Definition Induction log HDIL	High Resolution Induction HRI	
Curves	Induction Deep Resistivity ILD Induction Medium Resistivity ILM Spherically Focussed Log (Un)Averaged SFLU/SFLA Short Normal Resistivity SN	RILD RILM RFOC	R(ILD) R(ILM) R(LL)	Array Induction Sonde AIS RILD RILM Shallow Focussed Electric RSFE
Laterolog tools	Dual Laterolog Tool - DST Azimuthal Laterolog ALAT Laterolog deep resistivity LLD Laterolog shallow resistivity LLS High Resolution Resistivity LLHR	Dual Laterolog DLL High Definition Lateral log HDLL RD RS High Definition Resistivity SFR/5, SFR/3, SFR/2	Dual Laterolog DLL R(LLD) R(LLS)	Dual Laterolog Sonde DLS Deep Laterolog DLL Shallow Laterolog SLL
Curves	Micro SFL Resistivity Log MSFL Microlaterolog MLL	Microspherical Laterolog MSL Thin bed resistivity TBRT Microlaterolog MLL	MSF	Micro-Spherically Focussed MSF
Micro-resistivity tools				
Curves	Microlog Tool MLT Caliper CAL Microinverse resistivity BMIN Proximity log PML	Minilog ML Caliper CAL Proximity log PROX		Microlog Sonde MLS Microresistivity Sonde MRS Microlog Caliper CADF Micro-Inverse 1 1/2" MR1F/2F
Electromagnetic tool	Electromagnetic Propagation Tool EPT Attenuation EATT Hole Diameter HD Propagation Time TPL Electromagnetic Propagation Porosity EPHI	Dielectric Log DEL2 Attenuation A2TN Resistivity R2SL Propagation Time T2PL Water filled porosity P2DC	High Frequency Dielectric Log HFDT	
Curves				

COMMON LOGGING TOOLS/OUTPUT CURVES MNEMONICS (2)

Tool/Curves	Schlumberger	Western Atlas	Halliburton	BPB
Gamma Ray Tools	Gamma Ray Tool GFA Slim Hole GR Tool GR Scintillation Gamma Ray Tool SGT Natural Gamma Ray	Spectralog SL Digital Spectralog DSL	Natural GR Spectral Log	Spectral Gamma Sonde SGS
Curves	Spectrometry Tool NGT Gamma Ray GR Computed Gamma Ray CGR Spectroscopy Gamma Ray SGR Potassium concentration POTA Thorium concentration THOR Uranium concentration URAN Thorium/Potassium ratio TPRA Thorium/Uranium ratio TURA Uranium/potassium ratio UPRA Volume of Shale from CGR/SGR/ Potassium/Thorium/Uranium VSCG/VSSG/VSPC/VSTC/VSUC Neutron Porosity NPHI Neutron Porosity Limestone NPL	Potassium content K Thorium content Th Uranium content U Thorium/Potassium ratio RTHK Thorium/Uranium ratio RTHU Uranium/potassium ratio RUK	Gamma Ray Log GR	Spectral Gamma Ray GRSG Thorium/Potassium ratio RAKT Thorium/Uranium ratio RAUT Uranium/potassium ratio RAKU
Density Tools	Compensated Formation Density Log FDC	Compensated Density Log CDL	SLD	Compensated Density Sonde CDS
Curves	Litho Density Tool LDT Bulk Density RHOB Density correction DRHO Density Porosity DPHI High Res. Bulk Density HRHO High Res. Density Porosity HDPH Enhanced Bulk Density NRHO Porosity from Enhanced Density PHND High Res. Enhanced Bulk Density HNRH High Res. Porosity from Enhanced Bulk Density HPHN Photoelectric effect, long/short spacing) PEFL/PEFS	Compensated Z-Densilog ZDL Formation bulk density ZDEN Density correction ZCOR Porosity DPHI High resolution indicated by sample rate on log heading	$\rho(B)$ $\Delta\rho$	Photo Density Sonde PDS Density DEN Density correction DCORR Matrix Density MTXD High Resolution Density HDEN Limestone Density Por. DPRL Dolomite Density Porosity DPRD Sandstone Density Por. DPRS
		PE	PE	Far/near Photoelectric eff. PEDF/N

COMMON LOGGING TOOLS/OUTPUT CURVES MNEMONICS (3)

Tool/Curves	Schlumberger	Western Atlas	Halliburton	BPB
Neutron Tools Curves	Compensated Neutron Tool CNT Neutron Porosity NPHI Neutron Porosity Limestone NPL Thermal Neutron Porosity TNPH Epithermal Neutron Por. ENPH	Compensated Neutron Log CN Neutron Porosity NPHI	Compensated Neutron CNS	Compensated Neutron CNS Epithermal Neutron Sonde ENS
Pulsed Neutron Tools (Cased Hole) Curves	Thermal Decay Time Tool TDT Induced GR Spectroscopy Tool GST (Average) Carbon/Oxygen Ratio (A)COR Form. Iron Indicator Ratio FIIR Form. Lithology Ind. Ratio FLIR Form. Porosity Ind. Ratio FPIR Form. Salinity Ind. Ratio FSIR	Neutron Lifetime Log PDK-100 MSI Carbon/Oxygen Log MSI Magnetic Resonance Imaging Log MRIL Free Fluid Index MBVM Array of porosity MPHI Irreducible Fluid Index MBVI	TMD-L Pulsed Spectral Gamma Tool PSGT Reservoir Monitoring Tool RMT	Thermal N'tron Decay Sonde TDS TDT Potrosity TPOR
Nuclear Magnetism Tool Curves	Nuclear Magnetism Tool NMT Free Fluid Index FFI Free Porosity FPH	BHC Acoustilog AC Long Spaced BHC Acoustilog ACL Circumferential Acoustilog CAC Digital Array Acoustilog DAC Circumferential Borehole Imaging Log CBIL	BHC Sonic BCS Long Spaced Sonic LSS	Compensated Sonic Sonde CSS Long Spaced CSS - LCS
Sonic Tools Curves	Sonic Logging Tool SLT Long Spaced Sonic Tool LSS Sonic Digital Tool SDT Ultrasonic Borehole Imager UBI Dipole Shear Sonic Imager DSSI Digital Waveform Sonic Tool DWST Delta-T (μs/ft) DT Delta-T Long Spacing DTL Transit Time TT (ms) Integrated Transit Time (ms) ITT	Multipole Acoustic Tool MAC Delta-T AC TT	Full Wave Sonic FWS BCS ΔT ITT	Delta T1 etc. DT1 etc. Transit Times T1R2 etc. Sonic Porosity SPOR

COMMON LOGGING TOOLS/OUTPUT CURVES MNEMONICS (4)

Tool/Curves	Schlumberger	Western Atlas	Halliburton	BPB
Dipmeter & Directional Tools	High Resolution Dipmeter HDT Stratigraphic High Resolution	High Resolution 4-arm Diplog DIP Hexagonal Diplog HDIP Simultaneous Acoustic and Resistivity Imager STAR	Six Electrode Dipmeter SED	Multi Button Dipmeter MBD Precision Strata Dipmeter PSD Acoustic Scanning Tool AST
Curves	Measured Azimuth AZIM Calipers C1, C2 Deviation DEVI Relative bearing RB Hole Azimuth HAZI	Calipers CAL1, CAL2, CAL3 Deviation Dev Relative bearing RB Deviation Azimuth DAZ Acoustic Image Resistivity Image		Apparent Azimuth AAZD Calipers CALX, CALY Borehole Tilt TILD Relative Bearing RBAD Borehole Azimuth (Mag) AZID Borehole Azimuth (True) TAZI Dip Angle DIPC True Depth TDEP Temperature TEXTF
Seismic Tools	Temperature TEMP Well Seismic Tool WST/SAT Downhole seismic array DSA Vertical Seismic Profile Tool VSP Combined Seismic Acquisition Tool CSAT	Standard Velocity Survey VLS Multi-level Receiver MLR VSP	Well Seismic VSP	Seismic Reference Sonde SRS
Sampling/Testing Tools	Core Sample Taker CST Mechanical Sidewall Coring Tool MSCT	Sidewall Core SWC Rotary Sidewall Coring Tool RCOR Reservoir Characterisation Instrument RCI	Sidewall Core Gun SCG	Sidewall Core Gun SCG
Miscellaneous Tools	Repeat Formation Tester RFT Borehole Geometry Tool BGT Borehole Televiewer BHTV	Formation Multiester FMT Borehole Geometry 4CAL Borehole Video Camera BHVC	Selective Formation Tester SFT X-Y Caliper log	Repeat Formation Sampler RFS Borehole Geometry Sonde BGS

COMMON LOGGING TOOLS/OUTPUT CURVES MNEMONICS (5)

Tool/Curves	Schlumberger	Western Atlas	Halliburton	BPB
Positioning Tools	Casing Collar Locator CAL, CCL	Casing Collar Locator CCL	Casing Collar Locator CCL	
Curves	Digital Casing Collar Locator DCAL CCL Amplitude CCL			
Cement evaluation tools	Sonic Logging Tool SLT Cement Bond Log CBL Variable Density Log VDL Cement Evaluation Tool CET	Acoustic Cement Bond Log CBL CBL-Variable Density Log CBLV Segmented Bond Tool SBT Attenuation pads ATC-1-ATC6	Seismic Spectrum Pulse Echo Tool PET	
Curves	Caliper CAL Delta-T DT Transit Time TT Deviation DEVI CBL Amplitude CBL Bond Index BI	Transit Time PPT Single Receiver Bond Log SRB Dual Receiver Bond Log DRB Bond attenuation Log BAL		
Production Testing tools	Compact Production Logging Tool CPLT Production Sampling Tool SPST	Production Logging Tools PLT Through-tubing Borehole Fluid Sampler TBFS	Production Logging Tools PLT	Fluid Conductivity Sonde FCS Fluid Density Sonde FDS Fullbore Flowmeter Sonde FFS In-line Flowmeter Sonde IFS
Curves	Caliper CALI Deviation DEVI Well Fluid Density WFDE Well Pressure WPPE Well Pressure Gradient WPGR Well Temperature WTPE Well Temp. Gradient WTGR Total Estimated Downhole Flow Rate QTDE Flowmeter spinner speed SPIN	Caliper TCAL Fluid Density FDN Surface Rec. Pressure Log SRPL Diff. Press. Fluid Density FDDP Differential Temperature TEMP Spinner count rate FMTR		Fluid Density FDN Strain Gauge Pressure SGF Borehole Temperature TEXP Differential Temperature DTEP Fullbore Flow Rate FBFR Total Flow FTOT Fullbore Flowmeter Revolution Time FLTf
Corrosion Logging	Casing Inspection Tool CIT Pipe Analysis Tool PAT Electromag. Thickness Tool ETT Multifinger Caliper Tool MFCT Ultrasonic Imaging Tool USIT Tubing Geometry Tool FTGT	Vertilog DVRT Magnetlog (Multifrequency) DMA Multifinger Caliper Tool MFCT Circumferential Borehole Imaging Log CBIL		

K – BOPs AND OPERATING SYSTEMS

Clickable list

(Use the hierarchical list under "Bookmarks" to access individual tables and/or sub-topics)

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BLOW-OUT PREVENTERS – HYDRAULIC FLUID VOLUME REQUIREMENTS

Size and working pressure	Cameron Type A		Cameron Type U		Cameron Type F W2 Oper.		Cameron Type SS		Cameron Type QRC		Hydril Type GL		Hydril Type GK		Hydril Type MSP		Shaffer Type LWS with Posilock		Shaffer Type LWS Manual screw		Shaffer Spherical	
	Close	Open	Close	Open	Close	Open	Close	Open	Close	Open	Close	Open	Close	Open	Close	Open	Close	Open	Close	Open	Close	Open
4" 10,000																					5.0	4.7
6" 2,000																						
6" 3,000																						
6" 5,000	2.2	1.9	1.3	1.3	1.5	2.3	0.8	0.7	0.8	0.95			2.9	2.9					1.2	1.0	4.6	3.2
7 1/16" 10,000			1.3	1.3	1.5	2.3							3.9	3.9					1.2	1.0	4.6	3.2
7 1/16" 15,000			1.3	1.3	1.5	2.3							9.4	9.4					6.4	5.9		
8" 2,000																			6.4	5.9		
8" 3,000							1.5	1.3	2.4	2.7			4.3	4.3					2.6	2.3	7.2	5.0
8" 5,000													6.8	6.8					2.6	2.3	11.1	8.7
9" 10,000							1.5	1.3	2.4	2.7			15.9	15.9					2.4	2.1		
10" 2,000																						
10" 3,000																						
10" 5,000	7.8	6.5	3.4	3.2	2.8	3.7	1.5	1.3	2.8	3.2			7.4	7.4					1.8	1.5	11.0	6.8
11" 10,000	12.1	10.5	3.4	3.2	2.8	3.7	1.5	1.3	2.8	3.2			9.8	9.8					3.0	2.6	18.7	14.6
12" 3,000			3.4	3.2	2.8	3.7	2.9	2.5	4.4	5.0			25.1	25.1					3.6	3.3		
12" 5,000			5.8	5.5	4.1	5.3							11.4	11.4					4.7	3.4	23.5	14.7
13 5/8" 5,000	15.5	13.9	5.8	5.5							19.8	13.2	18.0	18.0					5.3	4.7	23.6	17.4
13 5/8" 10,000	21.5	18.7	5.8	5.5									34.5	34.5					10.5	9.8	51.2	42.7
13 5/8" 15,000			11.7	11.3																		
14" 5,000																						
16" 2,000					4.1	5.3	2.9	2.5														
16" 3,000					5.0	6.0			6.0	7.0			17.4	17.4								
16 1/4" 5,000	33.0	29.0	10.6	9.8	5.0	6.0							21.0	21.0					4.7	4.1		
18" 2,000			10.6	9.8							33.8	23.5	28.7	28.7					6.6	6.0	33.3	25.6
183/4" 5,000									6.0	7.0			21.1	21.1								
183/4" 10,000			24.9	23.0							44.0	29.3										
20" 2,000			8.4	7.9	5.0	6.0																
20" 3,000			8.4	7.9	5.0	6.0																
21 1/4" 5,000											58.0	38.7										
21 1/4" 7,500			20.4	17.8																		
21 1/4" 10,000			26.5	24.1																		
26 1/4" 2,000			10.4	9.9																		
26 1/4" 3,000			10.4	9.9																		

BOP OPERATING PRESSURES

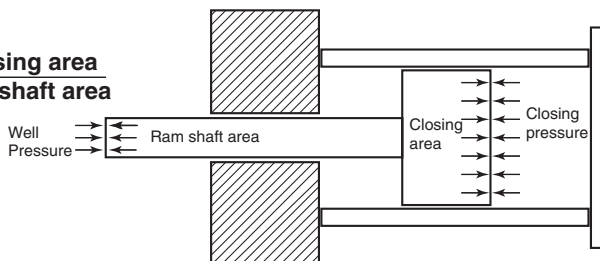
Operating pressures under various conditions are given in the Operator's Manual. However calculations can be made using closing and opening operating ratios as shown below and on the next page respectively - these ratios are very often given in catalogues.

The rated **continuous** working pressure for a Shaffer and Cameron ram type BOP is normally 10,340 kPa (1,500 psi) although some ram type BOPs have a working pressure of 15,169 kPa (2,200 psi). The rated **maximum** working pressure of ram type BOPs is normally 20,680 kPa (3,000 psi). When it is required to be able to operate BOPs under conditions of potentially high well pressures the rated working pressure of the operating system may be a limiting factor. This point is covered by Cameron in their Engineering Bulletin No.196D revision D1 (10th January 1966):

"The rated continuous working pressure for the Type 'U' B.O.P. operating system is 1,500 psi. Pressures of 300 to 500 psi normally provide a satisfactory operation. Pressures in excess of 1,500 psi may be required in high pressure BOP's (10,000 psi working pressure or more) to close the rams against high well pressures. In emergencies, pressures up to 5,000 psi can be applied to the closing side of the operating system. For optimum seal life, the applied hydraulic pressure should be limited to 1,500 psi, especially when 'ram open' pressure is required to be held continuously. Accumulator units should be fitted with a pressure regulator to control the pressure applied to a BOP."

Although Cameron say that up to 5,000 psi can be applied in an emergency, this should only be done where both system and lines are rated at, and have been tested to, that pressure.

$$\text{Closing ratio} = \frac{\text{Closing area}}{\text{Ram shaft area}}$$



$$\text{Closing pressure required to close ram preventer with pressure in the well} = \frac{\text{Well pressure}}{\text{Closing ratio}}$$

Example : Hydril 18³/₄" 10,000 psi WP ram type BOP

Closing ratio (shear & pipe) = 10.56

What will be the closing pressure at the rated working pressure of the BOP ?

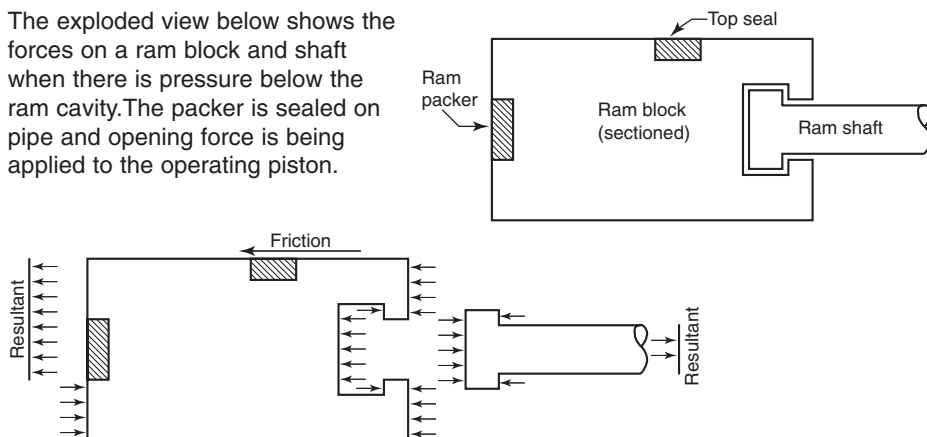
The required closing pressure = 10,000/10.56 = 947 psi.

Shearing operations

The closing ratio for a unit containing blind/shear rams refers only to the operating pressure required to move the rams into the well bore. If it is necessary to shear drill pipe or other tubulars an **additional** force will be necessary, its magnitude depending on the type of the tubular to be cut. The Operator's Manual should list the additional closing pressures required for common tubulars.

$$\text{Opening ratio} = \frac{\text{Opening area}}{\text{Resultant vertical areas exposed to well bore pressure}}$$

The exploded view below shows the forces on a ram block and shaft when there is pressure below the ram cavity. The packer is sealed on pipe and opening force is being applied to the operating piston.



$$\text{Opening pressure required to open rams with pressure in the well} = \frac{\text{Well pressure}}{\text{Opening ratio}}$$

Example 1 : NL Shaffer 18^{3/4}" 10,000 psi (68,950 kPa) WP ram type BOP

Opening ratio (shear & pipe) = 1.83

Assuming that the rated working pressure of the operating chamber is 3,000 psi (20,680 kPa), what is the maximum well bore pressure at which the rams could still be opened ?

The maximum well bore pressure = 3,000 x 1.83 = 5,490 psi (37,850 kPa)

Example 2 : Hydril 18^{3/4}" 15,000 psi (103,400 kPa) WP ram type BOP

Opening ratio (shear & pipe) = 2.15

What would be the opening pressure at the rated working pressure of the BOP ?

The required opening pressure = 15,000/2.15 = 6,977 psi (48,100 kPa)

This illustrates that opening the well under such a pressure would not be possible, given that the working pressure of the operating chambers is 3,000 psi (20,680 kPa).

Note:

Opening rams with the well under pressure will damage equipment and is not good safety practice. These examples are given to illustrate the principles.

Bag type preventers can be divided into two types:

- Well bore pressure assisted
- Non well bore pressure assisted.

Most Hydril bag type preventers are well bore pressure assisted. Cameron and NL Shaffer units are non-well bore pressure assisted. With increasing well bore pressure the hydraulic fluid pressure for a Hydril bag type preventer must be reduced; for Cameron and NL Shaffer units the hydraulic pressure must be increased.

Example :		Well bore pressure	Hydraulic pressure required	Well bore pressure	Hydraulic pressure required
SI Units :		(kPa)	(kPa)	(kPa)	(kPa)
NL Shaffer	13 ⁵ / ₈ " 5,000 psi	3,450	±4,500	13,800	±5,860
Cameron Type D	13 ⁵ / ₈ " 5,000 psi	3,450	±3,450	13,800	±5,170
Hydril	13 ⁵ / ₈ " 5,000 psi	3,450	±3,100	13,800	±690
Oilfield units :		(psi)	(psi)	(psi)	(psi)
NL Shaffer	13 ⁵ / ₈ " 5,000 psi	500	±650	2,000	±850
Cameron Type D	13 ⁵ / ₈ " 5,000 psi	500	±500	2,000	±750
Hydril	13 ⁵ / ₈ " 5,000 psi	500	±450	2,000	±100

Complete shut off will require higher hydraulic pressures. Closing a bag type preventer on larger sizes of pipe will in general require less hydraulic fluid pressure. An Operator's Manual **must** be available on the rig.

ACCUMULATORS

USABLE VOLUME REQUIREMENTS

The size of an accumulator installation is covered by two recommended practices - those issued by API and those incorporated in SIEP's Pressure Control Manual.

API

The API's recommended practice is published in API RP-53, third edition, March 1997, Chapters 14.2.2 and 14.2.3, which relate to closing units of sub-sea installations. It specifies that :

BOP systems should have sufficient usable hydraulic fluid volume (with pumps inoperative) to close and open one annular preventer and all ram-type preventers from a full open position against zero well bore pressure. After closing and opening one annular preventer and all ram-type preventers, the remaining pressure shall be 200 psi (1,380 kPa) above the minimum recommended pressure.

The sub-sea accumulator bottle capacity calculations should compensate hydrostatic pressure gradient at the rate of 0.445 psi/ft (10.067 kPa/m) of water depth.

Usable fluid volume is defined as the volume of fluid recoverable from an accumulator, between the accumulator operating pressure and 200 psi (1,380 kPa) above the precharge pressure.

Note: There is an inconsistency between the API recommended practice in the box above and their definition of usable volume. In one case they refer to 200 psi (1,380 kPa) above the minimum recommended pressure and in the other to 200 psi (1,380 kPa) above the precharge pressure.

SIEP

The SIEP recommended practice is given in EP 89-1500, Pressure Control Manual for Drilling and Workover Operations. It can be found in Section 3.2.5 as applied to surface BOP stacks and in Section 3.3.3 as applied to subsurface stacks. There is no practical difference between the two as far as volume requirements are concerned.

The relevant text, from the section dealing with sub-sea stacks, reads

Without recharging, the accumulator capacity shall be adequate for closing and opening all ram type preventers and one annular preventer around the drillpipe, and for closing again one ram type preventer and one annular preventer around the drillpipe and holding them closed against the rated working pressure of the preventers.

SIEP does not specifically define the usable volume.

ACCUMULATORS

VOLUME REQUIREMENTS

As an example we will take a sub-sea stack containing four 13⁵/₈", 10,000 psi (68,950 kPa) Shaffer Type LWS units with Poslock and one 13⁵/₈", 10,000 psi (68,950 kPa) Shaffer spherical BOP. The water depth is 305 m /1000 ft.

As recommended by SIEP there must be sufficient usable fluid to close and open each ram and the bag type BOP once, and then close one ram type preventer and one annular preventer.

Using the data from the table on page K-1, we get :

In Oilfield units:

BOP	Fluid requirements in gallons		
	Close	Open	Number
Ram units	11.7		5
		10.5	4
Spherical unit	51.2		2
		42.7	1
Total			245.6

In SI units:

BOP	Fluid requirements in litres		
	Close	Open	Number
Ram units	44.3		5
		39.7	4
Spherical unit	193.8		2
		161.6	1
Total			930

ACCUMULATORS

OPERATING PRESSURES

There are three pressures which have to be known - these are :

P_1 = Pressure of the accumulator when completely charged to its working pressure

P_2 = Minimum allowable operating pressure

P_3 = Nitrogen precharge pressure

For accumulator bottles the rated working pressure is normally 20,685 MPa (3,000 psi). For our example we will use this value.

The minimum allowable operating pressure is equal to the maximum closing pressure required by the the BOP stack when the well bore pressure inside it is equal to its rated working pressure. Note that the units making up the BOP stack will usually have different closing pressures due to their different closing ratios; the highest of these closing pressures must be used for calculating the minimum operating pressure.

The nitrogen precharge pressure for a 20,685 MPa (3,000 psi) accumulator on surface is normally 6,895 kPa (1,000 psi).

For the purposes of volume/pressure calculations using Boyle's Law P_1 , P_2 and P_3 must be absolute rather than gauge pressures, thus 100 kPa (15 psi) should added to the gauge pressures. In the following examples this point has been ignored for simplicity (in practical terms the error introduced is small).

Note also that for the use of Boyles Law the pressures P_1 , P_2 and P_3 must be those **in the accumulator** at its operating depth.

For sub-surface stacks the values of the pressures must be modified to allow for the effect of the hydrostatic head of the sea water. In our example the latter corresponds to 305 m (1,000 ft) water depth, i.e. $305 \times 10.1 = 3,080$ kPa ($1,000 \times 0.445 = 445$ psi). The required pressures are found as follows :

- The rated working pressure P_1 increases by an amount equal to the hydrostatic head of the sea-water column.

$$P_1 = 20,685 + 3,080 = 23,765 \text{ kPa} \quad (P_1 = 3,000 + 445 = 3,445 \text{ psi})$$

- The pre-charge pressure has to be increased by an amount equal to the hydrostatic head of the sea-water column.

$$P_3 = 6,895 + 3,080 = 9,975 \text{ kPa} \quad (P_3 = 1,000 + 445 = 1,445 \text{ psi})$$

- We shall take the minimum allowable operating pressure to be either the required closing pressure (the maximum internal pressure in the BOP stack divided by the closing ratio) or 1,380 kPa (200 psi) above the precharge pressure, whichever is greater. The maximum internal pressure in the BOP stack is equal to the rated surface working pressure of the BOP stack plus the hydrostatic head of the sea-water column.

For the example on the opposite page the closing ratios have to be found in the Operators Manuals for the BOPs - they are 7.1 for the ram units and 10.5 for the spherical unit. The required closing pressure is therefore defined by the ram units which have the higher closing pressure, and is equal to:

$$\frac{68,950 + 3,080}{7.1} = 10,140 \text{ kPa} \quad \left(\frac{10,000 + 445}{7.1} = 1,471 \text{ psi} \right)$$

This pressure is lower than 1,380 kPa (200 psi) above the precharge pressure, thus:

$$P_2 = 9,975 + 1,380 = 11,355 \text{ kPa} \quad (P_2 = 1,445 + 200 = 1,645 \text{ psi})$$

ACCUMULATORS

VOLUME CALCULATIONS

There are four volumes which have to be known - these are :

V_1 = Volume of Nitrogen in the accumulators at rated working pressure

V_2 = Volume of Nitrogen in the accumulators at minimum allowable pressure

V_3 = Total accumulator volume (Nitrogen + hydraulic fluid)
(= volume of Nitrogen in the accumulators at pre-charge pressure)

V_R = Total usable hydraulic fluid required

From Boyles Law : $P_1.V_1 = P_2.V_2 = P_3.V_3$ (assuming an isothermal expansion)
and, by definition $V_R = V_2 - V_1$

In case of the example, V_R is known and V_3 is the total accumulator volume which must be calculated.

From the above equations
$$V_R = \frac{P_3.V_3}{P_2} - \frac{P_3.V_3}{P_1} = V_3 \left(\frac{P_3}{P_2} - \frac{P_3}{P_1} \right)$$
$$V_3 = \frac{V_R}{\frac{P_3}{P_2} - \frac{P_3}{P_1}}$$

We have calculated the required volume V_R according to the recommended practice of SIEP on page K-6. This is 930 litres (245.6 gals).

If we substitute the values for P_1 , P_2 and P_3 obtained on the previous page into the above equation for V_3 we get :

$$V_3 = \frac{930}{\frac{9,975}{11,355} - \frac{9,975}{23,765}} = 2,027 \text{ litres} = \frac{245.6}{\frac{1,445}{1,645} - \frac{1,445}{3,445}} = 535 \text{ gallons}$$

The working capacity of a standard accumulator bottle is 10 gals (11 gals total capacity less 1 gal. for bladder/float displacement) which is 37.85 litres. Thus the number of bottles required, when rounded up to the next whole number, is 54 (2,027/37.85 or 535/10).

ACCUMULATORS

HIGH PRESSURE OPERATIONS

The rated working pressure of standard accumulator bottles is 21,000 kPa (3,000 psi). When it is required to be able to operate BOPs under conditions of potentially high well pressures this may be a limiting factor. It may affect their ability to apply sufficient pressure to the closing system of the BOP to close the rams, and will increase the number of accumulator bottles that are required to comply with recommended practices.

If the pressure itself is a limiting factor then there is no option but to change to bottles that have a rated working pressure of 35,000 kPa (5,000 psi).

ACCUMULATORS

TESTING

The following test procedure is taken from EP 89-1500 July 1989, Pressure Control Manual for Drilling and Workover Operations :

The accumulator bottles precharge pressure (nitrogen) shall be checked prior to drilling out cement in the casing shoe. Unless otherwise specified, the precharge pressure for a 20,685 kPa (3000 psi) WP system should be 6895 kPa (1000 psi) $\pm 10\%$.

Accumulator tests should be performed prior to first use of BOPs, or after repairs have been made to the accumulator system, i.e. bottles, bladders, pumps, etc.

The accumulator unit performance test is made by operating all BOPs on the stored energy in the accumulator, i.e. the pressure and the volume available without recharging.

The complete test procedure is as follows:

1. Check accumulator fluid pressure.
2. Check accumulator reservoir level.
3. Switch off accumulator pumps.
4. Close and open all preventers and check accumulator fluid pressure after each function and the volume of fluid used for each function for sub-sea units; record closing times. Adequate pressure and volume should still be present to close one annular and one ram type preventer. Precharge pressure should still be the same in all accumulator bottles.
5. Switch on accumulator pumps.
6. Record accumulator recharging time.

It is recommended to check the recharging capacity of the air pumps with the electric power switched off prior to start up of a newly contracted rig.

7. Check BOP closing times and accumulator recharge time with manufacturer's data for the system in use.
8. Cycle the annular preventer and check that the pumps will automatically start when the closing unit pressure has decreased to less than 90 percent of the accumulator operating pressure. This should be checked with only the electric pumps operative.
9. Should an emergency control system be employed, this should also be tested at the same time as the accumulator unit.
10. Results should be recorded on the daily tour sheets and the Blowout Prevention Equipment Checklist.

Note

It is of the utmost importance that the unit can be charged with only one of the two power systems operative.

NOTES ON BOP EQUIPMENT

RAMS

Pipe rams

Do not close pipe rams without a proper size mandrel or pipe in the hole. Thus closing around a tool joint should be avoided. Excessive packer wear can also result from closing pipe rams on themselves.

Standard ram packers are usually rated to a maximum temperature of 120°C (250°F), whereas packers for HP/HT applications are usually rated to a maximum temperature of 175°C (350°F).

Special hardened rams are required to hang off drill pipe. These rams are hardened around the top corner of the drill pipe cut out and will dig into and create a shoulder in the tool joint. Alternatively a special square shouldered hang-off tool can be used which eliminates the 18° tool joint taper. The maximum hang-off load for 5", 5 1/2" and 6 5/8" drill pipe is 265 kdaN (600,000 lbs).

Shearing blind rams

Shearing blind rams (SBR) for common sizes are meant to cut the pipe and then seal the well bore, whether the fish is suspended (hung off on e.g. tool joint some 2 1/2 ft below the SBR) or dropped. If the fish is not dropped, the lower shear ram will bend the cut pipe over a shoulder and away from the front face of the upper shear ram.

Large shear bonnets are standard on most present day ram preventers. Preventers with SBR but without (or even with) these large shear bonnets can also be fitted with tandem boosters to approximately double the applied shearing force in comparison with normal closing forces. They will usually apply this force during the cutting process, but disengage prior to energising the packers, in order to enhance their service life.

The API Spec 16-A for shear rams state that: "Each BOP equipped with shearing blind rams shall be subjected to a shearing test. The test requires shearing of 5" OD, 19.5 lbs/ft nominal, Grade E drill pipe for 11" BOPs and 5" OD, 19.5 lbs/ft nominal, Grade G for 13 5/8" and larger BOPs. The closing pressure required to achieve the above shall not exceed the hydraulic system rated pressure, and will usually be in the range of 2,700 to 2,800 psi (18.6 to 19.3 MPa)." It would be prudent to ensure that the shearing blind rams will function as envisaged and that nothing is left to chance.

Closing shear rams on drill collars or tool joints will generally destroy the sealing capability of the ram, without cutting the pipe. There are special rams available, such as the Super Shear Rams (SSR) from Cameron, which will shear drill collars, HWDP and large diameter casing, but they are usually non-sealing rams

Shearing blind rams with Super-Trim (H₂S resistant) are available but be aware that the hardened leading edges are highly susceptible to sulphide stress cracking).

There are many types of wedge locks, but they all should have a provision to prevent an accidental unlock; most types require the ram opening pressure to be activated (e.g. the NL Shaffer Ultralock or Hydril Multi-position Lock), others might use a wedge lock unlock pressure. In the latter case, one four-way valve and one pair of hydraulic lines are normally used to operate all the wedge locks.

Always lock the SBR in the closed position (wedge locks or locking screw).

Variable pipe rams

Avoid hanging off pipe on variable rams, particularly at the low end of the variable range.

NOTES ON BOP EQUIPMENT

OTHER EQUIPMENT

Annular Preventers

Three types of material are used in manufacturing annular packing elements and selection of the correct material is vitally important:

- Natural Rubber with good wear resistance, for use in water based drilling fluid environments only, at temperatures ranging from -28°C to 76°C (-20°F to 170°F).
- Nitrile Synthetic Compound, recommended for use in (pseudo) oil based mud environments at temperatures ranging from 4°C to 76°C (40°F to 170°F).
- Neoprene Synthetic Compound, recommended for use in (pseudo) oil based mud environments at temperatures ranging from -34°C to 76°C (-30°F to 170°F).

Due to the low collapse resistance of some casing strings, consideration has to be given to the initial closing pressure used when closing annular preventers on casing. Normal practice is to close the preventer with the minimum required closing pressure and then increase the closing pressure sufficiently to maintain a seal as well bore pressure increases. Use the appropriate manufacturers tables and diagrams to find the recommended initial closing pressure for the preventer in use, e.g. when large preventers are in use, these pressures could be as low as 180 psi for 18⁵/₈" casing and 475 psi for 13³/₈" casing.

It is not advisable to test Annular Preventers on open hole. If closed on open hole, apply the minimum required closing pressure to minimise damage to the packing element.

Choke and kill manifold

Gate valves used in choke and kill manifolds must be full opening, i.e. a 3¹/₁₆" valve should be fitted in a 3" line. A minimum of two valves are required upstream of the chokes and one valve downstream of the chokes.

The choke manifold piping should be designed with as few bends as is practical to avoid turbulence-induced wash-outs. Where bends are unavoidable they should be fitted with sacrificial lead targets to prevent wall thickness reduction.

Drilling chokes are usually meant to control back pressure from one direction only. They often include a positive sealing feature (positive shut-off). However, this sealing feature does not always allow sealing against downstream pressure, and if this is required one has to ensure that the choke is modified or manufactured accordingly.

L – DIRECTIONAL DRILLING

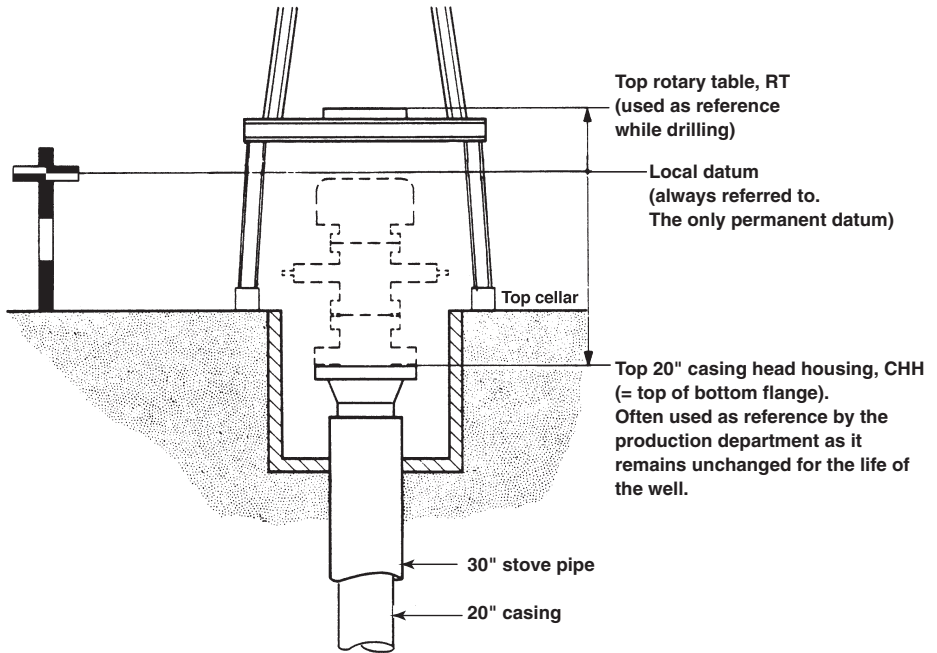
Clickable list

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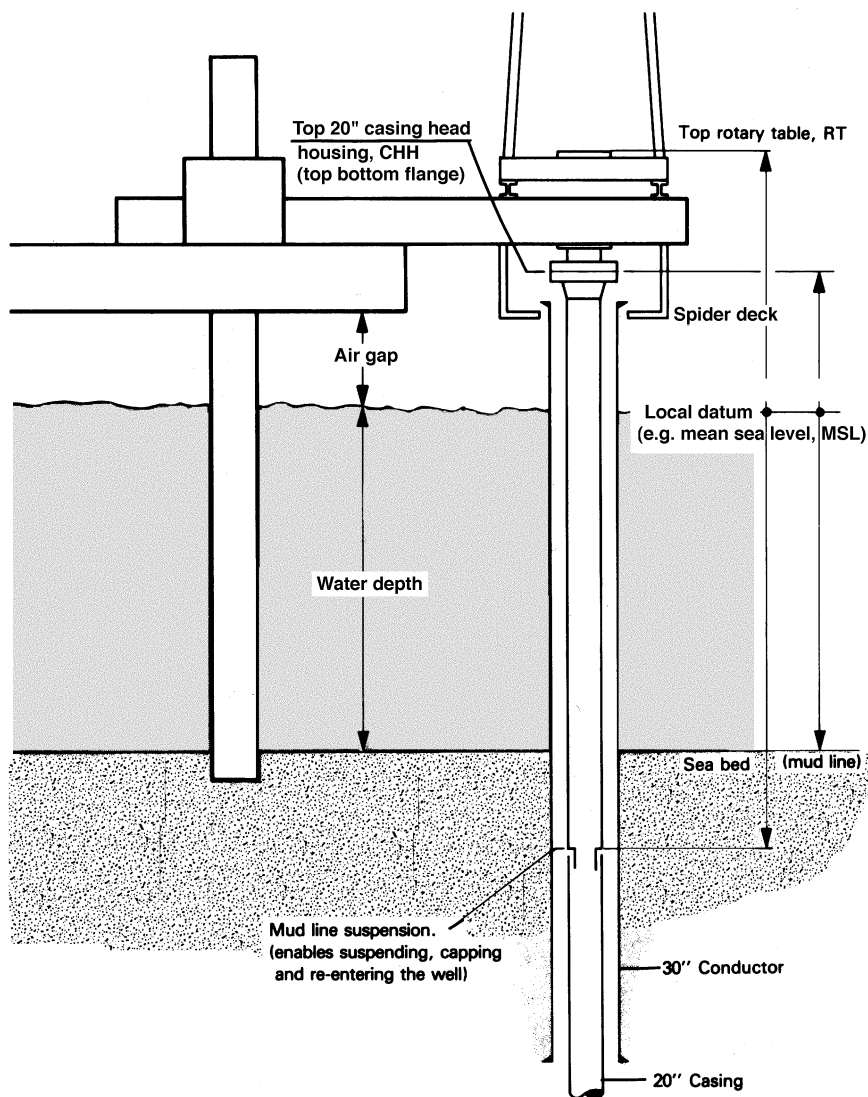
DEPTH REFERENCES

ONSHORE WELLS



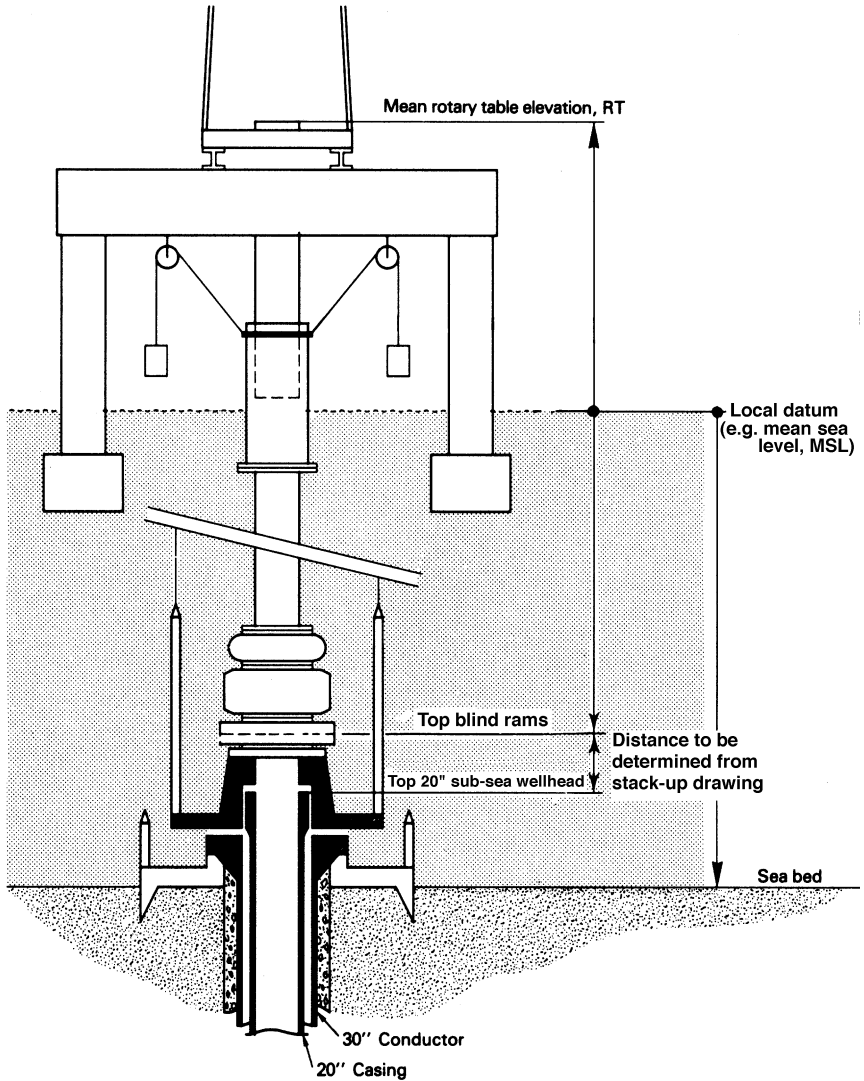
DEPTH REFERENCES

OFFSHORE WELLS DRILLED WITH SURFACE BOPS



DEPTH REFERENCES

OFFSHORE WELLS DRILLED WITH SUB-SEA BOPS



AZIMUTH - TRUE, MAGNETIC AND GRID

In the equations and diagrams below, which refer to a horizontal plane at the point in question :

A_{TN} = Azimuth with reference to True North

A_{MN} = Azimuth with reference to Magnetic North

A_{GN} = Azimuth with reference to Grid North

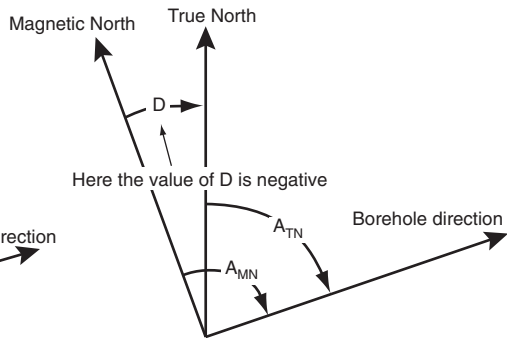
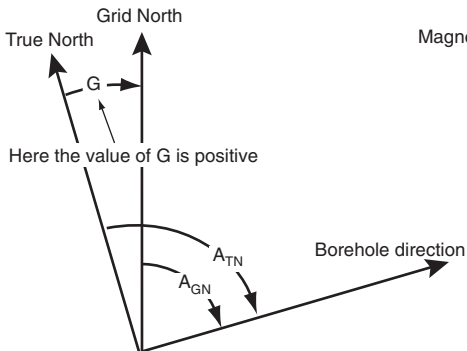
G = Grid Convergence, which is by definition positive when Grid North is East of True North

D = Magnetic Declination, which is by definition positive when Magnetic North is East of True North

Thus:

$$A_{TN} = A_{GN} + G$$

$$A_{TN} = A_{MN} + D$$



Be wary of the term “grid correction” which is used in a similar way to grid convergence but which is, by definition, the negative of grid convergence. Grid correction was the standard used in Well Engineering defined in a previous Borehole Surveying Manual (EP 59300). To comply with standards used in the survey industry and Topographic Departments, Grid convergence has now been adopted as the standard for Well Engineering.

Note also that not all OUs use the standard convention. Within an OU only the local convention should be used. These will be provided by the OU focal point.

DIRECTIONAL WELL PLAN EQUATIONS

With target co-ordinates of ΔN and ΔE relative to the surface position :

the horizontal displacement,

$$d = \sqrt{\Delta N^2 + \Delta E^2}$$

and the azimuth,

$$A_t = \tan^{-1} \frac{\Delta E}{\Delta N} (+ 180^\circ)$$

$$D = \text{TVD}_{\text{target}} - \text{TVD}_{\text{k.o.p.}}$$

From the build-up rate, BUR,

$$R = \frac{360}{2\pi} \times \frac{\text{characteristic length}}{\text{BUR}} = \frac{5,730}{\text{BUR}}$$

For the build-up section, with inclination α : $\Delta \text{AHD} = \frac{2\pi\alpha}{360} \times R$

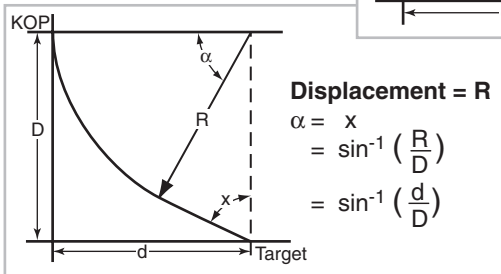
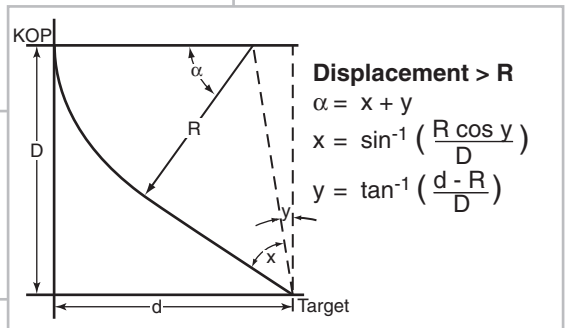
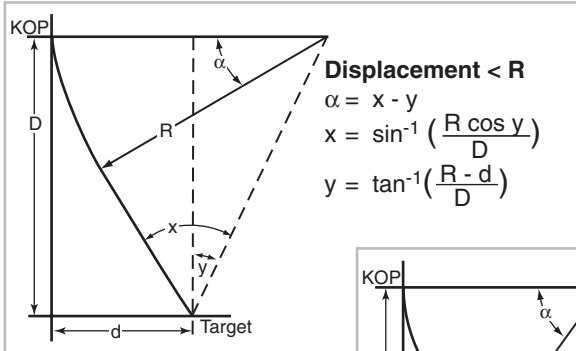
$$\Delta \text{TVD} = R \sin \alpha$$

$$\Delta d = R(1 - \cos \alpha)$$

For the tangent section :

$$\Delta \text{AHD} = \frac{\Delta \text{TVD}}{\cos \alpha}$$

$$\Delta d = \Delta \text{TVD} \tan \alpha$$



BOTTOM HOLE ASSEMBLIES

FUNDAMENTAL PRINCIPLES

Fundamentals of BHA Design

In all cases, the minimum practical amount of BHA should be run. By running the minimum amount of BHA the torque and drag will be reduced, this in turn will reduce the fatigue generated in the drill string and thereby increase the life of the drill string.

All BHAs place a side force at the drill bit. This side force affects the path followed by the drill bit and the rate of angle change, (dog leg severity), in the well bore.

By planning to minimise the rate of angle change and by selecting the minimum number of tools having the correct material properties and assembling them in the correct order, good BHA design can delay fatigue damage and reduce the severity of drill string failure.

To achieve correct BHA design, it is necessary to understand the basic principles and the effect of selected physical properties of the BHA components.

Factors Affecting BHA Behaviour

The directional behaviour of a rotary BHA is affected in three different ways: by the mechanical characteristics of the BHA, by the drilling parameters applied to the BHA, and by the formation being drilled – over which we have no control.

Characteristics affecting BHA behaviour can be summarised as follows:

- The gauge and placement of stabilisers and other BHA components
- The diameter, length and material of the BHA components
- Bit type

Drilling parameters affecting BHA behaviour are:

- Weight on bit
- Rotary speed
- Circulation or flow rate

Directional Control Principles

There are three basic principles used to control well bore direction.

- The fulcrum principle – used to increase the well bore inclination. Inclination is the angle, expressed in degrees, between the path of the well bore and vertical.
- The stabilisation principle – used to hold both inclination and azimuth. Azimuth is the direction, expressed in degrees, between the path of the well bore and true North, or grid North if specified.
- The pendulum principle – used to drop inclination.

Note :

This and the following eight pages about BHAs have been taken from Shell Expro's "Drillstring Failure Prevention - BottomHole Assembly Design Guidelines" (WEIN 553), also available as SIEP Report EP 94-1103.

BOTTOM HOLE ASSEMBLIES

THE FULCRUM PRINCIPLE

A BHA with a full gauge near bit stabiliser, and between 90 ft and 120 ft of drill collars before the first string stabiliser (or no string stabiliser at all) will build inclination when weight on bit is applied .

The drill collars above the near bit stabiliser bend due to their own weight and also due to the weight on bit. The near bit stabiliser acts as the fulcrum point of a lever transmitting this bending moment down to the bit and pushing the bit upwards, thus building angle.

The following factors act on the build-up rate of this type of drilling assembly:

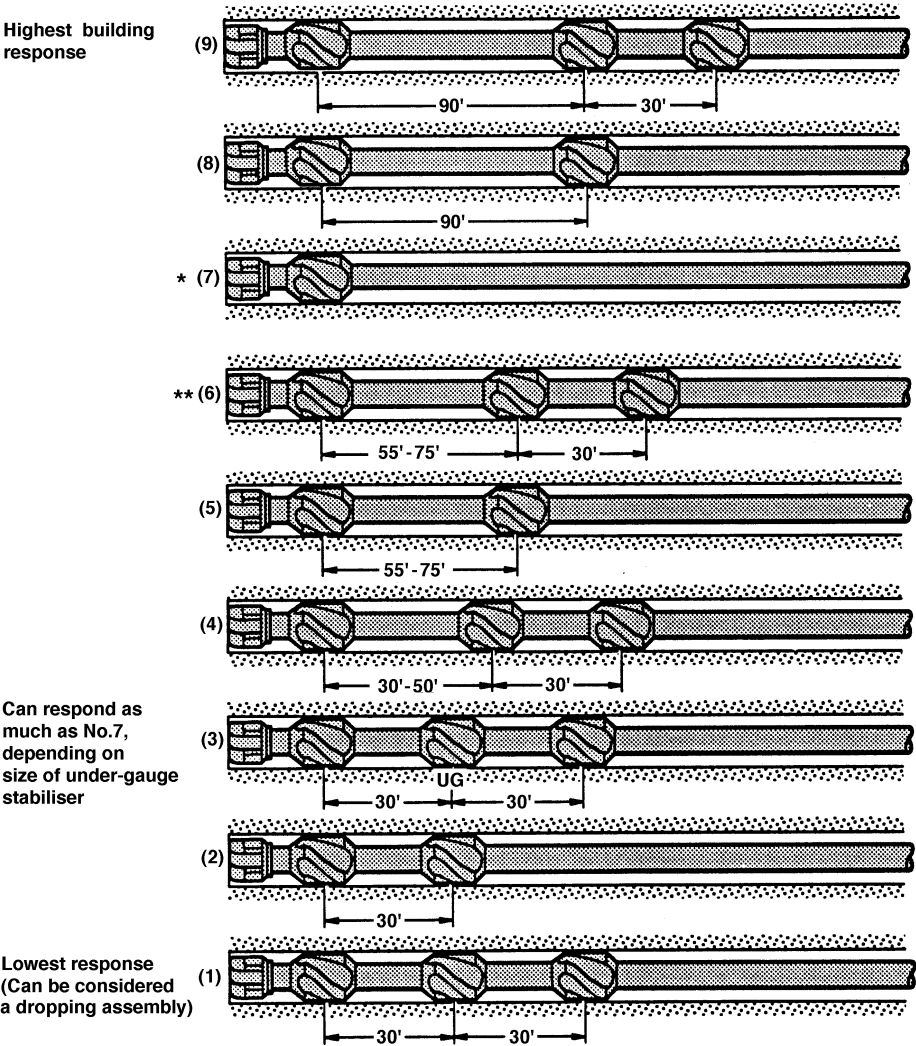
- Distance between the near bit stabiliser and the first string stabiliser.
As this distance increases, the build-up rate also increases. However, once the distance between the first two stabilisers reaches 120 feet any further increase in length has little or no effect and might allow the drill collar to touch the side of the hole.
- The outside diameter of the drill collars.
As the outside diameter increases, the collars become more rigid or "stiff" and the build-up rate decreases.
- Material of the drill collars.
In the field, a choice of material is seldom available, so options are not normally possible. In a critical well this option should be considered at the planning phase.
- Bit type e.g. Tri-cone, PDC etc.
The bit type has little effect on the build or drop rate, the exception being long gauge bits. The increase in gauge length decreases the build tendency. However the bit type does affect the "walk" or azimuth change, tricone bits tend to walk right whereas PDC bits exhibit little or no walk, but each bit does have its own characteristics.
- Weight on bit.
An increase of the weight on bit tends to increase the bending force on the collars above the near bit stabiliser and hence the build-up rate.
- Rotary speed
With an increase in rotating speed the BHA becomes effectively more rigid and the build-up rate decreases.
- Flow rate.
In soft formations, higher flow rates tend to decrease the building tendency due to the effect of the circulating fluid washing away the formation. This increases the hole size and decreases the support for the BHA.

Figure L-1 shows several BHAs which will exhibit a build tendency. They are graduated from highest to lowest tendency to build angle, and are typical for a 12¹/₄" hole.

BOTTOM HOLE ASSEMBLIES

THE FULCRUM PRINCIPLE (2)

Figure L-1 : BHAs for building inclination



* At lower inclinations this BHA is the most responsive

** The level of build tendency changes with inclination where BHA Nos. 6 & 7 generate more side force at higher angles

BOTTOM HOLE ASSEMBLIES

THE STABILISATION PRINCIPLE

By using three or more stabilisers with a short, large diameter drill collar between the near bit stabiliser and the first string stabiliser it is possible to reduce the transmission of bending moment to the bit, forcing it to follow a reasonably straight path. The BHAs that use this principle are called Packed Hole Assemblies and are used in vertical and deviated wells to maintain inclination and azimuth. Some bit walk may still be experienced when drilling with a packed hole assembly.

The following factors are of importance when designing stabilised BHAs:

- Stabiliser design.

In large diameter holes (i.e. greater than 17¹/₂") the use of straight bladed stabilisers is common. These are acceptable where the hole is vertical and the torque and drag when drilling is low. Due to its design, this style of stabiliser tends to dig into or "gouge" the well bore and will increase the torque and drag. For most hole sizes, stabilisers with 360° wall contact are available. These are of a long, wide, spiral blade design and provide full, effective support for the BHA without gouging the well bore.

- Near bit stabiliser.

In all packed drilling assemblies, the near bit stabiliser must be full gauge. The stabiliser type and the area of blade contact with the hole wall require careful consideration to match formation and hole conditions. In areas of severe tendencies, tandem stabilisers can be used at the near bit position when stabilisers with long and wide blades are not available.

- Stabiliser spacing.

The distance between the near bit stabiliser and the first string stabiliser, should be between 2 and 15 feet depending on hole size and hole condition. The shorter the spacing between the stabilisers the more rigid the assembly will be.

- First string stabiliser.

The gauge of the first string stabiliser is of great importance and for most cases the stabiliser must be full gauge. (In areas where the assembly tends to drop, e.g. for deviated wells, an under gauge stabiliser is used to help maintain inclination.)

- Bit type.

The two most commonly used bit types are tri-cone and PDC bits. The path drilled by a tri-cone bit will vary with applied weight on bit and rpm. PDC bits tend to drill straight holes regardless of weight on bit and RPM; long gauge PDC bits help to maintain a straight well path.

Where possible and depending on the formation, the use of PDC bits is recommended to help maintain a straight well path. With pendulum assemblies long gauge PDC bits can build angle as the long gauge acts as a near bit stabiliser.

- Rotary speed.

A higher rotating speed makes the BHA effectively stiffer and therefore less susceptible to deviate from the required well path.

BOTTOM HOLE ASSEMBLIES

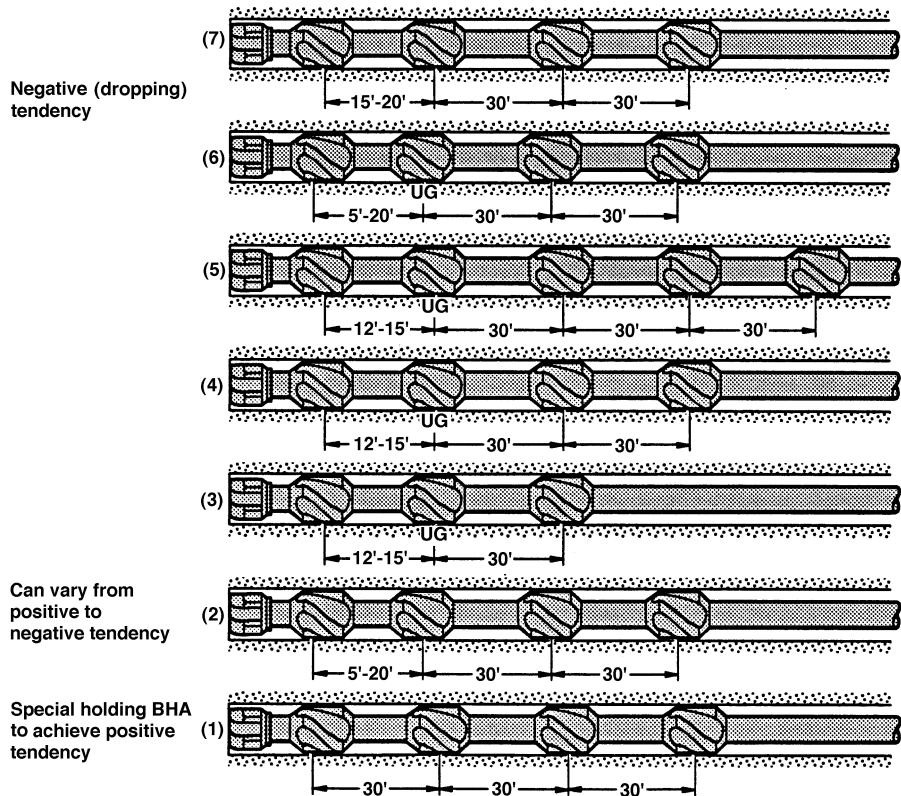
THE STABILISATION PRINCIPLE (2)

- Formation Effect

The formation being drilled will have an effect on the directional stability of the drilling assembly, however this effect is not the same for all assemblies. Action can be taken to mitigate the effect of formation characteristics and formation changes by studying the behaviour of BHAs in previous wells and catering for the effects observed. The greatest effect will be seen where no near bit stabiliser is in the BHA. Where a packed assembly is in use, the formation effect can take a BHA configured for a slight drop tendency and force it to drop heavily or even build angle.

Figure L-2 shows several packed hole assemblies. These are graduated from a slight building to a slight dropping tendency.

Figure L-2 : Packed hole assemblies for holding inclination angle



BOTTOM HOLE ASSEMBLIES

THE PENDULUM PRINCIPLE

The pendulum principle was originally used to drill vertical wells with slick (non stabilised) BHAs. It was modified to incorporate stabilisers and is still in use today to reduce inclination. The principle uses the weight of the BHA hanging below the tangent point to produce, via gravity, a force that pushes the bit to the low side of the hole. The effect of the pendulum varies with the length of the BHA below the tangent point.

The fundamental pendulum assembly increases the restoring force by increasing the pendulum length with a stabiliser in the proper position. The following are important factors to be considered in the design of pendulum drilling assemblies :

- Near bit stabiliser gauge.

All pendulum assemblies use either an under-gauge near bit stabiliser or omit the near bit stabiliser completely.

- Stabiliser spacing

The distance between the bit and the first string stabiliser controls the weight of the hanging portion of the BHA and therefore the pendulum force. If the first string stabiliser is placed too far away from the bit the tangent point will fall between the stabiliser and the bit, i.e. wall contact will take place, thereby reducing the effectiveness of the pendulum.

- Outside diameter of the drill collars.

Drill collar stiffness increases with the fourth power of the outside diameter. Stiffer drill collars will place the tangent point farther away from the bit and also increase the pendulum force. The weight per foot of the drill collars to be proportional to the second power of the outside diameter, i.e., heavier drill collars will produce a larger pendulum force.

In summary: For the portion of pendulum BHA below the tangent point or first drill string stabiliser, it is desirable to run drill collars with the largest possible outside diameter. **BUT potential problems associated with fishing the drill collars must be considered in the design stage.**

- Bit type.

To allow the pendulum force to work the bit must be free and unrestricted. Field experience has shown that tri-cone bits and short gauge, flat face PDC bits are the most effective with pendulum drilling assemblies.

- Weight on bit.

The higher the weight on bit, the more the assembly will bend. This can move the tangent point nearer to the bit and hence is detrimental to the effectiveness of the assembly. Furthermore, the side force at the bit, produced by the weight on bit, acts against the pendulum force. Weight on bit as low as possible is desirable for a pendulum assembly.

BOTTOM HOLE ASSEMBLIES

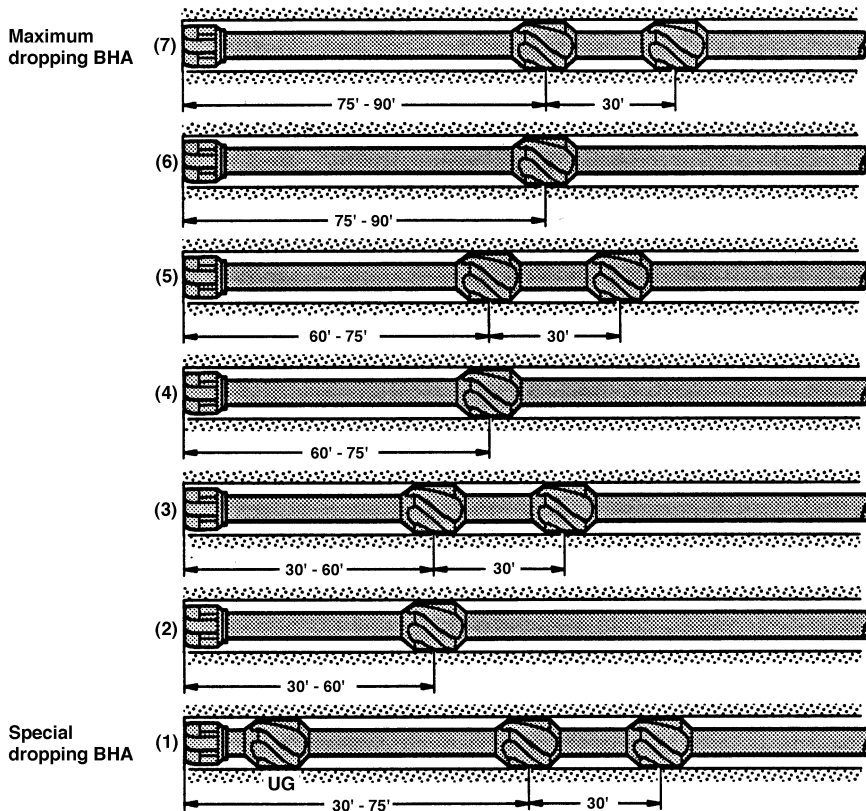
THE PENDULUM PRINCIPLE (2)

- Rotary speed.

A higher rotating speed makes the BHA effectively stiffer and therefore the tangent point moves farther away from the bit. As the assembly becomes stiffer, less bending (due to weight on bit) is transmitted to the bit. Higher rotating speeds will help to enhance the performance of pendulum assemblies, but will also tend to stiffen the pendulum thus increasing the drop. This is most noticeable on shorter pendulum assemblies. This tendency can be counteracted by increasing the length of the pendulum.

Figure L-3 shows a graduated series of pendulum assemblies used to drop inclination.

Figure L-3 : Pendulum assemblies for dropping inclination



BOTTOM HOLE ASSEMBLIES

VERTICAL WELLS

There is no such thing as a vertical well. All wells are deviated to some extent, the objective during drilling is to keep the well bore as close as possible to vertical. To achieve this objective the well is normally drilled with either a non stabilised slick assembly relying on the pendulum principle to keep the well pointing down, or it is drilled with a stabilised assembly. The principle then being that if it is properly stabilised it will not deviate from the desired path.

A typical method of drilling a vertical well is to use the special dropping assembly shown in Figure L-3. This assembly, when used in vertical holes with light weight on bit, acts as a minimum pendulum assembly but keeps any formation influenced building to a minimum. This type of assembly is mostly used with PDC bits which required low weight on bit.

In practice the wells are often drilled with a combination of both slick and stabilised assemblies.

Slick assemblies

When drilling in a vertical well with a slick assembly the pendulum principle applies.

An equation proposed by R. Hoch establishes a minimum drill collar outside diameter, OD_{dc} , to be run with a specific bit size, OD_b , into which a casing which has a coupling diameter of OD_{cc} is to be run.

$$OD_{dc} = 2 \times OD_{cc} - OD_b$$

Stabilised assemblies

In hard formations vertical wells are drilled using packed assemblies to allow maximum weight on bit to be run in order to drill faster. In soft and unconsolidated formations (normally shallow), pendulum BHAs are used to drill vertical wells.

As packed assemblies will bend slightly when used, there is sometimes a tendency to build angle. If this happens a pendulum assembly is used to drop the inclination, followed by a packed assembly to allow more weight to be applied to the bit and drilling to continue.

If the inclination is reduced by the pendulum assembly at too fast a rate, unacceptably large angle changes (dog-legs) can be created. These can prevent the following packed assembly from being successfully run in the hole without first having to ream to bottom. An even worse effect is that large angle changes speed up fatigue failure.

To avoid these problems, it is advisable to have the pendulum portion of the assembly below the packed BHA, so that any dog-legs are reamed as soon as they are created. A further advantage is that the pendulum becomes more efficient due to less bending being transmitted from the upper part of the BHA through the packed section down to the bit.

When drilling vertical wells with packed drilling assemblies the near bit stabiliser should be full gauge. In the event that the well starts to deviate from the vertical, the near bit stabiliser should be examined and replaced if it is found to be under gauge.

BOTTOM HOLE ASSEMBLIES

VERTICAL WELLS (2)

If the near bit stabiliser is full gauge, the width and length of the stabiliser blades should be checked, i.e. not too narrow or too short. If they are found to be acceptable then consideration should be given to either using a near bit stabiliser with wider and longer blades or by using tandem stabilisers in the near bit position.

Alternatively a "Big Bear" near bit stabiliser can be used. These are stabilisers of exceptional blade length, normally in the order of twice the blade length of that seen on a standard stabiliser (3 feet). They are therefore suitable to replace a tandem near bit stabiliser.

When applying any of these latter solutions, exceptional precautions have to be taken when running in hole. Due to the extreme stiffness of the near bit section great care should be taken not to mechanically stick the assembly, especially the first time such an assembly is run in the hole.

THE USE OF MUD MOTORS

GENERAL OPERATING PROCEDURES

Picking up a mud motor

Motors are generally supplied with a lifting or handling sub for transporting them to and from the rig floor. These lifting subs are normally rated to lift the motors only and should not be used for heavier lifts such as the complete drilling assembly.

Surface checks prior to running a mud motor in hole

Using the lifting sub, pick up the motor and set into the slips at the rotary table. Install the drill collar safety clamp below the dump valve ports, unlatch the elevators and remove the lifting sub. Check that the dump valve is free to move by pressing downwards with a hammer handle on the upper face of the piston, the piston should travel down two to three inches and return to the open position when the downwards pressure is released.

To check that the dump valve is not leaking, press on the piston again and, whilst holding the valve down in the closed position, fill the valve cavity with water. Release the downward pressure, the piston should return to the open position and the water in the valve cavity will drain out through the ports.

Using a cross over sub, connect the kelly or top drive to the motor. Remove the safety clamp and pick up the motor until the bit sub is above the rotary table. Measure the gap between the bit sub and the bearing housing. Set the motor down, making sure to protect the box shoulder by landing the bit box on wood or on a rubber mat over the rotary table. Measure the gap between the bit sub and the bearing housing again. Check that the measured play is within the specified tolerances for the motor.

Lower the motor so that the dump valve ports are below the rotary table. Start the pumps and, once there is no more flow through the ports, pick up the motor and observe the bit sub rotating. There should be flow between the bearing housing and the bit box. Lower the motor until the dump valve ports are below the rotary table and shut down the pumps.

Pick up the motor and attach the bit using a bit breaker while holding the bit sub with a tong.

Tripping into the hole

Run the tool in the hole carefully. Care should be taken not to run the motor into bridges, ledges or the bottom of the hole. Work through tight spots with the pump on and slow rotation. Should difficulty be experienced when reaming through tight spots care should be taken not to side-track the well through the application of high weight on bit or high rotary speeds.

When running in the hole if the drill string does not self fill, due to the properties of the drilling fluid preventing it from entering the drill string via the dump valve, periodically break circulation to fill the drill string. /...

Note :

These mud motor operating procedures have been taken from Shell Expro's "Drillstring Failure Prevention - BottomHole Assembly Design Guidelines" (WEIN 553), also available as SIEP Report EP 94-1103.

THE USE OF MUD MOTORS

GENERAL OPERATING PROCEDURES (2)

In hot wells, above 250°F bottom hole temperature, break circulation periodically while running in the hole to cool down the motor.

When using a PDC bit, avoid circulating inside the casing to prevent damage to the casing and to the bit.

Drilling

To commence drilling, with the bit two or three feet off bottom, start the pumps and slowly increase the flow rate to that required for drilling. Do not exceed the maximum flow rate for the motor. Once the pressure has stabilised make a note of the flow rate and the pump pressure, gently lower the bit to bottom and slowly increase the weight on bit, as the weight on bit increases there will be a corresponding increase in pump pressure.

For each motor there is a specified maximum differential pressure, the difference between the on bottom and off bottom pressure, this maximum should not be exceeded. It is good drilling practice to keep this differential pressure and the flow rate constant.

Tripping out

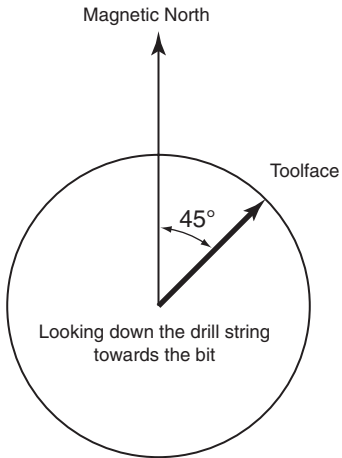
The procedures for tripping out of the hole are the same as when a rotary drilling assembly is in use.

However, once out of the hole, the bearing clearance should be checked in the same manner as it is checked prior to running in the hole. The motor should also be flushed with fresh water, and the bit removed. The same lift sub used to pick up the motor prior to running in the hole should be screwed in to the top of the motor and made up to a reduced torque valve. The lift sub should not be screwed in hand tight for lifting operations.

THE USE OF MUD MOTORS

STEERING BY MEANS OF “MAGNETIC TOOLFACE”

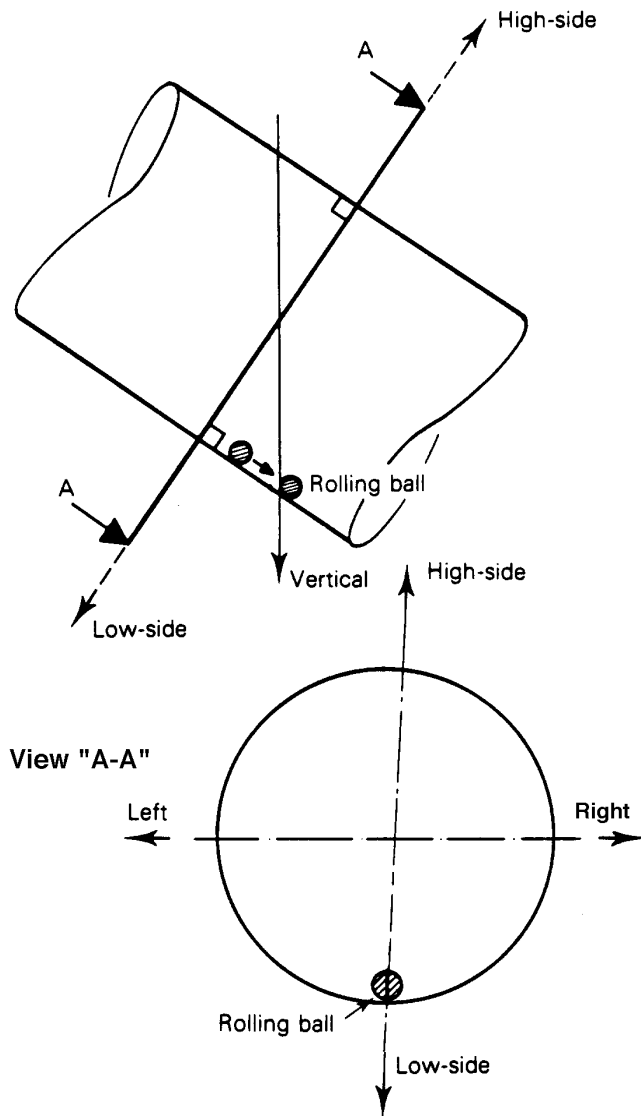
The magnetic toolface angle is the projection onto the horizontal plane of the angle between Magnetic North and the toolface. Steering tools are used in the magnetic toolface mode to change azimuth in near-vertical (less than about five degrees) wells.



Bit and mud motor trying to kick-off in azimuth 45° (Magnetic).

THE USE OF MUD MOTORS STEERING BY MEANS OF "HIGH-SIDE (GRAVITY) TOOLFACE

The high-side is the top of the hole viewed along the borehole axis. Assuming that the hole has inclination, the low side is the path a small, heavy, ball would follow if rolling slowly down the well. Steering tools are used in the high-side toolface mode to change azimuth in wells with an inclination of more than about five degrees..



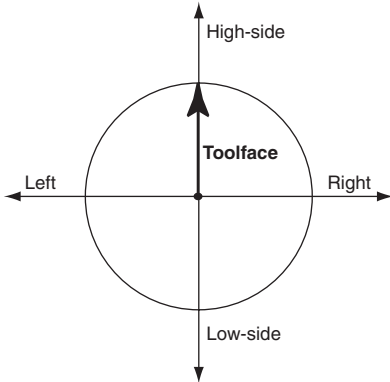
THE USE OF MUD MOTORS

STEERING BY MEANS OF “HIGH-SIDE (GRAVITY) TOOLFACE” (2)

Looking down the drill string towards the bit

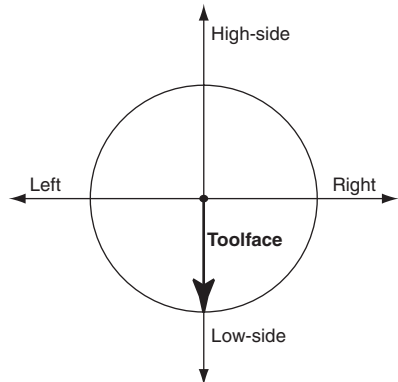
a) Toolface = 0°

Bit and mud-motor trying to build angle while maintaining azimuth



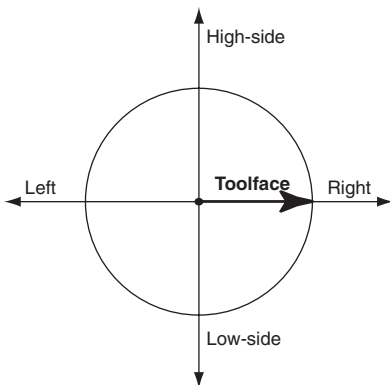
b) Toolface = 180°

Bit and mud-motor trying to drop angle while maintaining azimuth



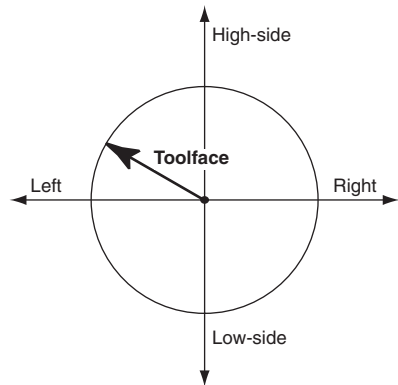
c) Toolface = 90°

Bit and mud-motor trying to maintain inclination and turn the well to the right



d) Toolface = 300° (60° left)

Bit and mud-motor trying to build angle and turn the well to the left



THE USE OF MUD MOTORS

REACTIVE TORQUE

A clockwise rotating downhole motor applies right-hand torque to the bit. There is therefore an equal and opposite torque applied by the bit to the stator housing, and thence to the string. Called 'reactive torque', this can easily be controlled by the operator, by controlling weight on bit. During directional drilling, this reactive torque must be taken into consideration, because it tends to turn the drill string to the left. The actual angle of twist created at the bottom of the string by reactive torque is governed by:

- The magnitude of the torque
- The length of drill pipe
- The torsional elasticity of the drill pipe
- The length and torsional elasticity of the HWDP and BHA.

The HWDP and BHA are both much shorter and much stiffer than the drill pipe and can therefore be neglected when estimating the BHA rotation due to reactive torque, given the accuracy to which the estimate is required.

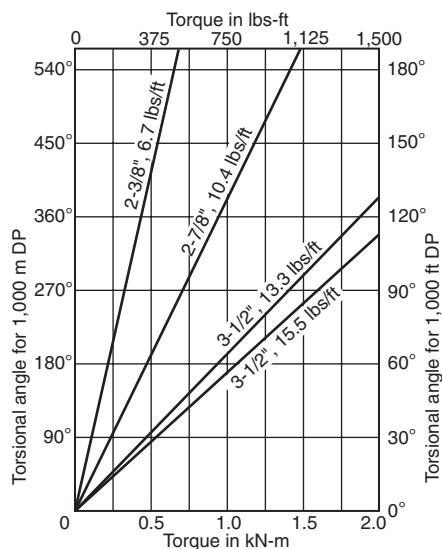
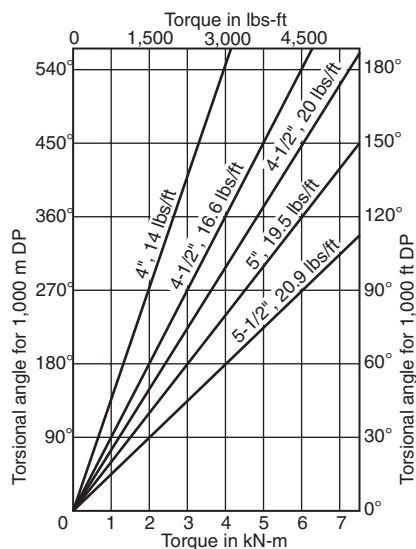
This BHA rotation in a drill string with a mud motor may be estimated as follows:

- Measure the standpipe pressure with the bit on bottom, when flow rate and weight on bit are adjusted to drilling conditions.
- Measure the standpipe pressure when the bit is lifted off bottom with the flow rate being kept constant.
- Calculate difference in standpipe pressure.
- If a diamond bit is in use, reduce the above value by the pressure drop at the bit.
- Read the reactive torque values for the calculated differential pressure from tables.
- Obtain the corresponding torsion angle per unit length for the drill pipe in use from the graphs on the [facing page](#).

After orientation by single shot measurement, the string has to be aligned to produce the required bore hole direction. To do so, the above calculated reactive torque angle is considered as a right-hand angle in addition to the direction change. Having applied the accumulated angle of the string with the rotary table, the string should be raised and lowered several times over a 30 ft interval.

Once a few feet/metres of hole have been made with the new settings, the result will be checked and the drill pipe alignment adjusted in light of the actual results. This is the reason why the preliminary estimate is only required to an "order of magnitude" accuracy.

THE USE OF MUD MOTORS
REACTIVE TORQUE CHARTS



NAVI-DRILL PERFORMANCE DATA

SI UNITS

Tool size Bit size	Power Section	Lobe config.	Flow rate l/min	Bit speed rpm	No load pressure (mean Q) kPa	Operating				Maximum			
						Diff. pressure kPa	Torque N-m	Power output kW	WOB kN	Diff. pressure kPa	Torque N-m	Power output kW	WOB kN
3 1/8" — 3 1/2" - 4 3/4"	MIX	5:6	300-600	180-365	1,000	3,000	520	10-20	30	4,800	830	16-32	45
	M1XL	5:6	300-600	180-365	2,000	6,000	975	18-37	30	9,600	1,560	29-60	45
	M1ADM	5:6	300-600	65-125	1,300	2,000	960	7-13	30	3,200	1,540	10-20	45
	*DDSII	5:6	300-600	180-365	1,300	4,000	600	11-23	--	6,400	960	18-37	--
3 3/4" 4 1/2" - 4 3/4" — 4 3/4" - 5 7/8"	*DDS	7:8	500-700	260-370	900	4,800	920	25-36	--	7,700	1,470	40-57	--
	M1C	5:6	250-700	120-340	1,600	5,500	1,200	15-43	30	8,800	1,920	24-68	55
	M2	1:2	250-800	250-800	900	5,000	650	17-54	29	8,000	1,040	27-87	55
	MIX	5:6	400-1,200	110-325	1,200	5,000	1,850	21-63	48	8,000	2,960	34-101	100
4 3/4" — 5 7/8" - 7 7/8"	M1XL	5:6	400-1,200	110-325	2,100	9,500	3,530	41-120	48	15,200	5,650	65-192	100
	M2PXL	2:3	300-1,000	180-600	1,300	11,000	1,950	37-123	48	17,600	3,120	59-196	100
	M1C	5:6	300-900	100-300	1,000	5,000	1,600	17-50	48	8,000	2,560	27-80	100
	M1P/HF	5:6	600-1,200	105-210	1,300	4,000	2,300	25-51	48	6,400	3,680	40-81	100
	M1ADM	5:6	600-1,200	55-110	1,300	2,000	2,200	13-25	48	3,200	3,520	20-41	100
	M2	1:2	300-1,000	195-650	800	5,000	1,000	20-68	48	8,000	1,600	33-109	100
4 3/4" 5 7/8" - 6 1/2"	*DDSII	7:8	5-850	150-255	800	2,400	820	13-22	--	3,800	1,310	21-35	--
	*DDSIII	7:8	5-850	150-255	1,000	4,200	1,420	22-38	--	6,700	2,270	36-61	--
6 3/4" — 8 3/8" - 9 7/8"	MIX	5:6	1,000-2,500	90-220	1,500	3,200	3,650	34-84	101	5,100	5,840	55-135	170
	M1XL	5:6	1,000-2,500	90-220	2,400	6,000	6,850	65-158	101	9,600	10,960	103-252	170
	M2PXL	2:3	700-2,000	235-430	1,400	8,000	3,650	90-164	101	12,800	5,840	144-263	170
	M1C	5:6	700-1,800	100-260	1,100	5,000	3,800	40-103	101	8,000	6,080	64-166	170
	M1P	7:8	1,000-1,800	110-200	1,200	6,000	5,800	67-121	101	9,600	9,280	107-194	170
	M1P/HF	7:8	1,300-2,300	100-180	1,700	5,000	6,500	68-123	101	8,000	10,400	109-196	170
8 3/8" - 9 7/8"	M1ADM	7:8	1,300-2,300	55-95	1,400	2,500	5,800	33-58	101	4,000	9,280	53-92	170
	M2	1:2	700-2,000	190-550	800	5,000	2,500	50-144	101	8,000	4,000	80-230	170
	*DDS	7:8	1,000-1,800	110-200	1,200	3,000	2,900	33-61	--	4,600	4,640	53-97	--

Tool_size Bit size	Power Section	Lobe config.	Flow rate l/min	Bit speed rpm	No load pressure (mean Q) kPa	Operating				Maximum			
						Diff. pressure kPa	Torque N-m	Power output kW	WOB kN	Diff. pressure kPa	Torque N-m	Power output kW	WOB kN
8" — 9 1/2"-12 1/4"	M1C	5:6	1,200-2,600	85-190	1,300	4,000	6,100	54-121	155	6,400	9,760	87-194	300
	M1P	7:8	1,500-2,500	90-150	900	6,000	10,500	99-165	155	9,600	16,800	158-264	300
	M1P/HF	9:10	2,000-3,400	90-150	2,000	5,000	11,500	108-181	155	8,000	18,400	173-289	300
	M1ADM	7:8	2,000-3,400	50-80	1,500	2,500	10,100	53-85	155	4,000	16,160	85-135	300
	M2	1:2	900-2,600	155-450	800	4,000	3,250	53-153	155	6,400	5,200	84-245	300
9 1/2" — 12 1/4"- 17 1/2"	M1XL	5:6	2,000-4,000	80-165	1,800	6,000	14,600	122-252	214	9,600	23,360	196-404	400
	M1C	5:6	1,500-2,800	100-190	1,000	5,500	9,300	97-185	214	8,800	14,880	156-296	400
	M1P	7:8	1,800-3,000	80-130	700	6,000	15,000	126-204	214	9,600	24,000	201-327	400
	M1P/HF	9:10	2,500-4,200	80-130	1,600	5,000	17,000	142-231	214	8,000	27,200	228-370	400
	M1ADM	7:8	2,500-4,200	40-70	1,300	2,500	15,000	63-110	214	4,000	24,000	101-176	400
	M2	1:2	1,500-3,000	200-400	900	6,000	6,450	135-270	214	9,600	10,320	216-432	400
11 1/4" — 16"- 26"	M1C	5:6	2,000-4,300	80-170	1,600	4,500	13,200	111-235	227	7,200	21,120	177-376	500
	M1P	9:10	3,000-4,800	70-110	1,600	5,000	24,000	176-276	227	8,000	38,400	281-442	500
	M2	1:2	2,000-4,300	155-330	800	4,000	7,500	122-259	227	6,400	12,000	195-415	500

Dog-leg capabilities

The dog-leg capabilities of assemblies incorporating the above motor sections vary with the hole size, the motor diameter, the motor type, the AKO setting, the stabiliser configuration and the drilling parameters. The figures in the table alongside have been taken from BHI's Nav-i-Drill Motor Handbook (1996) as a guide to the ranges available using the standard series of motor sections. The service company should be contacted for recommendations for particular cases.

* Motor section types DDS, DDSII and DDSIII are specialised motors used for drilling short radius build-up sections. These can be used to drill sections with a radius of curvature of 12 - 50 m (1.1 - 4.8 ° per metre).

Tool diameter	Dog-leg capability (°/30 m)
3 1/8"	3 - 40
4 3/4"	0.2 - 26
6 3/4"	0 - 19
9 1/2"	0 - 9.5
11 1/4"	1 - 11

NAVI-DRILL PERFORMANCE DATA

OILFIELD UNITS

Tool size Bit size	Power Section	Lobe config.	Flow rate gals/min	Bit speed rpm	No load pressure (mean Q) psi	Operating			Maximum				
						Diff. pressure psi	Torque lbs-ft	Power output HP	WOB lbs x 10 ³	Diff. pressure psi	Torque lbs-ft	Power output HP	WOB lbs x 10 ³
3 1/8" — 3 1/2" - 4 3/4"	MIX	5:6	80-160	180-365	145	435	385	13-27	67	695	620	21-43	102
	M1XL	5:6	80-160	180-365	290	870	720	25-50	67	1,390	1,150	39-80	102
	M1ADM	5:6	80-160	65-125	190	290	710	9-17	67	465	1,140	14-27	102
	*DDSII	5:6	80-160	180-365	190	580	440	15-31	--	930	700	24-49	--
3 3/4" 4 1/2" - 4 3/4" 3 3/4" 4 3/4" - 5 7/8"	*DDS	7:8	130-185	260-370	130	695	680	34-48	--	1,110	1,090	54-76	--
	M1C	5:6	65-185	120-340	230	800	885	20-57	67	1,280	1,420	32-92	122
	M2	1:2	65-210	250-800	130	725	480	23-73	65	1,160	770	37-117	122
	MIX	5:6	105-315	110-325	175	725	1,365	29-84	108	1,160	2,180	46-135	222
4 3/4" — 5 7/8" - 7 7/8"	M1XL	5:6	105-315	110-325	305	1,380	2,605	55-161	108	2,210	4,170	87-258	222
	M2PXL	2:3	80-265	180-600	190	1,595	1,440	49-165	108	2,550	2,300	79-263	222
	M1C	5:6	80-240	100-300	145	725	1,180	22-67	108	1,160	1,890	36-108	222
	M1P/HF	5:6	160-315	105-210	190	580	1,695	34-68	108	930	2,710	54-108	222
4 3/4" — 5 7/8" - 6 1/2"	M1ADM	5:6	160-315	55-110	190	290	1,625	17-34	108	465	2,600	27-54	222
	M2	1:2	80-265	195-650	115	725	740	27-92	108	1,160	1,180	44-146	222
	*DDSII	7:8	130-225	250-255	115	350	600	17-29	--	560	960	28-47	--
	*DDSIII	7:8	130-225	150-255	145	610	1,050	30-51	--	975	1,680	48-81	--
6 3/4" — 8 3/8" - 9 7/8"	MIX	5:6	265-660	90-220	220	465	2,690	46-113	228	745	4,300	74-180	382
	M1XL	5:6	265-660	90-220	350	870	5,050	87-212	228	1,390	8,080	138-338	382
	M2PXL	2:3	1,85-530	235-430	205	1,160	2690	120-220	228	1,855	4,300	192-352	382
	M1C	5:6	185-475	100-260	160	725	2,805	53-139	228	1,160	4,490	85-222	382
	M1P	7:8	265-475	110-200	175	870	4,280	90-163	228	1,390	6,850	143-261	382
	M1P/HF	7:8	345-610	100-180	245	725	4,795	91-164	228	1,160	7,670	146-263	382
	M1ADM	7:8	345-610	55-95	205	365	4,280	45-77	228	585	6,850	72-124	382
	M2	1:2	185-530	190-550	115	725	1,845	67-193	228	1,160	2,950	107-309	382

Tool size Bit size	Power Section	Lobe config.	Flow rate gals/min	Bit speed rpm	No load pressure (mean Q) psi	Operating			Maximum				
						Diff. pressure psi	Torque lbs-ft	Power output HP	WOB lbs x 10 ³	Diff. pressure psi	Torque lbs-ft	Power output HP	WOB lbs x 10 ³
8" 9 1/2"-12 1/4"	M1C	5:6	315-685	85-190	190	580	4,500	73-163	348	930	7,200	117-260	674
	M1P	7:8	395-660	90-150	130	870	7,745	133-221	348	1,390	12,390	212-354	674
	M1P/HF	9:10	530-900	90-150	290	725	8,480	145-242	348	1,160	13,570	233-388	674
	M1ADM	7:8	530-900	50-80	220	365	7,450	71-113	348	585	11,920	113-182	674
	M2	1:2	240-685	155-450	115	580	2,395	71-205	348	930	3,830	113-328	674
9 1/2" — 12 1/4" - 17 1/2"	M1XL	5:6	530-1,055	80-165	260	870	10,770	164-338	488	1,390	17,230	262-541	894
	M1C	5:6	395-740	100-190	145	800	6,860	131-248	488	1,280	10,980	209-397	894
	M1P	7:8	475-795	80-130	100	870	11,065	169-274	488	1,390	17,700	270-438	894
	M1P/HF	9:10	660-1,110	80-130	230	725	12,540	191-310	488	1,160	20,060	306-497	894
	M1ADM	7:8	660-1,110	40-70	190	365	11,065	84-147	488	585	17,700	135-236	894
	M2	1:2	395-795	200-400	130	870	4,755	181-362	488	1,390	7,610	290-580	894
11 1/4" — 16"- 26"	M1C	5:6	530-1,135	80-170	230	655	9,735	148-315	510	1,050	15,580	237-504	1124
	M1P	9:10	795-1,270	70-110	230	725	17,700	236-371	510	1,160	28,320	377-593	1124
	M2	1:2	530-1,135	155-330	115	580	5,530	163-347	510	930	8,850	261-556	1124

Dog-leg capabilities

The dog-leg capabilities of assemblies incorporating the above motor sections vary with the hole size, the motor diameter, the motor type, the AKO setting, the stabiliser configuration and the drilling parameters. The figures in the table alongside have been taken from the BHI's Navi-Drill Motor Handbook (1996) as a guide to the ranges available using the standard series of motor sections. The service company should be contacted for recommendations for particular cases.

* Motor section types DDS, DDSII and DDSIII are specialised motors used for drilling short radius build-up sections. These can be used to drill sections with a radius of curvature of 40 - 165 ft (0.35 - 1.45 ° per foot).

Tool diameter	Dog-leg capability (°/100 ft)
3 1/8"	3 - 40
4 3/4"	0.2 - 26
6 3/4"	0 - 19
9 1/2"	0 - 9.5
11 1/4"	1 - 11

NEYRFOR TURBINE DATA

Turbodrills for straight holes				Turbodrills for deviated holes							
Nominal size	5"	7 1/4"	7 1/4"	9 1/2"	9 1/2"	3 3/8"	4 3/4"	4 3/4"	6 5/8"	9 1/2"	9 1/2"
Type	T2	T2	T3	T2	T3	FBS	FBS	MK2	FBS	SBS Standard	SBS High flow
OD	5"	7 3/8"	7 3/8"	9 17/32"	9 17/32"	3 3/8"	4 3/4"	4 3/4"	6 5/8"	9 1/2"	9 1/2"
Bit size	5 5/8"-6 3/4"	8 1/2"-9 5/8"	8 1/2"-9 5/8"	11"-15"	11"-15"	3 3/4"-5 3/8"	5 5/8"-6 3/4"	5 5/8"-6 3/4"	7 5/8"-9 7/8"	12 1/4"-17 1/2"	12 1/4"-17 1/2"
Speed range (rpm)						800-1,800	700-1,400	700-1,400	400-1000	300-700	300-700
Bent housing angle :	Standard Available										
Dog-leg angle capability with standard bent housing (°/100ft - °/30 m)											
Nominal flow rate (gpm) (l/sec)	160 10	475 30	475 30	650 41	650 41	100 6.3	160 10	200 12.6	475 30	650 41	650 41
Pressure drop (psi) (kPa)	1,435 9,900	1,510 10,400	2,150 14,800	1,525 10,500	2,210 15,200	1,537 10,600	1,415 9,800	1,598 11,000	1,875 12,900		
Power (HP) (kW)	78 58	243 181	365 272	379 283	568 424	51 38	74 55	104 78	280 209	520 388	520 388
Torque Maximum drlg (lbs-ft) (N-m)								1,475 2,000	2,460 3,350	5,000 6,780	5,000 6,780
Stalling (lbs-ft) (N-m)						325 440	860 1,160				

Note : The pressure drop, power and torque figures given above are valid for the nominal flow rate, and for a drilling fluid density of 0.52 psi/ft or 11.75 kPa/m

FREQUENCY AND TYPES OF SURVEYS TO BE TAKEN

	Platform/Cluster/Template Wells and Other Wells During Drilling		Verification Survey		Isolated vertical wells During Drilling		Verification Survey	
	Type of Survey	Survey Interval**	Type of Survey	Survey Interval**	Type of Survey	Survey Interval**	Type of Survey	Survey Interval**
Stove Pipe/Foundation Pile*	None		None		None		None	
Marine Conductor*	GSS	30 ft/ 10 m	GMS	25 ft 8m	ESS 6,13	at section TD	GMS/EMS 1,7,8,14	50 ft 15 m
Conductor String (20/18 ⁵ / ₈ " Casing)	GSS/MWD/ST/ESS 6,10,12	30-90 ft 10-30 m	GMS	100 ft 25 m	MWD/ST/ESS 6,10,12,13	300 ft 100 m	GMS/EMS 1,7,8,9,14	100 ft 25 m
Surface String (13 ³ / ₈ " Casing)	MWD/ST/ESS 6,10,12	30-90 ft 10-30 m	GMS	100 ft 25 m	MWD/ST/ESS 6,10,12,13	300 ft 100 m	GMS/EMS 1,7,8,9,14	100 ft 25 m
Intermediate String (9 ⁵ / ₈ " casing)	MWD/ST/ESS 6,10,12	30-90 ft 10-30 m	GMS	100 ft 25 m	MWD/ST/ESS 6,10,12,13	300 ft 100 m	GMS/EMS 1,7,9,14	100 ft 25 m
Production String (7" Casing/Liner)	MWD/ST/ESS 6,10,12	30-90 ft 10-30 m	GMS	100 ft 25 m	MWD/ST/ESS 6,10,12	300 ft 100 m	GMS/EMS 4,9,14	100 ft 25 m
Production Liner (4 1/2" Liner)	ESS/EMS 4,6	as required	GMS/EMS 4,5,9	100 ft 25 m	ESS/EMS/DIP 4,6	300 ft 100 m	EMS/DIP 4	100 ft 25 m

Notes on table

* Whichever is applicable

** If a wireline survey is not made from surface, it should overlap at least 1000 ft (300 m) of the previous survey. Magnetic surveys should be taken into the last casing shoe.

1. IN (FINDS or RIGS) to replace GMS if available.
2. When conductors have been batch installed, all should be cleaned out and surveyed prior to drilling the first well.
3. Can use MWD and EMS when there are problems of getting gyro down. Run EMS prior to running casing.
4. EMS or Dip Meter survey allowed below the top of the lowest hydrocarbon bearing zone (in open hole).
5. In hot wells >120°C (250°F) there may not be enough room for the gyro heat shield. Run EMS prior to running casing.
6. ESS is preferred but MSS may be used.

7. GMS/EMS must be taken prior to entering any potential zone that could blow out.

8. GMS/EMS may be omitted where it is proven that the well will not penetrate a potential blow out zone in the next open hole section and (1) there is a GMS/EMS in the previous section and (2) the open hole magnetic survey of that section is good.

9. GMS surveys should be every 100 ft (25m), but this should be reduced to every 50 ft (15m) through sections with doglegs over 2.5°/100 ft (2.5°/30m).

10. Survey every stand (90 ft) when using MWD. Interval may be increased to 300 ft (100 m) when using an ESS.

11. IN (RIGS) may be considered in special cases of high accuracy requirements.

12. EMS to be taken when MWD/ST/ESS surveys have questionable quality.

13. Where criteria for relief well drilling have been relaxed, inclination only surveys may be considered.

14. Survey every stand (90 ft) when using EMS, but this should be reduced to every single (30 ft) through sections with doglegs over 2.5°/100 ft (2.5°/30 m).

GSS = Gyro Single Shot (Surface read-out preferred)

MWD = Measurement While Drilling

ST = Steering Tool

ESS = Electronic Magnetic Single Shot

MSS = Magnetic Single Shot (ESS preferred)

DIP = Dip Meter Log (which gives good survey results)

EMS = Electronic Magnetic Multi-Shot

GMS = Gyro Multi-Shot (North Seeking Gyro preferred)

IN = Inertial Navigation

MMS = Magnetic Multi-Shot (EMS preferred)

PRE-SURVEY CHECKLISTS

Totcos

Totcos are normally run by the driller. The pre-survey checklist and running procedure are given below. (When using a magnetic single shot tool for inclination only surveys follow the running procedure for MSS.)

1. Check that the instrument landing assembly will seat correctly in the landing ring (Totco ring), and not jam or land eccentrically.
2. Install the landing ring in the proper place when making up the BHA.
3. Avoid landing the instrument directly on top of bit, mud motor or turbine. The instrument could get stuck and furthermore make circulation impossible.
4. Check that the fishing tool will fit over the fishing neck.
5. Check that the instrument will pass the BHA above the landing ring and not hang up (e.g. in the jar).
6. Check that the instrument kit box is complete and that the angle units have been checked in the workshop before delivery to the well site. Check that no angle unit has been used more than 25 times after calibration.
7. Use sinker bars if the drilling fluid has a high density and/or is viscous.
8. Before surveying circulate sufficiently to avoid a back flow of cuttings into the BHA.
9. Estimate the time lapse. This should be equal to the sum of the times required to:
 - mount the instrument in the barrel
 - run the instrument through the drill string
 - provide a safety margin of a few minutes (3-5 minutes) in case of any delays.

Magnetic Single Shots

An OU representative should ensure that the following is carried out :

1. Check that the required length of NMDC is available.
2. Check that the instrument landing assembly will seat correctly in the landing ring (TOTCO ring), and will not jam or land eccentrically.
3. Check whether the instrument is to be top or bottom landed or used with a mule shoe.
4. Install the landing ring in the proper place when making up the BHA. Avoid landing the instrument directly on top of bit, mud motor or turbine. The instrument could get stuck and furthermore make it impossible to circulate.
5. Check that the instrument will pass the rest of the BHA above the landing ring and not hang up (e.g. in the jar).
6. Check that the instrument kit box is complete and that the angle units have been checked in the workshop before delivery to the well site. Check that no angle unit has been used more than 25 times after calibration.
Specifically check that the kit box includes:
 - two angle units of each range, which should be used alternately
 - batteries specified for the instrument
 - film discs
 - developing chemicalsand ensure that the film is kept dry before the survey is run.
7. Check the angle unit in the field test stand. Ensure that the angle unit inclination readings agree with the field test stand inclinations.

MINIMUM CURVATURE METHOD EQUATIONS

The directional surveys at consecutive stations (at AHD₁ and AHD₂) measure values of A₁, A₂, I₁ and I₂. These values are then substituted into the equations given below to yield values of dog-leg angle, ΔN, ΔE, ΔTVD and ΔPHD.

$$\cos DL = \cos (I_2 - I_1) - \sin I_1 \cdot \sin I_2 (1 - \cos (A_2 - A_1))$$

$$DLS = \frac{DL}{\Delta AHD} \times \text{characteristic length}$$

$$\Delta N = \frac{\Delta AHD}{2} (\sin I_1 \cdot \cos A_1 + \sin I_2 \cdot \cos A_2) \cdot RF$$

$$\Delta E = \frac{\Delta AHD}{2} (\sin I_1 \cdot \sin A_1 + \sin I_2 \cdot \sin A_2) \cdot RF$$

$$\Delta TVD = \frac{\Delta AHD}{2} (\cos I_1 + \cos I_2) \cdot RF$$

$$\Delta PHD = \Delta N \cdot \cos A_t + \Delta E \cdot \sin A_t$$

Where: AHD/TVD = Along hole / True vertical depths

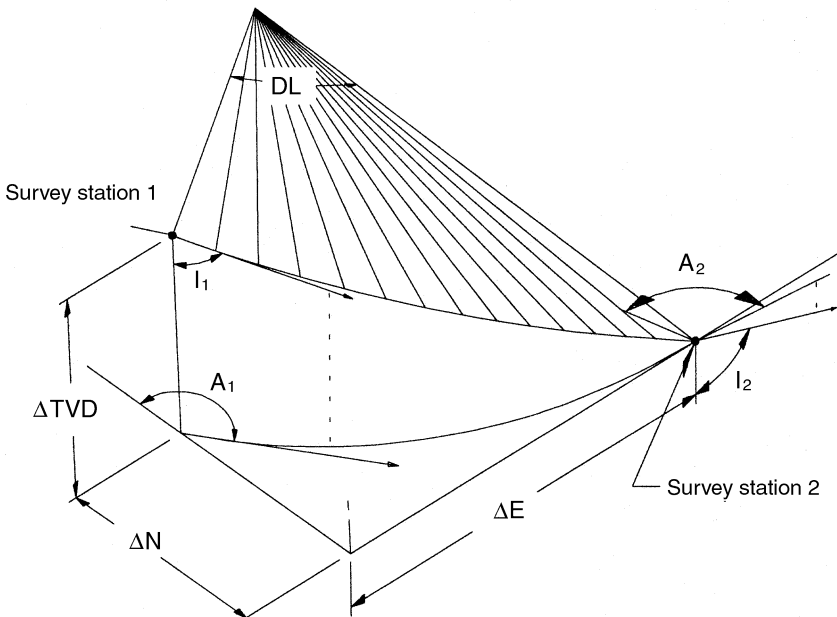
$$RF = \text{Ratio Factor} = \frac{180}{\pi \cdot DL} \times 2 \times \tan \frac{DL}{2}$$

DL = Dog-leg angle in degrees

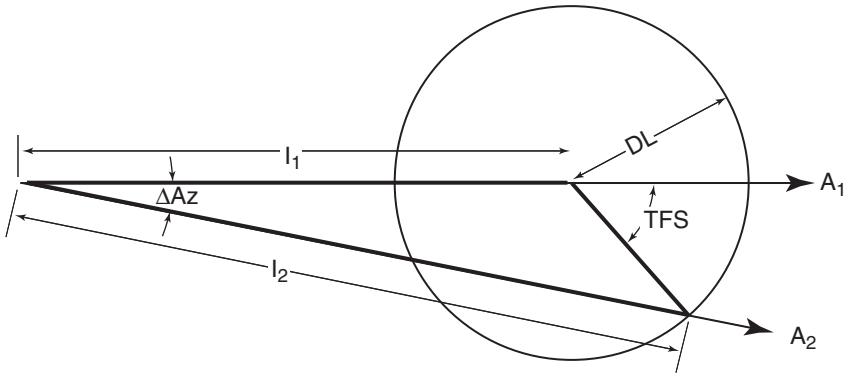
DLS = Dog-leg severity in degrees per characteristic length
(usually °/100 ft or °/30 m)

A_t = target Azimuth

PHD = Projected horizontal distance (in direction A_t)



BASIC VECTOR DIAGRAM



I_1 , I_2 , A_1 , A_2 and DL are as given on the [previous page](#)

TFS = Tool Face Setting angle, positive in the sense shown to give increasing azimuth

Knowing the value of any three of the sides/angles of the triangle allows the other three to be calculated using the standard equations :

If two angles and a side are known :

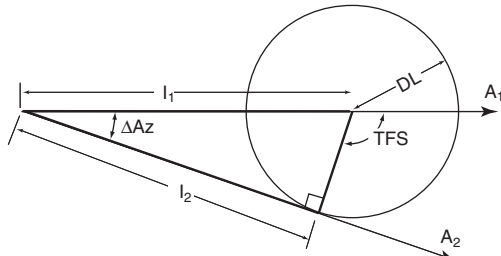
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

If two sides and the included angle are known : $a^2 = b^2 + c^2 - 2bc \cdot \cos A$ etc.

If two sides and a non-included angle are known : $a = c \cdot \cos B \pm \sqrt{b^2 - c^2 \cdot \sin^2 B}$
or $a = b \cdot \cos C \pm \sqrt{c^2 - b^2 \cdot \sin^2 C}$

Note :

For maximum change in Azimuth the vector representing A_2 is tangent to the circle whose radius represents DL .



M – SAFETY

Clickable list

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Royal Dutch/Shell Group Commitment to Health, Safety and the Environment

In the Group we are all committed to

- pursue the goal of no harm to people
- protect the environment
- use material and energy efficiently to provide our products and services
- develop energy resources, products and services consistent with these aims
- publicly report on our performance
- play a leading role in promoting best practice in our industries
- manage HSE matters as any other critical business activity
- promote a culture in which all Shell employees share this commitment.

In this way we aim to have an HSE performance we can be proud of, to earn the confidence of customers, shareholders and society at large, to be a good neighbour and to contribute to sustainable development.

Royal Dutch/Shell Group Health, Safety and Environmental Policy

Every Shell company

- has a systematic approach to HSE management designed to ensure compliance with the law and to achieve continuous performance improvement
- sets targets for improvement and measures, appraises and reports performance
- requires contractors to manage HSE in line with this policy
- requires joint ventures under its operational control to apply this policy and uses its influence to promote it in its other ventures
- includes HSE performance in the appraisal of all staff and rewards accordingly.

Endorsed by the Committee of Managing Directors. March 1997



INTRODUCTION

The Exploration and Production HSE manual, report EP 95-0000, is a structured collection of guidelines on HSE matters in all areas of EP operations. It incorporates the previous EP 55000 Safety Manual and a number of other EP reports on Health, Safety and the Environment published separately.

The guideline for managing HSE in drilling operations is EP 95-0210. It is essential that all those concerned with the management and supervision of drilling operations make themselves familiar with the document so that they are at least aware of what advice is available within it. To that end, the contents list of that report is reproduced on the next two pages.

Reproduced thereafter are the contents of three appendices from the same report :

- Appendix I : Policies
- Appendix II : Responsibilities of Key Staff
- Appendix IV : Classification of Hazardous Areas

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*Rev. 0 of 16/10/95

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POLICY GUIDELINES ON HEALTH, SAFETY AND THE ENVIRONMENT

It is the policy of Shell companies to conduct their activities in such a way as to take foremost account of the health and safety of their employees and of other persons, and to give proper regard to the conservation of the environment. They aim to be among the leaders in their respective industries in these matters.

Health

Shell companies seek to conduct their activities in such a way as to avoid harm to the health of their employees and others, and to promote, as appropriate, the health of their employees.

Safety

Shell companies work on the principle that all injuries should be prevented and actively promote amongst all those associated with their activities the high standards of safety consciousness and discipline that this principle demands.

Environment

Shell companies:

- pursue in their operations progressive reductions of emissions, effluents and discharges of waste materials that are known to have a negative impact on the environment, with the ultimate aim of eliminating them
- aim to provide products and services supported with practical advice which, when used in accordance with this advice, will not cause injury or undue effect on the environment
- promote protection of environments which may be affected by the development of their activities and seek continuous improvement in efficiency of use of natural resources and energy.

Common HSE aspects

Shell companies:

- assess health, safety and environmental aspects before entering into new activities and reassess them in case of significant change in circumstances
- require contractors working on their behalf to apply health, safety and environmental standards fully compatible with their own
- recognise the concerns of shareholders, employees and society on health, safety and environmental matters, provide them with relevant information and discuss with them related Company policies and practices
- develop and maintain contingency procedures, in co-operation with authorities and emergency services, in order to minimise harm from any accidents
- work with government and others in the development of improved regulations and industry standards which relate to health, safety and environmental matters
- conduct or support research towards the improvement of health, safety and environmental aspects of their products, processes and operations
- facilitate the transfer to others, freely or on a commercial basis, of know-how developed by Shell companies in these fields.

Endorsed by the Committee of Managing Directors - June 1991.

Reproduced from EP 95-0210 - Appendix 1

POLICY ON SUBSTANCE ABUSE

Definition

Substance is defined as any substance which chemically modifies the body's function resulting in psychological or behavioural change. In this context substance includes but is not limited to alcohol, intoxicating products and medication. Substance abuse is the use of these substances in a harmful or improper way.

Background

The Company conducts its business against high standards of safety and concern for the environment. In all areas of activity it pursues the reduction of risk to both. Also, the Company is committed to maintaining a healthy and productive workplace. All employees are expected to share in these objectives.

The abuse of substances in any quantity however small can impair performance at work, and can be a serious threat to safety and environment, health and productivity. The Company wishes to ensure that all employees recognise this threat and aims at minimising the risks involved. In order to achieve this, the following policy will apply and will be part of the employee's conditions of employment.

Policy

1. The Company recognises alcohol or drugs dependence as a treatable condition. Employees who have an alcohol or drugs dependence are encouraged to seek medical advice, and to follow appropriate treatment promptly. The Company will assist an employee to obtain treatment and employees who seek such help will not place employment in jeopardy by doing so, although alternative work might be considered. The normal Company benefits which apply in the case of any illness will be available.
2. Being at work while impaired by drugs or alcohol is strictly prohibited.
3. The illicit use of legal substances or the use, possession, distribution or sale of illegal substances on Company business or locations is strictly prohibited.
4. Preceding employment, the Company will test for substance abuse.
5. The Company may conduct unannounced searches for drugs and alcohol or any other substance on Company locations. It may also require employees to submit to alcohol and drugs testing where a good faith reason exists to suspect alcohol or drug abuse. Unannounced, periodic or random testing will be conducted when an employee meets any one of the following conditions:
 - holds a safety and environmentally sensitive position
 - holds a dedicated management position
 - holds a position where testing is required by law
 - holds a position where the individual acts alone or unsupervised.
6. If a test result is positive, in most cases, on a first time basis only, the employee will be allowed to continue in employment provided there is compliance with the appropriate rehabilitation procedures (eg education, counselling, treatment and unannounced testing).
7. Dismissal will normally occur in the following circumstances:
 - failure to co-operate with the implementation of this policy
 - failure to comply with the appropriate rehabilitation procedures
 - the use, possession, distribution or sale of illegal drugs or substances on Company business or locations
 - the use or possession of alcohol on Company business or locations unless previously authorised, and the use or possession of alcohol in safety or environmentally sensitive positions
 - a second positive test result following a prior positive result from a Company initiated test where employment has been continued, or after an earlier identification of an abuse problem.
8. All contractors are required to ensure that their employees do not create a presence of substance abuse on Company business or locations. In addition, contractors who perform safety or environmentally sensitive work are required to provide evidence of a comprehensive substance abuse policy and practices at least equivalent to those in force within the Company.

Reproduced from EP 95-0210 - Appendix 1

RESPONSIBILITIES OF COMPANY OPERATIONS STAFF

Head of Drilling Engineering

The Head of Drilling Engineering shall be responsible for ensuring that appropriate technical and operating standards are in place and to provide cohesion, direction and consistency throughout his area of responsibility for HSE such that staff discharge their duties in a professional manner and to a consistent standard. Core activities will include:

- the specification, maintenance and monitoring of policies, procedures and standards
- the harmonisation of Company and contractor policies, procedures and standards to a consistent and unambiguous approach
- the dissemination of technical information
- the maintaining of appropriate contacts in the Shell Group and with external resources.
- the provision of guidelines to his subordinate supervisors
- maintaining an awareness of the professional competence of all staff and co-ordinating their development through appropriate operational exposure and training.

Company Drilling Supervisor

The Company Drilling Supervisor is the Company's senior representative on site. His role with regard to HSE is to verify that the drilling contractor and service and subcontractors perform work, under their respective contracts in a manner which assures the health and safety of staff and avoids harmful emissions to the environment. As such he should be familiar with the provisions of the various contracts and be competent to verify correct implementation. His specific responsibilities relating to HSE include:

- verifying the implementation of hazards and effects management controls
- making quality assurance checks on contractors inspections
- taking part in accident investigations as dictated by the application of the 'Incident Potential Matrix'
- participating in HSE meetings
- making structured inspections of the facility in conjunction with the senior contractor representative and following up on corrective actions
- verifying that well integrity is being properly maintained
- verifying that effective lines of communication between the various contractors are being maintained
- alerting base supervisors to any changes which have a significant negative impact on well or operational HSE
- keeping themselves fully appraised of ongoing operations.

RESPONSIBILITIES OF COMPANY OPERATIONS STAFF (2)

Wellsite Drilling Engineer

The Wellsite Drilling Engineer's HSE responsibilities include observing that the following activities are performed safely and without endangering the health of personnel or damaging the environment by verifying that:

- electric logging operations are conducted such that:
 - radioactive sources are handled in a manner that avoids non-logging contractor staff being exposed to levels of radioactive emissions above 2.5 micro-Sieverts/hr
 - logging contractor staff wear their film badges;
 - radio silence procedures are observed during pertinent operations;
 - hazardous areas are prohibited to non-essential staff.
- radioactive sources are stored such that:
 - the area in which radioactive emissions exceed 2.5 micro-sieverts/hr is barriered
 - the area where radioactive emissions fall between 2.5 - 1.0 micro-sieverts/hr is designated as 'no stay'
- the radioactive source register is kept up to date
- primary and secondary explosives are stored separately either in an area protected by a deluge system or on a jettisonable platform
- the explosives register is kept up to date
- mud chemicals and mud testing chemicals are stored and handled in a manner that assures the safety of staff
- chemical safety data sheets are posted and a copy kept by the medic

Reproduced from EP 95-0210 - Appendix 2

RESPONSIBILITIES OF CONTRACTOR LINE STAFF

Contractor Rig Manager

The contractor Rig Manager is accountable for the following HSE matters:

- liaising with the Company's Head of Drilling Engineering to assure compatibility between Company and contractor safety systems, plans and objectives
- developing HSE objectives and plans to meet those objectives which derive from the contract, his company's corporate policy and the drilling programme
- maintaining the rig HSE Case(s) for the rig(s) under his control
- establishing the organisation and controls which ensure that all activity, including those performed by service and subcontractors, is conducted in accordance with the HSE Case
- demonstrating his commitment to high HSE standards by making regular structured visits to the rig with specific HSE objectives and through providing the resources to effect recommended improvements
- ensuring that staff are trained such that they develop the necessary competence to enable them to work safely and avoid damage to the environment
- liaising with the Company, to select service and subcontractors who can meet the same standards as themselves and monitor their work to confirm these standards are being maintained
- making suitable arrangements for consultation with line supervisors, employees and service and subcontractors' representatives on health, safety and environmental matters
- making certain that all incidents involving injury to persons, damage to property or the environment, and those having potential for serious effect are thoroughly investigated and that effective follow-up action is taken by:
 - establishing remedial action requirements
 - identifying action parties
 - establishing completion targets
 - regularly reviewing progress.
- establishing and discussing with subordinates individual responsibilities, targets and accountabilities for health, safety and the protection of the environment and confirm these during performance appraisal
- setting a clear leadership example by his own actions.

RESPONSIBILITIES OF CONTRACTOR LINE STAFF (2)

Contractor Rig Superintendent (Senior Toolpusher)

On an offshore rig the contractor Rig Superintendent or Senior Toolpusher will often also be the OIM with responsibilities defined by legislation and/or Company policy.

The contractor Rig Superintendent is responsible for the execution of all well and associated work programmes. This includes, rig moving, the drilling, completion, perforation and testing of new wells, the repair of existing wells by workover and the maintenance of the drilling facility, safety of the installation and all personnel on board.

Key safety responsibilities include assurance that:

- hazards are identified, assessed and controlled and plans for recovery are effectively in place
- injury to personnel, assets or the environment, is prevented
- the emergency/contingency plan is operable and tested and all site staff are competent to perform their assigned duties
- safe working codes and practices are implemented for all operations in accordance with recognised policies, standards and procedures as agreed by the Company
- prompt action is taken to rectify any deficiencies in working practices or conditions
- all employees receive appropriate induction and training in all aspects of their work and observe such safety requirements as the work situation warrants
- safety rules and procedures are followed and should transgressions be observed, corrective action is taken to ensure future compliance
- HSE meetings are held as follows:
 - weekly for all personnel with records being kept of attendees, topics discussed, action items arising, action parties responsible for close out and target date for completion
 - daily with work teams (crews) to discuss the shift work plan and any expected hazards. This should be logged, in the daily report
 - prior to non-routine operations, with all involved personnel, to ensure the job and its inherent hazards are understood, controls are in place, the tools and work practices are appropriate, relevant expertise is available and permit requirements are understood and verified as being in place.
- drilling and associated equipment is inspected and maintained in accordance with the inspection programme and the preventive maintenance system
- all accidents causing injury to personnel or damage to equipment and all significant near misses are reported in accordance with procedures and are investigated at the appropriate level, in the appropriate depth and that remedial actions are implemented
- employees use personal protective equipment as necessary
- hazardous work is performed under the permit-to-work system
- all relevant information is communicated between personnel at shift change.

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RESPONSIBILITIES OF CONTRACTOR LINE STAFF (3)

Night Toolpusher

The contractor's Night Toolpusher is responsible, during his shift, for the safe execution of all well work programmes issued through the contractor's Rig Superintendent. This requires that he:

- enforces the provisions of the drilling contractor's HSE policy, procedures and plan
- verifies that staff under his authority are knowledgeable of their role and competent to perform it
- ensures drilling equipment is maintained in a safe and operable condition
- where necessary applies for work permits and verifies that their provisions are followed
- ensures that all accidents and significant near misses are reported and takes part in their investigation. Disseminates findings amongst all staff in order to avoid recurrence
- verifies the quality of safety inspections performed by subordinates
- regularly monitors well conditions by liaising with relevant staff and ensures that proactive steps are taken to maintain primary well control
- provides emergency response support, both personally and together with drilling crews, and conducts a regular programme of exercises
- acts as the link between senior and junior rig supervision by attendance at both groups meetings and disseminating information as appropriate.

RESPONSIBILITIES OF CONTRACTOR LINE STAFF (4)

Driller

As the first line in the supervision of personnel, the Driller's prime objective is to ensure that instructions are carried out competently and therefore safely.

He is to verify that crew personnel are competent to carry out their work and use safe working practices.

He disseminates to his crew information on HSE and new safety procedures. Additionally he is to inform senior staff of safe working procedures suggested by his crew and other personnel. The introduction of any consequent change in procedure should be implemented under the direction of the Driller with guidance and approval from the Rig Superintendent if appropriate.

The Driller is instrumental for the following:

- seeing that all instructions of the contractor's Toolpusher concerning work methods and equipment are carried out
- ensuring that crew members fully understand their duties when carrying out a job
- taking necessary steps to correct hazardous conditions and incorrect practices and checking that protective devices are in good condition and used when needed
- anticipating hazardous conditions and remove the cause of possible accidents
- ensuring that crew members complete each job in an orderly way and leave no hazardous conditions behind
- promptly reporting any unsafe equipment that cannot be corrected by the drilling crew
- encouraging all crew members to make HSE suggestions and recognise their ability to contribute to accident prevention
- assisting in the investigation of all accidents in his line of responsibility
- seeing that all crew members are trained in correct operating procedures and policies. He is to make a particular effort to make new crew members HSE conscious and verify their job knowledge
- training the Assistant Driller so that he can competently perform the Driller's duties when necessary
- being conversant with the Company well control methods and be able to react accordingly
- setting an example to the crew by observing all HSE regulations
- adjusting the pace of operations to meet the competence of his crew
- ensuring proper use is made of the 'Permit-to-work System'
- preparing an adequate handover to ensure continuity during shift changes
- holding a pre-shift safety meeting to appraise crews of planned operations.

Reproduced from EP 95-0210 - Appendix 2

CLASSIFICATION OF HAZARDOUS AREAS

The classification of hazardous areas with respect to electrical equipment shall be in accordance with the Institute of Petroleum (IP) Area classification Code for Petroleum Installations. The following is only a summary of the requirements of the Code and is provided to give a ready appreciation but should not be used as a substitute for the Code.

Hazardous zones defined under the IP or any similar code should not be confused with any other type of hazardous area established, eg sour gas, high tension (HT) overhead no-go areas, radioactive store hazardous area.

Grades of Flammable Gas or Vapour Release

Continuous sources are where flammable fluids (gases) are normally present or present for more than 1,000 hours per year. Such atmospheres are normally present only in fixed roof tanks and at process vents. Continuous grade sources are not part of the drilling fluid circulation, wellhead or BOP system.

Primary sources are those which can release flammable vapours or gases in normal operation. Primary sources include vents and active mud tanks, ditches and mud treating equipment. Particular caution in the mud-gas separator piping is necessary due to the potential of high volumes of primary gas released both through the vent pipe outlet and via the mud drain.

Secondary sources are those which do not release flammable gases or vapours normally but can do so under abnormal (ie failure) circumstances. This includes minor and temporary containment failures such as occur from day to day, not catastrophic failure such as vessel rupture, burst pipes or blowouts.

Classification of Hazardous Zones

The hazardous zone resulting from a continuous source will be a greater hazard than the zone resulting from a primary source, because the probability is higher than it will contain a flammable mixture. To show this, hazardous zones are classified according to the type of source of flammable vapour or gas:

- the hazardous zone resulting from a continuous source is normally classified as Zone 0
- the hazardous zone resulting from a primary source is normally classified as Zone 1
- the hazardous zone resulting from a secondary source is normally classified as Zone 2.

The parts of the facility which are not classified as hazardous zones can be designated non-hazardous but may still contain a flammable mixture under calamity conditions.

Hazardous zone classification depends on the grade of release and the ventilation available as shown in the table overleaf.

For land rigs, the open air situation is the norm, with restricted ventilation only present where the drill floor is shrouded, or inside the free space of active mud tanks, ditches and well cellar areas. There should be no Zone 0 areas on any drilling rig installation. Outside Zone 0, 1 or 2 the worksite is 'non-hazardous'.

Reproduced from EP 95-0210 - Appendix 4

CLASSIFICATION OF HAZARDOUS AREAS (2)

Hazardous zone classification and impact of ventilation

Grades of release	Open air situation and adequately ventilated spaces with unrestricted air movement, ie at least 12 changes per hour	Restricted ventilation, eg inside modules with ventilation stopped or less than 12 changes per hour	No ventilation, e.g. inside tank
Continuous	Zone 0	Zone 0	Zone 0
Primary	Zone 1	Zone 1	Zone 0
Secondary	Zone 2	Zone 1	Zone 0

Hazardous zone dimensions

According to the IP Code (1990) for Drilling and Workover Installations (where diagrams are provided), the Zone 2 hazardous zones around the rig equipment extend to:

1. A cylinder 7.5 m around the bell nipple extending 9 m below the wellhead deck (offshore) or to ground level. The upper extent of the Zone 2 is 7.5 m above the rig floor, extending to the top of any existing wind break around the derrick area. Only the wellhead cellar and sunken ditches within the Zone 2 are classified as Zone 1.
2. A space around active mud tanks 3 m from the top and sides of each tank to ground level extending to 7.5 m from the sides of each tank at a height of 3 m. Inside the tank walls is Zone 1. Enclosures around the tank, unless adequately ventilated are classified as hazardous Zone 1 with Zone 2 extending 3 m from openings to the enclosure.
3. Around the shale shaker Zone 2 extends 7.5 m above and around the exterior surface of the shaker, and Zone 1 extends 1.5 m from the outer surface . If enclosed the enclosed space shall all be classified as Zone 1, with Zone 2 areas extending 7.5 m from any openings.
4. For any gas vent outlets, the extent of the hazardous zone is based on guidelines provided in Chapter 5 of the IP Code. If flow rates and type of effluent figures are not known the hazardous (Zone 2) should extend at least 15 m from the vent outlet in all directions.
5. For wireline operations, the point of reference is not the bell nipple but the stuffing box with other dimensions and zone classifications the same as with drilling rigs on land and to the main deck offshore.

For the purpose of ignition protection against small releases of flammable fluids around the rig floor area, the interior of the derrick or mast structure is classified as Zone 2. All purge air, cooling air and internal combustion engine air intake shall be taken from well outside Zone 1 and 2, ie from a designated non-hazardous zone. Equally, all electrical equipment in the derrick shall be suitably protected. Requirements are defined in the IP Code.

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CLASSIFICATION OF HAZARDOUS AREAS (3)

Cellars or pits below ground level in a Zone 2 space should be classified as Zone 1. Any enclosed premises, containing source of hazard which may give rise to a dangerous atmosphere under abnormal conditions should be classified as follows:

The interior of the enclosure Zone 1; the surrounding space in open air within a 7.5 m radius from any point of egress from the premises as Zone 2. Any enclosed premises not containing a source of hazard but located in a Zone 2 space should be classified as Zone 1, unless entry of a dangerous atmosphere is prevented by, eg fire walls, ventilation, etc where the enclosure may be classified as a Zone 2 or even as a safe zone if the space is ventilated and over-pressurised.

In naturally well-ventilated conditions (eg offshore) outside the limits of the derrick or mast, the vertical extent of the 'hazardous zone' above the highest source of hazard may be reduced to 3 m and extends over the whole classified area and below the source of hazard to ground level, except as described in the cases above. For full details refer to IP15 Chapter 6.

It must be clearly emphasised that the dimensions and conditions quoted are to be considered as the minimum case, and where any doubt exists, the dimensions (or even classification) of the hazardous zone should be increased by appropriate degree.

FIRE PREVENTION

SAFE DISTANCES FOR FIRE PREVENTION

Definitions

Dangerous areas are areas within 50 feet from :-

- inflammable products in open storage or open system
- gas-rich areas (in general)
- producing wells (open system)
- wildcat or exploration drilling
- exploitation drilling (possibility of abnormal pressures)

Remotely dangerous areas are areas within 25 feet from :-

- light products and crude oil pumps
- compressors of inflammable gases
- floating roof tanks and gas/oil separators
- producing well (closed system)
- exploitation drilling (only normal pressures expected)

Electrical Code

- Explosion-proof equipment to be used in dangerous areas.
- Non-sparking equipment to be used in remotely dangerous areas.
- Normal industrial electric equipment may be used without special precautions against explosion in safe areas.

Naked lights, open fires and all other sources of fire are NOT allowed

- within 100 feet from all locations and installations which are considered hazardous
- within 150 feet from places where explosives are being handled, used or stored within 25 feet of any acetylene generator or generator house.

Oil Well Sites

- Should be located at least 100 feet from railways, public works etc.
- Should be cleared from materials such as trees and undergrowth, which can create a fire hazard, for a radius of at least 50 feet from the well head.

Boilers

Should be located at least 150 feet from the well head.

Fracturing Equipment

Should be located at least 100 feet from the well head.

Flares and Flare Pits

- Flare pits and extremities of flare lines should be located at least 300 feet from railways, public works, processing units, tanks or other important plant equipment items or their boundaries. They should be at least 100 feet from a well, gas/oil separator or other unprotected source of ignitable vapours.

FIRE PREVENTION

SAFE DISTANCES FOR FIRE PREVENTION (2)

Storage of Corrosive and/or Toxic Chemicals and Packaged Oil Products

- To be located at least 50 feet from all buildings or the plot boundary.

Storage of Explosives

- Field stores to be located at least 500 feet away from any place, where drilling or production operations are being carried out.
- Stores containing blasting caps are to be located at least 100 feet from stores containing explosives.
- The land surrounding explosive stores should be kept clear from trees for at least 100 feet, and from brush, dried grass, leaves or other fire hazards for at least 25 feet.

Aircraft Fuelling

- No smoking or naked lights are permitted within 50 feet of the aircraft.

FIRE PREVENTION

RECOMMENDATIONS FOR FIRE EXTINGUISHERS

Type	Type of fire for which suitable	Type of fire for which NOT suitable	Frequency of testing	Method of testing	Recharging	Precautions	Capacity of standard units
Water	Wood, paper, textiles, rubbish	Electrical fires, or oil, paint, grease				Protect from frost by adding 10% calcium chloride	2-3 gallon fire buckets
Sand	Small surface fires of oil, grease or paint. Also for covering and absorbing spills	Electrical fires, or fires in tanks or containers				Keep dry. Protect from frost by adding 1-2% calcium chloride	Locally made sand bins
Soda/Acid	Wood, paper, textiles, rubbish, etc., in offices, stores and houses	Electrical fires, or oil, paint, grease	Every 6 months	Visually	According to directions on the extinguisher	Protect against temperatures below 40°F (4°C)	2 gallons
Carbon Dioxide (CO ₂)	Electrical: also in vehicles, small craft and at service stations	Deep seated fires of wood, paper, textiles, etc., unless used in conjunction with water	Every 6 months	By weight	CO ₂ cartridges or cylinders may be recharged at commercial CO ₂ bottling plants		2½, 5 and 10 lbs
Dry Chemical	Electrical: L.P.G. solvents, also in vehicles small craft and at service stations	Deep seated fires of wood, paper, textiles, etc., unless used in conjunction with water	Every 6 months	By weight. Check weight of CO ₂ cartridge or cylinder separately	CO ₂ cartridges or cylinders may be recharged at commercial CO ₂ bottling plants	Should not be used prior to foam on an oil fire	25 lbs and 150 lbs
Chemical Foam	Oil, grease, paint	Electrical fires, solvent fires	Every 6 months	(1) Follow makers' instructions or (2) mix 10 ml from inner container with 40 ml from outer container	If volume of foam produced in test is less than 350 ml the extinguisher should be recharged	Protect against temperatures below 40°F (4°C). Check nozzles frequently for blockage. Protect hose from exposure to sun. Do not omit to top-up after testing chemical foam extinguishers	2 gallons and 34 gallons
Mechanical Foam (AIf)	Oil, grease, paint	Electrical fires, solvent fires	Every 6 months	Check weight of gas cylinders	Replace the water/foam compound mixture every two years		30 gallons

Notes :

- (1) Where fire service mains are available, consideration should be given to the supply of a limited quantity of foam making compound, to be used with small foam making branch pipes, as an alternative, or in addition, to the larger sizes of foam extinguisher.
- (2) Recharges: two spare charges should be available for every foam extinguisher installed.

N – TRAINING

Clickable list

Training courses

N-1

TRAINING COURSES

Foundation Learning (charged to the Service Fee arrangement)

EP00	Introducing the E&P Business	28 days
W181	Preparation for the Wellsite	15 days
WEDLP	Well Engineering Distance Learning Package	
	* Leading to Round I & Round II exams	
Round 1 Examination (optional)		1 day
P130	Basic Well Engineering	24 days
G180	Subsurface Integration	27 days
Round II Examination		3 days

P130 takes place during study of the WEDLP

Supplementary & Advanced Learning

EP01	Developing E&P Business Skills	10 days
EP02	E&P Business Economics	5 days
EP03	Managing the E&P Business	10 days
EP04	Auditing in a Technical Environment	5 days
MHSE	Managing HSE in the Business	5 days
BTSA	Negotiating skills	4 days
IM25	IT for Business staff	5 days
P212	Petrophysics for other disciplines	5 days
P213	Production Geology for other disciplines	5 days
P214	Reservoir Engineering for other disciplines	5 days
P215	Production Technology for other disciplines	5 days
P282	Leading Hazops	5 days
P285	Quantitative Risk Assessment	5 days
P288	Fire & Explosion Hazard Management	5 days
P289	HSE Tools & Techniques	5 days

"Your future is in your hands" . While mentors will supply guidance, it is your responsibility to make learning a key priority.
Find out course details, location and availability at <http://sww.shell.com/openuniversity>.